# PILOT'S OPERATING HANDBOOK

Cessna® 1978

# Skyhawk

CESSNA MODEL 172N

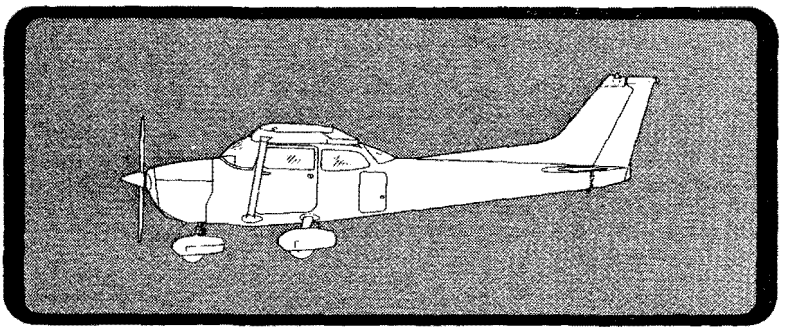


# PERFORMANCE - SPECIFICATIONS

| | | |
|---|---|---|
| SPEED: | | |
| Maximum at Sea Level | | 125 KNOTS |
| Cruise, 75% Power at 8000 Ft | | 122 KNOTS |
| CRUISE: Recommended lean mixture with fuel allowance for engine start, taxi, takeoff, climb and 45 minutes reserve at 45% power. | | |
| 75% Power at 8000 Ft | Range | 485 NM |
| 40 Gallons Usable Fuel | Time | 4.1 HRS |
| 75% Power at 8000 Ft | Range | 630 NM |
| 50 Gallons Usable Fuel | Time | 5.3 HRS |
| Maximum Range at 10,000 Ft | Range | 575 NM |
| 40 Gallons Usable Fuel | Time | 5.7 HRS |
| Maximum Range at 10,000 Ft | Range | 750 NM |
| 50 Gallons Usable Fuel | Time | 7.4 HRS |
| RATE OF CLIMB AT SEA LEVEL | | 770 FPM |
| SERVICE CEILING | | 14,200 FT |
| TAKEOFF PERFORMANCE: | | |
| Ground Roll | | 805 FT |
| Total Distance Over 50-Ft Obstacle | | 1440 FT |
| LANDING PERFORMANCE: | | |
| Ground Roll | | 520 FT |
| Total Distance Over 50-Ft Obstacle | | 1250 FT |
| STALL SPEED (CAS): | | |
| Flaps Up, Power Off | | 50 KNOTS |
| Flaps Down, Power Off | | 44 KNOTS |
| MAXIMUM WEIGHT | | 2300 LBS |
| STANDARD EMPTY WEIGHT: | | |
| Skyhawk | | 1393 LBS |
| Skyhawk II | | 1419 LBS |
| MAXIMUM USEFUL LOAD: | | |
| Skyhawk | | 907 LBS |
| Skyhawk II | | 881 LBS |
| BAGGAGE ALLOWANCE | | 120 LBS |
| WING LOADING: Pounds/Sq Ft | | 13.2 |
| POWER LOADING: Pounds/HP | | 14.4 |
| FUEL CAPACITY: Total | | |
| Standard Tanks | | 43 GAL. |
| Long Range Tanks | | 54 GAL. |
| OIL CAPACITY | | 6 QTS |
| ENGINE: Avco Lycoming | | O-320-H2AD |
| 160 BHP at 2700 RPM | | |
| PROPELLER: Fixed Pitch, Diameter | | 75 IN. |

# PILOT'S OPERATING HANDBOOK

Cessna®



# SKYHAWK

## 1978 MODEL 172N

Serial No._____

Registration No._____

THIS HANDBOOK INCLUDES THE MATERIAL REQUIRED TO BE FURNISHED TO THE PILOT BY CAR PART 3

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CESSNA AIRCRAFT COMPANY

WICHITA, KANSAS, USA

# CONGRATULATIONS ....

Welcome to the ranks of Cessna owners! Your Cessna has been designed and constructed to give you the most in performance, economy, and comfort. It is our desire that you will find flying it, either for business or pleasure, a pleasant and profitable experience.

This Pilot's Operating Handbook has been prepared as a guide to help you get the most pleasure and utility from your airplane. It contains information about your Cessna's equipment, operating procedures, and performance; and suggestions for its servicing and care. We urge you to read it from cover to cover, and to refer to it frequently.

Our interest in your flying pleasure has not ceased with your purchase of a Cessna. World-wide, the Cessna Dealer Organization backed by the Cessna Customer Services Department stands ready to serve you. The following services are offered by most Cessna Dealers:

- THE CESSNA WARRANTY, which provides coverage for parts and labor, is available at Cessna Dealers worldwide. Specific benefits and provisions of warranty, plus other important benefits for you, are contained in your Customer Care Program book, supplied with your airplane. Warranty service is available to you at authorized Cessna Dealers throughout the world upon presentation of your Customer Care Card which establishes your eligibility under the warranty.

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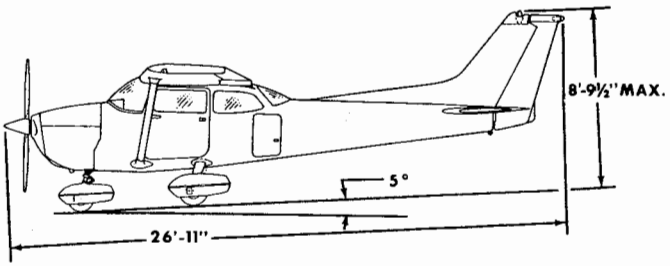
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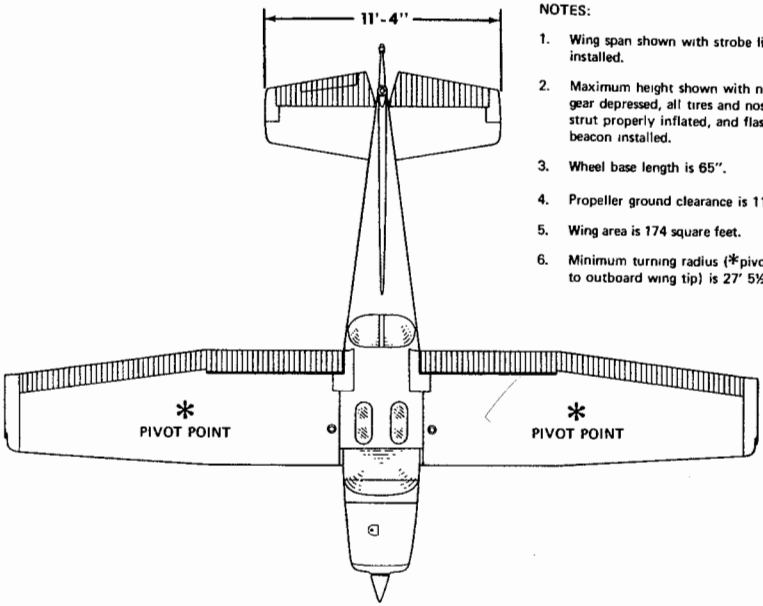
# SECTION 1
# GENERAL

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NOTES:

1. Wing span shown with strobe lights installed.
2. Maximum height shown with nose gear depressed, all tires and nose strut properly inflated, and flashing beacon installed.
3. Wheel base length is 65".
4. Propeller ground clearance is 11 3/4".
5. Wing area is 174 square feet.
6. Minimum turning radius (*pivot point to outboard wing tip) is 27' 5½".

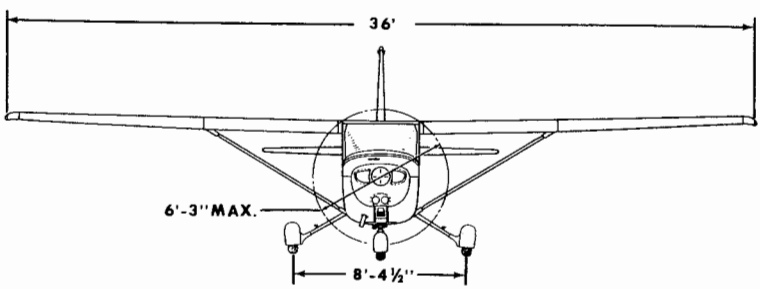



Figure 1-1. Three View

# INTRODUCTION

This handbook contains 9 sections, and includes the material required to be furnished to the pilot by CAR Part 3. It also contains supplemental data supplied by Cessna Aircraft Company.

Section 1 provides basic data and information of general interest. It also contains definitions or explanations of symbols, abbreviations, and terminology commonly used.

# DESCRIPTIVE DATA

## ENGINE

Number of Engines: 1.
Engine Manufacturer: Avco Lycoming.
Engine Model Number: O-320-H2AD.
Engine Type: Normally-aspirated, direct-drive, air-cooled, horizontally-opposed, carburetor equipped, four-cylinder engine with 320 cu. in. displacement.
Horsepower Rating and Engine Speed: 160 rated BHP at 2700 RPM.

## PROPELLER

Propeller Manufacturer: McCauley Accessory Division.
Propeller Model Number: 1C160/DTM7557.
Number of Blades: 2.
Propeller Diameter, Maximum: 75 inches.
Minimum: 74 inches.
Propeller Type: Fixed pitch.

## FUEL

Approved Fuel Grades (and Colors):
100LL Grade Aviation Fuel (Blue).
100 (Formerly 100/130) Grade Aviation Fuel (Green).

Fuel Capacity:

Standard Tanks:

Total Capacity: 43 gallons.
Total Capacity Each Tank: 21.5 gallons.
Total Usable: 40 gallons.

Long Range Tanks:

Total Capacity: 54 gallons.
Total Capacity Each Tank: 27 gallons.
Total Usable: 50 gallons.

NOTE

To ensure maximum fuel capacity when refueling, place the fuel selector valve in either LEFT or RIGHT position to prevent cross-feeding.

## OIL

Oil Grade (Specification):

MIL-L-6082 Aviation Grade Straight Mineral Oil: Use to replenish supply during first 25 hours and at the first 25-hour oil change. Continue to use until a total of 50 hours has accumulated or oil consumption has stabilized.

NOTE

The airplane was delivered from the factory with a corrosion preventive aircraft engine oil. This oil should be drained after the first 25 hours of operation.

MIL-L-22851 Ashless Dispersant Oil: This oil **must be used** after first 50 hours or consumption has stabilized.

Recommended Viscosity for Temperature Range:

MIL-L-6082 Aviation Grade Straight Mineral Oil:

SAE 50 above 16°C (60°F).
SAE 40 between -1°C (30°F) and 32°C (90°F).
SAE 30 between -18°C (0°F) and 21°C (70°F).
SAE 20 below -12°C (10°F).

MIL-L-22851 Ashless Dispersant Oil:

SAE 40 or SAE 50 above 16°C (60°F).
SAE 40 between -1°C (30°F) and 32°C (90°F).
SAE 30 or SAE 40 between -18°C (0°F) and 21°C (70°F).
SAE 30 below -12°C (10°F).

Oil Capacity:

Sump: 6 Quarts.
Total: 7 Quarts (if oil filter installed).

## MAXIMUM CERTIFICATED WEIGHTS

Takeoff, Normal Category: 2300 lbs.
Utility Category: 2000 lbs.
Landing, Normal Category: 2300 lbs.
Utility Category: 2000 lbs.
Weight in Baggage Compartment, Normal Category:
Baggage Area 1 (or passenger on child's seat) - Station 82 to 108: 120 lbs. See note below.
Baggage Area 2 - Station 108 to 142: 50 lbs. See note below.

NOTE

The maximum combined weight capacity for baggage areas 1 and 2 is 120 lbs.

Weight in Baggage Compartment, Utility Category: In this category, the baggage compartment and rear seat must not be occupied.

## STANDARD AIRPLANE WEIGHTS

Standard Empty Weight, Skyhawk: 1393 lbs.
Skyhawk II: 1419 lbs.

Maximum Useful Load:

| | Normal Category | Utility Category |
|---|---|---|
| Skyhawk: | 907 lbs. | 607 lbs. |
| Skyhawk II: | 881 lbs. | 581 lbs. |

## CABIN AND ENTRY DIMENSIONS

Detailed dimensions of the cabin interior and entry door openings are illustrated in Section 6.

## BAGGAGE SPACE AND ENTRY DIMENSIONS

Dimensions of the baggage area and baggage door opening are illustrated in detail in Section 6.

## SPECIFIC LOADINGS

Wing Loading: 13.2 lbs./sq. ft.
Power Loading: 14.4 lbs./hp.

# SYMBOLS, ABBREVIATIONS AND TERMINOLOGY

## GENERAL AIRSPEED TERMINOLOGY AND SYMBOLS

KCAS — **Knots Calibrated Airspeed** is indicated airspeed corrected for position and instrument error and expressed in knots. Knots calibrated airspeed is equal to KTAS in standard atmosphere at sea level.

KIAS — **Knots Indicated Airspeed** is the speed shown on the airspeed indicator and expressed in knots.

KTAS — **Knots True Airspeed** is the airspeed expressed in knots relative to undisturbed air which is KCAS corrected for altitude and temperature.

$V_A$ — **Manuevering Speed** is the maximum speed at which you may use abrupt control travel.

$V_{FE}$ — **Maximum Flap Extended Speed** is the highest speed permissible with wing flaps in a prescribed extended position.

$V_{NO}$ — **Maximum Structural Cruising Speed** is the speed that should not be exceeded except in smooth air, then only with caution.

$V_{NE}$ — **Never Exceed Speed** is the speed limit that may not be exceeded at any time.

$V_S$ — **Stalling Speed or the minimum steady flight speed** at which the airplane is controllable.

$V_{S_o}$ — **Stalling Speed or the minimum steady flight speed** at which the airplane is controllable in the landing configuration at the most forward center of gravity.

$V_X$ — **Best Angle-of-Climb Speed** is the speed which results in the greatest gain of altitude in a given horizontal distance.

$V_Y$ — **Best Rate-of-Climb Speed** is the speed which results in the greatest gain in altitude in a given time.

## METEOROLOGICAL TERMINOLOGY

OAT — **Outside Air Temperature** is the free air static temperature.

It is expressed in either degrees Celsius (formerly Centigrade) or degrees Fahrenheit.

Standard Temperature — **Standard Temperature** is 15°C at sea level pressure altitude and decreases by 2°C for each 1000 feet of altitude.

Pressure Altitude — **Pressure Altitude** is the altitude read from an altimeter when the altimeter's barometric scale has been set to 29.92 inches of mercury (1013 mb).

## ENGINE POWER TERMINOLOGY

BHP — **Brake Horsepower** is the power developed by the engine.

RPM — **Revolutions Per Minute** is engine speed.

Static RPM — **Static RPM** is engine speed attained during a full-throttle engine runup when the airplane is on the ground and stationary.

## AIRPLANE PERFORMANCE AND FLIGHT PLANNING TERMINOLOGY

Demonstrated Crosswind Velocity — **Demonstrated Crosswind Velocity** is the velocity of the crosswind component for which adequate control of the airplane during takeoff and landing was actually demonstrated during certification tests. The value shown is not considered to be limiting.

Usable Fuel — **Usable Fuel** is the fuel available for flight planning.

Unusable Fuel — **Unusable Fuel** is the quantity of fuel that can not be safely used in flight.

GPH — **Gallons Per Hour** is the amount of fuel (in gallons) consumed per hour.

NMPG — **Nautical Miles Per Gallon** is the distance (in nautical miles) which can be expected per gallon of fuel consumed at a specific engine power setting and/or flight configuration.

g — g is acceleration due to gravity.

## WEIGHT AND BALANCE TERMINOLOGY

| | |
|---|---|
| Reference Datum | **Reference Datum** is an imaginary vertical plane from which all horizontal distances are measured for balance purposes. |
| Station | **Station** is a location along the airplane fuselage given in terms of the distance from the reference datum. |
| Arm | **Arm** is the horizontal distance from the reference datum to the center of gravity (C.G.) of an item. |
| Moment | **Moment** is the product of the weight of an item multiplied by its arm. (Moment divided by the constant 1000 is used in this handbook to simplify balance calculations by reducing the number of digits.) |
| Center of Gravity (C.G.) | **Center of Gravity** is the point at which an airplane, or equipment, would balance if suspended. Its distance from the reference datum is found by dividing the total moment by the total weight of the airplane. |
| C.G. Arm | **Center of Gravity Arm** is the arm obtained by adding the airplane's individual moments and dividing the sum by the total weight. |
| C.G. Limits | **Center of Gravity Limits** are the extreme center of gravity locations within which the airplane must be operated at a given weight. |
| Standard Empty Weight | **Standard Empty Weight** is the weight of a standard airplane, including unusable fuel, full operating fluids and full engine oil. |
| Basic Empty Weight | **Basic Empty Weight** is the standard empty weight plus the weight of optional equipment. |
| Useful Load | **Useful Load** is the difference between takeoff weight and the basic empty weight. |
| Gross (Loaded) Weight | **Gross (Loaded) Weight** is the loaded weight of the airplane. |
| Maximum Takeoff Weight | **Maximum Takeoff Weight** is the maximum weight approved for the start of the takeoff run. |

Maximum Landing Weight

**Maximum Landing Weight** is the maximum weight approved for the landing touchdown.

Tare

**Tare** is the weight of chocks, blocks, stands, etc. used when weighing an airplane, and is included in the scale readings. Tare is deducted from the scale reading to obtain the actual (net) airplane weight.

# SECTION 2
# LIMITATIONS

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# INTRODUCTION

Section 2 includes operating limitations, instrument markings, and basic placards necessary for the safe operation of the airplane, its engine, standard systems and standard equipment. The limitations included in this section have been approved by the Federal Aviation Administration. When applicable, limitations associated with optional systems or equipment are included in Section 9.

NOTE

The airspeeds listed in the Airspeed Limitations chart (figure 2-1) and the Airspeed Indicator Markings chart (figure 2-2) are based on Airspeed Calibration data shown in Section 5 with the normal static source. If the alternate static source is being used, ample margins should be observed to allow for the airspeed calibration variations between the normal and alternate static sources as shown in Section 5.

Your Cessna is certificated under FAA Type Certificate No. 3A12 as Cessna Model No. 172N.

# AIRSPEED LIMITATIONS

Airspeed limitations and their operational significance are shown in figure 2-1. Maneuvering speeds shown apply to normal category operations. The utility category maneuvering speed is shown on the operational limitations placard.

| | SPEED | KCAS | KIAS | REMARKS |
|---|---|---|---|---|
| $V_{NE}$ | Never Exceed Speed | 158 | 160 | Do not exceed this speed in any operation. |
| $V_{NO}$ | Maximum Structural Cruising Speed | 126 | 128 | Do not exceed this speed except in smooth air, and then only with caution. |
| $V_A$ | Maneuvering Speed:<br>2300 Pounds<br>1950 Pounds<br>1600 Pounds | <br>96<br>88<br>80 | <br>97<br>89<br>80 | Do not make full or abrupt control movements above this speed. |
| $V_{FE}$ | Maximum Flap Extended Speed | 86 | 85 | Do not exceed this speed with flaps down. |
| | Maximum Window Open Speed | 158 | 160 | Do not exceed this speed with windows open. |

Figure 2-1. Airspeed Limitations

# AIRSPEED INDICATOR MARKINGS

Airspeed indicator markings and their color code significance are shown in figure 2-2.

| MARKING | KIAS VALUE OR RANGE | SIGNIFICANCE |
|---|---|---|
| White Arc | 41 - 85 | Full Flap Operating Range. Lower limit is maximum weight $V_{S_0}$ in landing configuration. Upper limit is maximum speed permissible with flaps extended. |
| Green Arc | 47 - 128 | Normal Operating Range. Lower limit is maximum weight $V_S$ at most forward C.G. with flaps retracted. Upper limit is maximum structural cruising speed. |
| Yellow Arc | 128 - 160 | Operations must be conducted with caution and only in smooth air. |
| Red Line | 160 | Maximum speed for all operations. |

Figure 2-2. Airspeed Indicator Markings

# POWER PLANT LIMITATIONS

Engine Manufacturer: Avco Lycoming.
Engine Model Number: O-320-H2AD.
Engine Operating Limits for Takeoff and Continuous Operations:
Maximum Power: 160 BHP.
Maximum Engine Speed: 2700 RPM.

NOTE

The static RPM range at full throttle (carburetor heat off and full rich mixture) is 2280 to 2400 RPM.

Maximum Oil Temperature: 118°C (245°F).
Oil Pressure, Minimum: 25 psi.
Maximum: 100 psi.
Propeller Manufacturer: McCauley Accessory Division.
Propeller Model Number: 1C160/DTM7557.
Propeller Diameter, Maximum: 75 inches.
Minimum: 74 inches.

# POWER PLANT INSTRUMENT MARKINGS

Power plant instrument markings and their color code significance are shown in figure 2-3.

| INSTRUMENT | RED LINE MINIMUM LIMIT | GREEN ARC NORMAL OPERATING | YELLOW ARC CAUTION RANGE | RED LINE MAXIMUM LIMIT |
|---|---|---|---|---|
| Tachometer | - - - | 2200 - 2700 RPM | - - - | 2700 RPM |
| Oil Temperature | - - - | 100°-245°F | - - - | 245°F |
| Oil Pressure | 25 psi | 60-90 psi | - - - | 100 psi |
| Carburetor Air Temperature | - - - | - - - | -15° to 5°C | - - - |

Figure 2-3. Power Plant Instrument Markings

# WEIGHT LIMITS

## NORMAL CATEGORY

Maximum Takeoff Weight: 2300 lbs.
Maximum Landing Weight: 2300 lbs.
Maximum Weight in Baggage Compartment:
Baggage Area 1 (or passenger on child's seat) - Station 82 to 108: 120 lbs. See note below.
Baggage Area 2 - Station 108 to 142: 50 lbs. See note below.

NOTE

The maximum combined weight capacity for baggage areas 1 and 2 is 120 lbs.

### UTILITY CATEGORY

Maximum Takeoff Weight: 2000 lbs.
Maximum Landing Weight: 2000 lbs.
Maximum Weight in Baggage Compartment: In the utility category, the baggage compartment and rear seat must be not occupied.

## CENTER OF GRAVITY LIMITS

### NORMAL CATEGORY

Center of Gravity Range:
Forward: 35.0 inches aft of datum at 1950 lbs. or less, with straight line variation to 38.5 inches aft of datum at 2300 lbs.
Aft: 47.3 inches aft of datum at all weights.
Reference Datum: Lower portion of front face of firewall.

### UTILITY CATEGORY

Center of Gravity Range:
Forward: 35.0 inches aft of datum at 1950 lbs. or less, with straight line variation to 35.5 inches aft of datum at 2000 lbs.
Aft: 40.5 inches aft of datum at all weights.
Reference Datum: Lower portion of front face of firewall.

## MANEUVER LIMITS

### NORMAL CATEGORY

This airplane is certificated in both the normal and utility category. The normal category is applicable to aircraft intended for non-aerobatic operations. These include any maneuvers incidental to normal flying, stalls (except whip stalls), lazy eights, chandelles, and turns in which the angle of bank is not more than 60°. Aerobatic maneuvers, including spins, are not approved.

### UTILITY CATEGORY

This airplane is not designed for purely aerobatic flight. However, in the acquisition of various certificates such as commercial pilot and flight instructor, certain maneuvers are required by the FAA. All of these maneuvers are permitted in this airplane when operated in the utility category.

In the utility category, the baggage compartment and rear seat must not be occupied. No aerobatic maneuvers are approved except those listed below:

| MANEUVER | RECOMMENDED ENTRY SPEED* |
|---|---|
| Chandelles | 105 knots |
| Lazy Eights | 105 knots |
| Steep Turns | 95 knots |
| Spins | Slow Deceleration |
| Stalls (Except Whip Stalls) | Slow Deceleration |

*Abrupt use of the controls is prohibited above 97 knots.

Aerobatics that may impose high loads should not be attempted. The important thing to bear in mind in flight maneuvers is that the airplane is clean in aerodynamic design and will build up speed quickly with the nose down. Proper speed control is an essential requirement for execution of any maneuver, and care should always be exercised to avoid excessive speed which in turn can impose excessive loads. In the execution of all maneuvers, avoid abrupt use of controls. Intentional spins with flaps extended are prohibited.

# FLIGHT LOAD FACTOR LIMITS

## NORMAL CATEGORY

Flight Load Factors (Gross Weight - 2300 lbs.):

| | |
|---|---|
| *Flaps Up | +3.8g, -1.52g |
| *Flaps Down | +3.0g |

*The design load factors are 150% of the above, and in all cases, the structure meets or exceeds design loads.

## UTILITY CATEGORY

Flight Load Factors (Gross Weight - 2000 lbs.):

| | |
|---|---|
| *Flaps Up | +4.4g, -1.76g |
| *Flaps Down | +3.0g |

*The design load factors are 150% of the above, and in all cases, the structure meets or exceeds design loads.

## KINDS OF OPERATION LIMITS

The airplane is equipped for day VFR and may be equipped for night VFR and/or IFR operations. FAR Part 91 establishes the minimum required instrumentation and equipment for these operations. The reference to types of flight operations on the operating limitations placard reflects equipment installed at the time of Airworthiness Certificate issuance.

Flight into known icing conditions is prohibited.

## FUEL LIMITATIONS

2 Standard Tanks: 21.5 U.S. gallons each.
Total Fuel: 43 U.S. gallons.
Usable Fuel (all flight conditions): 40 U.S. gallons.
Unusable Fuel: 3 U.S. gallons.
2 Long Range Tanks: 27 U.S. gallons each.
Total Fuel: 54 U.S. gallons.
Usable Fuel (all flight conditions): 50 U.S. gallons.
Unusable Fuel: 4 U.S. gallons.

NOTE

To ensure maximum fuel capacity when refueling, place the fuel selector valve in either LEFT or RIGHT position to prevent cross-feeding.

NOTE

Takeoff and land with the fuel selector valve handle in the BOTH position.

Approved Fuel Grades (and Colors):
100LL Grade Aviation Fuel (Blue).
100 (Formerly 100/130) Grade Aviation Fuel (Green).

# PLACARDS

The following information is displayed in the form of composite or individual placards.

1. In full view of the pilot: (The "DAY-NIGHT-VFR-IFR" entry, shown on the example below, will vary as the airplane is equipped.)

This airplane must be operated in compliance with the operating limitations as stated in the form of placards, markings, and manuals.

MAXIMUMS

| | Normal Category | Utility Category |
|---|---|---|
| MANEUVERING SPEED (IAS) | 97 knots | 97 knots |
| GROSS WEIGHT | 2300 lbs. | 2000 lbs. |
| FLIGHT LOAD FACTOR | | |
| Flaps Up | +3.8, -1.52 | +4.4, -1.76 |
| Flaps Down | +3.0 | +3.0 |

Normal Category - No Acrobatic maneuvers including spins approved.
Utility Category - Baggage compartment and rear seat must not be occupied.

NO ACROBATIC MANEUVERS APPROVED EXCEPT THOSE LISTED BELOW

| Maneuver | Recm. Entry Speed | Maneuver | Recm. Entry Speed |
|---|---|---|---|
| Chandelles | 105 knots | Spins | Slow Deceleration |
| Lazy Eights | 105 knots | Stalls (except whip stalls) | Slow Deceleration |
| Steep Turns | 95 knots | | |

Altitude loss in stall recovery -- 180 feet.
Abrupt use of the controls prohibited above 97 knots.
Spin Recovery: opposite rudder - forward elevator - neutralize controls. Intentional spins with flaps extended are prohibited. Flight into known icing conditions prohibited. This airplane is certified for the following flight operations as of date of original airworthiness certificate:

DAY - NIGHT - VFR - IFR

2. Forward of fuel selector valve:

| |
|---|
| BOTH TANKS ON FOR<br>TAKEOFF & LANDING |

3. On the fuel selector valve (standard tanks):

| |
|---|
| BOTH - 40 GAL. ALL FLIGHT ATTITUDES<br>LEFT - 20 GAL. LEVEL FLIGHT ONLY<br>RIGHT - 20 GAL. LEVEL FLIGHT ONLY<br>OFF |

On the fuel selector valve (long range tanks):

| |
|---|
| BOTH - 50 GAL. ALL FLIGHT ATTITUDES<br>LEFT - 25 GAL. LEVEL FLIGHT ONLY<br>RIGHT - 25 GAL. LEVEL FLIGHT ONLY<br>OFF |

4. Near fuel tank filler cap (standard tanks):

| |
|---|
| FUEL<br>100LL/100 MIN. GRADE AVIATION GASOLINE<br>CAP. 21.5 U.S. GAL. |

Near fuel tank filler cap (long range tanks):

| |
|---|
| FUEL<br>100LL/100 MIN. GRADE AVIATION GASOLINE<br>CAP. 27 U.S. GAL. |

5. Near flap indicator:

AVOID SLIPS WITH FLAPS EXTENDED

6. In baggage compartment:

120 POUNDS MAXIMUM
BAGGAGE AND/OR AUXILIARY PASSENGER
FORWARD OF BAGGAGE DOOR LATCH

50 POUNDS MAXIMUM
BAGGAGE AFT OF BAGGAGE DOOR LATCH

MAXIMUM 120 POUNDS COMBINED

FOR ADDITIONAL LOADING INSTRUCTIONS
SEE WEIGHT AND BALANCE DATA

# SECTION 3
# EMERGENCY PROCEDURES

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# INTRODUCTION

Section 3 provides checklist and amplified procedures for coping with emergencies that may occur. Emergencies caused by airplane or engine malfunctions are extremely rare if proper preflight inspections and maintenance are practiced. Enroute weather emergencies can be minimized or eliminated by careful flight planning and good judgment when unexpected weather is encountered. However, should an emergency arise, the basic guidelines described in this section should be considered and applied as necessary to correct the problem. Emergency procedures associated with ELT and other optional systems can be found in Section 9.

# AIRSPEEDS FOR EMERGENCY OPERATION

| | |
|---|---|
| Engine Failure After Takeoff: | |
| Wing Flaps Up | 65 KIAS |
| Wing Flaps Down | 60 KIAS |
| Maneuvering Speed: | |
| 2300 Lbs | 97 KIAS |
| 1950 Lbs | 89 KIAS |
| 1600 Lbs | 80 KIAS |
| Maximum Glide: | |
| 2300 Lbs | 65 KIAS |
| Precautionary Landing With Engine Power | 60 KIAS |
| Landing Without Engine Power: | |
| Wing Flaps Up | 65 KIAS |
| Wing Flaps Down | 60 KIAS |

# OPERATIONAL CHECKLISTS

## ENGINE FAILURES

### ENGINE FAILURE DURING TAKEOFF RUN

1. Throttle -- IDLE.
2. Brakes -- APPLY.
3. Wing Flaps -- RETRACT.
4. Mixture -- IDLE CUT-OFF.
5. Ignition Switch -- OFF.
6. Master Switch -- OFF.

### ENGINE FAILURE IMMEDIATELY AFTER TAKEOFF

1. Airspeed -- 65 KIAS (flaps UP).
   60 KIAS (flaps DOWN).
2. Mixture -- IDLE CUT-OFF.
3. Fuel Selector Valve -- OFF.
4. Ignition Switch -- OFF.
5. Wing Flaps -- AS REQUIRED.
6. Master Switch -- OFF.

### ENGINE FAILURE DURING FLIGHT

1. Airspeed -- 65 KIAS.
2. Carburetor Heat -- ON.
3. Fuel Selector Valve -- BOTH.
4. Mixture -- RICH.
5. Ignition Switch -- BOTH (or START if propeller is stopped).
6. Primer -- IN and LOCKED.

## FORCED LANDINGS

### EMERGENCY LANDING WITHOUT ENGINE POWER

1. Airspeed -- 65 KIAS (flaps UP).
   60 KIAS (flaps DOWN).
2. Mixture -- IDLE CUT-OFF.
3. Fuel Selector Valve -- OFF.
4. Ignition Switch -- OFF.
5. Wing Flaps -- AS REQUIRED (40° recommended).
6. Master Switch -- OFF.
7. Doors -- UNLATCH PRIOR TO TOUCHDOWN.
8. Touchdown -- SLIGHTLY TAIL LOW.
9. Brakes -- APPLY HEAVILY.

### PRECAUTIONARY LANDING WITH ENGINE POWER

1. Wing Flaps -- 20°.
2. Airspeed -- 60 KIAS.
3. Selected Field -- FLY OVER, noting terrain and obstructions, then retract flaps upon reaching a safe altitude and airspeed.
4. Avionics Power Switch and Electrical Switches -- OFF.
5. Wing Flaps -- 40° (on final approach).
6. Airspeed -- 60 KIAS.
7. Master Switch -- OFF.
8. Doors -- UNLATCH PRIOR TO TOUCHDOWN.

9. Touchdown -- SLIGHTLY TAIL LOW.
10. Ignition Switch -- OFF.
11. Brakes -- APPLY HEAVILY.

### DITCHING

1. Radio -- TRANSMIT MAYDAY on 121.5 MHz, giving location and intentions.
2. Heavy Objects (in baggage area) -- SECURE OR JETTISON.
3. Approach -- High Winds, Heavy Seas -- INTO THE WIND.
   Light Winds, Heavy Swells -- PARALLEL TO SWELLS.
4. Wing Flaps -- 20° - 40°.
5. Power -- ESTABLISH 300 FT/MIN DESCENT AT 55 KIAS.

NOTE

If no power is available, approach at 65 KIAS with flaps up or at 60 KIAS with 10° flaps.

6. Cabin Doors -- UNLATCH.
7. Touchdown -- LEVEL ATTITUDE AT ESTABLISHED RATE OF DESCENT.
8. Face -- CUSHION at touchdown with folded coat.
9. Airplane -- EVACUATE through cabin doors. If necessary, open window and flood cabin to equalize pressure so doors can be opened.
10. Life Vests and Raft -- INFLATE.

## FIRES

### DURING START ON GROUND

1. Cranking -- CONTINUE, to get a start which would suck the flames and accumulated fuel through the carburetor and into the engine.

If engine starts:

2. Power -- 1700 RPM for a few minutes.
3. Engine -- SHUTDOWN and inspect for damage.

If engine fails to start:

4. Throttle -- FULL OPEN.
5. Mixture -- IDLE CUT-OFF.

6. Cranking -- CONTINUE.
7. Fire Extinguisher -- OBTAIN (have ground attendants obtain if n installed).
8. Engine -- SECURE.
   a. Master Switch -- OFF.
   b. Ignition Switch -- OFF.
   c. Fuel Selector Valve -- OFF.
9. Fire -- EXTINGUISH using fire extinguisher, wool blanket, or dir
10. Fire Damage -- INSPECT, repair damage or replace damage components or wiring before conducting another flight.

## ENGINE FIRE IN FLIGHT

1. Mixture -- IDLE CUT-OFF.
2. Fuel Selector Valve -- OFF.
3. Master Switch -- OFF.
4. Cabin Heat and Air -- OFF (except overhead vents).
5. Airspeed -- 100 KIAS (If fire is not extinguished, increase glid speed to find an airspeed which will provide an incombustibl mixture).
6. Forced Landing -- EXECUTE (as described in Emergency Landin Without Engine Power).

## ELECTRICAL FIRE IN FLIGHT

1. Master Switch -- OFF.
2. Avionics Power Switch -- OFF.
3. All Other Switches (except ignition switch) -- OFF.
4. Vents/Cabin Air/Heat -- CLOSED.
5. Fire Extinguisher -- ACTIVATE (if available).

**WARNING**

After discharging an extinguisher within a closed cabin, ventilate the cabin.

If fire appears out and electrical power is necessary for continuance o flight:

6. Master Switch -- ON.
7. Circuit Breakers -- CHECK for faulty circuit, do not reset.
8. Radio Switches -- OFF.
9. Avionics Power Switch -- ON.
10. Radio/Electrical Switches -- ON one at a time, with delay afte each until short circuit is localized.

11. Vents/Cabin Air/Heat -- OPEN when it is ascertained that fire is completely extinguished.

### CABIN FIRE

1. Master Switch -- OFF.
2. Vents/Cabin Air/Heat -- CLOSED (to avoid drafts).
3. Fire Extinguisher -- ACTIVATE (if available).

**WARNING**

After discharging an extinguisher within a closed cabin, ventilate the cabin.

4. Land the airplane as soon as possible to inspect for damage.

### WING FIRE

1. Navigation Light Switch -- OFF.
2. Pitot Heat Switch (if installed) -- OFF.
3. Strobe Light Switch (if installed) -- OFF.

NOTE

Perform a sideslip to keep the flames away from the fuel tank and cabin, and land as soon as possible using flaps only as required for final approach and touchdown.

## ICING

### INADVERTENT ICING ENCOUNTER

1. Turn pitot heat switch ON (if installed).
2. Turn back or change altitude to obtain an outside air temperature that is less conducive to icing.
3. Pull cabin heat control full out and open defroster outlet to obtain maximum windshield defroster airflow. Adjust cabin air control to get maximum defroster heat and airflow.
4. Open the throttle to increase engine speed and minimize ice build-up on propeller blades.
5. Watch for signs of carburetor air filter ice and apply carburetor

heat as required. An unexplained loss in engine speed could be caused by carburetor ice or air intake filter ice. Lean the mixture for maximum RPM, if carburetor heat is used continuously.

6. Plan a landing at the nearest airport. With an extremely rapid ice build-up, select a suitable "off airport" landing site.
7. With an ice accumulation of 1/4 inch or more on the wing leading edges, be prepared for significantly higher stall speed.
8. Leave wing flaps retracted. With a severe ice build-up on the horizontal tail, the change in wing wake airflow direction caused by wing flap extension could result in a loss of elevator effectiveness.
9. Open left window and, if practical, scrape ice from a portion of the windshield for visibility in the landing approach.
10. Perform a landing approach using a forward slip, if necessary, for improved visibility.
11. Approach at 65 to 75 KIAS depending upon the amount of the accumulation.
12. Perform a landing in level attitude.

### STATIC SOURCE BLOCKAGE
### (Erroneous Instrument Reading Suspected)

1. Alternate Static Source Valve -- PULL ON.
2. Airspeed -- Consult appropriate calibration tables in Section 5.

## LANDING WITH A FLAT MAIN TIRE

1. Approach -- NORMAL.
2. Touchdown -- GOOD TIRE FIRST, hold airplane off flat tire as long as possible.

## ELECTRICAL POWER SUPPLY SYSTEM MALFUNCTIONS

### OVER-VOLTAGE LIGHT ILLUMINATES

1. Avionics Power Switch -- OFF.
2. Master Switch -- OFF (both sides).
3. Master Switch -- ON.
4. Over-Voltage Light -- OFF.
5. Avionics Power Switch -- ON.

If over-voltage light illuminates again:

6. Flight -- TERMINATE as soon as possible.

## AMMETER SHOWS DISCHARGE

1. Alternator -- OFF.
2. Nonessential Radio/Electrical Equipment -- OFF.
3. Flight -- TERMINATE as soon as practical.

# AMPLIFIED PROCEDURES

## ENGINE FAILURE

If an engine failure occurs during the takeoff run, the most important thing to do is stop the airplane on the remaining runway. Those extra items on the checklist will provide added safety after a failure of this type.

Prompt lowering of the nose to maintain airspeed and establish a glide attitude is the first response to an engine failure after takeoff. In most cases, the landing should be planned straight ahead with only small changes in direction to avoid obstructions. Altitude and airspeed are seldom sufficient to execute a 180° gliding turn necessary to return to the runway. The checklist procedures assume that adequate time exists to secure the fuel and ignition systems prior to touchdown.

After an engine failure in flight, the best glide speed as shown in figure 3-1 should be established as quickly as possible. While gliding toward a suitable landing area, an effort should be made to identify the cause of the failure. If time permits, an engine restart should be attempted as shown in the checklist. If the engine cannot be restarted, a forced landing without power must be completed.

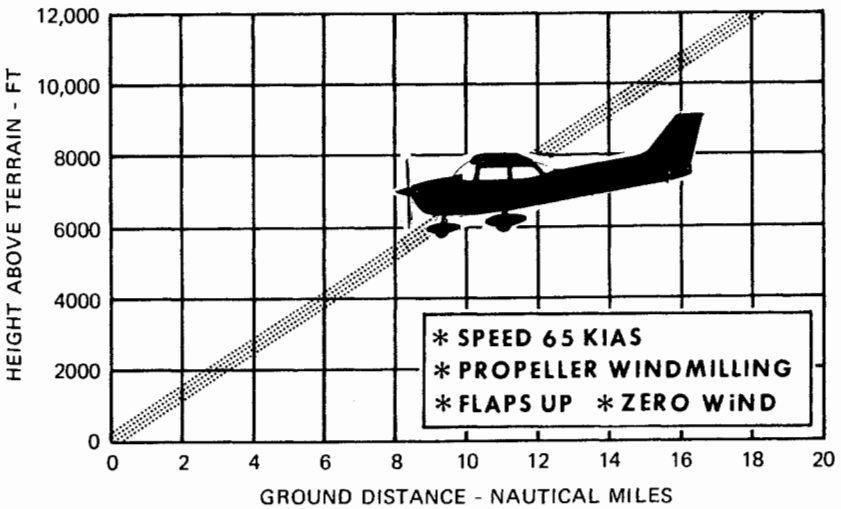



Figure 3-1. Maximum Glide

## FORCED LANDINGS

If all attempts to restart the engine fail and a forced landing is imminent, select a suitable field and prepare for the landing as discussed under the Emergency Landing Without Engine Power checklist.

Before attempting an "off airport" landing with engine power available, one should fly over the landing area at a safe but low altitude to inspect the terrain for obstructions and surface conditions, proceeding as discussed under the Precautionary Landing With Engine Power checklist.

Prepare for ditching by securing or jettisoning heavy objects located in the baggage area and collect folded coats for protection of occupants' face at touchdown. Transmit Mayday message on 121.5 MHz giving location and intentions. Avoid a landing flare because of difficulty in judging height over a water surface.

## LANDING WITHOUT ELEVATOR CONTROL

Trim for horizontal flight (with an airspeed of approximately 60 KIAS and flaps set to 20°) by using throttle and elevator trim controls. Then **do not change the elevator trim control setting**; control the glide angle by adjusting power exclusively.

At flareout, the nose-down moment resulting from power reduction is an adverse factor and the airplane may hit on the nose wheel. Consequently, at flareout, the elevator trim control should be adjusted toward the full nose-up position and the power adjusted so that the airplane will rotate to the horizontal attitude for touchdown. Close the throttle at touchdown.

## FIRES

Although engine fires are extremely rare in flight, the steps of the appropriate checklist should be followed if one is encountered. After completion of this procedure, execute a forced landing. Do not attempt to restart the engine.

The initial indication of an electrical fire is usually the odor of burning insulation. The checklist for this problem should result in elimination of the fire.

# EMERGENCY OPERATION IN CLOUDS
## (Vacuum System Failure)

In the event of a vacuum system failure during flight, the directional indicator and attitude indicator will be disabled, and the pilot will have to rely on the turn coordinator if he inadvertently flies into clouds. The following instructions assume that only the electrically-powered turn coordinator is operative, and that the pilot is not completely proficient in instrument flying.

## EXECUTING A 180° TURN IN CLOUDS

Upon inadvertently entering the clouds, an immediate plan should be made to turn back as follows:

1. Note the compass heading.
2. Note the time of the minute hand and observe the position of the sweep second hand on the clock.
3. When the sweep second hand indicates the nearest half-minute, initiate a standard rate left turn, holding the turn coordinator symbolic airplane wing opposite the lower left index mark for 60 seconds. Then roll back to level flight by leveling the miniature airplane.
4. Check accuracy of the turn by observing the compass heading which should be the reciprocal of the original heading.
5. If necessary, adjust heading primarily with skidding motions rather than rolling motions so that the compass will read more accurately.
6. Maintain altitude and airspeed by cautious application of elevator control. Avoid overcontrolling by keeping the hands off the control wheel as much as possible and steering only with rudder.

## EMERGENCY DESCENT THROUGH CLOUDS

If conditions preclude reestablishment of VFR flight by a 180° turn, a descent through a cloud deck to VFR conditions may be appropriate. If possible, obtain radio clearance for an emergency descent through clouds. To guard against a spiral dive, choose an easterly or westerly heading to minimize compass card swings due to changing bank angles. In addition, keep hands off the control wheel and steer a straight course with rudder control by monitoring the turn coordinator. Occasionally check the compass heading and make minor corrections to hold an approximate course. Before descending into the clouds, set up a stabilized let-down condition as follows:

1. Apply full rich mixture.
2. Use full carburetor heat.
3. Reduce power to set up a 500 to 800 ft/min rate of descent.
4. Adjust the elevator trim and rudder trim (if installed) for a stabilized descent at 70-80 KIAS.
5. Keep hands off the control wheel.
6. Monitor turn coordinator and make corrections by rudder alone.
7. Check trend of compass card movement and make cautious corrections with rudder to stop the turn.
8. Upon breaking out of clouds, resume normal cruising flight.

### RECOVERY FROM A SPIRAL DIVE

If a spiral is encountered, proceed as follows:

1. Close the throttle.
2. Stop the turn by using coordinated aileron and rudder control to align the symbolic airplane in the turn coordinator with the horizon reference line.
3. Cautiously apply elevator back pressure to slowly reduce the airspeed to 80 KIAS.
4. Adjust the elevator trim control to maintain an 80 KIAS glide.
5. Keep hands off the control wheel, using rudder control to hold a straight heading. Adjust rudder trim (if installed) to relieve unbalanced rudder force.
6. Apply carburetor heat.
7. Clear engine occasionally, but avoid using enough power to disturb the trimmed glide.
8. Upon breaking out of clouds, resume normal cruising flight.

## FLIGHT IN ICING CONDITIONS

Flight into icing conditions is prohibited. An inadvertent encounter with these conditions can best be handled using the checklist procedures. The best procedure, of course, is to turn back or change altitude to escape icing conditions.

### STATIC SOURCE BLOCKED

If erroneous readings of the static source instruments (airspeed, altimeter and rate-of-climb) are suspected, the alternate static source valve should be pulled on, thereby supplying static pressure to these instruments from the cabin.

NOTE

In an emergency on airplanes not equipped with an alternate static source, cabin pressure can be supplied to the static pressure instruments by breaking the glass in the face of the rate-of-climb indicator.

With the alternate static source on, adjust indicated airspeed slightly during climb or approach according to the alternate static source airspeed calibration table in Section 5, appropriate to vent/window(s) configuration, causing the airplane to be flown at the normal operating speeds.

Maximum airspeed and altimeter variation from normal is 4 knots and 30 feet over the normal operating range with the window(s) closed. With window(s) open, larger variations occur near stall speed. However, maximum altimeter variation remains within 50 feet of normal.

## SPINS

Should an inadvertent spin occur, the following recovery procedure should be used:

1. RETARD THROTTLE TO IDLE POSITION.
2. PLACE AILERONS IN NEUTRAL POSITION.
3. APPLY AND **HOLD** FULL RUDDER OPPOSITE TO THE DIRECTION OF ROTATION.
4. JUST **AFTER** THE RUDDER REACHES THE STOP, MOVE THE CONTROL WHEEL **BRISKLY** FORWARD FAR ENOUGH TO BREAK THE STALL. Full down elevator may be required at aft center of gravity loadings to assure optimum recoveries.
5. **HOLD** THESE CONTROL INPUTS UNTIL ROTATION STOPS. Premature relaxation of the control inputs may extend the recovery.
6. AS ROTATION STOPS, NEUTRALIZE RUDDER, AND MAKE A SMOOTH RECOVERY FROM THE RESULTING DIVE.

NOTE

If disorientation precludes a visual determination of the direction of rotation, the symbolic airplane in the turn coordinator may be referred to for this information.

For additional information on spins and spin recovery, see the discussion under SPINS in Normal Procedures (Section 4).

# ROUGH ENGINE OPERATION OR LOSS OF POWER

## CARBURETOR ICING

A gradual loss of RPM and eventual engine roughness may result from the formation of carburetor ice. To clear the ice, apply full throttle and pull the carburetor heat knob full out until the engine runs smoothly; then remove carburetor heat and readjust the throttle. If conditions require the continued use of carburetor heat in cruise flight, use the minimum amount of heat necessary to prevent ice from forming and lean the mixture for smoothest engine operation.

## SPARK PLUG FOULING

A slight engine roughness in flight may be caused by one or more spark plugs becoming fouled by carbon or lead deposits. This may be verified by turning the ignition switch momentarily from BOTH to either L or R position. An obvious power loss in single ignition operation is evidence of spark plug or magneto trouble. Assuming that spark plugs are the more likely cause, lean the mixture to the recommended lean setting for cruising flight. If the problem does not clear up in several minutes, determine if a richer mixture setting will produce smoother operation. If not, proceed to the nearest airport for repairs using the BOTH position of the ignition switch unless extreme roughness dictates the use of a single ignition position.

## MAGNETO MALFUNCTION

A sudden engine roughness or misfiring is usually evidence of magneto problems. Switching from BOTH to either L or R ignition switch position will identify which magneto is malfunctioning. Select different power settings and enrichen the mixture to determine if continued operation on BOTH magnetos is practicable. If not, switch to the good magneto and proceed to the nearest airport for repairs.

## LOW OIL PRESSURE

If low oil pressure is accompanied by normal oil temperature, there is a possibility the oil pressure gage or relief valve is malfunctioning. A leak in the line to the gage is not necessarily cause for an immediate precautionary landing because an orifice in this line will prevent a sudden loss of oil from the engine sump. However, a landing at the nearest airport would be advisable to inspect the source of trouble.

If a total loss of oil pressure is accompanied by a rise in oil temperature, there is good reason to suspect an engine failure is imminent. Reduce

engine power immediately and select a suitable forced landing field. Use only the minimum power required to reach the desired touchdown spot.

# ELECTRICAL POWER SUPPLY SYSTEM MALFUNCTIONS

Malfunctions in the electrical power supply system can be detected by periodic monitoring of the ammeter and over-voltage warning light; however, the cause of these malfunctions is usually difficult to determine. A broken alternator drive belt or wiring is most likely the cause of alternator failures, although other factors could cause the problem. A damaged or improperly adjusted voltage regulator can also cause malfunctions. Problems of this nature constitute an electrical emergency and should be dealt with immediately. Electrical power malfunctions usually fall into two categories: excessive rate of charge and insufficient rate of charge. The following paragraphs describe the recommended remedy for each situation.

## EXCESSIVE RATE OF CHARGE

After engine starting and heavy electrical usage at low engine speeds (such as extended taxiing) the battery condition will be low enough to accept above normal charging during the initial part of a flight. However, after thirty minutes of cruising flight, the ammeter should be indicating less than two needle widths of charging current. If the charging rate were to remain above this value on a long flight, the battery would overheat and evaporate the electrolyte at an excessive rate. Electronic components in the electrical system could be adversely affected by higher than normal voltage if a faulty voltage regulator is causing the overcharging. To preclude these possibilites, an over-voltage sensor will automatically shut down the alternator and the over-voltage warning light will illuminate if the charge voltage reaches approximately 31.5 volts. Assuming that the malfunction was only momentary, an attempt should be made to reactivate the alternator system. To do this, turn the avionics power switch off, then turn both sides of the master switch off and then on again. If the problem no longer exists, normal alternator charging will resume and the warning light will go off. The avionics power switch should then be turned on. If the light comes on again, a malfunction is confirmed. In this event, the flight should be terminated and/or the current drain on the battery minimized because the battery can supply the electrical system for only a limited period of time. If the emergency occurs at night, power must be conserved for later use of the landing lights and flaps during landing.

## INSUFFICIENT RATE OF CHARGE

If the ammeter indicates a continuous discharge rate in flight, the

alternator is not supplying power to the system and should be shut down since the alternator field circuit may be placing an unnecessary load on the system. All nonessential equipment should be turned off and the flight terminated as soon as practical.

# SECTION 4
# NORMAL PROCEDURES

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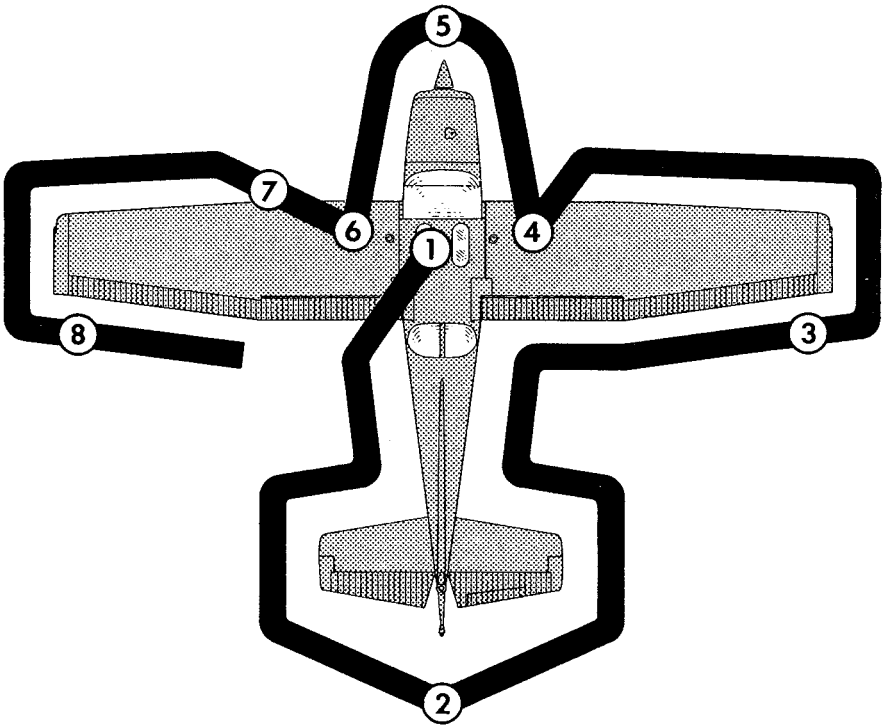
# INTRODUCTION

Section 4 provides checklist and amplified procedures for the conduct of normal operation. Normal procedures associated with optional systems can be found in Section 9.

# SPEEDS FOR NORMAL OPERATION

Unless otherwise noted, the following speeds are based on a maximum weight of 2300 pounds and may be used for any lesser weight. However, to achieve the performance specified in Section 5 for takeoff distance, the speed appropriate to the particular weight must be used.

| | |
|---|---|
| Takeoff, Flaps Up: | |
| Normal Climb Out | 70-80 KIAS |
| Short Field Takeoff, Flaps Up, Speed at 50 Feet | 59 KIAS |
| Enroute Climb, Flaps Up: | |
| Normal, Sea Level | 75-85 KIAS |
| Normal, 10,000 Feet | 70-80 KIAS |
| Best Rate of Climb, Sea Level | 73 KIAS |
| Best Rate of Climb, 10,000 Feet | 68 KIAS |
| Best Angle of Climb, Sea Level | 59 KIAS |
| Best Angle of Climb, 10,000 Feet | 61 KIAS |
| Landing Approach: | |
| Normal Approach, Flaps Up | 60-70 KIAS |
| Normal Approach, Flaps 40° | 55-65 KIAS |
| Short Field Approach, Flaps 40° | 60 KIAS |
| Balked Landing: | |
| Maximum Power, Flaps 20° | 55 KIAS |
| Maximum Recommended Turbulent Air Penetration Speed: | |
| 2300 Lbs | 97 KIAS |
| 1950 Lbs | 89 KIAS |
| 1600 Lbs | 80 KIAS |
| Maximum Demonstrated Crosswind Velocity: | |
| Takeoff or Landing | 15 KNOTS |




NOTE

Visually check airplane for general condition during walk-around inspection. In cold weather, remove even small accumulations of frost, ice or snow from wing, tail and control surfaces. Also, make sure that control surfaces contain no internal accumulations of ice or debris. Prior to flight, check that pitot heater (if installed) is warm to touch within 30 seconds with battery and pitot heat switches on. If a night flight is planned, check operation of all lights, and make sure a flashlight is available.

Figure 4-1. Preflight Inspection

# CHECKLIST PROCEDURES

## PREFLIGHT INSPECTION

### (1) CABIN

1. Control Wheel Lock -- REMOVE.
2. Ignition Switch -- OFF.
3. Avionics Power Switch -- OFF.
4. Master Switch -- ON.
5. Fuel Quantity Indicators -- CHECK QUANTITY.
6. Master Switch -- OFF.
7. Baggage Door -- CHECK, lock with key if child's seat is to be occupied.

### (2) EMPENNAGE

1. Rudder Gust Lock -- REMOVE.
2. Tail Tie-Down -- DISCONNECT.
3. Control Surfaces -- CHECK freedom of movement and security.

### (3) RIGHT WING Trailing Edge

1. Aileron -- CHECK freedom of movement and security.

### (4) RIGHT WING

1. Wing Tie-Down -- DISCONNECT.
2. Main Wheel Tire -- CHECK for proper inflation.
3. Before first flight of the day and after each refueling, use sampler cup and drain small quantity of fuel from fuel tank sump quick-drain valve to check for water, sediment, and proper fuel grade.
4. Fuel Quantity -- CHECK VISUALLY for desired level.
5. Fuel Filler Cap -- SECURE.

### (5) NOSE

1. Engine Oil Level -- CHECK, do not operate with less than four quarts. Fill to six quarts for extended flight.
2. Before first flight of the day and after each refueling, pull out strainer drain knob for about four seconds to clear fuel strainer of possible water and sediment. Check strainer drain closed. If water is observed, the fuel system may contain additional water, and further draining of the system at the strainer, fuel tank sumps, and fuel selector valve drain plug will be necessary.

3. Propeller and Spinner -- CHECK for nicks and security.
4. Landing Light(s) -- CHECK for condition and cleanliness.
5. Carburetor Air Filter -- CHECK for restrictions by dust or other foreign matter.
6. Nose Wheel Strut and Tire -- CHECK for proper inflation.
7. Nose Tie-Down -- DISCONNECT.
8. Static Source Opening (left side of fuselage) -- CHECK for stoppage.

### (6) LEFT WING

1. Main Wheel Tire -- CHECK for proper inflation.
2. Before first flight of the day and after each refueling, use sampler cup and drain small quantity of fuel from fuel tank sump quick-drain valve to check for water, sediment and proper fuel grade.
3. Fuel Quantity -- CHECK VISUALLY for desired level.
4. Fuel Filler Cap -- SECURE.

### (7) LEFT WING Leading Edge

1. Pitot Tube Cover -- REMOVE and check opening for stoppage.
2. Fuel Tank Vent Opening -- CHECK for stoppage.
3. Stall Warning Opening -- CHECK for stoppage. To check the system, place a clean handkerchief over the vent opening and apply suction; a sound from the warning horn will confirm system operation.
4. Wing Tie-Down -- DISCONNECT.

### (8) LEFT WING Trailing Edge

1. Aileron -- CHECK for freedom of movement and security.

## BEFORE STARTING ENGINE

1. Preflight Inspection -- COMPLETE.
2. Seats, Belts, Shoulder Harnesses -- ADJUST and LOCK.
3. Fuel Selector Valve -- BOTH.
4. Avionics Power Switch, Autopilot (if installed), Electrical Equipment -- OFF.

**CAUTION**

The avionics power switch must be OFF during engine start to prevent possible damage to avionics.

5. Brakes -- TEST and SET.
6. Circuit Breakers -- CHECK IN.

# STARTING ENGINE

1. Mixture -- RICH.
2. Carburetor Heat -- COLD.
3. Master Switch -- ON.
4. Prime -- AS REQUIRED (2 to 6 strokes; none if engine is warm).
5. Throttle -- OPEN 1/8 INCH.
6. Propeller Area -- CLEAR.
7. Ignition Switch -- START (release when engine starts).
8. Oil Pressure -- CHECK.

# BEFORE TAKEOFF

1. Parking Brake -- SET.
2. Cabin Doors and Window(s) -- CLOSED and LOCKED.
3. Flight Controls -- FREE and CORRECT.
4. Flight Instruments -- SET.
5. Fuel Selector Valve -- BOTH.
6. Mixture -- RICH (below 3000 feet).
7. Elevator Trim and Rudder Trim (if installed) -- TAKEOFF.
8. Throttle -- 1700 RPM.
   a. Magnetos -- CHECK (RPM drop should not exceed 125 RPM on either magneto or 50 RPM differential between magnetos).
   b. Carburetor Heat -- CHECK (for RPM drop).
   c. Engine Instruments and Ammeter -- CHECK.
   d. Suction Gage -- CHECK.
9. Avionics Power Switch -- ON.
10. Radios -- SET.
11. Autopilot (if installed) -- OFF.
12. Air Conditioner (if installed) -- OFF.
13. Flashing Beacon, Navigation Lights and/or Strobe Lights -- ON as required.
14. Throttle Friction Lock -- ADJUST.
15. Brakes -- RELEASE.

# TAKEOFF

## NORMAL TAKEOFF

1. Wing Flaps -- UP.
2. Carburetor Heat -- COLD.
3. Throttle -- FULL OPEN.
4. Elevator Control -- LIFT NOSE WHEEL (at 55 KIAS).
5. Climb Speed -- 70-80 KIAS.

### SHORT FIELD TAKEOFF

1. Wing Flaps -- UP.
2. Carburetor Heat -- COLD.
3. Brakes -- APPLY.
4. Throttle -- FULL OPEN.
5. Mixture -- RICH (above 3000 feet, LEAN to obtain maximum RPM).
6. Brakes -- RELEASE.
7. Elevator Control -- SLIGHTLY TAIL LOW.
8. Climb Speed -- 59 KIAS (until all obstacles are cleared).

## ENROUTE CLIMB

1. Airspeed -- 70-85 KIAS.

NOTE

If a maximum performance climb is necessary, use speeds shown in the Rate Of Climb chart in Section 5.

2. Throttle -- FULL OPEN.
3. Mixture -- RICH (above 3000 feet, LEAN to obtain maximum RPM).

## CRUISE

1. Power -- 2200-2700 RPM (no more than 75% is recommended).
2. Elevator and Rudder Trim (if installed) -- ADJUST.
3. Mixture -- LEAN.

## DESCENT

1. Mixture -- ADJUST for smooth operation (full rich for idle power).
2. Power -- AS DESIRED.
3. Carburetor Heat -- AS REQUIRED (to prevent carburetor icing).

## BEFORE LANDING

1. Seats, Belts, Harnesses -- SECURE.
2. Fuel Selector Valve -- BOTH.
3. Mixture -- RICH.
4. Carburetor Heat -- ON (apply full heat before closing throttle).
5. Autopilot (if installed) -- OFF.
6. Air Conditioner (if installed) -- OFF.

# LANDING

## NORMAL LANDING

1. Airspeed -- 60-70 KIAS (flaps UP).
2. Wing Flaps -- AS DESIRED (below 85 KIAS).
3. Airspeed -- 55-65 KIAS (flaps DOWN).
4. Touchdown -- MAIN WHEELS FIRST.
5. Landing Roll -- LOWER NOSE WHEEL GENTLY.
6. Braking -- MINIMUM REQUIRED.

## SHORT FIELD LANDING

1. Airspeed -- 60-70 KIAS (flaps UP).
2. Wing Flaps -- FULL DOWN (40°).
3. Airspeed -- 60 KIAS (until flare).
4. Power -- REDUCE to idle after clearing obstacle.
5. Touchdown -- MAIN WHEELS FIRST.
6. Brakes -- APPLY HEAVILY.
7. Wing Flaps -- RETRACT.

## BALKED LANDING

1. Throttle -- FULL OPEN.
2. Carburetor Heat -- COLD.
3. Wing Flaps -- 20° (immediately).
4. Climb Speed -- 55 KIAS.
5. Wing Flaps -- 10° (until obstacles are cleared).
   RETRACT (after reaching a safe altitude and 60 KIAS).

# AFTER LANDING

1. Wing Flaps -- UP.
2. Carburetor Heat -- COLD.

# SECURING AIRPLANE

1. Parking Brake -- SET.
2. Avionics Power Switch, Electrical Equipment, Autopilot (if installed) -- OFF.
3. Mixture -- IDLE CUT-OFF (pulled full out).
4. Ignition Switch -- OFF.
5. Master Switch -- OFF.
6. Control Lock -- INSTALL.

# AMPLIFIED PROCEDURES

## STARTING ENGINE

During engine starting, open the throttle approximately 1/8 inch. In warm temperatures, one or two strokes of the primer should be sufficient. In cold weather, up to six strokes of the primer may be necessary. If the engine is warm, no priming will be required. In extremely cold temperatures, it may be necessary to continue priming while cranking the engine.

Weak intermittent firing followed by puffs of black smoke from the exhaust stack indicates overpriming or flooding. Excess fuel can be cleared from the combustion chambers by the following procedure: set the mixture control full lean and the throttle full open; then crank the engine through several revolutions with the starter. Repeat the starting procedure without any additional priming.

If the engine is underprimed (most likely in cold weather with a cold engine) it will not fire at all, and additional priming will be necessary. As soon as the cylinders begin to fire, open the throttle slightly to keep it running.

After starting, if the oil gage does not begin to show pressure within 30 seconds in the summertime and about twice that long in very cold weather, stop engine and investigate. Lack of oil pressure can cause serious engine damage. After starting, avoid the use of carburetor heat unless icing conditions prevail.

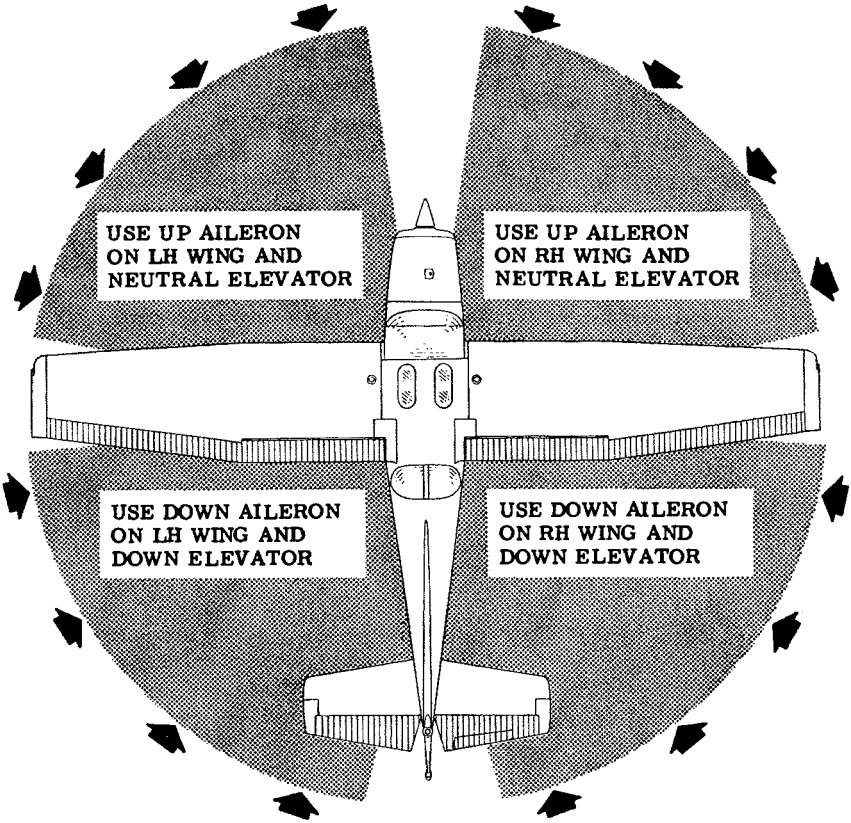
NOTE

Additional details concerning cold weather starting and operation may be found under COLD WEATHER OPERATION paragraphs in this section.

## TAXIING

When taxiing, it is important that speed and use of brakes be held to a minimum and that all controls be utilized (see Taxiing Diagram, figure 4-2) to maintain directional control and balance. Taxiing over loose gravel or cinders should be done at low engine speed to avoid abrasion and stone damage to the propeller tips.

The carburetor heat control knob should be pushed full in during all ground operations unless heat is absolutely necessary. When the knob is pulled out to the heat position, air entering the engine is not filtered.




CODE

WIND DIRECTION

NOTE

Strong quartering tail winds require caution. Avoid sudden bursts of the throttle and sharp braking when the airplane is in this attitude. Use the steerable nose wheel and rudder to maintain direction.

Figure 4-2. Taxiing Diagram

Taxiing over loose gravel or cinders should be done at low engine speed to avoid abrasion and stone damage to the propeller tips.

# BEFORE TAKEOFF

## WARM-UP

If the engine accelerates smoothly, the airplane is ready for takeoff. Since the engine is closely cowled for efficient in-flight engine cooling, precautions should be taken to avoid overheating during prolonged engine operation on the ground. Also, long periods of idling may cause fouled spark plugs.

## MAGNETO CHECK

The magneto check should be made at 1700 RPM as follows. Move ignition switch first to R position and note RPM. Next move switch back to BOTH to clear the other set of plugs. Then move switch to the L position, note RPM and return the switch to the BOTH position. RPM drop should not exceed 125 RPM on either magneto or show greater than 50 RPM differential between magnetos. If there is a doubt concerning operation of the ignition system, RPM checks at higher engine speeds will usually confirm whether a deficiency exists.

An absence of RPM drop may be an indication of faulty grounding of one side of the ignition system or should be cause for suspicion that the magneto timing is set in advance of the setting specified.

## ALTERNATOR CHECK

Prior to flights where verification of proper alternator and voltage regulator operation is essential (such as night or instrument flights), a positive verification can be made by loading the electrical system momentarily (3 to 5 seconds) with the landing light or by operating the wing flaps during the engine runup (1700 RPM). The ammeter will remain within a needle width of its initial reading if the alternator and voltage regulator are operating properly.

# TAKEOFF

## POWER CHECK

It is important to check full-throttle engine operation early in the

takeoff run. Any sign of rough engine operation or sluggish engine acceleration is good cause for discontinuing the takeoff. If this occurs, you are justified in making a thorough full-throttle static runup before another takeoff is attempted. The engine should run smoothly and turn approximately 2280 to 2400 RPM with carburetor heat off and mixture full rich.

NOTE

Carburetor heat should not be used during takeoff unless it is absolutely necessary for obtaining smooth engine acceleration.

Full-throttle runups over loose gravel are especially harmful to propeller tips. When takeoffs must be made over a gravel surface, it is very important that the throttle be advanced slowly. This allows the airplane to start rolling before high RPM is developed, and the gravel will be blown back of the propeller rather than pulled into it. When unavoidable small dents appear in the propeller blades, they should be immediately corrected as described in Section 8 under Propeller Care.

Prior to takeoff from fields above 3000 feet elevation, the mixture should be leaned to give maximum RPM in a full-throttle, static runup.

After full throttle is applied, adjust the throttle friction lock clockwise to prevent the throttle from creeping back from a maximum power position. Similar friction lock adjustments should be made as required in other flight conditions to maintain a fixed throttle setting.

## WING FLAP SETTINGS

Normal and short field takeoffs are performed with flaps up. Flap settings greater than 10° are not approved for takeoff.

Use of 10° flaps is reserved for takeoff from soft or rough fields. Use of 10° flaps allows safe use of approximately 5 KIAS lower takeoff speeds than with flaps up. The lower speeds result in shortening takeoff distances up to approximately 10%. However, this advantage is lost if flaps up speeds are used, or in high altitude takeoffs at maximum weight where climb performance would be marginal with 10° flaps. Therefore, use of 10° flaps is not recommended for takeoff over an obstacle at high altitude in hot weather.

## SHORT FIELD TAKEOFF

If an obstruction dictates the use of a steep climb angle, after liftoff accelerate to and climb out at an obstacle clearance speed of 59 KIAS with flaps retracted. This speed provides the best overall climb speed to clear

obstacles when taking into account the turbulence often found near ground level. The takeoff performance data provided in Section 5 is based on the flaps up configuration.

If 10° of flaps are used on soft or rough fields with obstacles ahead, it is normally preferable to leave them extended rather than retract them in the climb to the obstacle. With 10° flaps, use an obstacle clearance speed of 55 KIAS. As soon as the obstacle is cleared, the flaps may be retracted as the airplane accelerates to the normal flaps-up climb-out speed.

### CROSSWIND TAKEOFF

Takeoffs into strong crosswinds normally are performed with the minimum flap setting necessary for the field length, to minimize the drift angle immediately after takeoff. The airplane is accelerated to a speed slightly higher than normal, then pulled off abruptly to prevent possible settling back to the runway while drifting. When clear of the ground, make a coordinated turn into the wind to correct for drift.

## ENROUTE CLIMB

Normal climbs are performed with flaps up and full throttle and at speeds 5 to 10 knots higher than best rate-of-climb speeds for the best combination of performance, visibility and engine cooling. The mixture should be full rich below 3000 feet and may be leaned above 3000 feet for smoother operation or to obtain maximum RPM. For maximum rate of climb, use the best rate-of-climb speeds shown in the Rate-of-Climb chart in Section 5. If an obstruction dictates the use of a steep climb angle, the best angle-of-climb speed should be used with flaps up and maximum power. Climbs at speeds lower than the best rate-of-climb speed should be of short duration to improve engine cooling.

## CRUISE

Normal cruising is performed between 55% and 75% power. The engine RPM and corresponding fuel consumption for various altitudes can be determined by using your Cessna Power Computer or the data in Section 5.

NOTE

Cruising should be done at 65% to 75% power until a total of 50 hours has accumulated or oil consumption has stabil-

ized. This is to ensure proper seating of the rings and is applicable to new engines, and engines in service following cylinder replacement or top overhaul of one or more cylinders.

The Cruise Performance Table, figure 4-3, illustrates the true airspeed and nautical miles per gallon during cruise for various altitudes and percent powers. This table should be used as a guide, along with the available winds aloft information, to determine the most favorable altitude and power setting for a given trip. The selection of cruise altitude on the basis of the most favorable wind conditions and the use of low power settings are significant factors that should be considered on every trip to reduce fuel consumption.

To achieve the recommended lean mixture fuel consumption figures shown in Section 5, the mixture should be leaned until engine RPM peaks and drops 25-50 RPM. At lower powers it may be necessary to enrichen the mixture slightly to obtain smooth operation.

Should it be necessary to cruise at higher than 75% power, the mixture should not be leaned more than is required to provide peak RPM.

Carburetor ice, as evidenced by an unexplained drop in RPM, can be removed by application of full carburetor heat. Upon regaining the original RPM (with heat off), use the minimum amount of heat (by trial and error) to prevent ice from forming. Since the heated air causes a richer mixture, readjust the mixture setting when carburetor heat is to be used continuously in cruise flight.

| | 75% POWER | | 65% POWER | | 55% POWER | |
|---|---|---|---|---|---|---|
| ALTITUDE | KTAS | NMPG | KTAS | NMPG | KTAS | NMPG |
| Sea Level | 114 | 13.5 | 107 | 14.8 | 100 | 16.1 |
| 4000 Feet | 118 | 14.0 | 111 | 15.3 | 103 | 16.6 |
| 8000 Feet | 122 | 14.5 | 115 | 15.8 | 106 | 17.1 |
| Standard Conditions | | | | | | Zero Wind |

Figure 4-3. Cruise Performance Table

The use of full carburetor heat is recommended during flight in heavy rain to avoid the possibility of engine stoppage due to excessive water ingestion or carburetor ice. The mixture setting should be readjusted for smoothest operation. Power changes should be made cautiously, followed by prompt adjustment of the mixture for smoothest operation.

## STALLS

The stall characteristics are conventional and aural warning is provided by a stall warning horn which sounds between 5 and 10 knots above the stall in all configurations.

Power-off stall speeds at maximum weight for both forward and aft C.G. positions are presented in Section 5.

## SPINS

Intentional spins are approved in this airplane within certain restricted loadings. Spins with baggage loadings or occupied rear seat(s) are not approved.

However, before attempting to perform spins several items should be carefully considered to assure a safe flight. No spins should be attempted without first having received dual instruction both in spin entries and spin recoveries from a qualified instructor who is familiar with the spin characteristics of the Cessna 172N.

The cabin should be clean and all loose equipment (including the microphone and rear seat belts) should be stowed or secured. For a solo flight in which spins will be conducted, the copilot's seat belt and shoulder harness should also be secured. The seat belts and shoulder harnesses should be adjusted to provide proper restraint during all anticipated flight conditions. However, care should be taken to ensure that the pilot can easily reach the flight controls and produce maximum control travels.

It is recommended that, where feasible, entries be accomplished at high enough altitude that recoveries are completed 4000 feet or more above ground level. At least 1000 feet of altitude loss should be allowed for a 1-turn spin and recovery, while a 6-turn spin and recovery may require somewhat more than twice that amount. For example, the recommended entry altitude for a 6-turn spin would be 6000 feet above ground level. In any case, entries should be planned so that recoveries are completed well above the minimum 1500 feet above ground level required by FAR 91.71.

Another reason for using high altitudes for practicing spins is that a greater field of view is provided which will assist in maintaining pilot orientation.

The normal entry is made from a power-off stall. As the stall is approached, the elevator control should be smoothly pulled to the full aft position. Just prior to reaching the stall "break", rudder control in the desired direction of the spin rotation should be applied so that full rudder deflection is reached almost simultaneously with reaching full aft elevator. A slightly greater rate of deceleration than for normal stall entries, application of ailerons in the direction of the desired spin, and the use of power at the entry will assure more consistent and positive entries to the spin. As the airplane begins to spin, reduce the power to idle and return the ailerons to neutral. Both elevator and rudder controls should be held full with the spin until the spin recovery is initiated. An inadvertent relaxation of either of these controls could result in the development of a nose-down spiral.

For the purpose of training in spins and spin recoveries, a 1 or 2 turn spin is adequate and should be used. Up to 2 turns, the spin will progress to a fairly rapid rate of rotation and a steep attitude. Application of recovery controls will produce prompt recoveries (within 1/4 turn). During extended spins of two to three turns or more, the spin will tend to change into a spiral, particularly to the right. This will be accompanied by an increase in airspeed and gravity loads on the airplane. If this occurs, recovery should be accomplished quickly by leveling the wings and recovering from the resulting dive.

Regardless of how many turns the spin is held or how it is entered, the following recovery technique should be used:

1. VERIFY THAT THROTTLE IS IN IDLE POSITION AND AILERONS ARE NEUTRAL.
2. APPLY AND **HOLD** FULL RUDDER OPPOSITE TO THE DIRECTION OF ROTATION.
3. JUST **AFTER** THE RUDDER REACHES THE STOP, MOVE THE CONTROL WHEEL **BRISKLY** FORWARD FAR ENOUGH TO BREAK THE STALL.
4. **HOLD** THESE CONTROL INPUTS UNTIL ROTATION STOPS.
5. AS ROTATION STOPS, NEUTRALIZE RUDDER, AND MAKE A SMOOTH RECOVERY FROM THE RESULTING DIVE.

NOTE

If disorientation precludes a visual determination of the direction of rotation, the symbolic airplane in the turn coordinator may be referred to for this information.

Variations in basic airplane rigging or in weight and balance due to installed equipment or right seat occupancy can cause differences in behavior, particularly in extended spins. These differences are normal and will result in variations in the spin characteristics and in the spiraling tendencies for spins of more than 2 turns. However, the recovery technique should always be used and will result in the most expeditious recovery from any spin.

Intentional spins with flaps extended are prohibited, since the high speeds which may occur during recovery are potentially damaging to the flap/wing structure.

# LANDING

## NORMAL LANDING

Normal landing approaches can be made with power-on or power-off with any flap setting desired. Surface winds and air turbulence are usually the primary factors in determining the most comfortable approach speeds. Steep slips should be avoided with flap settings greater than 20° due to a slight tendency for the elevator to oscillate under certain combinations of airspeed, sideslip angle, and center of gravity loadings.

NOTE

Carburetor heat should be applied prior to any significant reduction or closing of the throttle.

Actual touchdown should be made with power-off and on the main wheels first to reduce the landing speed and subsequent need for braking the landing roll. The nose wheel is lowered to the runway gently after the speed has diminished to avoid unnecessary nose gear loads. This procedure is especially important in rough or soft field landings.

## SHORT FIELD LANDING

For a short field landing in smooth air conditions, make an approach at the minimum recommended airspeed with full flaps using enough power to control the glide path. (Slightly higher approach speeds should be used under turbulent air conditions.) After all approach obstacles are cleared, progressively reduce power and maintain the approach speed by lowering the nose of the airplane. Touchdown should be made with power off and on the main wheels first. Immediately after touchdown, lower the nose wheel and apply heavy braking as required. For maximum brake effectiveness, retract the flaps, hold the control wheel full back, and apply maximum brake pressure without sliding the tires.

### CROSSWIND LANDING

When landing in a strong crosswind, use the minimum flap setting required for the field length. If flap settings greater than 20° are used in sideslips with full rudder deflection, some elevator oscillation may be felt at normal approach speeds. However, this does not affect control of the airplane. Although the crab or combination method of drift correction may be used, the wing-low method gives the best control. After touchdown, hold a straight course with the steerable nose wheel and occasional braking if necessary.

The maximum allowable crosswind velocity is dependent upon pilot capability as well as aircraft limitations. With average pilot technique, direct crosswinds of 15 knots can be handled with safety.

### BALKED LANDING

In a balked landing (go-around) climb, reduce the flap setting to 20° immediately after full power is applied. If obstacles must be cleared during the go-around climb, reduce the wing flap setting to 10° and maintain a safe airspeed until the obstacles are cleared. Above 3000 feet, lean the mixture to obtain maximum RPM. After clearing any obstacles, the flaps may be retracted as the airplane accelerates to the normal flaps-up climb speed.

## COLD WEATHER OPERATION

### STARTING

Prior to starting on cold mornings, it is advisable to pull the propeller through several times by hand to "break loose" or "limber" the oil, thus conserving battery energy.

NOTE

When pulling the propeller through by hand, treat it as if the ignition switch is turned on. A loose or broken ground wire on either magneto could cause the engine to fire.

In extremely cold (-18°C and lower) weather, the use of an external preheater and an external power source are recommended whenever possible to obtain positive starting and to reduce wear and abuse to the engine and electrical system. Pre-heat will thaw the oil trapped in the oil cooler, which probably will be congealed prior to starting in extremely cold temperatures. When using an external power source, the position of the master switch is important. Refer to Section 7 under Ground Service Plug Receptacle for operating details.

Cold weather starting procedures are as follows:

**With Preheat:**

1. With ignition switch OFF and throttle closed, prime the engine four to eight strokes as the propeller is being turned over by hand.

NOTE

Use heavy strokes of primer for best atomization of fuel. After priming, push primer all the way in and turn to locked position to avoid possibility of engine drawing fuel through the primer.

2. Propeller Area -- CLEAR.
3. Avionics Power Switch -- OFF.
4. Master Switch -- ON.
5. Mixture -- FULL RICH.
6. Throttle -- OPEN 1/8 INCH.
7. Ignition Switch -- START.
8. Release ignition switch to BOTH when engine starts.
9. Oil Pressure -- CHECK.

**Without Preheat:**

1. Prime the engine six to ten strokes while the propeller is being turned by hand with the throttle closed. Leave the primer charged and ready for a stroke.
2. Propeller Area -- CLEAR.
3. Avionics Power Switch -- OFF.
4. Master Switch -- ON.
5. Mixture -- FULL RICH.
6. Ignition Switch -- START.
7. Pump throttle rapidly to full open twice. Return to 1/8 inch open position.
8. Release ignition switch to BOTH when engine starts.
9. Continue to prime engine until it is running smoothly, or alternately, pump throttle rapidly over first 1/4 of total travel.
10. Oil Pressure -- CHECK.
11. Pull carburetor heat knob full on after engine has started. Leave on until engine is running smoothly.
12. Primer -- LOCK.

NOTE

If the engine does not start during the first few attempts, or if engine firing diminishes in strength, it is probable that the spark plugs have been frosted over. Preheat must be used before another start is attempted.

**CAUTION**

Pumping the throttle may cause raw fuel to accumulate in the intake air duct, creating a fire hazard in the event of a backfire. If this occurs, maintain a cranking action to suck flames into the engine. An outside attendant with a fire extinguisher is advised for cold starts without preheat.

During cold weather operations no indication will be apparent on the oil temperature gage prior to takeoff if outside air temperatures are very cold. After a suitable warm-up period (2 to 5 minutes at 1000 RPM), accelerate the engine several times to higher engine RPM. If the engine accelerates smoothly and the oil pressure remains normal and steady, the airplane is ready for takeoff.

### FLIGHT OPERATIONS

Takeoff is made normally with carburetor heat off. Avoid excessive leaning in cruise.

Carburetor heat may be used to overcome any occasional engine roughness due to ice.

When operating in temperatures below -18°C, avoid using partial carburetor heat. Partial heat may increase the carburetor air temperature to the 0° to 21°C range, where icing is critical under certain atmospheric conditions.

## HOT WEATHER OPERATION

Refer to the general warm temperature starting information under Starting Engine in this section. Avoid prolonged engine operation on the ground.

## NOISE ABATEMENT

Increased emphasis on improving the quality of our environment requires renewed effort on the part of all pilots to minimize the effect of airplane noise on the public.

We, as pilots, can demonstrate our concern for environmental improvement, by application of the following suggested procedures, and thereby tend to build public support for aviation:

1. Pilots operating aircraft under VFR over outdoor assemblies of persons, recreational and park areas, and other noise-sensitive areas should make every effort to fly not less than 2000 feet above the surface, weather permitting, even though flight at a lower level may be consistent with the provisions of government regulations.
2. During departure from or approach to an airport, climb after takeoff and descent for landing should be made so as to avoid prolonged flight at low altitude near noise-sensitive areas.

NOTE

The above recommended procedures do not apply where they would conflict with Air Traffic Control clearances or instructions, or where, in the pilot's judgment, an altitude of less than 2000 feet is necessary for him to adequately exercise his duty to see and avoid other aircraft.

The certificated noise level for the Model 172N at 2300 pounds maximum weight is 73.8 dB(A). No determination has been made by the Federal Aviation Administration that the noise levels of this airplane are or should be acceptable or unacceptable for operation at, into, or out of, any airport.

# SECTION 5
# PERFORMANCE

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# INTRODUCTION

Performance data charts on the following pages are presented so that you may know what to expect from the airplane under various conditions, and also, to facilitate the planning of flights in detail and with reasonable accuracy. The data in the charts has been computed from actual flight tests with the airplane and engine in good condition and using average piloting techniques.

It should be noted that the performance information presented in the range and endurance profile charts allows for 45 minutes reserve fuel based on 45% power. Fuel flow data for cruise is based on the recommended lean mixture setting. Some indeterminate variables such as mixture leaning technique, fuel metering characteristics, engine and propeller condition, and air turbulence may account for variations of 10% or more in range and endurance. Therefore, it is important to utilize all available information to estimate the fuel required for the particular flight.

# USE OF PERFORMANCE CHARTS

Performance data is presented in tabular or graphical form to illustrate the effect of different variables. Sufficiently detailed information is provided in the tables so that conservative values can be selected and used to determine the particular performance figure with reasonable accuracy.

# SAMPLE PROBLEM

The following sample flight problem utilizes information from the various charts to determine the predicted performance data for a typical flight. The following information is known:

| | |
|---|---|
| AIRPLANE CONFIGURATION | |
| Takeoff weight | 2250 Pounds |
| Usable fuel | 40 Gallons |
| TAKEOFF CONDITIONS | |
| Field pressure altitude | 1500 Feet |
| Temperature | 28°C (16°C above standard) |
| Wind component along runway | 12 Knot Headwind |
| Field length | 3500 Feet |

| | |
|---|---|
| CRUISE CONDITIONS | |
| Total distance | 460 Nautical Miles |
| Pressure altitude | 5500 Feet |
| Temperature | 20°C (16°C above standard) |
| Expected wind enroute | 10 Knot Headwind |
| LANDING CONDITIONS | |
| Field pressure altitude | 2000 Feet |
| Temperature | 25°C |
| Field length | 3000 Feet |

## TAKEOFF

The takeoff distance chart, figure 5-4, should be consulted, keeping in mind that the distances shown are based on the short field technique. Conservative distances can be established by reading the chart at the next higher value of weight, altitude and temperature. For example, in this particular sample problem, the takeoff distance information presented for a weight of 2300 pounds, pressure altitude of 2000 feet and a temperature of 30°C should be used and results in the following:

| | |
|---|---|
| Ground roll | 1075 Feet |
| Total distance to clear a 50-foot obstacle | 1915 Feet |

These distances are well within the available takeoff field length. However, a correction for the effect of wind may be made based on Note 3 of the takeoff chart. The correction for a 12 knot headwind is:

$$\frac{12 \text{ Knots}}{9 \text{ Knots}} \times 10\% = 13\% \text{ Decrease}$$

This results in the following distances, corrected for wind:

| | |
|---|---|
| Ground roll, zero wind | 1075 |
| Decrease in ground roll (1075 feet × 13%) | 140 |
| Corrected ground roll | 935 Feet |

| | |
|---|---|
| Total distance to clear a 50-foot obstacle, zero wind | 1915 |
| Decrease in total distance (1915 feet × 13%) | 249 |
| Corrected total distance to clear 50-foot obstacle | 1666 Feet |

## CRUISE

The cruising altitude should be selected based on a consideration of trip length, winds aloft, and the airplane's performance. A typical cruising altitude and the expected wind enroute have been given for this sample problem. However, the power setting selection for cruise must be determined based on several considerations. These include the cruise performance characteristics presented in figure 5-7, the range profile chart presented in figure 5-8, and the endurance profile chart presented in figure 5-9.

The relationship between power and range is illustrated by the range profile chart. Considerable fuel savings and longer range result when lower power settings are used.

The range profile chart indicates that use of 65% power at 5500 feet yields a predicted range of 523 nautical miles with no wind. The endurance profile chart, figure 5-9, shows a corresponding 4.7 hours.

The range figure of 523 nautical miles is corrected to account for the expected 10 knot headwind at 5500 feet.

| | |
|---|---|
| Range, zero wind | 523 |
| Decrease in range due to wind (4.7 hours × 10 knot headwind) | 47 |
| Corrected range | 476 Nautical Miles |

This indicates that the trip can be made without a fuel stop using approximately 65% power.

The cruise performance chart, figure 5-7, is entered at 6000 feet altitude and 20°C above standard temperature. These values most nearly correspond to the planned altitude and expected temperature conditions. The engine speed chosen is 2500 RPM, which results in the following:

| | |
|---|---|
| Power | 64% |
| True airspeed | 114 Knots |
| Cruise fuel flow | 7.1 GPH |

The power computer may be used to determine power and fuel consumption more accurately during the flight.

## FUEL REQUIRED

The total fuel requirement for the flight may be estimated using the performance information in figures 5-6 and 5-7. For this sample problem, figure 5-6 shows that a climb from 2000 feet to 6000 feet requires 1.3 gallons

of fuel. The corresponding distance during the climb is 9 nautical miles. These values are for a standard temperature and are sufficiently accurate for most flight planning purposes. However, a further correction for the effect of temperature may be made as noted on the climb chart. The approximate effect of a non-standard temperature is to increase the time, fuel, and distance by 10% for each 10°C above standard temperature, due to the lower rate of climb. In this case, assuming a temperature 16°C above standard, the correction would be:

$$\frac{16°C}{10°C} \times 10\% = 16\% \text{ Increase}$$

With this factor included, the fuel estimate would be calculated as follows:

| | |
|---|---|
| Fuel to climb, standard temperature | 1.3 |
| Increase due to non-standard temperature (1.3 × 16%) | 0.2 |
| Corrected fuel to climb | 1.5 Gallons |

Using a similar procedure for the distance to climb results in 10 nautical miles.

The resultant cruise distance is:

| | |
|---|---|
| Total distance | 460 |
| Climb distance | -10 |
| Cruise distance | 450 Nautical Miles |

With an expected 10 knot headwind, the ground speed for cruise is predicted to be:

114
-10
104 Knots

Therefore, the time required for the cruise portion of the trip is:

$$\frac{450 \text{ Nautical Miles}}{104 \text{ Knots}} = 4.3 \text{ Hours}$$

The fuel required for cruise is:

4.3 hours × 7.1 gallons/hour = 30.5 Gallons

The total estimated fuel required is as follows:

| | |
|---|---|
| Engine start, taxi, and takeoff | 1.1 |
| Climb | 1.5 |
| Cruise | 30.5 |
| Total fuel required | 33.1 Gallons |

This will leave a fuel reserve of:

40.0
-33.1
6.9 Gallons

Once the flight is underway, ground speed checks will provide a more accurate basis for estimating the time enroute and the corresponding fuel required to complete the trip with ample reserve.

## LANDING

A procedure similar to takeoff should be used for estimating the landing distance at the destination airport. Figure 5-10 presents landing distance information for the short field technique. The distances corresponding to 2000 feet and 30°C are as follows:

| | |
|---|---|
| Ground roll | 590 Feet |
| Total distance to clear a 50-foot obstacle | 1370 Feet |

A correction for the effect of wind may be made based on Note 2 of the landing chart using the same procedure as outlined for takeoff.

# AIRSPEED CALIBRATION

## NORMAL STATIC SOURCE

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| FLAPS UP | | | | | | | | | | | |
| KIAS | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |
| KCAS | 49 | 55 | 62 | 70 | 80 | 89 | 99 | 108 | 118 | 128 | 138 |
| FLAPS 10° | | | | | | | | | | | |
| KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| KCAS | 49 | 55 | 62 | 71 | 80 | 85 | --- | --- | --- | --- | --- |
| FLAPS 40° | | | | | | | | | | | |
| KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| KCAS | 47 | 54 | 62 | 71 | 81 | 86 | --- | --- | --- | --- | --- |

Figure 5-1. Airspeed Calibration (Sheet 1 of 2)

# AIRSPEED CALIBRATION
## ALTERNATE STATIC SOURCE

### HEATER/VENTS AND WINDOWS CLOSED

| FLAPS UP | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |
| ALTERNATE KIAS | 39 | 51 | 61 | 71 | 82 | 91 | 101 | 111 | 121 | 131 | 141 |
| FLAPS 10° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 40 | 51 | 61 | 71 | 81 | 85 | --- | --- | --- | --- | --- |
| FLAPS 40° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 38 | 50 | 60 | 70 | 79 | 83 | --- | --- | --- | --- | --- |

### HEATER/VENTS OPEN AND WINDOWS CLOSED

| FLAPS UP | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |
| ALTERNATE KIAS | 36 | 48 | 59 | 70 | 80 | 89 | 99 | 108 | 118 | 128 | 139 |
| FLAPS 10° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 38 | 49 | 59 | 69 | 79 | 84 | --- | --- | --- | --- | --- |
| FLAPS 40° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 34 | 47 | 57 | 67 | 77 | 81 | --- | --- | --- | --- | --- |

### WINDOWS OPEN

| FLAPS UP | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 | 130 | 140 |
| ALTERNATE KIAS | 26 | 43 | 57 | 70 | 82 | 93 | 103 | 113 | 123 | 133 | 143 |
| FLAPS 10° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 25 | 43 | 57 | 69 | 80 | 85 | --- | --- | --- | --- | --- |
| FLAPS 40° | | | | | | | | | | | |
| NORMAL KIAS | 40 | 50 | 60 | 70 | 80 | 85 | --- | --- | --- | --- | --- |
| ALTERNATE KIAS | 25 | 41 | 54 | 67 | 78 | 84 | --- | --- | --- | --- | --- |

Figure 5-1. Airspeed Calibration (Sheet 2 of 2)

# TEMPERATURE CONVERSION CHART

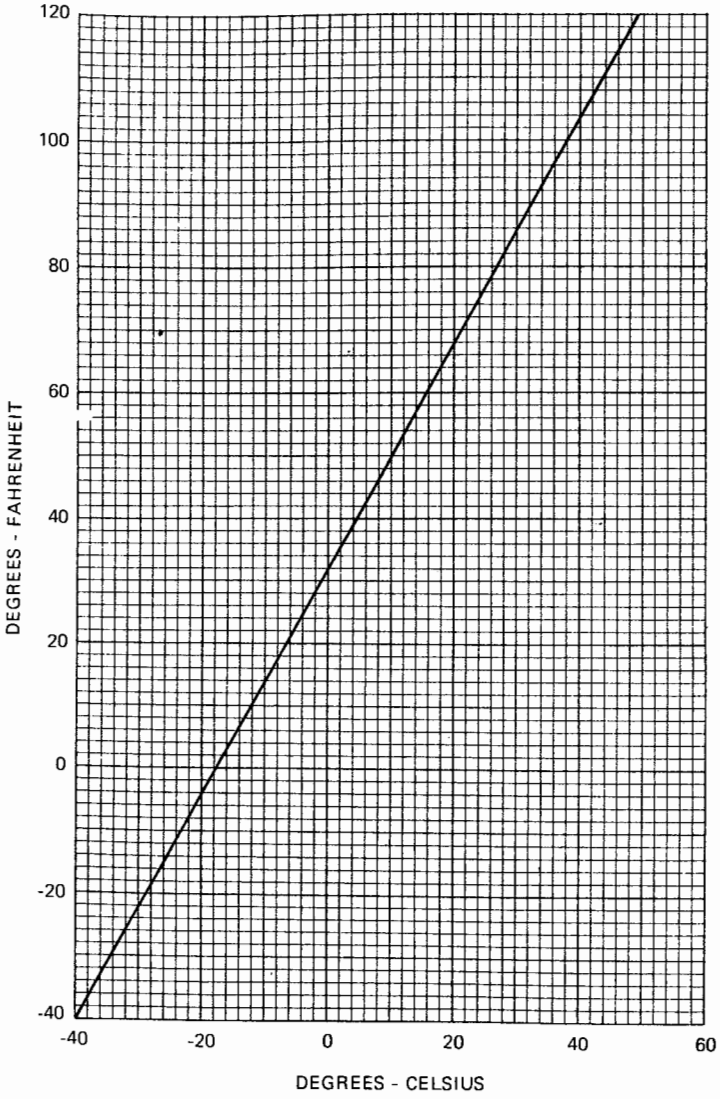



Figure 5-2. Temperature Conversion Chart

# STALL SPEEDS

CONDITIONS:
Power Off

NOTES:
1. Maximum altitude loss during a stall recovery may be as much as 180 feet.
2. KIAS values are approximate.

## MOST REARWARD CENTER OF GRAVITY

| WEIGHT LBS | FLAP DEFLECTION | ANGLE OF BANK | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 0° | | 30° | | 45° | | 60° | |
| | | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS |
| 2300 | UP | 42 | 50 | 45 | 54 | 50 | 59 | 59 | 71 |
| | 10° | 38 | 47 | 40 | 51 | 45 | 56 | 54 | 66 |
| | 40° | 36 | 44 | 38 | 47 | 43 | 52 | 51 | 62 |

## MOST FORWARD CENTER OF GRAVITY

| WEIGHT LBS | FLAP DEFLECTION | ANGLE OF BANK | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | 0° | | 30° | | 45° | | 60° | |
| | | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS | KIAS | KCAS |
| 2300 | UP | 47 | 53 | 51 | 57 | 56 | 63 | 66 | 75 |
| | 10° | 44 | 51 | 47 | 55 | 52 | 61 | 62 | 72 |
| | 40° | 41 | 47 | 44 | 51 | 49 | 56 | 58 | 66 |

Figure 5-3. Stall Speeds

# TAKEOFF DISTANCE
## MAXIMUM WEIGHT 2300 LBS

**SHORT FIELD**

CONDITIONS:
Flaps Up
Full Throttle Prior to Brake Release
Paved, Level, Dry Runway
Zero Wind

NOTES:
1. Short field technique as specified in Section 4.
2. Prior to takeoff from fields above 3000 feet elevation, the mixture should be leaned to give maximum RPM in a full throttle, static runup.
3. Decrease distances 10% for each 9 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2 knots.
4. For operation on a dry, grass runway, increase distances by 15% of the "ground roll" figure.

| WEIGHT LBS | TAKEOFF SPEED KIAS | | PRESS ALT FT | 0°C | | 10°C | | 20°C | | 30°C | | 40°C | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | LIFT OFF | AT 50 FT | | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS |
| 2300 | 52 | 59 | S.L. | 720 | 1300 | 775 | 1390 | 835 | 1490 | 895 | 1590 | 960 | 1700 |
| | | | 1000 | 790 | 1420 | 850 | 1525 | 915 | 1630 | 980 | 1745 | 1050 | 1865 |
| | | | 2000 | 865 | 1555 | 930 | 1670 | 1000 | 1790 | 1075 | 1915 | 1155 | 2055 |
| | | | 3000 | 950 | 1710 | 1025 | 1835 | 1100 | 1970 | 1185 | 2115 | 1270 | 2265 |
| | | | 4000 | 1045 | 1880 | 1125 | 2025 | 1210 | 2175 | 1300 | 2335 | 1400 | 2510 |
| | | | 5000 | 1150 | 2075 | 1240 | 2240 | 1335 | 2410 | 1435 | 2595 | 1540 | 2795 |
| | | | 6000 | 1265 | 2305 | 1365 | 2485 | 1475 | 2680 | 1585 | 2895 | 1705 | 3125 |
| | | | 7000 | 1400 | 2565 | 1510 | 2770 | 1630 | 3000 | 1755 | 3245 | 1890 | 3515 |
| | | | 8000 | 1550 | 2870 | 1675 | 3110 | 1805 | 3375 | 1945 | 3670 | 2095 | 3990 |

Figure 5-4. Takeoff Distance (Sheet 1 of 2)



# TAKEOFF DISTANCE
## 2100 LBS AND 1900 LBS

**SHORT FIELD**

REFER TO SHEET 1 FOR APPROPRIATE CONDITIONS AND NOTES.

| WEIGHT LBS | TAKEOFF SPEED KIAS | | PRESS ALT FT | 0°C | | 10°C | | 20°C | | 30°C | | 40°C | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | LIFT OFF | AT 50 FT | | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS |
| 2100 | 50 | 56 | S.L. | 585 | 1070 | 630 | 1140 | 680 | 1220 | 725 | 1300 | 780 | 1390 |
| | | | 1000 | 640 | 1165 | 690 | 1245 | 740 | 1330 | 795 | 1420 | 850 | 1520 |
| | | | 2000 | 700 | 1270 | 755 | 1360 | 810 | 1455 | 870 | 1555 | 935 | 1665 |
| | | | 3000 | 770 | 1390 | 830 | 1490 | 890 | 1595 | 955 | 1710 | 1025 | 1830 |
| | | | 4000 | 845 | 1525 | 910 | 1640 | 980 | 1755 | 1050 | 1880 | 1130 | 2015 |
| | | | 5000 | 930 | 1680 | 1000 | 1805 | 1075 | 1935 | 1155 | 2075 | 1240 | 2230 |
| | | | 6000 | 1025 | 1850 | 1100 | 1990 | 1185 | 2140 | 1275 | 2300 | 1370 | 2475 |
| | | | 7000 | 1130 | 2050 | 1215 | 2210 | 1310 | 2380 | 1410 | 2560 | 1515 | 2755 |
| | | | 8000 | 1245 | 2275 | 1345 | 2460 | 1450 | 2655 | 1560 | 2865 | 1680 | 3090 |
| 1900 | 47 | 54 | S.L. | 470 | 865 | 505 | 920 | 540 | 985 | 580 | 1045 | 620 | 1115 |
| | | | 1000 | 515 | 940 | 550 | 1005 | 590 | 1070 | 635 | 1140 | 680 | 1215 |
| | | | 2000 | 560 | 1025 | 605 | 1095 | 645 | 1170 | 695 | 1245 | 745 | 1330 |
| | | | 3000 | 615 | 1115 | 660 | 1195 | 710 | 1275 | 760 | 1365 | 815 | 1455 |
| | | | 4000 | 670 | 1220 | 725 | 1305 | 780 | 1400 | 835 | 1495 | 895 | 1595 |
| | | | 5000 | 740 | 1340 | 795 | 1435 | 855 | 1535 | 920 | 1640 | 985 | 1755 |
| | | | 6000 | 810 | 1470 | 875 | 1575 | 940 | 1690 | 1010 | 1810 | 1085 | 1940 |
| | | | 7000 | 895 | 1620 | 965 | 1740 | 1035 | 1865 | 1115 | 2000 | 1195 | 2145 |
| | | | 8000 | 985 | 1790 | 1065 | 1925 | 1145 | 2065 | 1230 | 2220 | 1320 | 2385 |

Figure 5-4. Takeoff Distance (Sheet 2 of 2)

# RATE OF CLIMB

## MAXIMUM

CONDITIONS:
Flaps Up
Full Throttle

NOTE:
Mixture leaned above 3000 feet for maximum RPM.

| WEIGHT LBS | PRESS ALT FT | CLIMB SPEED KIAS | RATE OF CLIMB - FPM | | | |
|---|---|---|---|---|---|---|
| | | | -20°C | 0°C | 20°C | 40°C |
| 2300 | S.L. | 73 | 875 | 815 | 755 | 695 |
| | 2000 | 72 | 765 | 705 | 650 | 590 |
| | 4000 | 71 | 655 | 600 | 545 | 485 |
| | 6000 | 70 | 545 | 495 | 440 | 385 |
| | 8000 | 69 | 440 | 390 | 335 | 280 |
| | 10,000 | 68 | 335 | 285 | 230 | --- |
| | 12,000 | 67 | 230 | 180 | --- | --- |

Figure 5-5. Rate of Climb

# TIME, FUEL, AND DISTANCE TO CLIMB

## MAXIMUM RATE OF CLIMB

CONDITIONS:
Flaps Up
Full Throttle
Standard Temperature

NOTES:
1. Add 1.1 gallons of fuel for engine start, taxi and takeoff allowance.
2. Mixture leaned above 3000 feet for maximum RPM.
3. Increase time, fuel and distance by 10% for each 10°C above standard temperature.
4. Distances shown are based on zero wind.

| WEIGHT LBS | PRESSURE ALTITUDE FT | TEMP °C | CLIMB SPEED KIAS | RATE OF CLIMB FPM | FROM SEA LEVEL | | |
|---|---|---|---|---|---|---|---|
| | | | | | TIME MIN | FUEL USED GALLONS | DISTANCE NM |
| 2300 | S.L. | 15 | 73 | 770 | 0 | 0.0 | 0 |
| | 1000 | 13 | 73 | 725 | 1 | 0.3 | 2 |
| | 2000 | 11 | 72 | 675 | 3 | 0.6 | 3 |
| | 3000 | 9 | 72 | 630 | 4 | 0.9 | 5 |
| | 4000 | 7 | 71 | 580 | 6 | 1.2 | 8 |
| | 5000 | 5 | 71 | 535 | 8 | 1.6 | 10 |
| | 6000 | 3 | 70 | 485 | 10 | 1.9 | 12 |
| | 7000 | 1 | 69 | 440 | 12 | 2.3 | 15 |
| | 8000 | -1 | 69 | 390 | 15 | 2.7 | 19 |
| | 9000 | -3 | 68 | 345 | 17 | 3.2 | 22 |
| | 10,000 | -5 | 68 | 295 | 21 | 3.7 | 27 |
| | 11,000 | -7 | 67 | 250 | 24 | 4.2 | 32 |
| | 12,000 | -9 | 67 | 200 | 29 | 4.9 | 38 |

Figure 5-6. Time, Fuel, and Distance to Climb

# CRUISE PERFORMANCE

CONDITIONS:
2300 Pounds
Recommended Lean Mixture

| PRESSURE ALTITUDE FT | RPM | 20°C BELOW STANDARD TEMP | | | STANDARD TEMPERATURE | | | 20°C ABOVE STANDARD TEMP | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | % BHP | KTAS | GPH | % BHP | KTAS | GPH | % BHP | KTAS | GPH |
| 2000 | 2500 | --- | --- | --- | 75 | 116 | 8.4 | 71 | 115 | 7.9 |
| | 2400 | 72 | 111 | 8.0 | 67 | 111 | 7.5 | 63 | 110 | 7.1 |
| | 2300 | 64 | 106 | 7.1 | 60 | 105 | 6.7 | 56 | 105 | 6.3 |
| | 2200 | 56 | 101 | 6.3 | 53 | 100 | 6.1 | 50 | 99 | 5.8 |
| | 2100 | 50 | 95 | 5.8 | 47 | 94 | 5.6 | 45 | 93 | 5.4 |
| 4000 | 2550 | --- | --- | --- | 75 | 118 | 8.4 | 71 | 118 | 7.9 |
| | 2500 | 76 | 116 | 8.5 | 71 | 115 | 8.0 | 67 | 115 | 7.5 |
| | 2400 | 68 | 111 | 7.6 | 64 | 110 | 7.1 | 60 | 109 | 6.7 |
| | 2300 | 60 | 105 | 6.8 | 57 | 105 | 6.4 | 54 | 104 | 6.1 |
| | 2200 | 54 | 100 | 6.1 | 51 | 99 | 5.9 | 48 | 98 | 5.7 |
| | 2100 | 48 | 94 | 5.6 | 46 | 93 | 5.5 | 44 | 92 | 5.3 |
| 6000 | 2600 | --- | --- | --- | 75 | 120 | 8.4 | 71 | 120 | 7.9 |
| | 2500 | 72 | 116 | 8.1 | 67 | 115 | 7.6 | 64 | 114 | 7.1 |
| | 2400 | 64 | 110 | 7.2 | 60 | 109 | 6.8 | 57 | 109 | 6.4 |
| | 2300 | 57 | 105 | 6.5 | 54 | 104 | 6.2 | 52 | 103 | 5.9 |
| | 2200 | 51 | 99 | 5.9 | 49 | 98 | 5.7 | 47 | 97 | 5.5 |
| | 2100 | 46 | 93 | 5.5 | 44 | 92 | 5.4 | 42 | 91 | 5.2 |
| 8000 | 2650 | --- | --- | --- | 75 | 122 | 8.4 | 71 | 122 | 7.9 |
| | 2600 | 76 | 120 | 8.6 | 71 | 120 | 8.0 | 67 | 119 | 7.5 |
| | 2500 | 68 | 115 | 7.7 | 64 | 114 | 7.2 | 60 | 113 | 6.8 |
| | 2400 | 61 | 110 | 6.9 | 58 | 109 | 6.5 | 55 | 108 | 6.2 |
| | 2300 | 55 | 104 | 6.2 | 52 | 103 | 6.0 | 50 | 102 | 5.8 |
| | 2200 | 49 | 98 | 5.7 | 47 | 97 | 5.5 | 45 | 96 | 5.4 |
| 10,000 | 2650 | 76 | 122 | 8.5 | 71 | 122 | 8.0 | 67 | 121 | 7.5 |
| | 2600 | 72 | 120 | 8.1 | 68 | 119 | 7.6 | 64 | 118 | 7.1 |
| | 2500 | 65 | 114 | 7.3 | 61 | 114 | 6.8 | 58 | 112 | 6.5 |
| | 2400 | 58 | 109 | 6.5 | 55 | 108 | 6.2 | 52 | 107 | 6.0 |
| | 2300 | 52 | 103 | 6.0 | 50 | 102 | 5.8 | 48 | 101 | 5.6 |
| | 2200 | 47 | 97 | 5.6 | 45 | 96 | 5.4 | 44 | 95 | 5.3 |
| 12,000 | 2600 | 68 | 119 | 7.7 | 64 | 118 | 7.2 | 61 | 117 | 6.8 |
| | 2500 | 62 | 114 | 6.9 | 58 | 113 | 6.5 | 55 | 111 | 6.2 |
| | 2400 | 56 | 108 | 6.3 | 53 | 107 | 6.0 | 51 | 106 | 5.8 |
| | 2300 | 50 | 102 | 5.8 | 48 | 101 | 5.6 | 46 | 100 | 5.5 |
| | 2200 | 46 | 96 | 5.5 | 44 | 95 | 5.4 | 43 | 94 | 5.3 |

Figure 5-7. Cruise Performance

# RANGE PROFILE

## 45 MINUTES RESERVE
## 40 GALLONS USABLE FUEL

CONDITIONS:
2300 Pounds
Recommended Lean Mixture for Cruise
Standard Temperature
Zero Wind

NOTES:
1. This chart allows for the fuel used for engine start, taxi, takeoff and climb, and the distance during climb as shown in figure 5-6.
2. Reserve fuel is based on 45 minutes at 45% BHP and is 4.1 gallons.

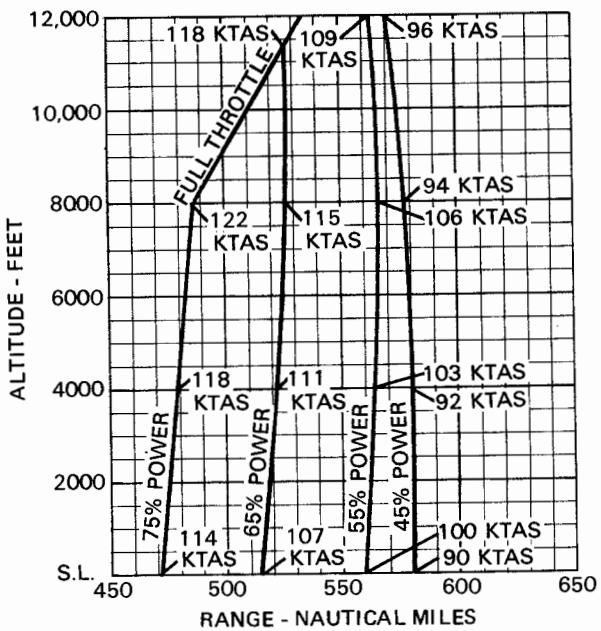



Figure 5-8. Range Profile (Sheet 1 of 2)

# RANGE PROFILE

## 45 MINUTES RESERVE
## 50 GALLONS USABLE FUEL

CONDITIONS:
2300 Pounds
Recommended Lean Mixture for Cruise
Standard Temperature
Zero Wind

NOTES:
1. This chart allows for the fuel used for engine start, taxi, takeoff and climb, and the distance during climb as shown in figure 5-6.
2. Reserve fuel is based on 45 minutes at 45% BHP and is 4.1 gallons.

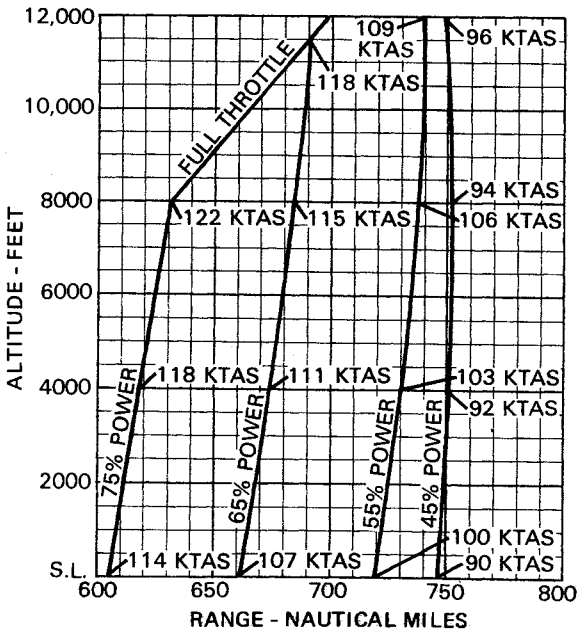



Figure 5-8. Range Profile (Sheet 2 of 2)

# ENDURANCE PROFILE
## 45 MINUTES RESERVE
## 40 GALLONS USABLE FUEL

CONDITIONS:
2300 Pounds
Recommended Lean Mixture for Cruise
Standard Temperature

NOTES:
1. This chart allows for the fuel used for engine start, taxi, takeoff and climb, and the time during climb as shown in figure 5-6.
2. Reserve fuel is based on 45 minutes at 45% BHP and is 4.1 gallons.

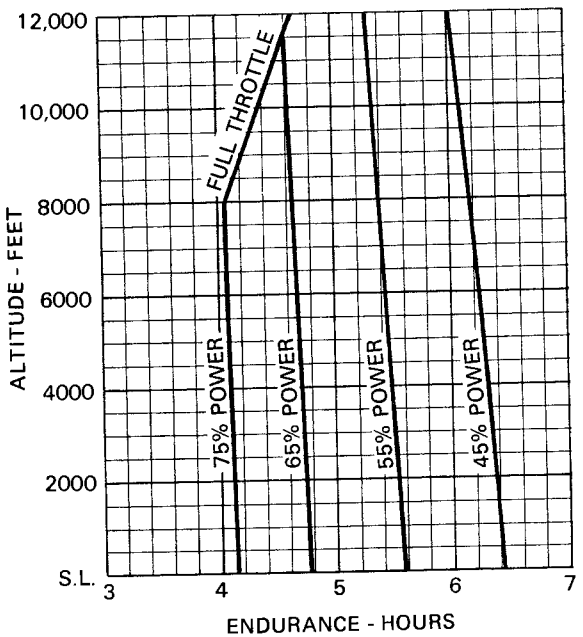



Figure 5-9. Endurance Profile (Sheet 1 of 2)

# ENDURANCE PROFILE
## 45 MINUTES RESERVE
## 50 GALLONS USABLE FUEL

CONDITIONS:
2300 Pounds
Recommended Lean Mixture for Cruise
Standard Temperature

NOTES:
1. This chart allows for the fuel used for engine start, taxi, takeoff and climb, and the time during climb as shown in figure 5-6.
2. Reserve fuel is based on 45 minutes at 45% BHP and is 4.1 gallons.

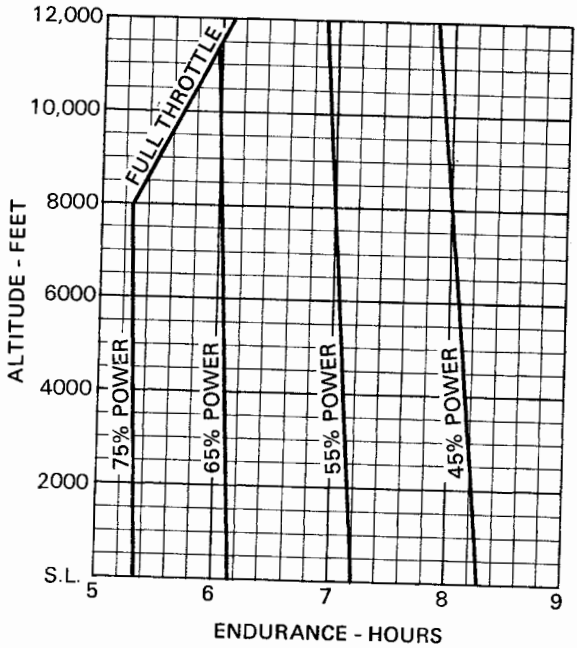



Figure 5-9. Endurance Profile (Sheet 2 of 2)

# LANDING DISTANCE

## SHORT FIELD

CONDITIONS:
Flaps 40°
Power Off
Maximum Braking
Paved, Level, Dry Runway
Zero Wind

NOTES:
1. Short field technique as specified in Section 4.
2. Decrease distances 10% for each 9 knots headwind. For operation with tailwinds up to 10 knots, increase distances by 10% for each 2 knots
3. For operation on a dry, grass runway, increase distances by 45% of the "ground roll" figure.

| WEIGHT LBS | SPEED AT 50 FT KIAS | PRESS ALT FT | 0°C | | 10°C | | 20°C | | 30°C | | 40°C | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS | GRND ROLL | TOTAL TO CLEAR 50 FT OBS |
| 2300 | 60 | S.L. | 495 | 1205 | 510 | 1235 | 530 | 1265 | 545 | 1295 | 565 | 1330 |
| | | 1000 | 510 | 1235 | 530 | 1265 | 550 | 1300 | 565 | 1330 | 585 | 1365 |
| | | 2000 | 530 | 1265 | 550 | 1300 | 570 | 1335 | 590 | 1370 | 610 | 1405 |
| | | 3000 | 550 | 1300 | 570 | 1335 | 590 | 1370 | 610 | 1405 | 630 | 1440 |
| | | 4000 | 570 | 1335 | 590 | 1370 | 615 | 1410 | 635 | 1445 | 655 | 1480 |
| | | 5000 | 590 | 1370 | 615 | 1415 | 635 | 1450 | 655 | 1485 | 680 | 1525 |
| | | 6000 | 615 | 1415 | 640 | 1455 | 660 | 1490 | 685 | 1535 | 705 | 1570 |
| | | 7000 | 640 | 1455 | 660 | 1495 | 685 | 1535 | 710 | 1575 | 730 | 1615 |
| | | 8000 | 665 | 1500 | 690 | 1540 | 710 | 1580 | 735 | 1620 | 760 | 1665 |

Figure 5-10. Landing Distance

# SECTION 6
# WEIGHT & BALANCE/ EQUIPMENT LIST

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# INTRODUCTION

This section describes the procedure for establishing the basic empty weight and moment of the airplane. Sample forms are provided for reference. Procedures for calculating the weight and moment for various operations are also provided. A comprehensive list of all Cessna equipment available for this airplane is included at the back of this section.

It should be noted that specific information regarding the weight, arm, moment and installed equipment list for this airplane can only be found in the appropriate weight and balance records carried in the airplane.

# AIRPLANE WEIGHING PROCEDURES

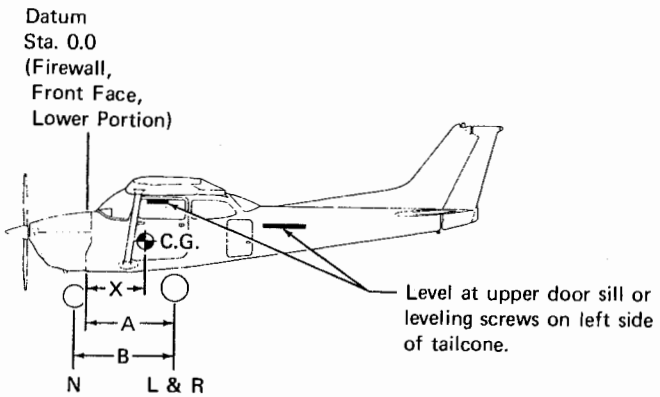
1. Preparation:
   a. Inflate tires to recommended operating pressures.
   b. Remove the fuel tank sump quick-drain fittings and fuel selector valve drain plug to drain all fuel.
   c. Remove oil sump drain plug to drain all oil.
   d. Move sliding seats to the most forward position.
   e. Raise flaps to the fully retracted position.
   f. Place all control surfaces in neutral position.

2. Leveling:
   a. Place scales under each wheel (minimum scale capacity, 500 pounds nose, 1000 pounds each main).
   b. Deflate the nose tire and/or lower or raise the nose strut to properly center the bubble in the level (see figure 6-1).

3. Weighing:
   a. With the airplane level and brakes released, record the weight shown on each scale. Deduct the tare, if any, from each reading.

4. Measuring:
   a. Obtain measurement A by measuring horizontally (along the airplane center line) from a line stretched between the main wheel centers to a plumb bob dropped from the firewall.
   b. Obtain measurement B by measuring horizontally and parallel to the airplane center line, from center of nose wheel axle, left side, to a plumb bob dropped from the line between the main wheel centers. Repeat on right side and average the measurements.

5. Using weights from item 3 and measurements from item 4, the




| Scale Position | Scale Reading | Tare | Symbol | Net Weight |
|---|---|---|---|---|
| Left Wheel | | | L | |
| Right Wheel | | | R | |
| Nose Wheel | | | N | |
| Sum of Net Weights (As Weighed) | | | W | |

$$X = ARM = (A) - \frac{(N) \times (B)}{W} \; ; \; X = (\quad) - \frac{(\quad) \times (\quad)}{(\quad)} = (\quad) \text{ IN.}$$

| Item | Weight (Lbs.) | X C.G. Arm (In.) = | Moment/1000 (Lbs.-In.) |
|---|---|---|---|
| Airplane Weight (From Item 5, page 6-3) | | | |
| Add Oil: No Oil Filter (6 Qts at 7.5 Lbs/Gal) | | -14.0 | |
| With Oil Filter (7 Qts at 7.5 Lbs/Gal) | | -14.0 | |
| Add Unusable Fuel: Std. Tanks (3 Gal at 6 Lbs/Gal) | | 46.0 | |
| L.R. Tanks (4 Gal at 6 Lbs/Gal) | | 46.0 | |
| Equipment Changes | | | |
| Airplane Basic Empty Weight | | | |

Figure 6-1. Sample Airplane Weighing

# SAMPLE WEIGHT AND BALANCE RECORD

(Continuous History of Changes in Structure or Equipment Affecting Weight and Balance)

| AIRPLANE MODEL | | | SERIAL NUMBER | | | | | | PAGE NUMBER | |
|---|---|---|---|---|---|---|---|---|---|---|
| DATE | ITEM NO. | | DESCRIPTION OF ARTICLE OR MODIFICATION | WEIGHT CHANGE | | | | | | RUNNING BASIC EMPTY WEIGHT | |
| | | | | ADDED (+) | | | REMOVED (-) | | | | |
| | In | Out | | Wt. (lb.) | Arm (In.) | Moment /1000 | Wt. (lb.) | Arm (In.) | Moment /1000 | Wt. (lb.) | Moment /1000 |
| | | | | | | | | | | | |
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Figure 6-2. Sample Weight and Balance Record

airplane weight and C.G. can be determined.

6. Basic Empty Weight may be determined by completing figure 6-1.

## WEIGHT AND BALANCE

The following information will enable you to operate your Cessna within the prescribed weight and center of gravity limitations. To figure weight and balance, use the Sample Problem, Loading Graph, and Center of Gravity Moment Envelope as follows:

Take the basic empty weight and moment from appropriate weight and balance records carried in your airplane, and enter them in the column titled YOUR AIRPLANE on the Sample Loading Problem.

NOTE

In addition to the basic empty weight and moment noted on these records, the C.G. arm (fuselage station) is also shown, but need not be used on the Sample Loading Problem. The moment which is shown must be divided by 1000 and this value used as the moment/1000 on the loading problem.

Use the Loading Graph to determine the moment/1000 for each additional item to be carried; then list these on the loading problem.

NOTE

Loading Graph information for the pilot, passengers and baggage is based on seats positioned for average occupants and baggage loaded in the center of the baggage areas as shown on the Loading Arrangements diagram. For loadings which may differ from these, the Sample Loading Problem lists fuselage stations for these items to indicate their forward and aft C.G. range limitations (seat travel and baggage area limitation). Additional moment calculations, based on the actual weight and C.G. arm (fuselage station) of the item being loaded, must be made if the position of the load is different from that shown on the Loading Graph.

Total the weights and moments/1000 and plot these values on the Center of Gravity Moment Envelope to determine whether the point falls within the envelope, and if the loading is acceptable.

## LOADING ARRANGEMENTS

* Pilot or passenger center of gravity on adjustable seats positioned for average occupant. Numbers in parentheses indicate forward and aft limits of occupant center of gravity range.

** Arm measured to the center of the areas shown.

NOTE: The rear cabin wall (approximate station 108) or aft baggage wall (approximate station 142) can be used as convenient interior reference points for determining the location of baggage area fuselage stations.

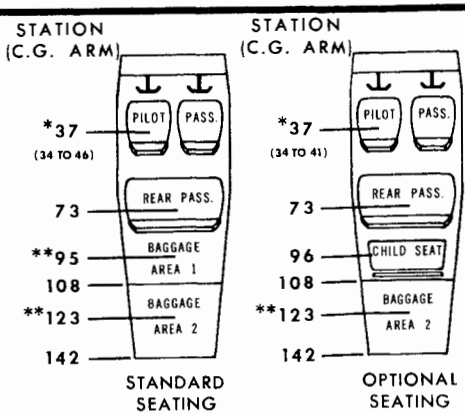
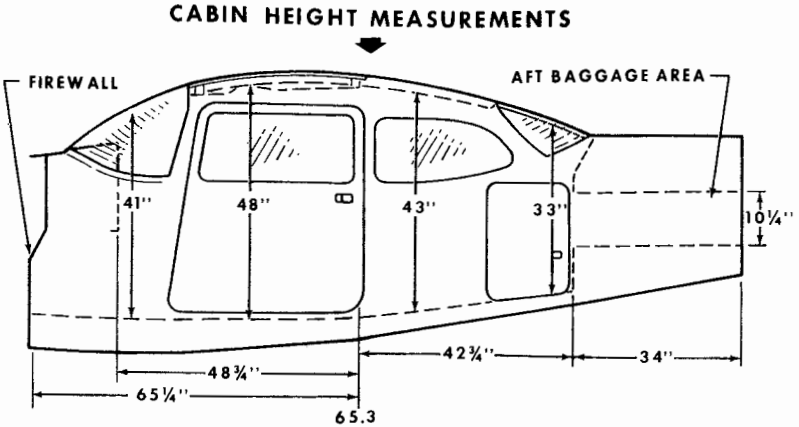



Figure 6-3. Loading Arrangements




**DOOR OPENING DIMENSIONS**

| | WIDTH (TOP) | WIDTH (BOTTOM) | HEIGHT (FRONT) | HEIGHT (REAR) |
|---|---|---|---|---|
| CABIN DOOR | 32" | 37" | 40" | 41" |
| BAGGAGE DOOR | 15¼" | 15¼" | 22" | 21" |

WIDTH
• LWR WINDOW LINE
* CABIN FLOOR

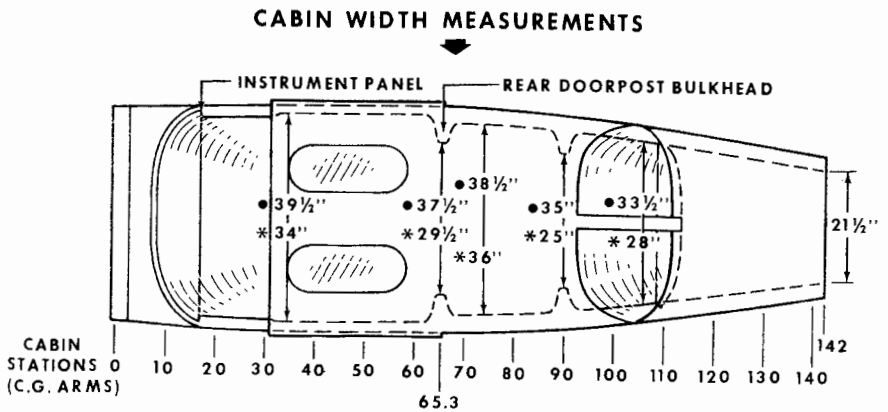



Figure 6-4. Internal Cabin Dimensions

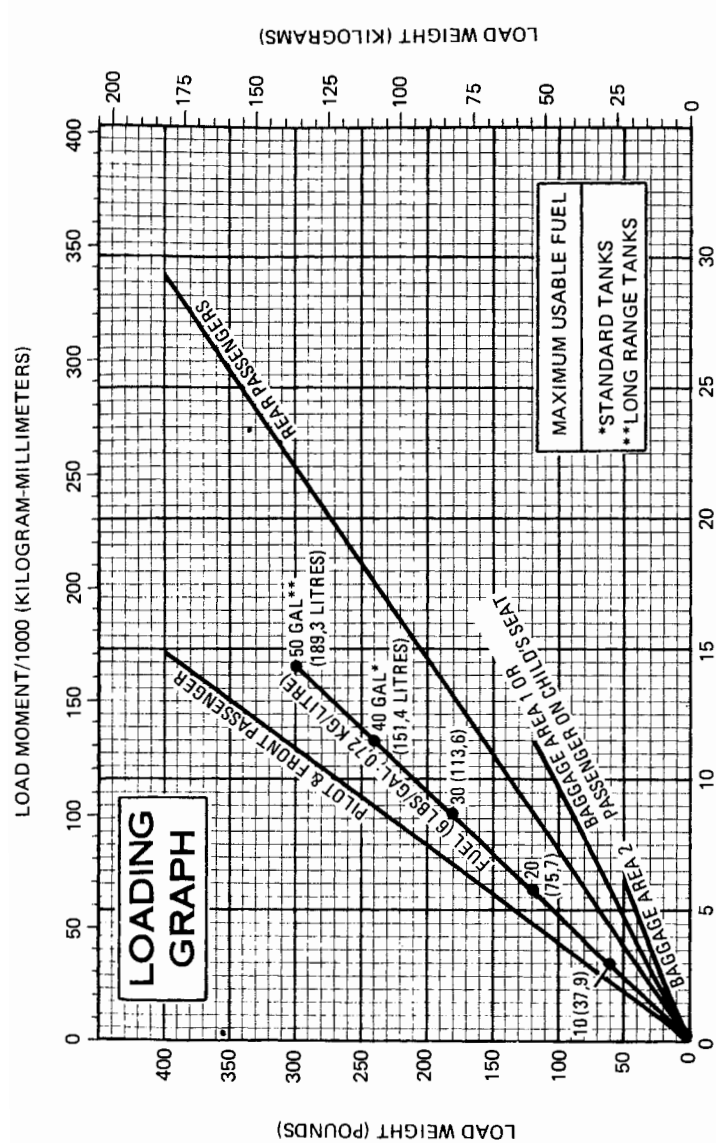
| SAMPLE LOADING PROBLEM | SAMPLE AIRPLANE | | YOUR AIRPLANE | |
|---|---|---|---|---|
| | Weight (lbs.) | Moment (lb. -ins. /1000) | Weight (lbs.) | Moment (lb. - ins. /1000) |
| 1. Basic Empty Weight (Use the data pertaining to your airplane as it is presently equipped. Includes unusable fuel and full oil) . . . . . | 1454 | 57.6 | | |
| 2. Usable Fuel (At 6 Lbs./Gal.) Standard Tanks (40 Gal. Maximum) . . . . . | 240 | 11.5 | | |
| Long Range Tanks (50 Gal. Maximum) . . . . . | | | | |
| 3. Pilot and Front Passenger (Station 34 to 46) . . . . . | 340 | 12.6 | | |
| 4. Rear Passengers . . . . . | 170 | 12.4 | | |
| 5. * Baggage Area 1 or Passenger on Child's Seat (Station 82 to 108) 120 Lbs. Max. . . . . | 96 | 9.1 | | |
| 6. * Baggage Area 2 (Station 108 to 142) 50 Lbs. Max . . . . . | | | | |
| 7. TOTAL WEIGHT AND MOMENT | 2300 | 103.2 | | |

8. Locate this point (2300 at 103.2) on the Center of Gravity Moment Envelope, and since this point falls within the envelope, the loading is acceptable.

**NOTE**

*** The maximum allowable combined weight capacity for baggage areas 1 and 2 is 120 lbs.**

Figure 6-5. Sample Loading Problem




NOTE: Line representing adjustable seats shows the pilot or passenger center of gravity on adjustable seats positioned for an average occupant. Refer to the Loading Arrangements diagram for forward and aft limits of occupant C.G. range.

Figure 6-6. Loading Graph

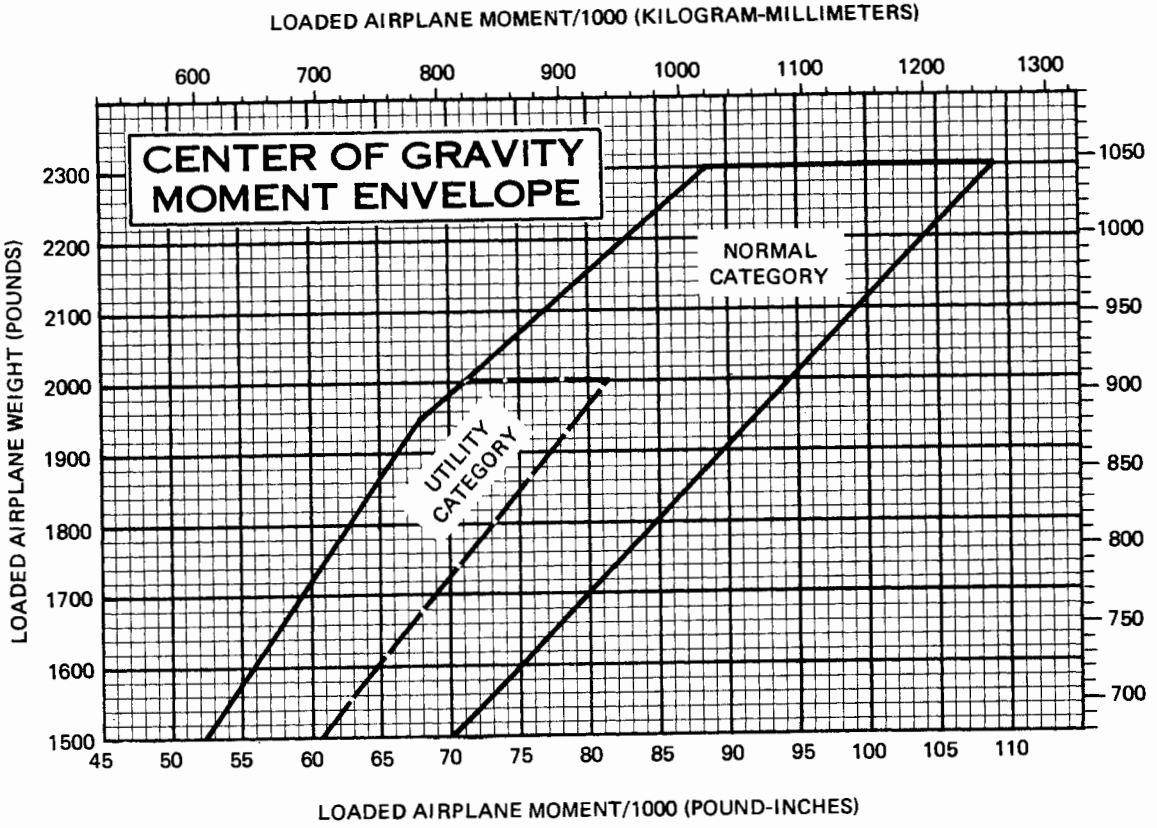



Figure 6-7. Center of Gravity Moment Envelope

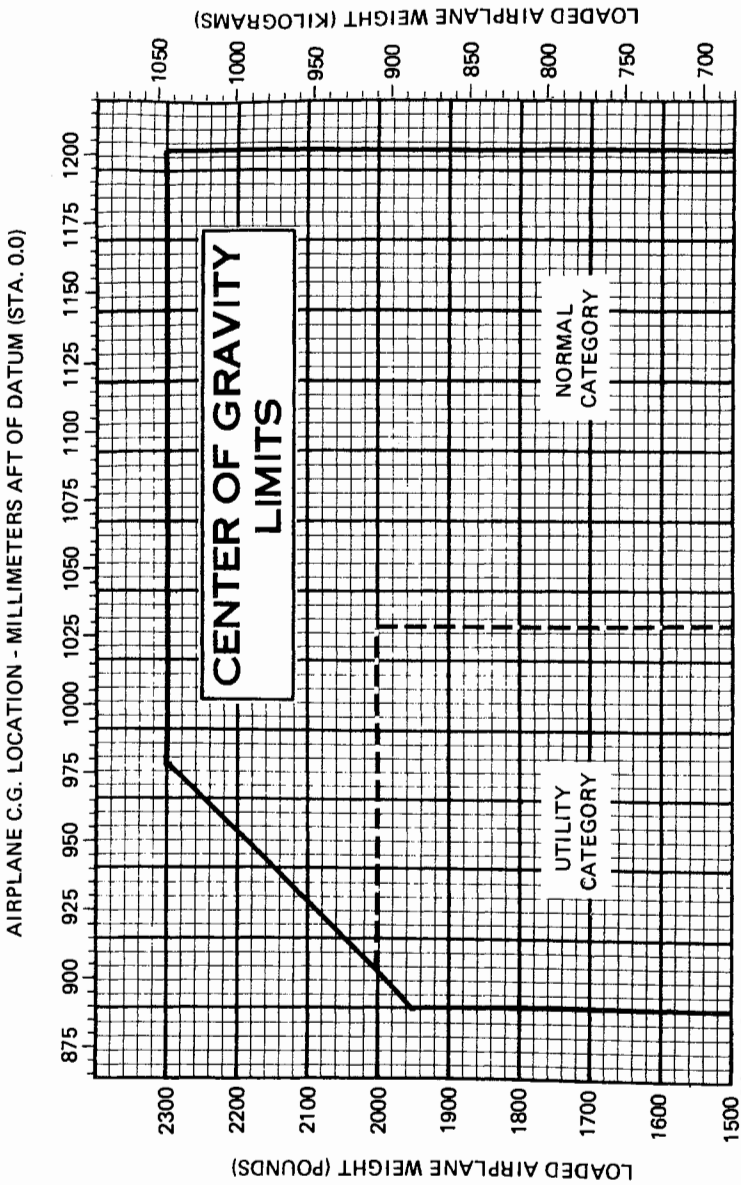



Figure 6-8. Center of Gravity Limits

# EQUIPMENT LIST

The following equipment list is a comprehensive list of all Cessna equipment available for this airplane. A separate equipment list of items installed in your specific airplane is provided in your aircraft file. The following list and the specific list for your airplane have a similar order of listing.

This equipment list provides the following information:

An **item number** gives the identification number for the item. Each number is prefixed with a letter which identifies the **descriptive** grouping (example: A. Powerplant & Accessories) under which it is listed. Suffix letters identify the equipment as a required item, a standard item or an optional item. Suffix letters are as follows:

-R = required items of equipment for FAA certification
-S = standard equipment items
-O = optional equipment items replacing required or standard items
-A = optional equipment items which are in addition to required or standard items

A **reference drawing** column provides the drawing number for the item.

NOTE

If additional equipment is to be installed, it must be done in accordance with the reference drawing, accessory kit instructions, or a separate FAA approval.

Columns showing **weight (in pounds)** and **arm (in inches)** provide the weight and center of gravity location for the equipment.

NOTE

Unless otherwise indicated, true values (not net change values) for the weight and arm are shown. Positive arms are distances aft of the airplane datum; negative arms are distances forward of the datum.

NOTE

Asterisks (*) after the item weight and arm indicate complete assembly installations. Some major components of the assembly are listed on the lines immediately following. The summation of these major components does not necessarily equal the complete assembly installation.

| ITEM NO | EQUIPMENT LIST DESCRIPTION | REF DRAWING | WT LBS | ARM INS |
|---|---|---|---|---|
| | J15-A INTERCONNECT SYSTEM (STOWED)<br>COWL ASSY, FLOATPLANE (NET CHG) | 0560012<br>0552162 | 0.4<br>NEGL | 95.0<br>- - |

# SECTION 7
# AIRPLANE & SYSTEMS DESCRIPTIONS

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# INTRODUCTION

This section provides description and operation of the airplane and its systems. Some equipment described herein is optional and may not be installed in the airplane. Refer to Section 9, Supplements, for details of other optional systems and equipment.

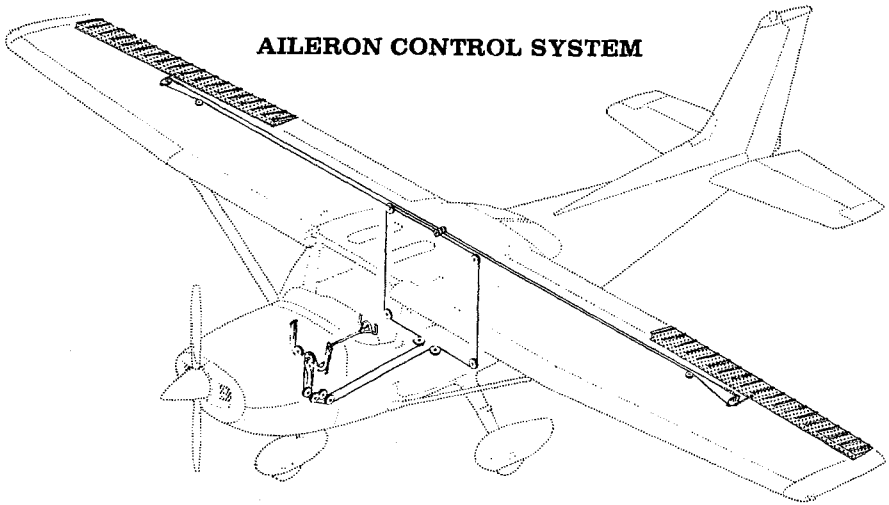
# AIRFRAME

The airplane is an all-metal, four-place, high-wing, single-engine airplane equipped with tricycle landing gear, and is designed for general utility purposes.

The construction of the fuselage is a conventional formed sheet metal bulkhead, stringer, and skin design referred to as semimonocoque. Major items of structure are the front and rear carry-through spars to which the wings are attached, a bulkhead and forgings for main landing gear attachment at the base of the rear door posts, and a bulkhead with attaching plates at the base of the forward door posts for the lower attachment of the wing struts. Four engine mount stringers are also attached to the forward door posts and extend forward to the firewall.

The externally braced wings, containing the fuel tanks, are constructed of a front and rear spar with formed sheet metal ribs, doublers, and stringers. The entire structure is covered with aluminum skin. The front spars are equipped with wing-to-fuselage and wing-to-strut attach fittings. The aft spars are equipped with wing-to-fuselage attach fittings, and are partial-span spars. Conventional hinged ailerons and single-slot type flaps are attached to the trailing edge of the wings. The ailerons are constructed of a forward spar containing a balance weight, formed sheet metal ribs and "V" type corrugated aluminum skin joined together at the trailing edge. The flaps are constructed basically the same as the ailerons, with the exception of the balance weights and the addition of a formed sheet metal leading edge section.

The empennage (tail assembly) consists of a conventional vertical stabilizer, rudder, horizontal stabilizer, and elevator. The vertical stabilizer consists of a spar, formed sheet metal ribs and reinforcements, a wrap-around skin panel, formed leading edge skin and a dorsal. The rudder is constructed of a formed leading edge skin containing hinge halves, a center wrap-around skin panel, ribs, an aft wrap-around skin panel which is joined at the trailing edge of the rudder by a filler strip, and a ground adjustable trim tab at the base of the trailing edge. The top of the rudder incorporates a leading edge extension which contains a balance weight.



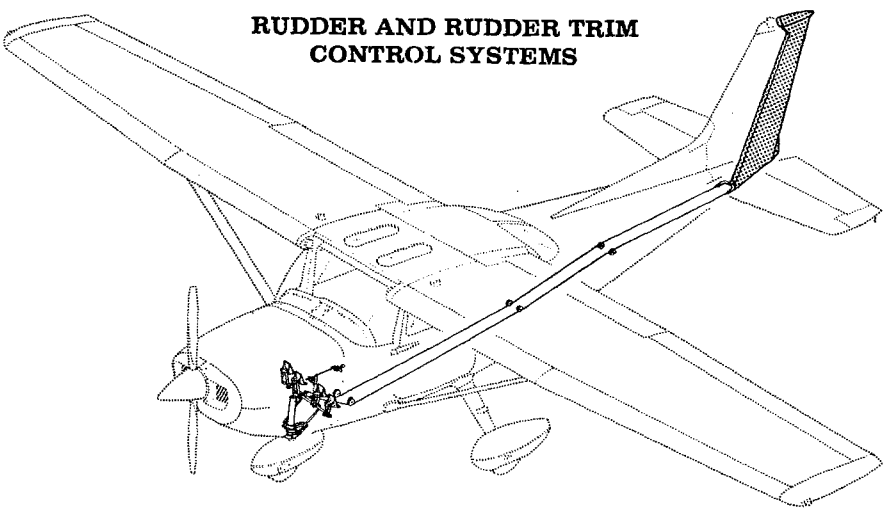




Figure 7-1. Flight Control and Trim Systems (Sheet 1 of 2)

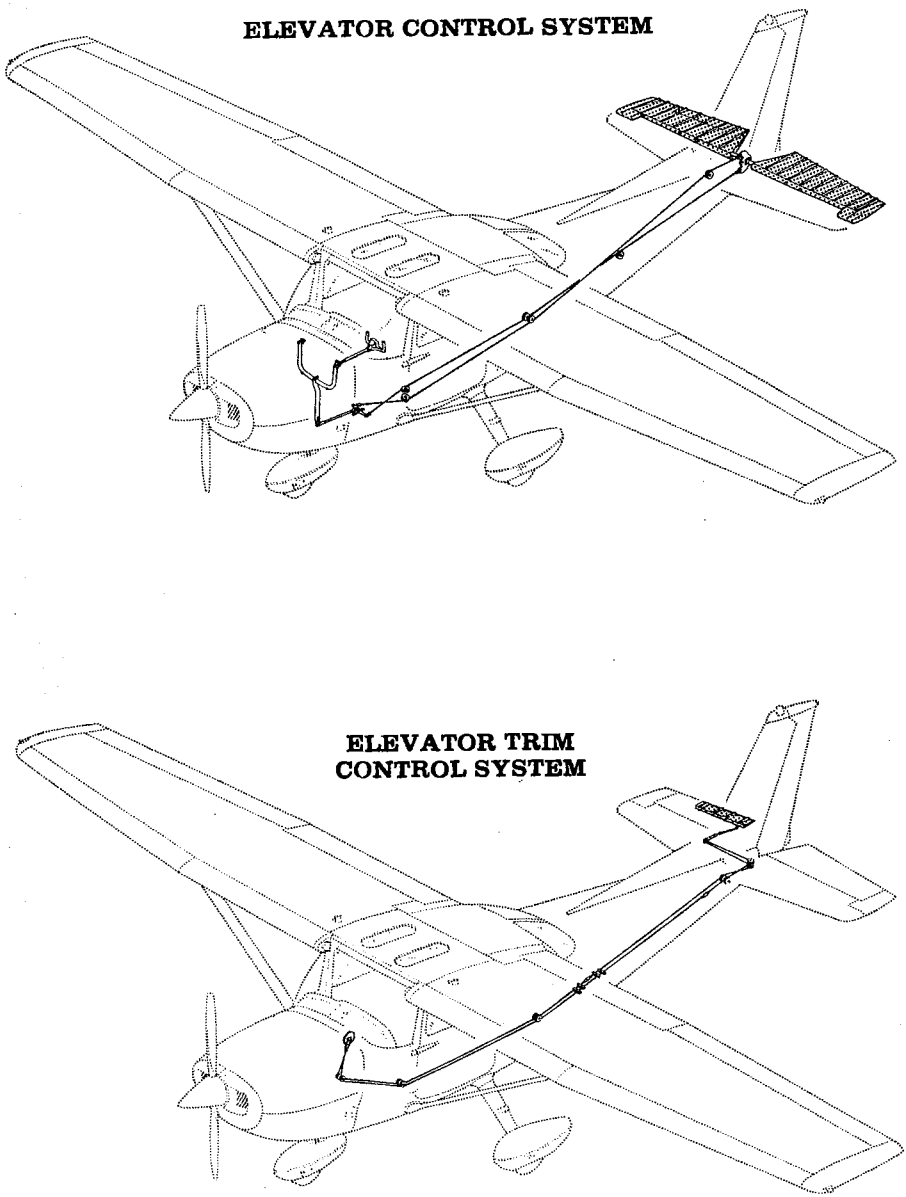



Figure 7-1. Flight Control and Trim Systems (Sheet 2 of 2)

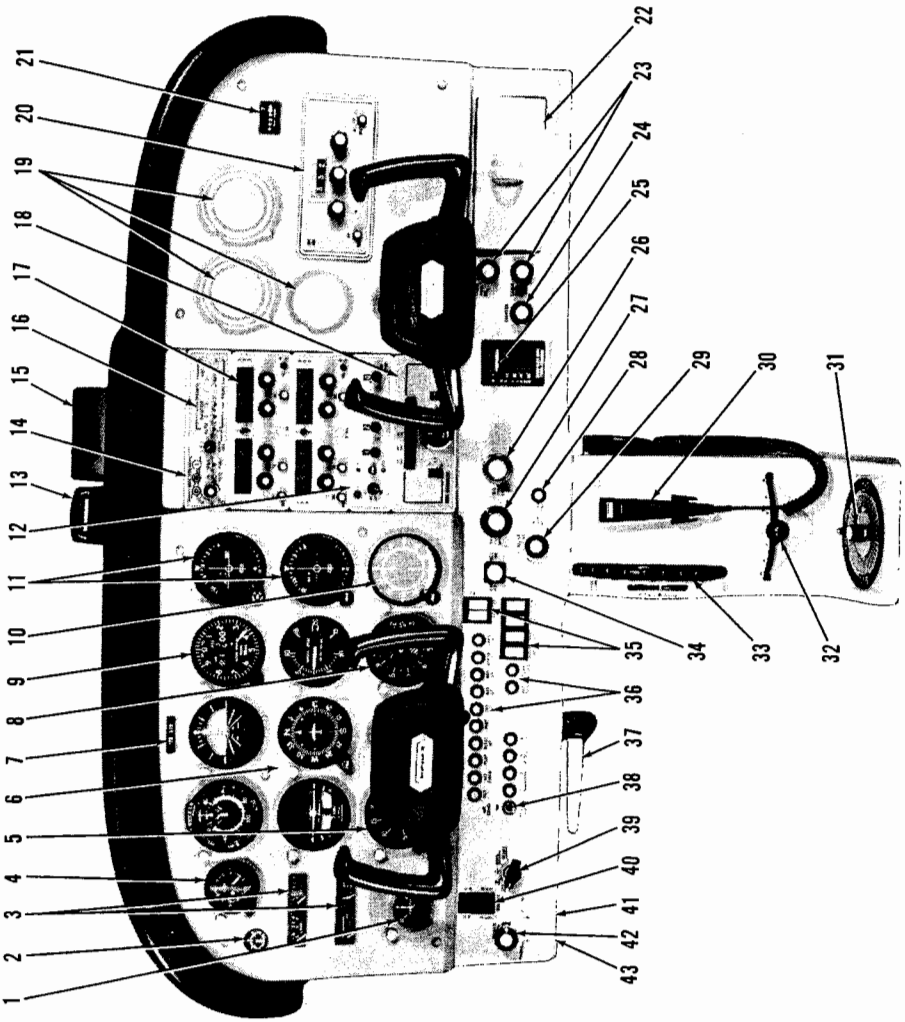



Figure 7-2. Instrument Panel (Sheet 1 of 2)

1. Ammeter
2. Suction Gage
3. Oil Temperature, Oil Pressure, and Left and Right Fuel Quantity Indicators
4. Clock
5. Tachometer
6. Flight Instrument Group
7. Airplane Registration Number
8. Secondary Altimeter
9. Encoding Altimeter
10. ADF Bearing Indicator
11. Omni Course Indicators
12. Transponder
13. Magnetic Compass
14. Marker Beacon Indicator Lights and Switches
15. Rear View Mirror
16. Audio Control Panel
17. Radios
18. Autopilot Control Unit
19. Additional Instrument Space
20. ADF Radio
21. Flight Hour Recorder
22. Map Compartment
23. Cabin Heat and Air Cotnrol Knobs
24. Lighter
25. Wing Flap Switch and Position Indicator
26. Mixture Control Knob
27. Throttle (With Friction Lock)
28. Static Pressure Alternate Source Valve
29. Instrument and Radio Dial Light Rheostat Control Knobs
30. Microphone
31. Fuel Selector Valve Handle
32. Rudder Trim Control Lever
33. Elevator Trim Control Wheel
34. Carburetor Heat Control Knob
35. Electrical Switches
36. Circuit Breakers
37. Parking Brake Handle
38. Avionics Power Switch
39. Ignition Switch
40. Master Switch
41. Auxiliary Mike Jack
42. Primer
43. Phone Jack

Figure 7-2. Instrument Panel (Sheet 2 of 2)

The horizontal stabilizer is constructed of a forward and aft spar, ribs and stiffeners, center, left, and right wrap-around skin panels, and formed leading edge skins. The horizontal stabilizer also contains the elevator trim tab actuator. Construction of the elevator consists of formed leading edge skins, a forward spar, aft channel, ribs, torque tube and bellcrank, left upper and lower "V" type corrugated skins, and right upper and lower "V" type corrugated skins incorporating a trailing edge cut-out for the trim tab. The elevator trim tab consists of a spar, rib, and upper and lower "V" type corrugated skins. The leading edge of both left and right elevator tips incorporate extensions which contain balance weights.

# FLIGHT CONTROLS

The airplane's flight control system (see figure 7-1) consists of conventional aileron, rudder, and elevator control surfaces. The control surfaces are manually operated through mechanical linkage using a control wheel for the ailerons and elevator, and rudder/brake pedals for the rudder.

Extensions are available for the rudder/brake pedals. They consist of a rudder pedal face, two spacers and two spring clips. To install an extension, place the clip on the bottom of the extension under the bottom of the rudder pedal and snap the top clip over the top of the rudder pedal. Check that the extension is firmly in place. To remove the extensions, reverse the above procedures.

## TRIM SYSTEM

A manually-operated elevator trim system is provided; a rudder trim system may also be installed (see figure 7-1). Elevator trimming is accomplished through the elevator trim tab by utilizing the vertically mounted trim control wheel. Forward rotation of the trim wheel will trim nose-down; conversely, aft rotation will trim nose-up. Rudder trimming is accomplished through a bungee connected to the rudder control system and a trim lever, mounted on the control pedestal. Rudder trimming is accomplished by lifting the trim lever up to clear a detent, then moving it either left or right to the desired trim position. Moving the trim lever to the right will trim the airplane nose-right; conversely, moving the lever to the left will trim the airplane nose-left.

# INSTRUMENT PANEL

The instrument panel (see figure 7-2) is designed around the basic "T" configuration. The gyros are located immediately in front of the pilot, and arranged vertically over the control column. The airspeed indicator and

altimeter are located to the left and right of the gyros, respectively. The remainder of the flight instruments are located around the basic "T". Engine instruments and fuel quantity indicators are near the left edge of the panel. Avionics equipment is stacked approximately on the centerline of the panel, with the right side of the panel containing space for additional instruments and avionics equipment. A subpanel under the primary instrument panel contains the primer, master and ignition switches, avionics power switch, circuit breakers, and electrical switches on the left side, with the engine controls, light intensity controls, and alternate static air control in the center, over the control pedestal. The right side of the subpanel contains the wing flap switch lever and position indicator, cabin heat and vent controls, cigar lighter, and map compartment. A pedestal, installed below the subpanel, contains the elevator trim control wheel and position indicator, and provides a bracket for the microphone. A rudder trim control lever may be installed below the trim wheel and microphone bracket, and the fuel selector valve handle is located at the base of the pedestal. A parking brake handle is mounted below the subpanel in front of the pilot.

For details concerning the instruments, switches, circuit breakers, and controls on this panel, refer in this section to the description of the systems to which these items are related.

## GROUND CONTROL

Effective ground control while taxiing is accomplished through nose wheel steering by using the rudder pedals; left rudder pedal to steer left and right rudder pedal to steer right. When a rudder pedal is depressed, a spring-loaded steering bungee (which is connected to the nose gear and to the rudder bars) will turn the nose wheel through an arc of approximately 10° each side of center. By applying either left or right brake, the degree of turn may be increased up to 30° each side of center.

Moving the airplane by hand is most easily accomplished by attaching a tow bar to the nose gear strut. If a tow bar is not available, or pushing is required, use the wing struts as push points. Do not use the vertical or horizontal surfaces to move the airplane. If the airplane is to be towed by vehicle, never turn the nose wheel more than 30° either side of center or structural damage to the nose gear could result.

The minimum turning radius of the airplane, using differential braking and nose wheel steering during taxi, is approximately 27 feet 5 and 1/2 inches. To obtain a minimum radius turn during ground handling, the airplane may be rotated around either main landing gear by pressing down on a tailcone bulkhead just forward of the horizontal stabilizer to raise the nose wheel off the ground.

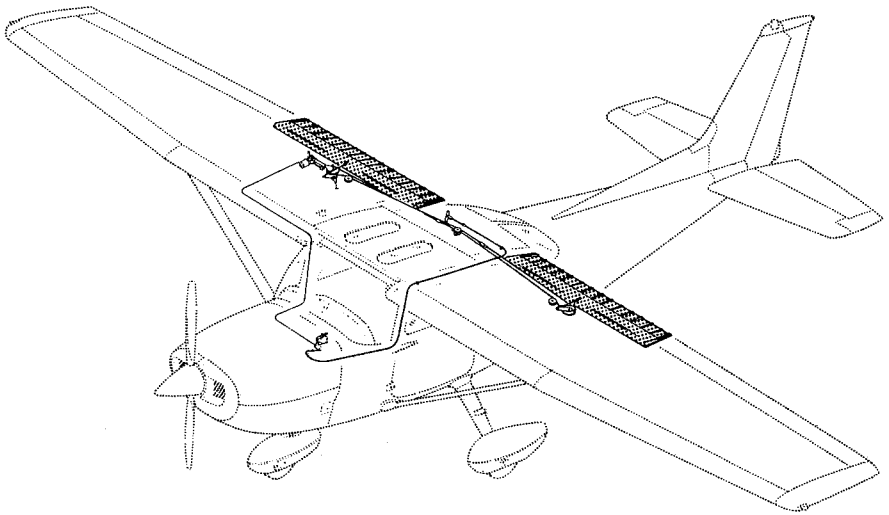


Figure 7-3. Wing Flap System

## WING FLAP SYSTEM

The wing flaps are of the single-slot type (see figure 7-3), and are extended or retracted by positioning the wing flap switch lever on the instrument panel to the desired flap deflection position. The switch lever is moved up or down in a slotted panel that provides mechanical stops at the 10° and 20° positions. For flap settings greater than 10°, move the switch lever to the right to clear the stop and position it as desired. A scale and pointer on the left side of the switch lever indicates flap travel in degrees. The wing flap system circuit is protected by a 15-ampere circuit breaker, labeled FLAP, on the left side of the instrument panel.

## LANDING GEAR SYSTEM

The landing gear is of the tricycle type with a steerable nose wheel, two main wheels, and wheel fairings. Shock absorption is provided by the tubular spring-steel main landing gear struts and the air/oil nose gear shock strut. Each main gear wheel is equipped with a hydraulically actuated disc-type brake on the inboard side of each wheel, and an aerodynamic fairing over each brake.

# BAGGAGE COMPARTMENT

The baggage compartment consists of two areas, one extending from the back of the rear passenger seats to the aft cabin bulkhead, and an additional area aft of the bulkhead. Access to both baggage areas is gained through a lockable baggage door on the left side of the airplane, or from within the airplane cabin. A baggage net with eight tie-down straps is provided for securing baggage and is attached by tying the straps to tie-down rings provided in the airplane. When loading the airplane, children should not be placed or permitted in the baggage compartment, unless a child's seat is installed, and any material that might be hazardous to the airplane or occupants should not be placed anywhere in the airplane. For baggage area and door dimensions, refer to Section 6.

# SEATS

The seating arrangement consists of two separate adjustable seats for the pilot and front passenger, a split-backed fixed seat in the rear, and a child's seat (if installed) aft of the rear seats. The pilot's and front passenger's seats are available in two different designs: four-way and six-way adjustable.

Four-way seats may be moved forward or aft, and the seat back angle changed. To position either seat, lift the tubular handle under the center of the seat, slide the seat into position, release the handle, and check that the seat is locked in place. The seat back is spring-loaded to the vertical position. To adjust its position, lift the lever under the right front corner of the seat, reposition the back, release the lever, and check that the back is locked in place. The seat backs will also fold full forward.

The six-way seats may be moved forward or aft, adjusted for height, and the seat back angle is infinitely adjustable. Position the seat by lifting the tubular handle, under the center of the seat bottom, and slide the seat into position; then release the lever and check that the seat is locked in place. Raise or lower the seat by rotating a large crank under the right corner of the left seat and the left corner of the right seat. Seat back angle is adjustable by rotating a small crank under the left corner of the left seat and the right corner of the right seat. The seat bottom angle will change as the seat back angle changes, providing proper support. The seat backs will also fold full forward.

The rear passenger's seats consist of a fixed one-piece seat bottom with individually adjustable seat backs. Two adjustment levers, under the left and right corners of the seat bottom, are used to adjust the angle of the respective seat backs. To adjust either seat back, lift the adjustment lever

and reposition the back. The seat backs are spring-loaded to the vertical position.

A child's seat may be installed aft of the rear passenger seats, and is held in place by two brackets mounted on the floorboard. The seat is designed to swing upward into a stowed position against the aft cabin bulkhead when not in use. To stow the seat, rotate the seat bottom up and aft as far as it will go. When not in use, the seat should be stowed.

Headrests are available for any of the seat configurations except the child's seat. To adjust the headrest, apply enough pressure to it to raise or lower it to the desired level. The headrest may be removed at any time by raising it until it disengages from the top of the seat back.

## SEAT BELTS AND SHOULDER HARNESSES

All seat positions are equipped with seat belts (see figure 7-4). The pilot's and front passenger's seats are also equipped with separate shoulder harnesses; shoulder harnesses are available for the rear seat positions. Integrated seat belt/shoulder harnesses with inertia reels can be furnished for the pilot's and front passenger's seat positions if desired.

### SEAT BELTS

All of the seat belts are attached to fittings on the floorboard. The buckle half is inboard of each seat and the link half is outboard of each seat.

To use the seat belts for the front seats, position the seat as desired, and then lengthen the link half of the belt as needed by grasping the sides of the link and pulling against the belt. Insert and lock the belt link into the buckle. Tighten the belt to a snug fit. Seat belts for the rear seats and the child's seat (if installed) are used in the same manner as the belts for the front seats. To release the seat belts, grasp the top of the buckle opposite the link and pull outward.

### SHOULDER HARNESSES

Each front seat shoulder harness (see figure 7-4) is attached to a rear doorpost above the window line and is stowed behind a stowage sheath above the cabin door. To stow the harness, fold it and place it behind the sheath. The rear seat shoulder harnesses are attached adjacent to the lower corners of the rear window. Each rear seat harness is stowed behind a

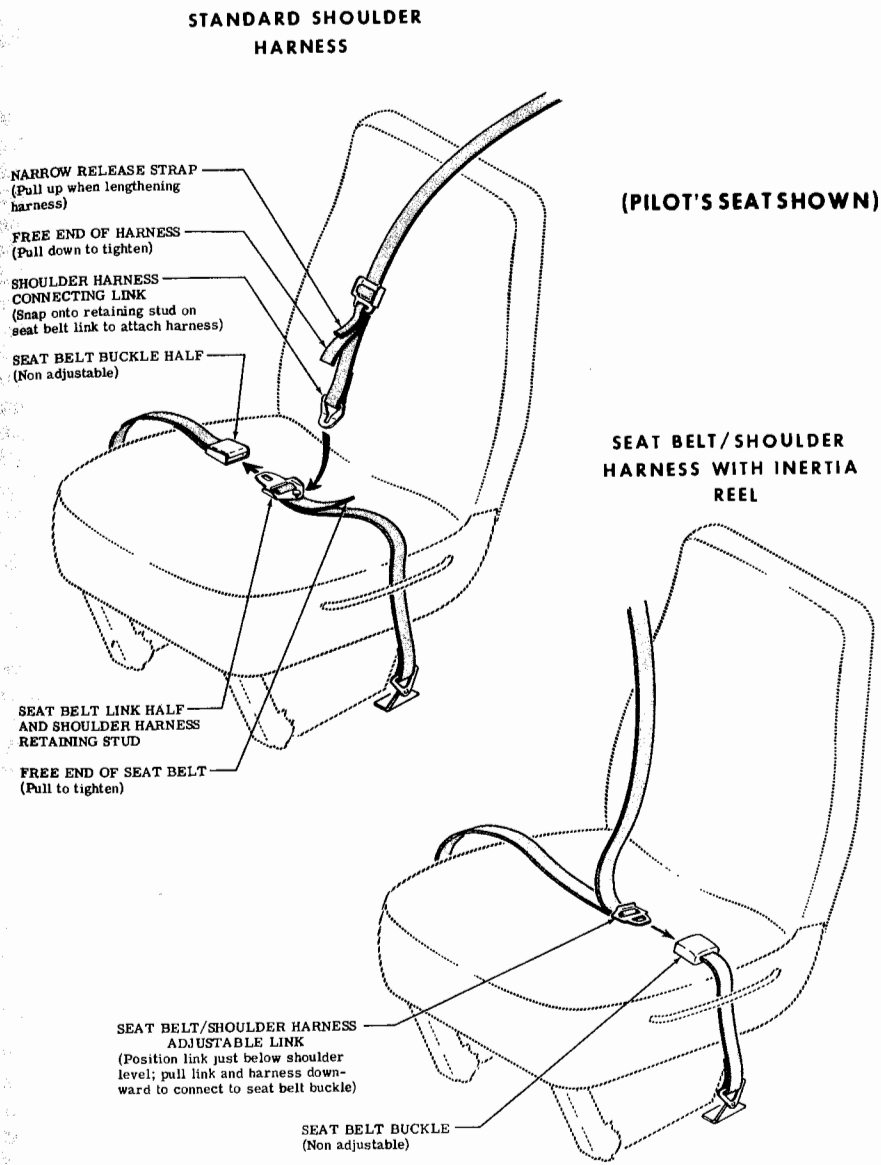



Figure 7-4. Seat Belts and Shoulder Harnesses

stowage sheath above an aft side window. No harness is available for the child's seat.

To use a front or rear seat shoulder harness fasten and adjust the seat belt first. Lengthen the harness as required by pulling on the connecting link on the end of the harness and the narrow release strap. Snap the connecting link firmly onto the retaining stud on the seat belt link half. Then adjust to length. A properly adjusted harness will permit the occupant to lean forward enough to sit completely erect, but prevent excessive forward movement and contact with objects during sudden deceleration. Also, the pilot will want the freedom to reach all controls easily.

Removing the shoulder harness is accomplished by pulling upward on the narrow release strap, and removing the harness connecting link from the stud on the seat belt link. In an emergency, the shoulder harness may be removed by releasing the seat belt first, and allowing the harness, still attached to the link half of the seat belt, to drop to the side of the seat.

## INTEGRATED SEAT BELT/SHOULDER HARNESSES WITH INERTIA REELS

Integrated seat belt/shoulder harnesses with inertia reels are available for the pilot and front seat passenger. The seat belt/shoulder harnesses extend from inertia reels located in the cabin ceiling to attach points inboard of the two front seats. A separate seat belt half and buckle is located outboard of the seats. Inertia reels allow complete freedom of body movement. However, in the event of a sudden deceleration, they will lock automatically to protect the occupants.

NOTE

> The inertia reels are located for maximum shoulder harness comfort and safe retention of the seat occupants. This location requires that the shoulder harnesses cross near the top so that the right hand inertia reel serves the pilot and the left hand reel serves the front passenger. When fastening the harness, check to ensure the proper harness is being used.

To use the seat belt/shoulder harness, position the adjustable metal link on the harness just below shoulder level, pull the link and harness downward, and insert the link into the seat belt buckle. Adjust belt tension across the lap by pulling upward on the shoulder harness. Removal is accomplished by releasing the seat belt buckle, which will allow the inertia reel to pull the harness inboard of the seat.

# ENTRANCE DOORS AND CABIN WINDOWS

Entry to, and exit from the airplane is accomplished through either of two entry doors, one on each side of the cabin at the front seat positions (refer to Section 6 for cabin and cabin door dimensions). The doors incorporate a recessed exterior door handle, a conventional interior door handle, a key-operated door lock (left door only), a door stop mechanism, and an openable window in the left door. An openable right door window is also available.

To open the doors from outside the airplane, utilize the recessed door handle near the aft edge of either door by grasping the forward edge of the handle and pulling outboard. To close or open the doors from inside the airplane, use the combination door handle and arm rest. The inside door handle has three positions and a placard at its base which reads OPEN, CLOSE, and LOCK. The handle is spring-loaded to the CLOSE (up) position. When the door has been pulled shut and latched, lock it by rotating the door handle forward to the LOCK position (flush with the arm rest). When the handle is rotated to the LOCK position, an over-center action will hold it in that position. Both cabin doors should be locked prior to flight, and should not be opened intentionally during flight.

NOTE

Accidental opening of a cabin door in flight due to improper closing does not constitute a need to land the airplane. The best procedure is to set up the airplane in a trimmed condition at approximately 75 knots, momentarily shove the door outward slightly, and forcefully close and lock the door.

Exit from the airplane is accomplished by rotating the door handle from the LOCK position, past the CLOSE position, aft to the OPEN position and pushing the door open. To lock the airplane, lock the right cabin door with the inside handle, close the left cabin door, and using the ignition key, lock the door.

The left cabin door is equipped with an openable window which is held in the closed position by a detent equipped latch on the lower edge of the window frame. To open the window, rotate the latch upward. The window is equipped with a spring-loaded retaining arm which will help rotate the window outward, and hold it there. An openable window is also available for the right door, and functions in the same manner as the left window. If required, either window may be opened at any speed up to 160 knots. The cabin top windows (if installed), rear side windows, and rear windows are of the fixed type and cannot be opened.

# CONTROL LOCKS

A control lock is provided to lock the ailerons and elevator control surfaces in a neutral position and prevent damage to these systems by wind buffeting while the airplane is parked. The lock consists of a shaped steel rod with a red metal flag attached to it. The flag is labeled CONTROL LOCK, REMOVE BEFORE STARTING ENGINE. To install the control lock, align the hole in the top of the pilot's control wheel shaft with the hole in the top of the shaft collar on the instrument panel and insert the rod into the aligned holes. Proper installation of the lock will place the red flag over the ignition switch. In areas where high or gusty winds occur, a control surface lock should be installed over the vertical stabilizer and rudder. The control lock and any other type of locking device should be removed prior to starting the engine.

# ENGINE

The airplane is powered by a horizontally-opposed, four-cylinder, overhead-valve, air-cooled, carbureted engine with a wet sump oil system. The engine is a Lycoming Model O-320-H2AD and is rated at 160 horsepower at 2700 RPM. Major accessories include a starter and belt-driven alternator mounted on the front of the engine, and dual magnetos and a vacuum pump which are mounted on an accessory drive pad on the rear of the engine. Provisions are also made for a full flow oil filter.

## ENGINE CONTROLS

Engine power is controlled by a throttle located on the lower center portion of the instrument panel. The throttle operates in a conventional manner; in the full forward position, the throttle is open, and in the full aft position, it is closed. A friction lock, which is a round knurled disk, is located at the base of the throttle and is operated by rotating the lock clockwise to increase friction or counterclockwise to decrease it.

The mixture control, mounted above the right corner of the control pedestal, is a red knob with raised points around the circumference and is equipped with a lock button in the end of the knob. The rich position is full forward, and full aft is the idle cut-off position. For small adjustments, the control may be moved forward by rotating the knob clockwise, and aft by rotating the knob counterclockwise. For rapid or large adjustments, the knob may be moved forward or aft by depressing the lock button in the end of the control, and then positioning the control as desired.

## ENGINE INSTRUMENTS

Engine operation is monitored by the following instruments: oil pressure gage, oil temperature gage, and a tachometer. A carburetor air temperature gage is also available.

The oil pressure gage, located on the left side of the instrument panel, is operated by oil pressure. A direct pressure oil line from the engine delivers oil at engine operating pressure to the oil pressure gage. Gage markings indicate that minimum idling pressure is 25 PSI (red line), the normal operating range is 60 to 90 PSI (green arc), and maximum pressure is 100 PSI (red line).

Oil temperature is indicated by a gage adjacent to the oil pressure gage. The gage is operated by an electrical-resistance type temperature sensor which receives power from the airplane electrical system. Oil temperature limitations are the normal operating range (green arc) which is 38°C (100°F) to 118°C (245°F), and the maximum (red line) which is 118°C (245°F).

The engine-driven mechanical tachometer is located near the lower portion of the instrument panel to the left of the pilot's control wheel. The instrument is calibrated in increments of 100 RPM and indicates both engine and propeller speed. An hour meter below the center of the tachometer dial records elapsed engine time in hours and tenths. Instrument markings include a normal operating range (green arc) of 2200 to 2700 RPM, and a maximum (red line) of 2700 RPM.

A carburetor air temperature gage may be installed on the right side of the instrument panel to help detect carburetor icing conditions. The gage is marked in 5° increments from -30°C to +30°C, and has a yellow arc between -15°C and +5°C which indicates the temperature range most conducive to icing in the carburetor. A placard on the lower half of the gage face reads KEEP NEEDLE OUT OF YELLOW ARC DURING POSSIBLE CARBURETOR ICING CONDITIONS.

## NEW ENGINE BREAK-IN AND OPERATION

The engine underwent a run-in at the factory and is ready for the full range of use. It is, however, suggested that cruising be accomplished at 65% to 75% power until a total of 50 hours has accumulated or oil consumption has stabilized. This will ensure proper seating of the rings.

The airplane is delivered from the factory with corrosion preventive oil in the engine. If, during the first 25 hours, oil must be added, use only aviation grade straight mineral oil conforming to Specification No. MIL-L-6082.

## ENGINE OIL SYSTEM

Oil for engine lubrication is supplied from a sump on the bottom of the engine. The capacity of the engine sump is six quarts (one additional quart is required if a full flow oil filter is installed). Oil is drawn from the sump through an oil suction strainer screen into the engine-driven oil pump. From the pump, oil is routed to a bypass valve. If the oil is cold, the bypass valve allows the oil to bypass the oil cooler and go directly from the pump to the oil pressure screen (full flow oil filter if installed). If the oil is hot, the bypass valve routes the oil out of the accessory housing and into a flexible hose leading to the oil cooler on the lower right side of the firewall. Pressure oil from the cooler returns to the accessory housing where it passes through the pressure strainer screen (full flow oil filter, if installed). The filter oil then enters a pressure relief valve which regulates engine oil pressure by allowing excessive oil to return to the sump while the balance of the oil is circulated to various engine parts for lubrication. Residual oil is returned sump by gravity flow.

An oil filler cap/oil dipstick is located at the rear of the engine near the center. The filler cap/dipstick is accessible through an access door in the engine cowling. The engine should not be operated on less than four quarts of oil. To minimize loss of oil through the breather, fill to five quarts for normal flights of less than three hours. For extended flight, fill to six quarts (dipstick indication only). For engine oil grade and specifications, refer to Section 8 of this handbook.

An oil quick-drain valve is available to replace the drain plug on the bottom of the oil sump, and provides quicker, cleaner draining of the engine oil. To drain the oil with this valve, slip a hose over the end of the valve and push upward on the end of the valve until it snaps into the open position. Spring clips will hold the valve open. After draining, use a suitable tool to snap the valve into the extended (closed) position and remove the drain hose.

## IGNITION-STARTER SYSTEM

Engine ignition is provided by an engine-driven dual magneto, and two spark plugs in each cylinder. The right magneto fires the lower right and upper left spark plugs, and the left magneto fires the lower left and upper right spark plugs. Normal operation is conducted with both magnetos due to the more complete burning of the fuel-air mixture with dual ignition.

Ignition and starter operation is controlled by a rotary type switch located on the left switch and control panel. The switch is labeled clockwise, OFF, R, L, BOTH, and START. The engine should be operated on both magnetos (BOTH position) except for magneto checks. The R and L

positions are for checking purposes and emergency use only. When the switch is rotated to the spring-loaded START position, (with the master switch in the ON position), the starter contactor is energized and the starter will crank the engine. When the switch is released, it will automatically return to the BOTH position.

## AIR INDUCTION

The engine air induction system receives ram air through an intake in the lower front portion of the engine cowling. The intake is covered by an air filter which removes dust and other foreign matter from the induction air. Airflow passing through the filter enters an airbox. After passing through the airbox, induction air enters the inlet in the carburetor which is under the engine, and is then ducted to the engine cylinders through intake manifold tubes. In the event carburetor ice is encountered or the intake filter becomes blocked, alternate heated air can be obtained from a shroud around an exhaust riser through a duct to a valve, in the airbox, operated by the carburetor heat control on the instrument panel. Heated air from the shroud is obtained from an unfiltered outside source. Use of full carburetor heat at full throttle will result in a loss of approximately 100 to 225 RPM.

## EXHAUST SYSTEM

Exhaust gas from each cylinder passes through riser assemblies to a muffler and tailpipe. The muffler is constructed with a shroud around the outside which forms a heating chamber for cabin heater air.

## CARBURETOR AND PRIMING SYSTEM

The engine is equipped with an up-draft, float-type, fixed jet carburetor mounted on the bottom of the engine. The carburetor is equipped with an enclosed accelerator pump, simplified fuel passages to prevent vapor locking, an idle cut-off mechanism, and a manual mixture control. Fuel is delivered to the carburetor by gravity flow from the fuel system. In the carburetor, fuel is atomized, proportionally mixed with intake air, and delivered to the cylinders through intake manifold tubes. The proportion of atomized fuel to air is controlled, within limits, by the mixture control on the instrument panel.

For easy starting in cold weather, the engine is equipped with a manual primer. The primer is actually a small pump which draws fuel from the fuel strainer when the plunger is pulled out, and injects it into the cylinder intake ports when the plunger is pushed back in. The plunger knob, on the instrument panel, is equipped with a lock and, after being pushed full in, must be rotated either left or right until the knob cannot be pulled out.

## COOLING SYSTEM

Ram air for engine cooling enters through two intake openings in the front of the engine cowling. The cooling air is directed around the cylinders and other areas of the engine by baffling, and is then exhausted through an opening at the bottom aft edge of the cowling. No manual cooling system control is provided.

A winterization kit is available and consists of two baffles which attach to the air intakes in the cowling nose cap, a restrictive cover plate for the oil cooler air inlet in the right rear vertical engine baffle, insulation for the crankcase breather line, and a placard to be installed on the instrument panel. This equipment should be installed for operations in temperatures consistently below -7°C (20°F). Once installed, the crankcase breather insulation is approved for permanent use in both hot and cold weather.

# PROPELLER

The airplane is equipped with a two-bladed, fixed-pitch, one-piece forged aluminum alloy propeller which is anodized to retard corrosion. The propeller is 75 inches in diameter.

# FUEL SYSTEM

The airplane may be equipped with either a standard fuel system or long range system (see figure 7-6). Both systems consist of two vented fuel tanks (one in each wing), a four-position selector valve, fuel strainer, manual primer, and carburetor. Refer to figure 7-5 for fuel quantity data for both systems.

| FUEL QUANTITY DATA (U. S. GALLONS) | | | |
|---|---|---|---|
| TANKS | TOTAL USABLE FUEL ALL FLIGHT CONDITIONS | TOTAL UNUSABLE FUEL | TOTAL FUEL VOLUME |
| STANDARD (21.5 Gal. Each) | 40 | 3 | 43 |
| LONG RANGE (27 Gal. Each) | 50 | 4 | 54 |

Figure 7-5. Fuel Quantity Data

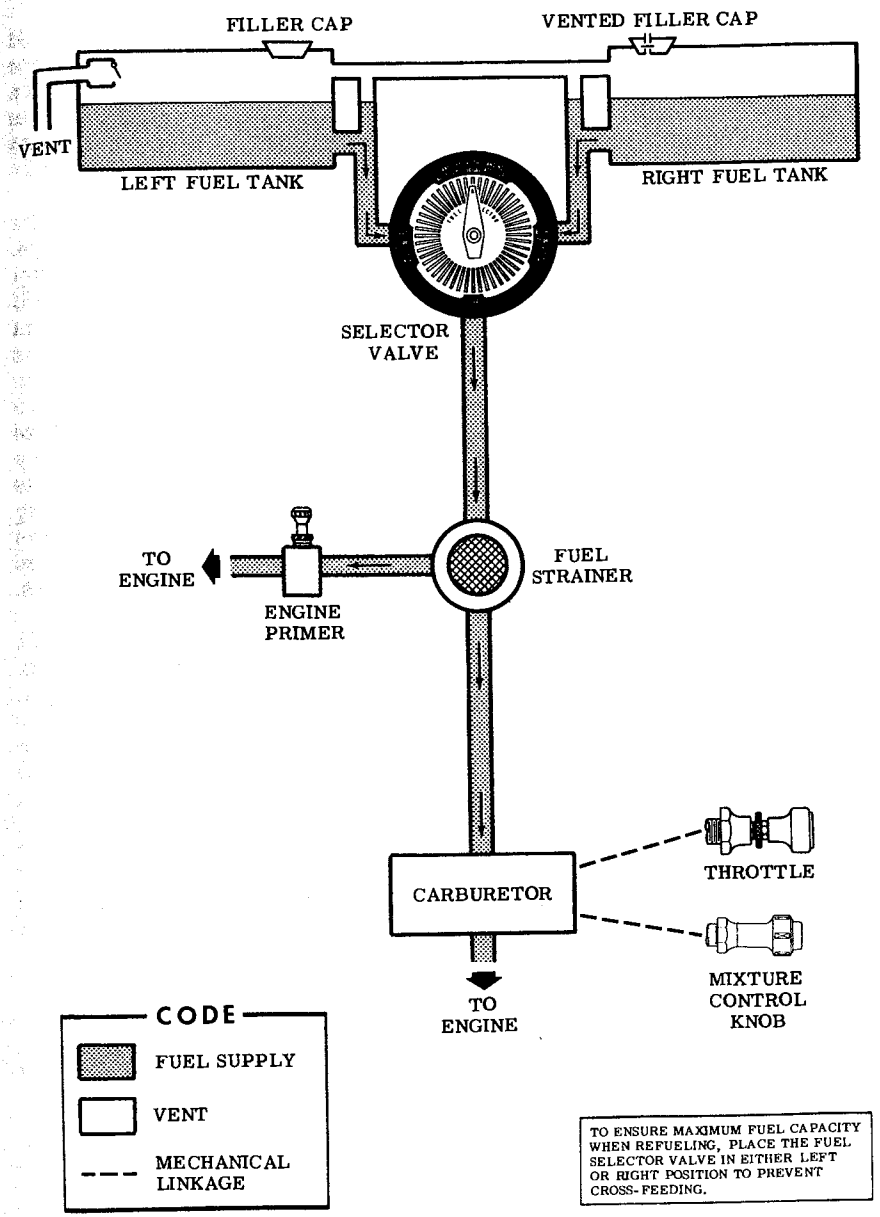



Figure 7-6. Fuel System (Standard and Long Range)

Fuel flows by gravity from the two wing tanks to a four-position selector valve, labeled BOTH, RIGHT, LEFT, and OFF. With the selector valve in either the BOTH, LEFT, or RIGHT position, fuel flows through a strainer to the carburetor. From the carburetor, mixed fuel and air flows to the cylinders through intake manifold tubes. The manual primer draws its fuel from the fuel strainer and injects it into the cylinder intake ports.

Fuel system venting is essential to system operation. Blockage of the system will result in decreasing fuel flow and eventual engine stoppage. Venting is accomplished by an interconnecting line from the right fuel tank to the left tank. The left fuel tank is vented overboard through a vent line, equipped with a check valve, which protrudes from the bottom surface of the left wing near the wing strut. The right fuel tank filler cap is also vented.

Fuel quantity is measured by two float-type fuel quantity transmitters (one in each tank) and indicated by two electrically-operated fuel quantity indicators on the left side of the instrument panel. An empty tank is indicated by a red line and the letter E. When an indicator shows an empty tank, approximately 1.5 gallons remain in a standard tank, and 2 gallons remain in a long range tank as unusuable fuel. The indicators cannot be relied upon for accurate readings during skids, slips, or unusual attitudes.

The fuel selector valve should be in the BOTH position for takeoff, climb, landing, and maneuvers that involve prolonged slips or skids. Operation from either LEFT or RIGHT tank is reserved for cruising flight.

NOTE

When the fuel selector valve handle is in the BOTH position in cruising flight, unequal fuel flow from each tank may occur if the wings are not maintained exactly level. Resulting wing heaviness can be alleviated gradually by turning the selector valve handle to the tank in the "heavy" wing.

NOTE

It is not practical to measure the time required to consume all of the fuel in one tank, and, after switching to the opposite tank, expect an equal duration from the remaining fuel. The airspace in both fuel tanks is interconnected by a vent line and, therefore, some sloshing of fuel between tanks can be expected when the tanks are nearly full and the wings are not level.

The fuel system is equipped with drain valves to provide a means for

the examination of fuel in the system for contamination and grade. The system should be examined before the first flight of every day and after each refueling, by using the sampler cup provided to drain fuel from the wing tank sumps, and by utilizing the fuel strainer drain under an access panel on the right side of the engine cowling. The fuel tanks should be filled after each flight to prevent condensation.

## BRAKE SYSTEM

The airplane has a single-disc, hydraulically-actuated brake on each main landing gear wheel. Each brake is connected, by a hydraulic line, to a master cylinder attached to each of the pilot's rudder pedals. The brakes are operated by applying pressure to the top of either the left (pilot's) or right (copilot's) set of rudder pedals, which are interconnected. When the airplane is parked, both main wheel brakes may be set by utilizing the parking brake which is operated by a handle under the left side of the instrument panel. To apply the parking brake, set the brakes with the rudder pedals, pull the handle aft, and rotate it 90° down.

For maximum brake life, keep the brake system properly maintained, and minimize brake usage during taxi operations and landings.

Some of the symptoms of impending brake failure are: gradual decrease in braking action after brake application, noisy or dragging brakes, soft or spongy pedals, and excessive travel and weak braking action. If any of these symptoms appear, the brake system is in need of immediate attention. If, during taxi or landing roll, braking action decreases, let up on the pedals and then re-apply the brakes with heavy pressure. If the brakes become spongy or pedal travel increases, pumping the pedals should build braking pressure. If one brake becomes weak or fails, use the other brake sparingly while using opposite rudder, as required, to offset the good brake.

## ELECTRICAL SYSTEM

Electrical energy (see figure 7-7) is supplied by a 28-volt, direct-current system powered by an engine-driven, 60-amp alternator and a 24-volt, 14-amp hour battery (or 17-amp hour battery, if installed) located on the left side of the firewall. Power is supplied to most general electrical and all avionics circuits through the primary bus bar and the avionics bus bar, which are interconnected by an avionics power switch. The primary bus is

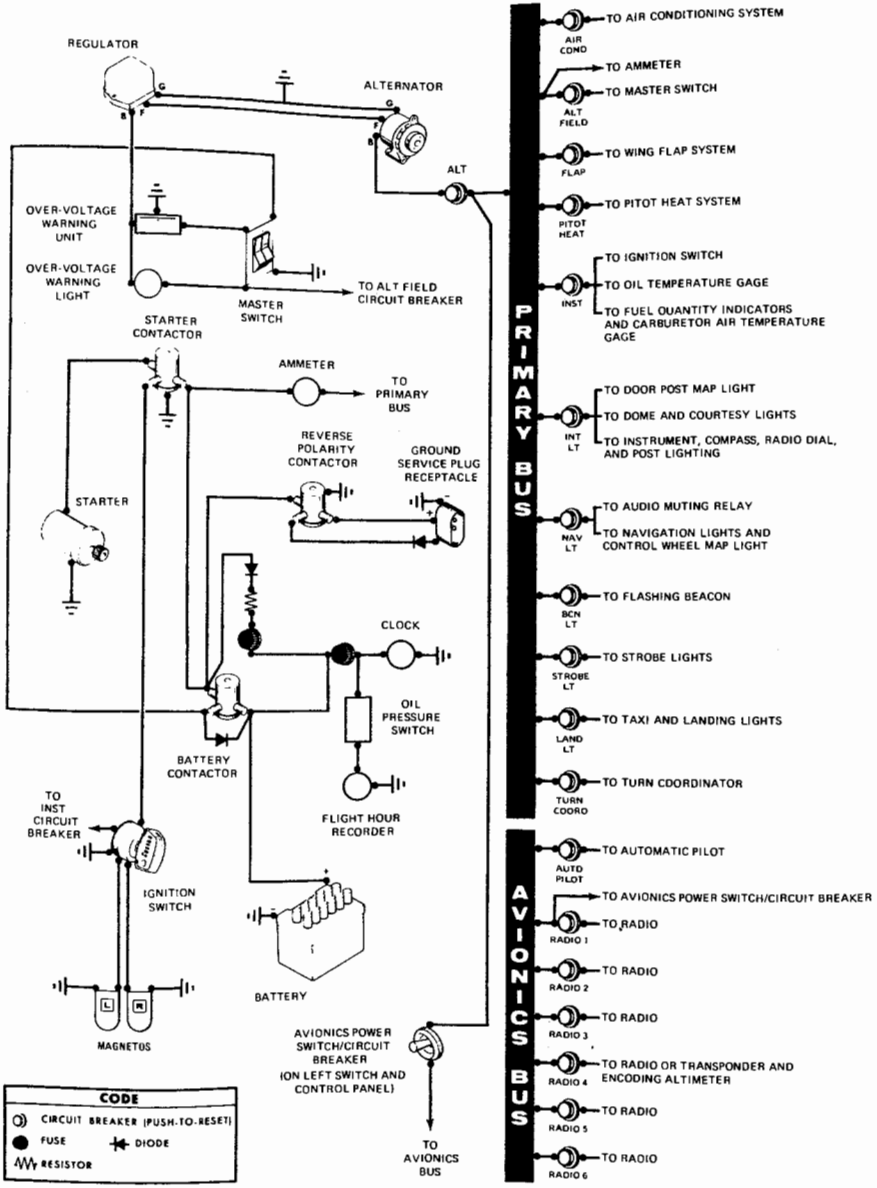



Figure 7-7. Electrical System

on anytime the master switch is turned on, and is not affected by starter or external power usage. Both bus bars are on anytime the master and avionics power switches are turned on.

**CAUTION**

Prior to turning the master switch on or off, starting the engine or applying an external power source, the avionics power switch, labeled AVIONICS POWER, should be turned off to prevent any harmful transient voltage from damaging the avionics equipment.

## MASTER SWITCH

The master switch is a split-rocker type switch labeled MASTER, and is ON in the up position and off in the down position. The right half of the switch, labeled BAT, controls all electrical power to the airplane. The left half, labeled ALT, controls the alternator.

Normally, both sides of the master switch should be used simultaneously; however, the BAT side of the switch could be turned on separately to check equipment while on the ground. To check or use avionics equipment or radios while on the ground, the avionics power switch must also be turned on. The ALT side of the switch, when placed in the off position, removes the alternator from the electrical system. With this switch in the off position, the entire electrical load is placed on the battery. Continued operation with the alternator switch in the off position will reduce battery power low enough to open the battery contactor, remove power from the alternator field, and prevent alternator restart.

## AVIONICS POWER SWITCH

Electrical power from the airplane primary bus to the avionics bus (see figure 7-7) is controlled by a toggle-type circuit breaker-switch labeled AVIONICS POWER. The switch is located on the left side of the switch and control panel and is on in the up position and off in the down position. With the switch in the off position, no electrical power will be applied to the avionics equipment, regardless of the position of the master switch or the individual equipment switches. The avionics power switch also functions as a circuit breaker. If an electrical malfunction should occur and cause the circuit breaker to open, electrical power to the avionics equipment will be interrupted and the switch toggle will automatically move to the off position. If this occurs, allow the circuit breaker approximately two minutes to cool before placing the toggle in the on position again. If the circuit breaker opens again, do not reset it. The avionics power switch should be placed in the off position prior to turning the master switch on or off, starting the engine, or applying an external power source,

and may be utilized in place of the individual avionics equipment switches.

## AMMETER

The ammeter indicates the flow of current, in amperes, from the alternator to the battery or from the battery to the airplane electrical system. When the engine is operating and the master switch is turned on, the ammeter indicates the charging rate applied to the battery. In the event the alternator is not functioning or the electrical load exceeds the output of the alternator, the ammeter indicates the battery discharge rate.

## OVER-VOLTAGE SENSOR AND WARNING LIGHT

The airplane is equipped with an automatic over-voltage protection system consisting of an over-voltage sensor behind the instrument panel and a red warning light, labeled HIGH VOLTAGE, adjacent to the ammeter.

In the event an over-voltage condition occurs, the over-voltage sensor automatically removes alternator field current and shuts down the alternator. The red warning light will then turn on, indicating to the pilot that the alternator is not operating and the battery is supplying all electrical power.

The over-voltage sensor may be reset by turning off the avionics power switch and then turning the master switch off and back on again. If the warning light does not illuminate, normal alternator charging has resumed; however, if the light does illuminate again, a malfunction has occurred, and the flight should be terminated as soon as practical. In either case, the avionics power switch may be turned on again if required.

The warning light may be tested by momentarily turning off the ALT portion of the master switch and leaving the BAT portion turned on.

## CIRCUIT BREAKERS AND FUSES

Most of the electrical circuits in the airplane are protected by "push-to-reset" circuit breakers mounted on the lower left side of the instrument panel. In addition to the individual circuit breakers, a toggle-type circuit breaker-switch, labeled AVIONICS POWER, on the left switch and control panel also protects the avionics systems. The cigar lighter is protected by a manually-reset type circuit breaker on the back of the lighter, and a fuse behind the instrument panel. The control wheel map light (if installed) is protected by the NAV LT circuit breaker and a fuse behind the instrument panel. Electrical circuits which are not protected by circuit breakers are the battery contactor closing (external power) circuit, clock circuit, and

flight hour recorder circuit. These circuits are protected by fuses mounted adjacent to the battery.

### GROUND SERVICE PLUG RECEPTACLE

A ground service plug receptacle may be installed to permit the use of an external power source (generator type or battery cart) for cold weather starting and during lengthy maintenance work on the airplane electrical system. The receptacle is located under a cover plate, on the lower left side of the cowling.

NOTE

If no avionics equipment is to be used or worked on, the avionics power switch should be turned off. If maintenance is required on the avionics equipment, it is advisable to utilize a battery cart external power source to prevent damage to the avionics equipment by transient voltage. Do not crank or start the engine with the avionics power switch turned on.

Just before connecting an external power source (generator type or battery cart), the avionics power switch should be turned off, and the master switch on.

The ground service plug receptacle circuit incorporates a polarity reversal protection. Power from the external power source will flow only if the ground service plug is correctly connected to the airplane. If the plug is accidentally connected backwards, no power will flow to the electrical system, thereby preventing any damage to electrical equipment.

The battery and external power circuits have been designed to completely eliminate the need to "jumper" across the battery contactor to close it for charging a completely "dead" battery. A special fused circuit in the external power system supplies the needed "jumper" across the contacts so that with a "dead" battery and an external power source applied, turning on the master switch will close the battery contactor.

## LIGHTING SYSTEMS

### EXTERIOR LIGHTING

Conventional navigation lights are located on the wing tips and top of

the rudder. A single landing light or dual landing/taxi lights are installed in the cowl nose cap, and a flashing beacon is mounted on top of the vertical fin. Additional lighting is available and includes a strobe light on each wing tip and two courtesy lights, one under each wing, just outboard of the cabin door. The courtesy lights are operated by the dome light switch on the overhead console. All exterior lights, except the courtesy lights, are controlled by rocker type switches on the left switch and control panel. The switches are ON in the up position and OFF in the down position.

The flashing beacon should not be used when flying through clouds or overcast; the flashing light reflected from water droplets or particles in the atmosphere, particularly at night, can produce vertigo and loss of orientation.

The high intensity strobe lights will enhance anti-collision protection. However, the lights should be turned off when taxiing in the vicinity of other aircraft, or during night flight through clouds, fog or haze.

## INTERIOR LIGHTING

Instrument and control panel lighting is provided by flood lighting, integral lighting, and post lighting (if installed). Two concentric rheostat control knobs below the engine controls, labeled PANEL LT and RADIO LT, control intensity of the instrument and control panel lighting. A slide-type switch (if installed) on the overhead console, labeled PANEL LIGHTS, is used to select flood lighting in the FLOOD position, post lighting in the POST position, or a combination of post and flood lighting in the BOTH position.

Instrument and control panel flood lighting consists of a single red flood light in the forward part of the overhead console. To use the flood lighting, rotate the PANEL LT rheostat control knob clockwise to the desired intensity.

The instrument panel may be equipped with post lights which are mounted at the edge of each instrument or control and provide direct lighting. The lights are operated by placing the PANEL LIGHTS selector switch in the POST position and adjusting light intensity with the PANEL LT rheostat control knob. By placing the PANEL LIGHTS selector switch in the BOTH position, the post lights can be used in combination with the standard flood lighting.

The engine instrument cluster (if post lighting is installed), radio equipment, and magnetic compass have integral lighting and operate independently of post or flood lighting. Light intensity of the engine instrument cluster and radio lighting is controlled by the RADIO LT

rheostat control knob. The integral compass light intensity is controlled by the PANEL LT rheostat control knob.

A cabin dome light, in the aft part of the overhead console, is operated by a switch near the light. To turn the light on, move the switch to the right.

A control wheel map light is available and is mounted on the bottom of the pilot's control wheel. The light illuminates the lower portion of the cabin just forward of the pilot and is helpful when checking maps and other flight data during night operations. To operate the light, first turn on the NAV LT switch; then adjust the map light's intensity with the knurled disk type rheostat control located at the bottom of the control wheel.

A doorpost map light is available, and is located on the left forward doorpost. It contains both red and white bulbs and may be positioned to illuminate any area desired by the pilot. The light is controlled by a switch, below the light, which is labeled RED, OFF, and WHITE. Placing the switch in the top position will provide a red light. In the bottom position, standard white lighting is provided. In the center position, the map light is turned off. Light intensity is controlled by the PANEL LT rheostat control knob.

The most probable cause of a light failure is a burned out bulb; however, in the event any of the lighting systems fail to illuminate when turned on, check the appropriate circuit breaker. If the circuit breaker has opened (white button popped out), and there is no obvious indication of a short circuit (smoke or odor), turn off the light switch of the affected lights, reset the breaker, and turn the switch on again. If the breaker opens again, do not reset it.

## CABIN HEATING, VENTILATING AND DEFROSTING SYSTEM

The temperature and volume of airflow into the cabin can be regulated to any degree desired by manipulation of the push-pull CABIN HT and CABIN AIR control knobs (see figure 7-8).

For cabin ventilation, pull the CABIN AIR knob out. To raise the air temperature, pull the CABIN HT knob out approximately 1/4 to 1/2 inch for a small amount of cabin heat. Additional heat is available by pulling the knob out farther; maximum heat is available with the CABIN HT knob pulled out and the CABIN AIR knob pushed full in. When no heat is desired in the cabin, the CABIN HT knob is pushed full in.

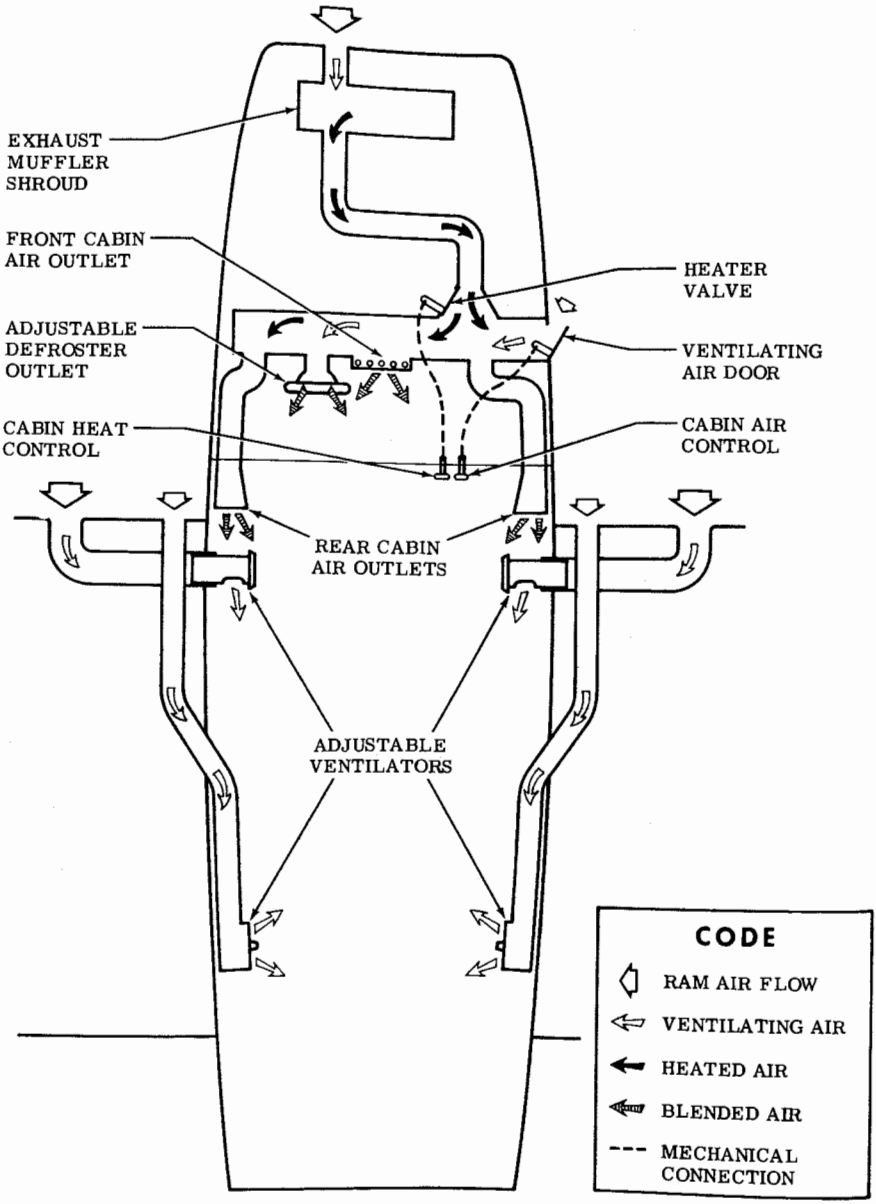



Figure 7-8. Cabin Heating, Ventilating, and Defrosting System

Front cabin heat and ventilating air is supplied by outlet holes spaced across a cabin manifold just forward of the pilot's and copilot's feet. Rear cabin heat and air is supplied by two ducts from the manifold, one extending down each side of the cabin to an outlet at the front doorpost at floor level. Windshield defrost air is also supplied by a duct leading from the cabin manifold. Two knobs control sliding valves in the defroster outlet and permit regulation of defroster airflow.

Separate adjustable ventilators supply additional air; one near each upper corner of the windshield supplies air for the pilot and copilot, and two ventilators are available for the rear cabin area to supply air to the rear seat passengers. The airplane may also be equipped with an air conditioning system. For operating instructions and details concerning this system, refer to Section 9, Supplements.

## PITOT-STATIC SYSTEM AND INSTRUMENTS

The pitot-static system supplies ram air pressure to the airspeed indicator and static pressure to the airspeed indicator, rate-of-climb indicator and altimeter. The system is composed of either an unheated or heated pitot tube mounted on the lower surface of the left wing, an external static port on the lower left side of the forward fuselage, and the associated plumbing necessary to connect the instruments to the sources.

The heated pitot system consists of a heating element in the pitot tube, a rocker-type switch labeled PITOT HT on the lower left side of the instrument panel, a 5-amp circuit breaker on the switch and control panel, and associated wiring. When the pitot heat switch is turned on, the element in the pitot tube is heated electrically to maintain proper operation in possible icing conditions. Pitot heat should be used only as required.

A static pressure alternate source valve may be installed adjacent to the throttle, and can be used if the external static source is malfunctioning. This valve supplies static pressure from inside the cabin instead of the external static port.

If erroneous instrument readings are suspected due to water or ice in the pressure line going to the standard external static pressure source, the alternate static source valve should be pulled on.

Pressures within the cabin will vary with open cabin ventilators and windows. Refer to Section 5 for the effect of varying cabin pressures on airspeed and altimeter readings.

### AIRSPEED INDICATOR

The airspeed indicator is calibrated in knots and miles per hour. Limitation and range markings include the white arc (41 to 85 knots), green arc (47 to 128 knots), yellow arc (128 to 160 knots), and a red line (160 knots).

If a true airspeed indicator is installed, it is equipped with a rotatable ring which works in conjunction with the airspeed indicator dial in a manner similar to the operation of a flight computer. To operate the indicator, first rotate the ring until **pressure** altitude is aligned with outside air temperature in degrees Fahrenheit. Pressure altitude should not be confused with indicated altitude. To obtain pressure altitude, momentarily set the barometric scale on the altimeter to 29.92 and read pressure altitude on the altimeter. Be sure to return the altimeter barometric scale to the original barometric setting after pressure altitude has been obtained. Having set the ring to correct for altitude and temperature, read the true airspeed shown on the rotatable ring by the indicator pointer. For best accuracy, the indicated airspeed should be corrected to calibrated airspeed by referring to the Airspeed Calibration chart in Section 5. Knowing the calibrated airspeed, read true airspeed on the ring opposite the calibrated airspeed.

### RATE-OF-CLIMB INDICATOR

The rate-of-climb indicator depicts airplane rate of climb or descent in feet per minute. The pointer is actuated by atmospheric pressure changes resulting from changes of altitude as supplied by the static source.

### ALTIMETER

Airplane altitude is depicted by a barometric type altimeter. A knob near the lower left portion of the indicator provides adjustment of the instrument's barometric scale to the current altimeter setting.

## VACUUM SYSTEM AND INSTRUMENTS

An engine-driven vacuum system (see figure 7-9) provides the suction necessary to operate the attitude indicator and directional indicator. The system consists of a vacuum pump mounted on the engine, a vacuum relief valve and vacuum system air filter on the aft side of the firewall below the instrument panel, and instruments (including a suction gage) on the left side of the instrument panel.

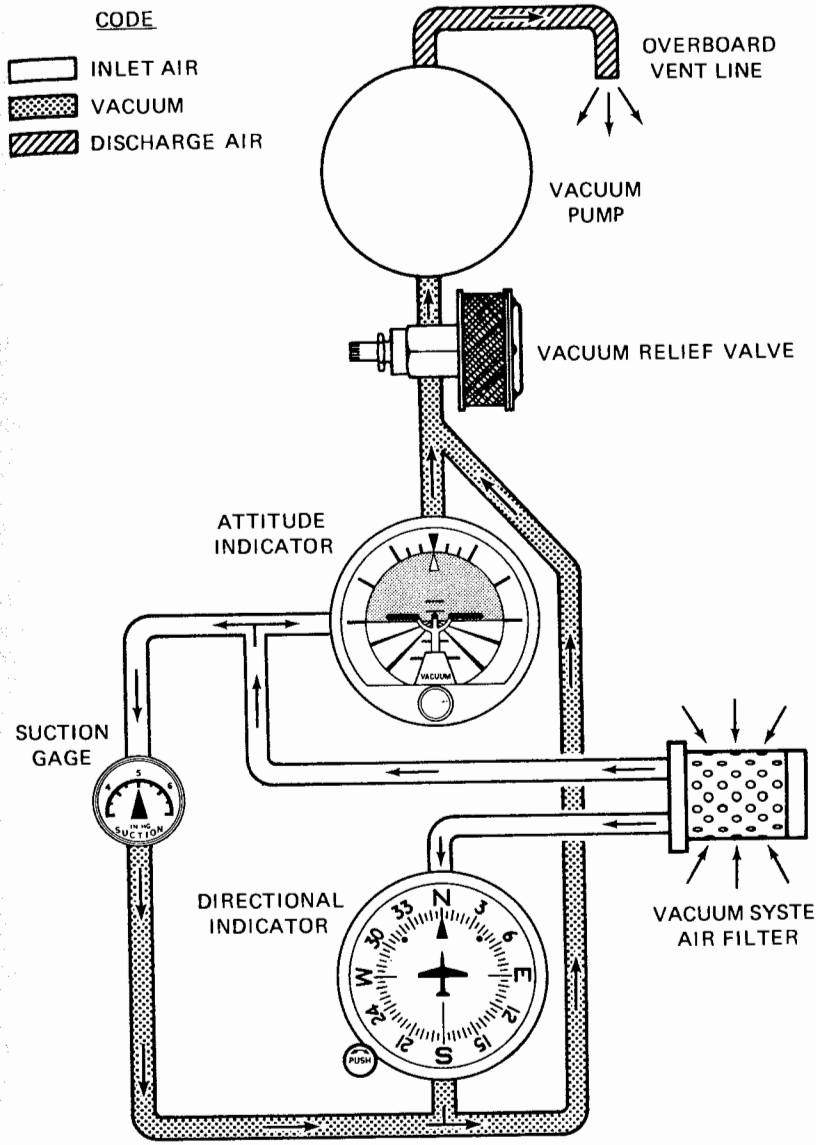



Figure 7-9. Vacuum System

### ATTITUDE INDICATOR

The attitude indicator gives a visual indication of flight attitude. Bank attitude is presented by a pointer at the top of the indicator relative to the bank scale which has index marks at 10°, 20°, 30°, 60°, and 90° either side of the center mark. Pitch and roll attitudes are presented by a miniature airplane in relation to the horizon bar. A knob at the bottom of the instrument is provided for in-flight adjustment of the miniature airplane to the horizon bar for a more accurate flight attitude indication.

### DIRECTIONAL INDICATOR

A directional indicator displays airplane heading on a compass card in relation to a fixed simulated airplane image and index. The indicator will precess slightly over a period of time. Therefore, the compass card should be set in accordance with the magnetic compass just prior to takeoff, and occasionally re-adjusted on extended flights. A knob on the lower left edge of the instrument is used to adjust the compass card to correct for precession.

### SUCTION GAGE

The suction gage, located on the left side of the instrument panel, is calibrated in inches of mercury and indicates suction available for operation of the attitude and directional indicators. The desired suction range is 4.6 to 5.4 inches of mercury. A suction reading below this range may indicate a system malfunction or improper adjustment, and in this case, the indicators should not be considered reliable.

## STALL WARNING SYSTEM

The airplane is equipped with a pneumatic-type stall warning system consisting of an inlet in the leading edge of the left wing, an air-operated horn near the upper left corner of the windshield, and associated plumbing. As the airplane approaches a stall, the low pressure on the upper surface of the wings moves forward around the leading edge of the wings. This low pressure creates a differential pressure in the stall warning system which draws air through the warning horn, resulting in an audible warning at 5 to 10 knots above stall in all flight conditions.

The stall warning system should be checked during the preflight inspection by placing a clean handkerchief over the vent opening and applying suction. A sound from the warning horn will confirm that the system is operative.

# AVIONICS SUPPORT EQUIPMENT

The airplane may, at the owner's discretion, be equipped with various types of avionics support equipment such as an audio control panel, microphone-headset, and static dischargers. The following paragraphs discuss these items.

# AUDIO CONTROL PANEL

Operation of radio equipment is covered in Section 9 of this handbook. When one or more radios are installed, a transmitter/audio switching system is provided (see figure 7-10). The operation of this switching system is described in the following paragraphs.

## TRANSMITTER SELECTOR SWITCH

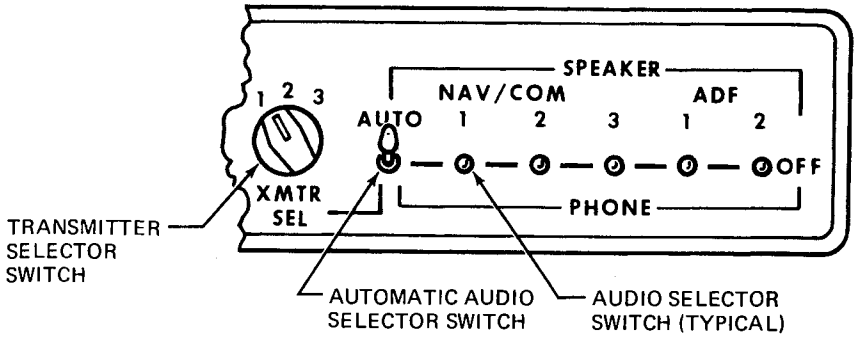
A rotary type transmitter selector switch, labeled XMTR SEL, is provided to connect the microphone to the transmitter the pilot desires to use. To select a transmitter, rotate the switch to the number corresponding to that transmitter. The numbers 1, 2 and 3 above the switch correspond to the top, second and third transceivers in the avionics stack.

The audio amplifier in the NAV/COM radio is required for speaker and transmitter operation. The amplifier is automatically selected, along with the transmitter, by the transmitter selector switch. As an example, if the number 1 transmitter is selected, the audio amplifier in the associated NAV/COM receiver is also selected, and functions as the amplifier for ALL speaker audio. In the event the audio amplifier in use fails, as evidenced by loss of all speaker audio and transmitting capability of the selected transmitter, select another transmitter. This should re-establish speaker audio and transmitter operation. Since headset audio is not affected by audio amplifier operation, the pilot should be aware that, while utilizing a headset, the only indication of audio amplifier failure is loss of the selected transmitter. This can be verified by switching to the speaker function.

## AUTOMATIC AUDIO SELECTOR SWITCH

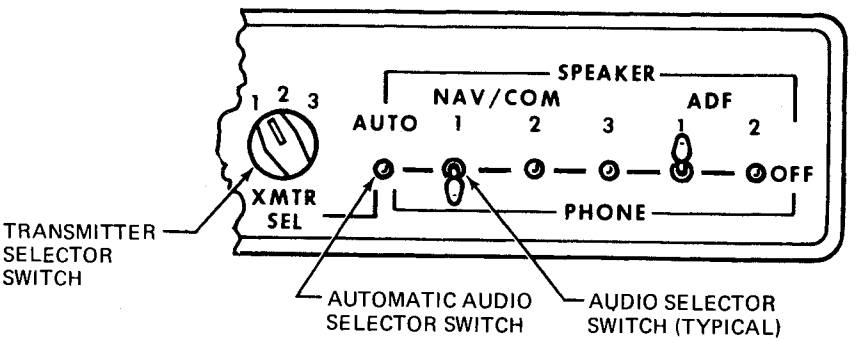
A toggle switch, labeled AUTO, can be used to automatically match the appropriate NAV/COM receiver audio to the transmitter being selected. To utilize this automatic feature, leave all NAV/COM receiver switches in the OFF (center) position, and place the AUTO selector switch in either the SPEAKER or PHONE position, as desired. Once the AUTO selector switch is positioned, the pilot may then select any transmitter and its associated NAV/COM receiver audio simultaneously with the transmitter selector

## AUTOMATIC AUDIO SELECTION




As illustrated, the number 1 transmitter is selected, the AUTO selector switch is in the SPEAKER position, and the NAV/COM 1, 2 and 3 and ADF 1 and 2 audio selector switches are in the OFF position. With the switches set as shown, the pilot will transmit on the number 1 transmitter and hear the number 1 NAV/COM receiver through the airplane speaker.

## INDIVIDUAL AUDIO SELECTION




As illustrated, the number 1 transmitter is selected, the AUTO selector switch is in the OFF position, the number 1 NAV/COM receiver is in the PHONE position, and the number 1 ADF is in the SPEAKER position. With the switches set as shown, the pilot will transmit on the number 1 transmitter and hear the number 1 NAV/COM receiver on a headset; while the passengers are listening to the ADF audio through the airplane speaker. If another audio selector switch is placed in either the PHONE or SPEAKER position, it will be heard simultaneously with either the number 1 NAV/COM or number 1 ADF respectively.

Figure 7-10. Audio Control Panel

switch. If automatic audio selection is not desired, the AUTO selector switch should be placed in the OFF (center) position.

NOTE

Cessna radios are equipped with sidetone capability (monitoring of the operator's own voice transmission). Sidetone will be heard on either the airplane speaker or a headset as selected with the AUTO selector switch. sidetone may be eliminated by placing the AUTO selector switch in the OFF position, and utilizing the individual radio selector switches.

### AUDIO SELECTOR SWITCHES

The audio selector switches, labeled NAV/COM 1, 2 and 3 and ADF 1 and 2, allow the pilot to initially pre-tune all NAV/COM and ADF receivers, and then individually select and listen to any receiver or combination of receivers. To listen to a specific receiver, first check that the AUTO selector switch is in the OFF (center) position, then place the audio selector switch corresponding to that receiver in either the SPEAKER (up) or PHONE (down) position. To turn off the audio of the selected receiver, place that switch in the OFF (center) position. If desired, the audio selector switches can be positioned to permit the pilot to listen to one receiver on a headset while the passengers listen to another receiver on the airplane speaker.

The ADF 1 and 2 switches may be used anytime ADF audio is desired. If the pilot wants only ADF audio, for station identification or other reasons, the AUTO selector switch (if in use) and all other audio selector switches should be in the OFF position. If simultaneous ADF and NAV/COM audio is acceptable to the pilot, no change in the existing switch positions is required. Place the ADF 1 or 2 switch in either the SPEAKER or PHONE position and adjust radio volume as desired.

NOTE

If the NAV/COM audio selector switch corresponding to the selected transmitter is in the PHONE position with the AUTO selector switch in the SPEAKER position, all audio selector switches placed in the PHONE position will automatically be connected to both the airplane speaker and any headsets in use.

## MICROPHONE-HEADSET

The microphone-headset combination consists of the microphone and headset combined in a single unit and a microphone keying switch located

on the left side of the pilot's control wheel. The microphone-headset permits the pilot to conduct radio communications without interrupting other control operations to handle a hand-held microphone. Also, passengers need not listen to all communications. The microphone and headset jacks are located near the lower left corner of the instrument panel.

## STATIC DISCHARGERS

If frequent IFR flights are planned, installation of wick-type static dischargers is recommended to improve radio communications during flight through dust or various forms of precipitation (rain, snow or ice crystals). Under these conditions, the build-up and discharge of static electricity from the trailing edges of the wings, rudder, elevator, propeller tips and radio antennas can result in loss of usable radio signals on all communications and navigation radio equipment. Usually the ADF is first to be affected and VHF communication equipment is the last to be affected.

Installation of static dischargers reduces interference from precipitation static, but it is possible to encounter severe precipitation static conditions which might cause the loss of radio signals, even with static dischargers installed. Whenever possible, avoid known severe precipitation areas to prevent loss of dependable radio signals. If avoidance is impractical, minimize airspeed and anticipate temporary loss of radio signals while in these areas.

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# Supplement
# Pilot´s Operating Handbook
## for the
# (Reims) Cessna (F) 172
# N & P
## Equipped with TAE 125-02-114 Installation

***Issue 2***

| | |
|---|---|
| MODEL No. | CESSNA 172 N |
| SERIAL No. | 17269394 |
| REGISTER No. | EC-NNA |

This supplement must be attached to the EASA approved Pilot´s Operating Handbook when the TAE 125-02-114 installation has been installed in accordance with EASA STC 10014287.

The information contained in this supplement supersede or add to the information published in the EASA approved Pilot´s Operating Handbook only as set forth herein.
For limitations, procedures, performance and loading information not contained in this supplement, consult the EASA approved Pilot´s Operating Handbook.

This supplement Pilot's Operating Handbook is approved with EASA AFM Approval 10036563.

Doc.-No.: 20-0310-20122

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# APPROVAL

The content of approved chapters is approved by EASA. All other content is approved by TAE under the authority of EASA DOA No. EASA.21J.010 in accordance with Part 21.

# LOG OF REVISIONS

| Revision | Section | Description | Approved | |
|---|---|---|---|---|
| | | | Date | Endorsed |
| 2/0 | all | new Issue | May 21, 2010 | EASA STC 10014287 |
| 2/1 | 1 | New oil, editorial changes | April 14, 2011 | Revision No. 1 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: April 14, 2011 Office of Airworthiness |
| | 2 | New oil, editorial changes | | |
| | 3 | Procedures updated | | |
| | 4 | Procedures updated | | |
| | 5 | Editorial changes | | |
| | 6 | Editorial changes | | |
| | 9 | New Section | | |
| 2/2 | 1 | New gearbox oil, editorial changes | Sept. 22, 2011 | EASA AFM Approval 10036563 |
| | 2 | New gearbox oil, Fuel capacity integral fuel tank | | |
| | 4 | Procedures updated | | |
| | 5 | Flight performance with integral fuel tanks | | |
| | 7 | Editorial changes | | |

| Revision | Section | Description | Approved | |
|---|---|---|---|---|
| | | | Date | Endorsed |
| 2/3 | 1 | New Fuel,<br>new gearbox oil | March 16 2012 | Revision No. 3 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: March 16, 2012 Office of Airworthiness |
| | 2 | New fuel,<br>new gearbox oil | | |
| | 4 | New fuel,<br>Procedures updated | | |
| | 5 | Procedures updated | | |
| | 6 | New fuel | | |
| 2/4 | 1 | New gearbox oil | March 11, 2013 | Revision No. 4 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: March 11, 2013 Office of Airworthiness |
| | 2 | New gearbox oil | | |
| | 5 | Procedures updated | | |
| 2/5 | --- | EASA STC / AFM numbers corrected on the cover | May 27, 2013 | Revision No. 5 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: May 27, 2013 Office of Airworthiness |

| Revision | Section | Description | Approved | |
|---|---|---|---|---|
| | | | Date | Endorsed |
| 2/6 | 1 | Safety Recommendation<br>New fuel<br>New gearbox oil<br>Note fuel additive | 03.09.2014 | Revision No. 6 to AFM supplement ref. 20-0310-22122 is approved under the authority of DOA ref. EASA.21J.010. Date: september, 03. 2014 Office of Airworthiness |
| | 2 | Note added<br>New fuel<br>New gearbox oil<br>Note fuel additive | 03.09.2014 | |
| | 3 | Description adapted wording | 03.09.2014 | |
| | 4 | Note added | 03.09.2014 | |
| | 5 | Wording | 03.09.2014 | |
| | 6 | Wording | 03.09.2014 | |
| | 7 | Wording | 03.09.2014 | |
| | 8 | Wording | 03.09.2014 | |
| | 9 | Wording | 03.09.2014 | |
| 2/7 | 4 | Procedure added | 26.01.2015 | Revision No. 7 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: January 26, 2015 Office of Airworthiness |

| Revision | Section | Description | Approved | |
|---|---|---|---|---|
| | | | Date | Endorsed |
| 2/8 | 1 | New propeller | April 08, 2015 | EASA STC 10014287, Rev. 8 |
| | 5 | splitted due to new propeller specs | | |
| | 5a | New section | | |
| | 5b | New section | | |
| 2/9 | 1 | Update Liquids | Jan. 22, 2018 | Revision No. 9 to AFM supplement ref. 20-0310-20122 is approved under the authority of DOA ref. EASA.21J.010. Date: Jan. 22, 2018 Office of Airworthiness |
| | 2 | Update liquids<br>Update Engine Instrument Markings | | |
| | 3 | various minor corrections | | |
| | 4 | Update FADEC Test above 5500ft | | |

Remark: The parts of the text which changed are marked with a vertical line on the margin of the page.

# LIST OF EFFECTIVE SECTIONS

| Sections | Issue/Revision | Date |
|---|---|---|
| 1 | 2/8 | Jan. 22, 2018 |
| 2 | 2/7 | Jan. 22, 2018 |
| 3 | 2/7 | Jan. 22, 2018 |
| 4 | 2/7 | Jan. 22, 2018 |
| 5 | 2/7 | April 08, 2015 |
| 5a | 2/0 | April 08, 2015 |
| 5b | 2/0 | April 08, 2015 |
| 6 | 2/6 | Sept. 03, 2014 |
| 7 | 2/6 | Sept. 03, 2014 |
| 8 | 2/6 | Sept. 03, 2014 |
| 9 | 2/6 | Sept. 03, 2014 |

# GENERAL REMARK

The content of this POH supplement is developed on basis of the EASA-approved POH.

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COVER SHEET

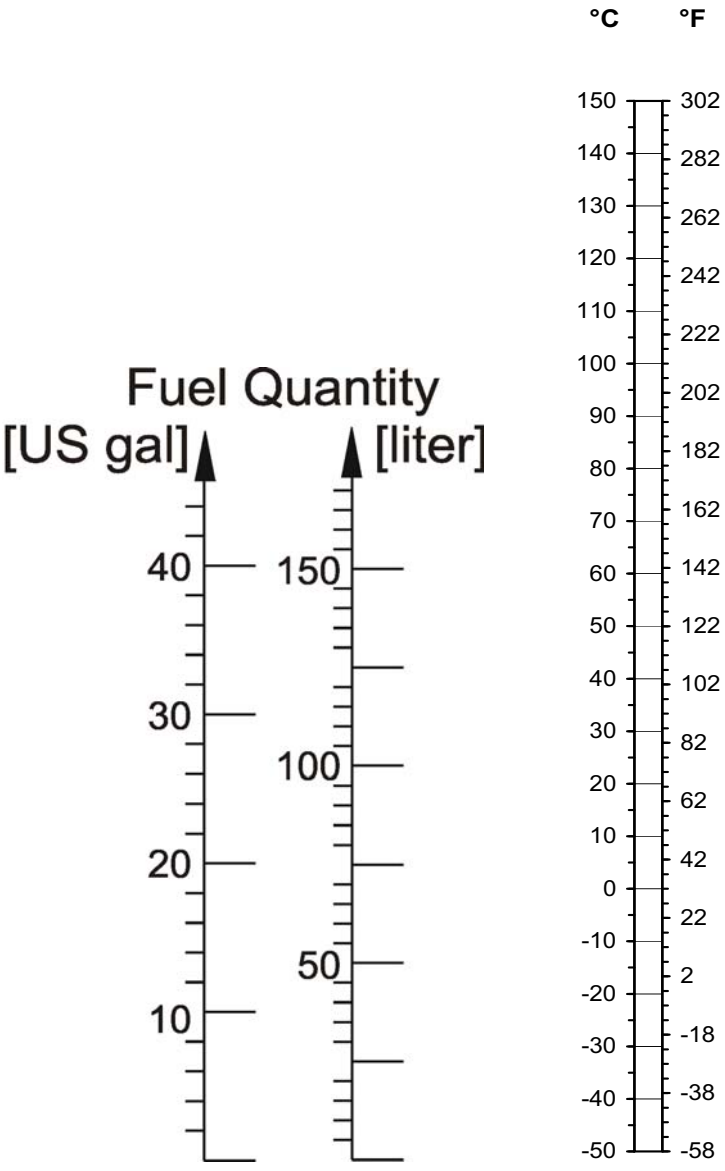
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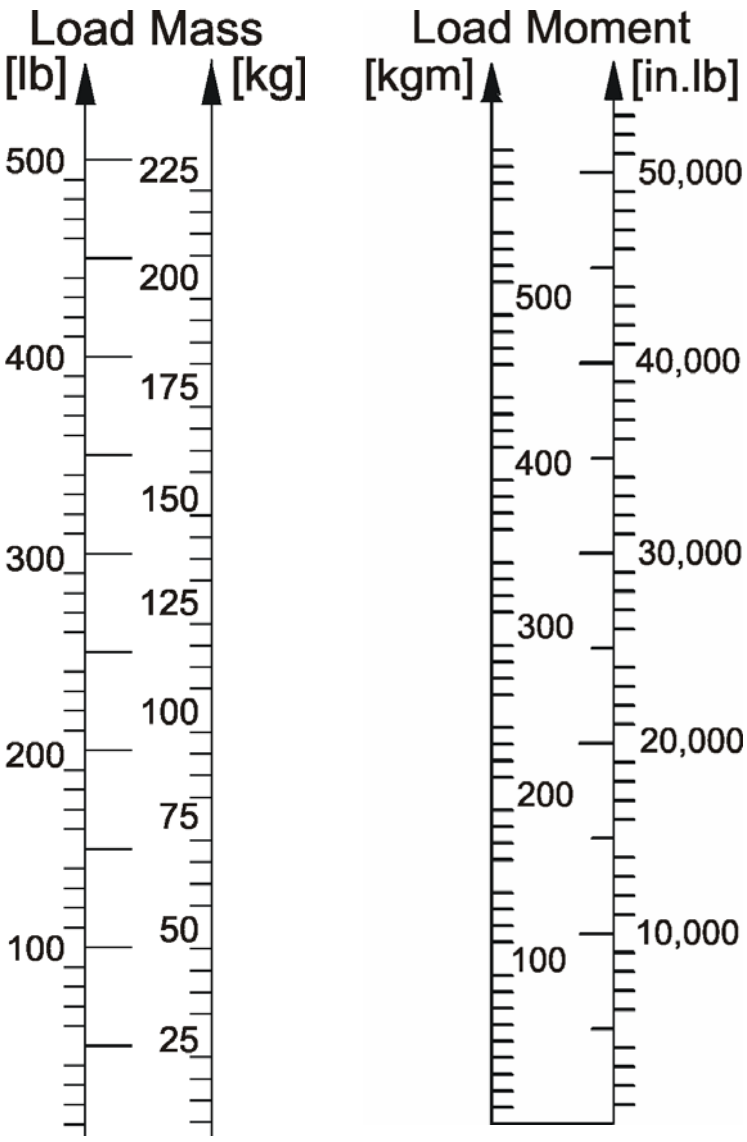
| | |
|---|---|
| SECTION 1 | GENERAL<br>(a non-approved chapter) |
| SECTION 2 | LIMITATIONS<br>(an approved chapter) |
| SECTION 3 | EMERGENCY PROCEDURES<br>(a non-approved chapter) |
| SECTION 4 | NORMALPROCEDURES<br>(a non-approved chapter) |
| SECTION 5 | PERFORMANCE<br>(a non-approved chapter) |
| SECTION 6 | HANDLING ON GROUND &<br>MAINTENANCE<br>(a non-approved chapter) |
| SECTION 7 | WEIGHT & BALANCE<br>(a non-approved chapter) |
| SECTION 8 | SPECIAL EQUIPMENT<br>(a non-approved chapter) |
| SECTION 9 | SUPPLEMENTS |

# CONVERSION TABLES

| VOLUME | | |
|---|---|---|
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Liter [l]<br><br><br><br>US gallon [US gal]<br>US quart [US qt]<br>Imperial gallon [Imp gal]<br>Cubic inch [in³] | [l] / 3.7854 = [US gal]<br>[l] / 0.9464 = [US qt]<br>[l] / 4.5459 = [[Imp gal]<br>[l] x 61.024 = [in³] | <br><br><br><br>[US gal] x 3.7854 = [l]<br>[[US qt] x 0.9464 = [l]<br>[[Imp gal] x 4.5459 = [l]<br>[in³] / 61.024 = [l] |
| **TORQUE** | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Kilopondmeter [kpm]<br><br><br>Foot pound [ft.lb]<br>Inch pound [in.lb] | [kpm] x 7.2331 = [ft.lb]<br>[kpm] x 86.7962 = [in.lb] | <br><br><br>[ft.lb] / 7.2331 = [kpm]<br>[in.lb] / 86.7962 = [kpm] |
| **TEMPERATURE** | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Degree Celsius [ºC]<br>Degree Fahrenheit [ºF] | [ºC] x 1.8 + 32 = [ºF] | <br>([ºF] - 32) / 1.8 = [ºC] |
| **SPEED** | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Kilometers per hour [km/h]<br><br>Meters per second [m/s]<br>Miles per hour [mph]<br>Knots [kts]<br>Feet per minute [fpm] | [km/h] / 1.852 = [kts]<br>[km/h] / 1.609 = [mph]<br>[m/s] x 196.85 = [fpm] | <br><br><br>[mph] x 1.609 = [km/h]<br>[kts] x 1.852 = [km/h]<br>[fpm] / 196.85 = [m/s] |

| PRESSURE | | |
|---|---|---|
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Bar [bar]<br>Hectopascal [hpa]<br>=Millibar [mbar]<br><br>Pounds per square inch [psi] inches of mercury column [inHg] | [bar] x 14.5038 = [psi]<br>[hpa] / 33.864= [inHg]<br><br>[mbar] / 33.864 = [inHg] | <br><br><br>psi] / 14.5038 = [bar]<br>[inHg] x 33.864 = [hPa]<br><br>[inHg] x 33.864 = [mbar] |
| MASS | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Kilogramm [kg]<br>Pound [lb] | [kg] / 0.45359 = [lb] | <br>[lb] x 0.45359 = [kg] |
| LENGTH | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Meter [m]<br>Millimeter [mm]<br>Kilometer [km]<br><br>Inch [in]<br>Foot [ft]<br>Nautical mile [nm]<br>Statute mile [sm] | [m] / = 0.3048 [ft]<br>[mm] / = 25.4 [in]<br>[km] / = 1.852 [nm]<br>[km] / = 1.609 [sm] | <br><br><br>[in] x 25.4 = [mm]<br>[ft] x 0.3048 = [m]<br>[nm] x 1.852 = [km]<br>[sm] x 1.609 = [km] |
| FORCE | | |
| Unit [Abbr.] | Conversion factor<br>SI to US / Imperial | Conversion factor<br>US / Imperial to Si |
| Newton [N]<br>Decanewton [daN]<br>Pound [lb] | [N] / 4.448 = [lb]<br>[daN] / 0.4448 = [lb] | <br><br>[lb] x 4.448 = [N]<br>[lb] x 0.4448 = [daN] |


°C
°F
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
0
-10
-20
-30
-40
-50
302
282
262
242
222
202
182
162
142
122
102
82
62
42
22
2
-18
-38
-58
Fuel Quantity
[US gal]
[liter]
40
30
20
10
150
100
50


Load Mass
[lb]
[kg]
500
400
300
200
100
225
200
175
150
125
100
75
50
25
Load Moment
[kgm]
[in.lb]
500
400
300
200
100
50,000
40,000
30,000
20,000
10,000

# ABBREVIATIONS

| | |
|---|---|
| **FADEC** | Full Authority Digital Engine Control |
| **CED 125** | Compact Engine Display<br>Multifunctional instrument for indication of engine data of the TAE 125-02-114 |
| **AED 125** | Auxiliary Engine Display<br>Multifunctional instrument for indication of engine and airplane data |

# SECTION 1
# GENERAL

### Safety Recommendations

The following symbols and warnings are used in this manual. They must be heeded strictly to prevent personal injury and material damage, to avoid impairment of the operational safety of the aircraft and to rule out any damage to the aircraft as a consequence of improper handling.

| | |
|---|---|
| ▲ **WARNING:** | Non-compliance with these safety rules could lead to injury or even death. |
| ■ CAUTION: | Non-compliance with these special notes and safety measures could cause damage to the engine or to the other components. |
| ◆ Note: | Information added for a better understanding of an instruction. |

### UPDATE AND REVISION OF THE MANUAL

| | |
|---|---|
| ▲ **WARNING:** | A safe operation is only assured with an up to date POH supplement. Information about actual POH supplement issues and revisions are published in the Service Bulletin TM TAE 000-0004. |
| ◆ Note: | The Doc.-No of this POH supplement is published on the cover sheet of this supplement. |

## ENGINE

▲ **WARNING:** The engine requires an electrical power source for operation. If the main battery and alternator fail, the engine will only operate for a maximum of 30 minutes on FADEC backup battery power. Therefore, it is important to pay attention to indications of alternator failure.

Engine manufacturer:..... Technify Motors GmbH
Engine model: .....TAE 125-02-114

The TAE 125-02-114 is a liquid cooled in-line four-stroke 4-cylinder turbocharged engine with DOHC (double overhead camshaft), direct fuel injection and common-rail technology. It has a displacement of 1991 ccm (121.5 $in^3$). The engine is controlled by a FADEC system. The propeller is driven by a built-in gearbox (i = 1.69) with mechanical vibration dampening and overload release. The engine has an electrical self starter and an alternator.

▲ **WARNING:** The engine requires an electrical power source for operation. If the main battery and alternator fail, the engine will only operate for a maximum of 30 minutes on FADEC backup battery power. Therefore, it is important to pay attention to indications of alternator failure.

Due to this specific characteristic, all of the information from the are no longer valid with reference to:

- carburetor and carburetor pre-heating
- ignition magnetos and spark plugs, and
- mixture control and priming system

## PROPELLER

Manufacturer: ..... MT Propeller Entwicklung GmbH
Model: ..... MTV-6-A/187-129
..... MTV-6-A/190-69
Number of blades: ..... 3
Diameter: ..... 1.87 m (MTV-6-A/187-129
..... 1.90 m (MTV-6-A/190-69)
Type: ..... constant speed

## FUELS and LIQUIDS

| | | |
|---|---|---|
| ▲ | **WARNING:** | The engine must not be started under any circumstances if any fluid level is too low. |
| ■ | CAUTION: | Use of unapproved fuels may result in damage to the engine and fuel system components, resulting in possible engine failure. |
| ■ | CAUTION: | Use approved oil with exact designation only! |
| ■ | CAUTION: | Normally it is not necessary to fill the cooling liquid or gearbox oil between maintenance intervals. If the level is too low, please notify the service center immediately. |

Fuel: ..... JET A-1 (ASTM 1655)
..... JET A (ASTM 1655)
..... Jet Fuel No.3 (GB 6537-2006)
..... JP-8 (MIL-DTL-83133E)
..... JP-8+100 (MIL-DTL-83133E)
..... TS-1 (GOST 10227-86)
..... TS-1 (GSTU 320.00149943.011-99
Alternative: ..... Diesel (**DIN** EN 590)
..... SASOL GTL Diesel

◆ Note: The liquid fuel additive Biobor JF can be used in jet and diesel fuel systems to eliminate growth of fungi. For further details refer to the manufacturer specifications.

Engine oil: ..... AeroShell Oil Diesel Ultra
..... AeroShell Oil Diesel 10W-40
.....Shell Helix Ultra 5W-30
.....Shell Helix Ultra 5W-40

Gearbox oil: .....Centurion Gearbox Oil N1
..... Shell Spirax S6 ATF ZM
..... Shell Spirax S6 GXME 75W-80, API GL-4
..... Shell Spirax S4 G 75W-90, API GL-4

Coolant:.....Water/Radiator Protection at a ratio of 50:50

Radiator Protection: ..... BASF Glysantin Protect Plus / G48
..... Valvoline/Zerex Glysantin G48
.....Mobil Antifreeze Extra (G48)
..... Comma Xstream Green - Concentrate/G48

◆ Note: The freezing point of the coolant is -36°C.

## INSTRUMENT PANEL

Components of the new installation can be seen as example in the following Figure

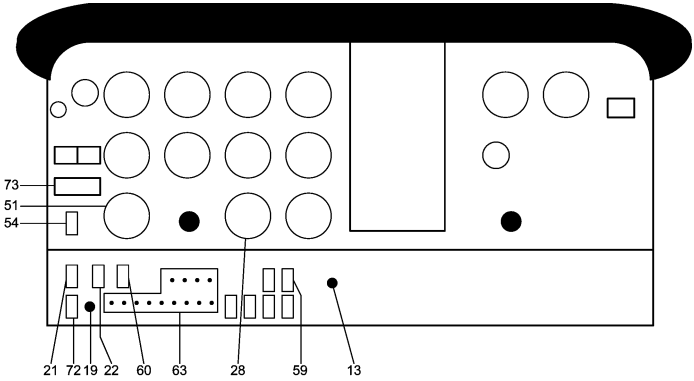



Figure 1-1 Example of Instrument panel

13. "Alt. Air Door" Alternate Air Door
    (Carburetor Heat Button **N/A**)
19. "Starter"-Push Button for Starter
21. "BAT"-Switch for Battery
22. "MAIN"-Switch for Main Bus
28. CED 125 (Tachometer **N/A**)
    The Compact Engine Display contains indication of Propeller Rotary Speed, Oil Pressure, Oil Temperature, Coolant Temperature, Gearbox Temperature and Load.
51. AED 125 SR (Voltmeter, Ammeter) with indication of Fuel-Temperature,
    Voltage and a caution light "Water Level" (amber) for low coolant level
54. "Force B"-Switch for manually switching the FADEC
59. "Fuel Pump"-Switch for the Electric Fuel Pump
60. "ALT"-Switch for Alternator
62. Fuse Electric Fuel Pump

63. Fuses, among other for Alternator Warning light, Starter, FADEC and Main Bus

72. "Engine Master"-Switch
electrical supply FADEC

73. Lightpanel with:
"FADEC" Test Knob
"A FADEC B" Warning Lights for FADEC A and B (red)
"Alt" Alternator Warning Light (red)
"AED" Caution Light (amber) for AED 125
"CED" Caution Light (amber) for CED 125
"CED/AED" - Test/Confirm Knob for CED 125, AED 125 and Caution Lights (amber)
"Fuel L";"Fuel R" Caution Lights for low fuel level (amber)
"Glow" Glow Control Light (amber)

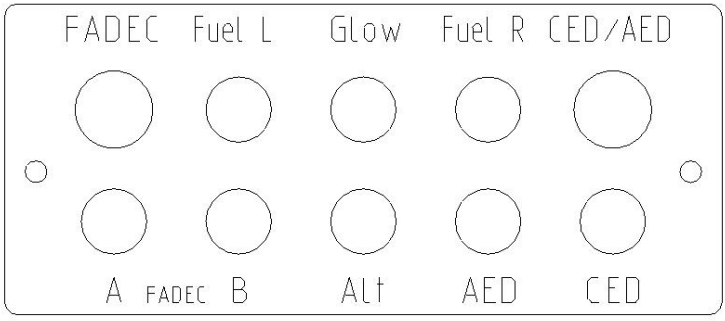



Figure 1-2 Lightpanel

## FUEL SYSTEM (Left, Right, Both)

The fuel system of the engine includes the original standard or long-range tanks of the Cessna 172. Additional sensors for fuel temperature and "Low Level" warning are installed.
The fuel flows out of the tanks to the fuel selector valve with the positions LEFT, RIGHT and BOTH, through a reservoir tank to the fuel shut-off valve and then via the electrically driven fuel pump to the fuel filter.
The electrically driven fuel pump supports the fuel flow to the filter module if required. Fuel can be shut off by the separate shut-off valve. The engine-driven feed pump and the high-pressure pump supply the rail, from where the fuel is injected into the cylinders depending upon the position of the thrust lever and regulation by the FADEC. Surplus fuel flows to the fuel cooler and through the fuel selector valve back into the pre-selected tank; if BOTH is selected the fuel returns to both tanks.
A temperature sensor in the filter module controls the heat exchange between the fuel feed and return. The fuel cooler reduces the fuel temperature in the return line.
The fuel cooler receives its cooling air through an inlet in the air duct to the heating radiator. This inlet is closed with a baffle, which must be removed at high outside air temperatures (OAT higher than 20 °C (68 °F), see also Section 4).
Figure 1-3Since the density of diesel and jet fuel (0.84 kg/l) is higher than AVGAS (0.715 kg/l), the usable fuel capacity was reduced by this factor through the fuel filler neck, to stay within the approved wing load

| Fuel Capacity | | | |
|---|---|---|---|
| Tanks | Total Capacity | Total Unusable Fuel | Total Usable Fuel |
| 2 Standard-Tanks: each 69.4l (18.30 US gal) | 138.8 l (36.6 US gal) | 11.4 l (3 US gal) | 127.4 l (33.6 US gal) |
| 2 Long-Range Tanks: each 86.8l (22.95 US gal) | 173.6 l (45.9 US gal) | 15.1 l (4 US gal) | 158.6 l (41.9 US gal) |
| 2 Integral Tanks (normal category): each 119.8 l (29 US gal) | 219.6 l (58 US gal) | 22.8 l (6 US Gal) | 196.8 l (52 US gal) |
| 2 Integral Tanks (utility category): each 90.7 l (23.95 US gal) | 181.4 l (47.9 US gal) | 22.8 l (6 US Gal) | 158.6 l (41.9 US gal) |

## FUEL SYSTEM (Left, Right, Both)

■ CAUTION: In flight conditions with downward pointing wing, switch the fuel selector to the upper fuel tank or to the BOTH position.

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

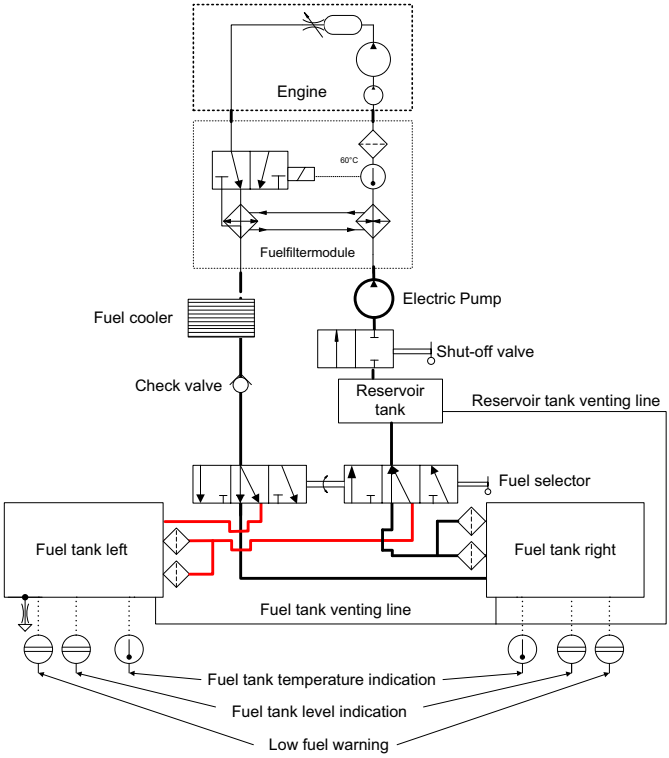



Figure 1-4 Scheme of the Fuel System (Left, Right, Both)

◆ Note: The handling of the fuel selector positions left, right and both are described in the original POH.

## ELECTRICAL SYSTEM

The electrical system differs from the previous installation and is equipped with the following operating and display elements:

1. "Main Bus" Switch
   This switch controls the Main Bus. The Main Bus is required to be able to run FADEC and engine with the Battery/Alternator in the event of electrical system malfunctions. In normal operation Alternator, Main Bus and Battery must be ON.

2. "Alternator" Switch
   Controls the alternator. Must be ON in normal operation.

3. "Battery" Switch
   Controls the battery.

4. "Starter" Push Button
   Controls the magneto switch of the starter.

5. Ammeter
   The Ammeter shows the alternator current. In case of battery discharge if alternator inoperative the alternator warning light will illuminate.

6. "Alternator" Warning light
   Illuminates when the power output of the alternator is too low or the alternator switch is switched off. Normally, this warning light always illuminates when the "Engine Master" is switched on without revolution and extinguishes immediately after starting the engine.

7. "Fuel Pump" Switch
   Controls the electric fuel pump.

8. Engine Master" Switch
   Controls the two redundant FADEC components and the alternator excitation battery with two independent contacts. The alternator excitation battery is used to ensure that the alternator continues to function properly even if the main battery fails.

▲ **WARNING:** If the "Engine Master" is switched off, the power supply to the FADEC is interrupted and the engine will shut down.

9. "Force B" Switch
   If the FADEC does not automatically switch from A-FADEC to B-FADEC in an emergency, this switch allows to manually switch to the B-FADEC.

▲ **WARNING:** When operating on FADEC backup battery only, the "Force B" switch must not be activated. This will shut down the engine.

10. FADEC Backup Battery
    The electrical system includes a FADEC backup battery to ensure power supply to A-FADEC in case the battery and alternator fail or are disconnected. The engine can be operated for a maximum of 30 minutes when powered by the FADEC backup battery only. Only A-FADEC is connected to the backup battery.

The basic wiring of the installation is available in 14V as well as 28V versions.

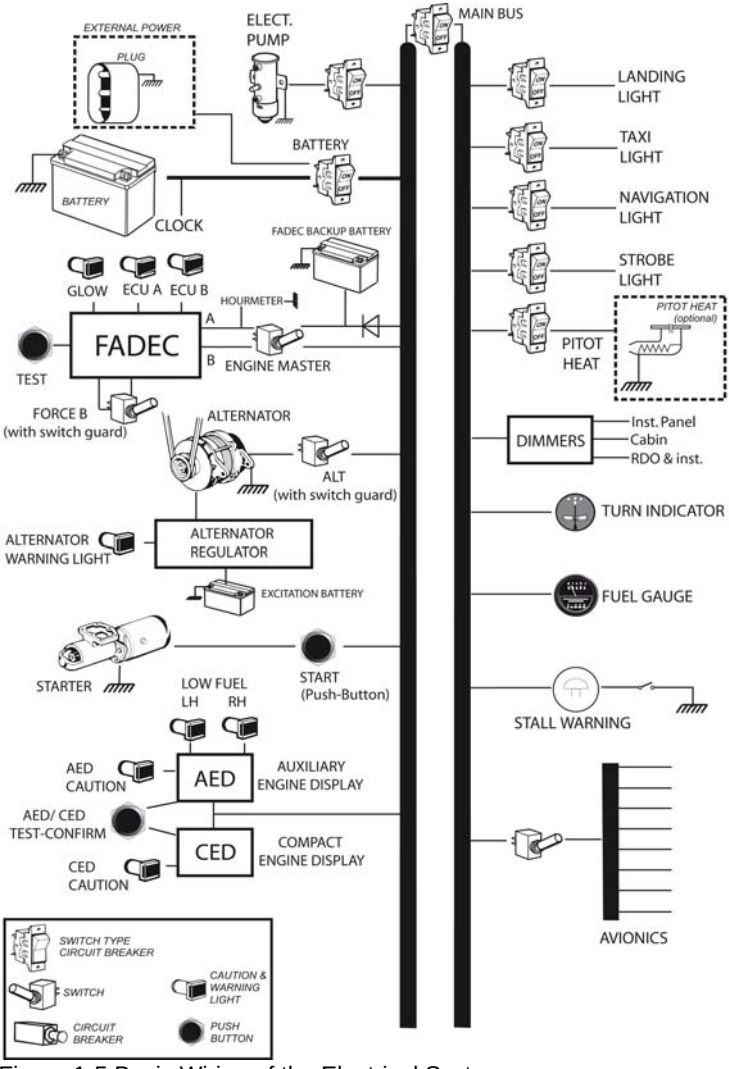



Figure 1-5 Basic Wiring of the Electrical System

**FADEC-RESET**

In case of a FADEC warning, one or both FADEC warning lights are flashing. If the "FADEC" test knob is pressed for at least 2 seconds,

a) the active warning lights will extinguish if it was a LOW category warning.

b) the active warning lights will be illuminated steady if it was a HIGH category warning.

■ CAUTION: If a FADEC warning occurred, contact your service center.

When a high category warning occurs the pilot should land as soon as possible, since the affected FADEC ECU has diagnosed a severe fault. A low category fault has no significant impact on engine operation.
Refer also to the engine OM-02-02 for additional information.

**COOLING**

The installation is fitted with a fluid-cooling system, whose three-way thermostat regulates the flow of coolant between the large and small cooling circuit.
At a coolant temperature of up to 84°C (183°F) the coolant flows exclusively through the small circuit, between 84°C (183°F) and 94°C (201°F) through the small and the large circuit simultaneously.
If the cooling water temperature rises above 94°C (201°F), the complete volume of coolant flows through the large circuit and therefore through the radiator. This ensures a maximum cooling water temperature of 105°C (221°F).
There is a sensor in the expansion reservoir, which sends a signal to the warning light "Water level" on the AED 125 in the instrument panel if the coolant level is low.
The cooling water temperature is measured in the cylinder head near the thermostat and passed on to the FADEC and CED 125.
The connection to the heat exchanger for cabin heating is always open; the warm air supply is regulated by the pilot over

the heating valve. See Figure 1-5.
The supply of warm air into the cabin is controlled through the cabin heat control knob. In normal operation, the cabin heat control knob must be in the OPEN position.
In case of certain emergencies (refer to Section 3), the control knob "Shut-off Cabin Heat" has to be closed according to the appropriate procedures.

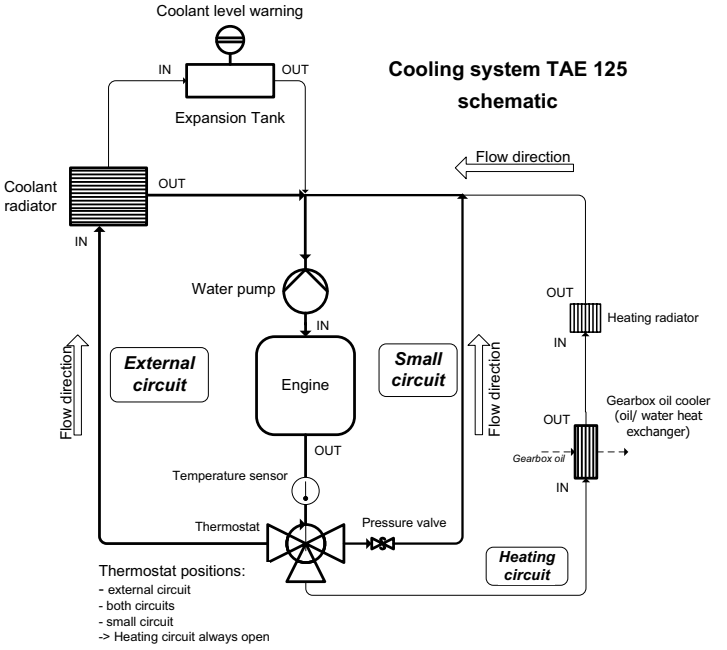



Figure 1-5 Cooling system

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# SECTION 2
# LIMITATIONS

▲ **WARNING:** It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

**Normal Category Cessna 172 N:**

Maximum Ramp Weight: ..... 1044 kg (2302 lbs)
Maximum Takeoff Weight: ..... 1043 kg (2300 lbs)
Maximum Landing Weight ..... 1043 kg (2300 lbs)

**Utility Category Cessna 172 N:**

Maximum Ramp Weight: ..... 908 kg (2002 lbs)
Maximum Takeoff Weight: ..... 907 kg (2000 lbs)
Maximum Landing Weight ..... 907 kg (2000 lbs)

**Normal Category Cessna 172 P:**

Maximum Ramp Weight: ..... 1090 kg (2402 lbs)
Maximum Takeoff Weight: ..... 1089 kg (2400 lbs)
Maximum Landing Weight ..... 1089 kg (2400 lbs)

**Utility Category Cessna 172 P:**

Maximum Ramp Weight: ..... 954 kg (2102 lbs)
Maximum Takeoff Weight: ..... 953 kg (2100 lbs)
Maximum Landing Weight ..... 953 kg (2100 lbs)

## MANEUVER LIMITS

| | |
|---|---|
| ■ CAUTION: | Intentionally initiating negative G maneuvers is prohibited |

Normal Category: No change

Utility Category: Intentionally initiating spins is prohibited

## FLIGHT LOAD FACTORS

No change

| | |
|---|---|
| ■ CAUTION: | Avoid extended negative g-loads duration. Extended negative g-loads can cause propeller control and engine problems. |

| | |
|---|---|
| ◆ Note: | The load factor limits for the engine must also be observed. Refer to the Operation & Maintenance Manual for the engine. |

## ENGINE OPERATING LIMITS

Engine manufacturer:..... Technify Motors GmbH

Engine model: .....TAE 125-02-114

Take-off and Max. continuous power:..... 114 kw (155 HP)

Take-off and Max. continuous RPM:..... 2300 $min^{-1}$

Max. recommended cruise.....85%

| | |
|---|---|
| ◆ Note: | In the absence of any other explicit statements, all of the information on RPM in this supplement to the Pilot´s Operating Handbook are propeller RPM. |

| | |
|---|---|
| ◆ Note: | This change of the original aircraft is certified up to an altitude of 18,000 ft. |

Engine operating limits for take-off and continuous operation**:**

| | |
|---|---|
| ▲ **WARNING:** | It is not allowed to start the engine outside of these temperature limits. |
| ◆ Note: | The operating limit temperature is a temperature limit below which the engine may be started, but not operated at the Take-off RPM. The warm-up RPM to be selected can be found in Section 4 of this supplement. |

**Oil temperature:**

Minimum engine starting temperature: ..... -32 °C
Minimum operating limit temperature: .....50 °C
Maximum operating limit temperature: .....140 °C

**Coolant temperature:**

Minimum engine starting temperature: ..... -32 °C
Minimum operating limit temperature: .....60 °C
Maximum operating limit temperature: .....105 °C

**Gearbox temperature:**

Mininum operating limit temperature: ..... -30 °C
Maximum operating limit temperature: .....120 °C

**Min. fuel temperature limits in the fuel tank:**

| Fuel | Minimum fuel temperature in the fuel tank before Take-off | Minimum fuel temperature in the fuel tank during the flight |
|---|---|---|
| JET A-1, JET A, Fuel No.3 JP-8, JP-8+100 TS-1 | -30°C | -35°C |
| Diesel Sasol GTL Diesel | greater than 0°C | -5°C |

Table 2-3a Minimum fuel temperature limits in the fuel tank

▲ **WARNING:** The fuel temperature of the fuel tank not used should be monitored if its later use is intended.

▲ **WARNING:** The following applies to Diesel and JET fuel mixtures in the tank:
As soon as the proportion of Diesel in the tank is more than 10% Diesel, the fuel temperature limits for Diesel operation must be monitored. If there is uncertainty about which fuel is in the tank, the assumption should be made that it is Diesel.

**Oil Pressure**

Minimum oil pressure: .....1.2 bar
Minimum oil pressure (at Take-off power) .....2.3 bar
Minimum oil pressure (in flight) .....2.3 bar
Maximum oil pressure.....6.0 bar
Maximum oil pressure (cold start < 20 sec.): .....6.5 bar
Maximum oil consumption: .....0.1 l/h (0.1 quart/h)

## ENGINE INSTRUMENT MARKINGS

The engine data to be monitored are integrated in the combined engine instrument CED 125.
The ranges of the individual engine monitoring parameters are shown in the following table.

◆ Note: "Load" describes the available percentage of maximum engine power.

| Instrument AED/CED | | Red range | Amber range | Green range | Amber range | Red range |
|---|---|---|---|---|---|---|
| Tachometer | [RPM] | -------- | -------- | 0-2300 | -------- | > 2300 |
| Oil pressure | [bar] | 0 - 1.1 | 1.2 - 2.2 | 2.3 - 5.1 | 5.2 - 6.5 | > 6.5 |
| | [psi] | 0 - 16 | 17.4 - 32 | 33.4 - 74 | 75.4 - 87.0 | > 87.0 |
| Coolant temperature | [°C] | < -32 | -32...+59 | 60 - 100 | 101 - 105 | > 105 |
| Oil temperature | [°C] | < -32 | -32...+49 | 50 - 129 | 130 - 140 | > 140 |
| Gearbox temperature | [°C] | -------- | -------- | < 115 | 115 - 120 | > 120 |
| Load | [%] | -------- | -------- | 0 - 100 | -------- | -------- |
| Fuel Temperature (left and right) | [°C] | < -30 | -30...-1 | 0 - 69 | 70 - 75 | > 75 |
| Alternator Current (14V) | [A] | -------- | -------- | 0 - 84 | 85 - 90 | >90 |
| Alternator Current (28V) | [A] | -------- | -------- | 0 - 52.4 | 52.5 - 60 | >60 |
| Electrical System Voltage (14V) | [V] | 0 - 10 | 11 - 12.5 | 12.6 - 14.0 | 15.0 | >15.0 |
| Electrical System Voltage (28V) | [V] | 0 - 21 | 22 - 24 | 25 - 29.4 | 29.5 - 30 | >30 |

Table 2-2 Markings of the engine instruments

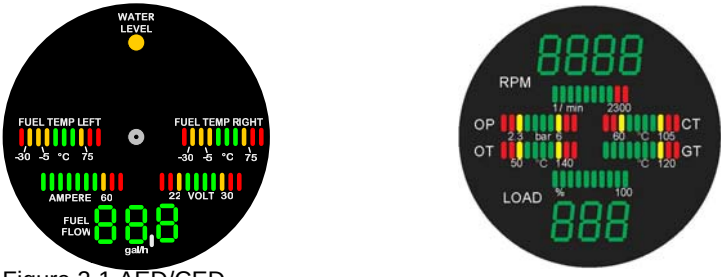



Figure 2-1 AED/CED

◆ Note: The AED/CED caution lamp is switched on if an engine reading is in the amber or red range.
The AED/CED caution lamp remains on even when the parameter returns to the green/normal operating range and must be confirmed by pressing the Confirm/Test knob.
After being confirmed, the AED/CED caution lamp will switch on again whenever another parameter enters amber/red range.
Pressing the Confirm/Test knob longer than one second will initiate the power-up test sequence.

## PERMISSIBLE FUEL GRADES

■ CAUTION: Using non-approved fuels and additives can lead to dangerous engine malfunctions.

Fuel: .....JET A-1 (ASTM 1655)
.....JET A (ASTM 1655)
..... Jet Fuel No.3 (GB 6537-2006)
.....JP-8 (MIL-DTL-83133E)
.....JP-8+100 (MIL-DTL-83133E)
..... TS-1 (GOST 10227-86)
.....TS-1 (GSTU 320.00149943.011-99

Alternative:
.....SASOL GTL Diesel

◆ Note: The liquid fuel additive Biobor JF can be used in jet and diesel fuel systems to eliminate growth of fungi. For further details refer to the manufacturer specifications.

## MAXIMUM FUEL QUANTITIES

Due to the higher specific density of Kerosene in comparison to Aviation Gasoline (AVGAS) the permissible tank capacity has been reduced.

| Fuel Capacity | | | |
|---|---|---|---|
| Tanks | Total Capacity | Total Unusable Fuel | Total Usable Fuel |
| 2 Standard-Tanks: each 69.4l (18.30 US gal) | 138.8 l (36.6 US gal) | 11.4 l (3 US gal) | 127.4 l (33.6 US gal) |
| 2 Long-Range Tanks: each 86.8 l (22.95 US gal) | 173.6 l (45.9 US gal) | 15.1 l (4 US gal) | 158.6 l (41.9 US gal) |
| 2 Integral Tanks (normal category): each 119.8 l (29 US gal) | 219.6 l (58 US gal) | 22.8 l (6 US gal) | 196.8 l (52 US gal) |
| 2 Integral Tanks (utility category): each 90.7 l (23.95 US gal) | 181.4 l (47.9 US gal) | 22.8 l (6 US gal) | 158.6 l (41.9 US gal) |

■ CAUTION: To prevent air from penetrating into the fuel system avoid running one tank dry. As soon as the "Low Level" caution light illuminates, switch to the tank with sufficient fuel or land as soon as possible.

■ CAUTION: With ¼ tank or less, prolonged uncoordinated flight is prohibited when operating on either left or right tank.

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

◆ Note: The tanks are equipped with a low fuel sensor. If the fuel level is below 10 l (2.6 US gal) usable fuel per tank, the "Fuel L" or "Fuel R" Caution light illuminates respectively.

## PERMISSIBLE OIL TYPES

Engine oil: ..... AeroShell Oil Diesel Ultra
..... AeroShell Oil Diesel 10W-40
..... Shell Helix Ultra 5W-30
..... Shell Helix Ultra 5W-40

Gearbox oil: .....Centurion Gearbox Oil N1
..... Shell Spirax S6 ATF ZM
..... Shell Spirax S6 GXME 75W-80, API GL-4
..... Shell Spirax S4 G 75W-90, API GL-4

■ CAUTION: Use approved oil with exact designation only!

## PERMISSIBLE COOLING LIQUID

Coolant: ..... Water/Radiator Protection at a ratio of 50:50

Radiator Protection: ..... BASF Glysantin Protect Plus / G48
..... Valvoline/Zerex Glysantin G48
.....Mobil Antifreeze Extra (G48)
.....Comma Xstream Green - Concentrate/G4 48

**PLACARDS**

Near the fuel tank caps:

With standard tanks:

JET FUEL ONLY
JET A-1 / DIESEL
CAP. 63.7 LITER (16.8 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

With long-range tanks:

JET FUEL ONLY
JET A-1 / DIESEL
CAP. 79.3 LITER (20.9 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

Normal category aircraft with integral fuel tanks:

JET FUEL ONLY
JET A-1/ DIESEL
CAP. 98.4 LITER (26 U.S. GAL.)
USABLE TO BOTTOM OF FILLER INDICATOR TAB

At the fuel selector valve:

With standard tanks:

Left and Right position: 63.7 Ltr/ 16.8 gal
Both position: 127.4 Ltr/ 33.6 gal

With long-range tanks:

Left and Right position: 79.3 Ltr/ 20.9 gal
Both position: 158.6 Ltr/ 41.9 gal

Normal category aircraft with integral fuel tanks:

Left and Right position: 98.4 Ltr/ 26 gal
Both position: 196.8 Ltr/ 52 gal

Utility category aircraft with integral fuel tanks:

Left and Right position: 79.3 Ltr/ 20.9 gal
Both position: 158.6 Ltr/ 41.9 gal

On the oil funnel or at the flap of the engine cowling:

"Oil, see POH supplement"

If installed, at the flap of the engine cowling to the External Power Receptacle**:**

"ATTENTION 12 V DC OBSERVE CORRECT POLARITY"
OR
"ATTENTION 24 V DC OBSERVE CORRECT POLARITY“

All further placards contained in this section remain valid.

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# SECTION 3
# EMERGENCY PROCEDURES

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**GENERAL**

In addition to the original AFM/POH, the following applies:

| | | |
|---|---|---|
| ▲ | **WARNING:** | Due to failures indicated by the FADEC warning lights there might be a loss propeller valve current which leads in a low pitch setting of the propeller. This might result in overspeed.<br>Airspeeds below 100 KIAS are suitable to avoid overspeed in failure case. If the propeller speed control fails, climbs can be performed at 65 KIAS and a powersetting of 100%. |

## EMERGENCY PROCEDURES CHECK LIST

### ENGINE MALFUNCTION

DURING TAKE-OFF (WITH SUFFICENT RUNWAY AHEAD)

(1) Thrust Lever - IDLE
(2) Brakes - APPLY
(3) Wing flaps (if extended) - RETRACT to increase the braking effect on the runway
(4) Engine Master - OFF
(5) AlternatorMain Bus and Battery switch - OFF

IMMEDIATELY AFTER TAKE-OFF

If there is an engine malfunction after take-off, at first lower the nose to keep the airspeed and attain gliding attitude. In most cases, landing should be executed straight ahead with only small corrections in direction to avoid obstacles.

▲ **WARNING:** Altitude and airspeed are seldom sufficient for a return to the airfield with a 180° turn while gliding.

(1) Airspeed..... 65 KIAS (wing flaps retracted)
.....60 KIAS (wing flaps extended)
(2) Fuel Shut-off Valve - CLOSED
(3) Engine Master - OFF
(4) Wing flaps - as required ( recommended)
(5) AlternatorMain Bus and Battery switch - OFF

DURING FLIGHT

◆ Note: Running a tank dry activates both FADEC warning lights flashing.

In case that one fuel tank was flown empty, at the first signs of insufficient fuel feed proceed as follows:

(1) Fuel Shut-off Valve - OPEN (push full in)
(2) Immediately switch the Fuel Selector to BOTH position
(3) Electric Fuel Pump - ON
(4) Check the engine (engine parameters, airspeed/altitude change, whether the engine responds to changes in the Thrust Lever position).
(5) If the engine acts normally, continue the flight and land as soon as possible.

▲ **WARNING:** The high-pressure pump must be checked by an authorized service center before the next flight.

RESTART AFTER ENGINE FAILURE

Whilst gliding to a suitable landing strip, try to determine the reason for the engine malfunction. If time permits and a restart of the engine is possible, proceed as follows:

(1) Airspeed between 65 and 85 KIAS
(2) Glide below 13,000 ft
(3) Fuel Shut-off Valve - OPEN (push full in)
(4) Fuel Selector switch toBOTH position
(5) Electric Fuel Pump - ON
(6) Thrust Lever - IDLE
(7) Engine Master OFF and then ON
(if the propeller does not turn, then additionally Starter ON)

| | |
|---|---|
| ◆ Note: | The propeller will normally continue to turn as long as the airspeed is above 65 KIAS. Should the propeller stop at an airspeed of more than 65 KIAS or more, the reason for this should be found out before attempting a restart.<br>If it is obvious that the engine or propeller is blocked, do not use the Starter. |

)

| | |
|---|---|
| ◆ Note: | If the Engine Master is in position OFF, the Load Display shows no value even if the propeller is turning. |

(8) Check the engine power: Thrust lever 100%, engine parameters, check altitude and airspeed.

**FADEC WARNING**

FADEC MALFUNCTION IN FLIGHT

◆ Note: The FADEC consists of two components that are independent of each other: FADEC A and FADEC B. In case of malfunctions in the active FADEC, it automatically switches to the other.

**a) One FADEC Light is flashing**

1. Press FADEC test knob at least 2 seconds
2. FADEC light extinguished (LOW category warning):
   a) Continue flight normally
   b) Inform service center after landing
3. FADEC light illuminated steady (HIGH category warning)
   a) Observe the other FADEC light
   b) Land as soon as possible
   c) Select an airspeed to avoid engine overspeed
   d) Inform service center after landing

**a) Both FADEC Lights are flashing**

◆ Note: CED load display should be considered unreliable with both FADEC lights illuminated. Use other indications to assess engine condition.

1. Press FADEC test knob at least 2 seconds
2. FADEC Lights extinguished (LOW category warning):
   a) Continue flight normally
   b) Inform service center after landing
3. Steady FADEC Lights (HIGH category warning):
   a) Check the available engine power
   b) Expect engine failure
   c) Flight can be continued, however the pilot should
      i) Select an appropriate airspeed to avoid engine overspeed.
      ii) Land as soon as possible
      iii) Be prepared for an emergency landing
   d) Inform service center after landing
4. In case a fuel tank was flown empty, proceed at the first signs of insufficient fuel feed as follows:
   a) Immediately switch the Fuel Selector to BOTH
   b) Electric Fuel Pump - ON
   c) Select an airspeed to avoid engine overspeed.
   d) Check the engine (engine parameters, airspeed/altitude change, whether the engine responds to changes in the Thrust Lever position).
   e) If the engine acts normally, continue the flight and land as soon as possible.

▲ **WARNING:** The high-pressure pump must be checked by an authorized service center before the next flight.

ABNORMAL ENGINE BEHAVIOR

If the engine acts abnormal during flight and the system does not automatically switch to the B-FADEC, it is possible switch to the B-FADEC manually.

▲ **WARNING:** It is only possible to switch from the automatic position to B-FADEC (A-FADEC is active in normal operation, B-FADEC is active in case of malfunction). This only becomes necessary when no automatic switching occurred in case of abnormal engine behavior.

(1) Select an appropriate airspeed to avoid engine overspeed.

▲ **WARNING:** When operating on FADEC backup battery only, the "Force B" switch MUST not be activated. This will shut down the engine.

(2) "FORCE-B" switch to B-FADEC

(3) Flight may be continued, but the pilot should:

i) Select an airspeed to avoid engine overspeed

ii) Land as soon as possible

iii) Be prepared for an emergency landing

## FIRES

ENGINE FIRE WHEN STARTING ENGINE ON GROUND

(1) Engine Master - OFF
(2) Fuel Shut-off Valve - CLOSED
(3) Electric Fuel Pump - OFF
(4) Battery Switch - OFF
(5) Extinguish the flames with a fire extinguisher, wool blankets or sand.
(6) Inform service center after landing for examination of fire damages.

ENGINE FIRE DURING TAKE-OFF (ON GROUND)

(1) Engine Master - OFF
(2) Fuel Selector - CLOSED
(3) Electric Fuel Pump - OFF
(4) Battery switch - OFF
(5) Extinguish the flames with a fire extinguisher, wool blankets or sand.
(6) Inform service center after landing for examination of fire damages.

ENGINE FIRE IN FLIGHT

(1) Engine Master - OFF
(2) Fuel Shut-off Valve - CLOSED
(3) Select an airspeed to avoid engine overspeed
(4) Electric Fuel Pump - OFF
(5) Cabin heat and ventilation OFF resp. CLOSE (except the fresh air nozzles on the ceiling)
(6) Perform emergency landing (as described in the procedure "Emergency Landing With Engine Out")

ELECTRICAL FIRE IN FLIGHT

The first sign of an electrical fire is an unmistakable sharp, acrid smell. As the fire grows, electrical load might be higher than normal or circuit breakers start to trip. In this event proceed as follows:

(1) Main Bus - OFF
(2) Avionics Master - OFF
(3) Fresh air nozzles, Cabin Heat and Ventilation - OFF (closed)
(4) Fire Extinguisher - Activate (if available)
(5) All electrical consumers - Switch OFF, leave Alternator, battery and Engine Master ON

▲ **WARNING:** After the fire extinguisher has been used, make sure that the fire is extinguished before exterior air is used to remove smoke from the cabin.

(6) If there is evidence of continued electrical fire, consider turning off battery and alternator.

▲ **WARNING:** If both alternator and main battery are turned OFF, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only.

(7) Fresh Air Nozzles, Cabin Heat and Ventilation - ON (open)
(8) Check Circuit Breakers, do not reset if open

If the fire has been extinguished:

(9) Main Bus - ON
(10) Avionics Master - ON

▲ **WARNING:** Turn on electrical equipment required to continue flight depending on the situation and land as soon as possible. Switch circuit breakers switch ON one at a time, with delay after each.

## ENGINE SHUT DOWN IN FLIGHT

If it is necessary to shut down the engine in flight (for instance, abnormal engine behavior does not allow continued flight or there is a fuel leak, etc.), proceed as follows:

(1) Select an airspeed to avoid engine overspeed (best glide recommended)
(2) Engine Master - OFF
(3) Fuel Shut-off Valve - CLOSED
(4) Electric Fuel Pump - OFF
(5) If the propeller also has to be stopped (for instance, due to excessive vibrations)
    i) Reduce airspeed below 55 KIAS
    ii) When the propeller is stopped, continue to glide at 65 KIAS

## EMERGENCY LANDING

EMERGENCY LANDING WITH ENGINE OUT

If all attempts to restart the engine fail and an emergency landing is immanent, select suitable site and proceed as follows:

(1) Airspeed
    i) 65 KIAS (flaps retracted)
    ii) 60 KIAS (flaps extended)
(2) Fuel Shut-off Valve - CLOSED,
(3) Engine Master - OFF
(4) Wing Flaps - as required ( recommended)
(5) AlternatorMain Bus and Battery switch - OFF
(6) Cabin Doors - unlock before touch-down
(7) Touch-down - slightly nose up attitude
(8) Brake firmly

◆ Note: Gliding Distance. Refer to "Maximum Glide" in the approved Pilot´s Operating Handbook“

## FLIGHT IN ICING CONDITIONS

▲ **WARNING:** It is prohibited to fly in known icing conditions.

In case of inadvertent icing encounter proceed as follows:

(1) Pitot Heat switch - ON (if installed)
(2) Turn back or change the altitude to obtain an outside air temperature that is less conducive to icing.
(3) Pull the cabin heat control full out and open defroster outlets to obtain maximum windshield defroster airflow. Adjust cabin air control to get maximum defroster heat and airflow.
(4) Advance the Thrust Lever to increase the propeller speed and keep ice accumulation on the propeller blades as low as possible.
(5) Watch for signs of air filter icing and pull the "Alternate Air Door" control if necessary. An unexplaned loss in engine power could be caused by ice blocking the air intake filter. Opening the "Alternate Air Door" allows preheated air from the engine compartment to be aspirated.
(6) Plan a landing at the nearest airfield. With an extremely rapid ice build up, select a suitable "off airfield" landing site.
(7) With an ice accumulation of 0.5 cm or more on the wing leading edges, a significantly higher stall speed should be expected.
(8) Leave wing flaps retracted. With a severe ice build up on the horizontal tail, the change in wing wake airflow direction caused by wing flap extension could result in a loss of elevator effectiveness.
(9) Perform a landing approach using a forward slip, if necessary, for improved visibility.
(10) Approach at 65 to 75 KIAS depending upon the amount of the accumulation.
(11) Perform a landing in level attitude.

## RECOVERY FROM SPIRAL DIVE

If a spiral is encountered in the clouds, proceed as follows:

(1) Retard Thrust Lever to idle position
(2) Stop the turn by using coordinated aileron and rudder control to align the symbolic airplane in the turn coordinator with the horizontal reference line.
(3) Cautiously apply elevator back pressure to slowly reduce the airspeed to 80 KIAS.
(4) Adjust the elevator trim control to maintain an 80 KIAS glide.
(5) Keep hands off the control wheel, using rudder control to hold a straight heading.
(6) Readjust the rudder trim (if installed) to relieve the rudder of asymmetric forces.
(7) Clear the engine occasionally, but avoid using enough power to disturb the trimmed glide.
(8) Upon breaking out of clouds, resume normal cruising flight and continue the flight.

## ELECTRICAL POWER SUPPLY SYSTEM MALFUNCTIONS

▲ **WARNING** If the power supply from both alternator and main battery is interrupted, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only. In this case, all electrical equipment will not operate:

- land as soon as possible
- do not switch the FORCE-B switch, this will shut down the engine

■ CAUTION: The TAE 125-02-114 requires an electrical power source for its operation. If the alternator fails, continued engine operation time is dependent upon the remaining capacity of the main battery, the FADEC backup battery and equipment powered. The engine has been demonstrated to continue operating for approximately 120 minutes based upon the following assumptions:

■ CAUTION: This table only gives a reference point. The pilot should turn off all nonessential items and supply power only to equipment which is absolutely necessary for continued flight depending upon the situation.

Deviating from this recommendation, the remaining engine operating time may change.

| Equipment | | Time switched on | |
|---|---|---|---|
| | | in [min] | in [%] |
| NAV/COM 1 receiving | ON | 120 | 100 |
| NAV/COM 1 transmitting | ON | 12 | 10 |
| NAV/COM 2 receiving | OFF | 0 | 0 |
| NAV/COM 2 transmitting | OFF | 0 | 0 |
| GPS | ON | 60 | 50 |
| Transponder | ON | 120 | 100 |
| Fuel Pump | OFF | 0 | 0 |
| AED-125 | ON | 120 | 100 |
| Battery | ON | 120 | 100 |
| CED-125 | ON | 120 | 100 |
| Landing Light | ON | 12 | 10 |
| Flood Light | ON | 1.2 | 1 |
| Pitot Heat | ON | 24 | 20 |
| Wing Flaps | ON | 1.2 | 1 |
| Interior Lighting | OFF | 0 | 0 |
| Nav Lights | OFF | 0 | 0 |
| Beacon | OFF | 0 | 0 |
| Strobes | OFF | 0 | 0 |
| ADF | OFF | 0 | 0 |
| Intercom | OFF | 0 | 0 |
| Engine Control | ON | 120 | 100 |
| | | | |

Table 3-1a

ALTERNATOR WARNING DURING NORMAL ENGINE OPERATION

(1) Ammeter - CHECK
(2) Alternator switch CHECK - ON
(3) Battery Switch CHECK - ON

■ CAUTION: If the FADEC was supplied by battery only until this point, the RPM can momentarily drop, when the alternator will be switched on. In any case: leave the alternator switched ON!

(4) Electrical load - REDUCE IMMEDIATELY as follows:

iii) Fuel Pump – OFF
iv) Landing Light – OFF (use as required for landing)
v) Taxi Light – OFF
vi) Strobe Light – OFF
vii) Nav Lights – OFF
viii) Beacon – OFF
ix) Interior Lights – OFF
x) Intercom – OFF
xi) Pitot Heat – OFF (use as required)
xii) Autopilot – OFF
xiii) Non-essential equipment – OFF

(5) The pilot should:

i) Land as soon as possible.
ii) Be prepared for an emergency landing.
iii) Expect an engine failure.

AMMETER SHOWS BATTERY DISCHARGE DURING NORMAL ENGINE OPERATION FOR MORE THAN 5 MINUTES

(1) Ammeter - CHECK
(2) Alternator switch CHECK - ON
(3) Battery Switch CHECK - ON

■ CAUTION: If the FADEC was supplied by battery only until this point, the RPM can momentarily drop, when the alternator will be switched on. In any case: leave the alternator switched ON!

(4) Electrical load - REDUCE IMMEDIATELY as follows:
   i) NAV/ COM 2 – OFF
   ii) Fuel Pump – OFF
   iii) Landing Light – OFF (use as required for landing)
   iv) Taxi Light – OFF
   v) Strobe Light – OFF
   vi) Nav Lights – OFF
   vii) Beacon – OFF
   viii) Interior Lights – OFF
   ix) Intercom – OFF
   x) Pitot Heat – OFF (use as required)
   xi) Autopilot – OFF
   xii) Non-essential equipment – OFF
(5) The pilot should:
   i) Land as soon as possible
   ii) Be prepared for an emergency landing
   iii) Expect an engine failure

TOTAL ELECTRICAL FAILURE
(all equipment inoperative, except engine)

▲ **WARNING:** If the power supply from both alternator and main battery is interrupted simultaneously, continued engine operation is dependent on the remaining capacity of the FADEC backup battery. The engine has been demonstrated to continue operating for a maximum of 30 minutes when powered by the FADEC backup battery only. In this case, all other electrical equipment will not operate.

▲ **WARNING:** If the aircraft was operated on battery power only until this point (alternator warning light illuminated), the remaining engine operating time may be less than 30 minutes.

▲ **WARNING:** Do not activate the FORCE-B switch, this will shut down the engine.

(1) Alternator switch CHECK - ON
(2) Battery Switch CHECK – ON
(3) Land as soon as possible
    i) Be prepared for an emergency landing
    ii) Expect an engine failure

**ROUGH ENGINE OPERATION OR LOSS OF POWER**

DECREASE IN POWER

(1) Push Thrust Lever full forward (take-off position)
(2) Fuel Selector to BOTH position
(3) Electric Fuel Pump - ON
(4) Reduce airspeed to 65-85 KIAS/75-98 mph (best glide recommended), (max. 100 KIAS/115 mph)
(5) Check engine parameters (FADEC lights, oil pressure and temperature, fuel quantity)

If normal engine power is not achieved, the pilot should:

i) Land as soon as possible
ii) Be prepared for an emergency landing
iii) Expect an engine failure

▲ **WARNING:** The high pressure pump must be checked by an authorized service center before the next flight.

ICE FORMATION IN THE CARBURETOR

- N/A, since this is a Diesel engine -

SOILED SPARK PLUGS

- N/A, since this is a Diesel engine -

IGNITION MAGNET MALFUNCTIONS

- N/A, since this is a Diesel engine -

iv)

OIL PRESSURE TOO LOW (< 2.3 BAR IN CRUISE (AMBER RANGE) OR < 1.2 BAR AT IDLE (RED RANGE)):

(1) Reduce power as quickly as possible
(2) Check oil temperature: If the oil temperature is high or near operating limits,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

◆ Note: During warm-weather operation or long climbs at low airspeed engine temperatures could rise into the amber range and trigger the "Caution" light. This indication allows the pilot to avoid overheating of the engine as follows:

(3) Increase the climbing airspeed, reduce angle of climb
(4) Reduce power, if the engine temperatures approach the red range

OIL TEMPERATURE TOO HIGH (RED RANGE):

(1) Increase airspeed and reduce power as quickly as possible
(2) Check oil pressure: if the oil pressure is lower than normal (< 2.3 bar in cruise or < 1.2 bar at idle),
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure
(3) If the oil pressure is in the normal range:
    i) Land as soon as possible

COOLANT TEMPERATURE TOO HIGH (RED RANGE)**:**

(1) Increase airspeed and reduce power as quickly as possible
(2) Cabin Heat - COLD
(3) If coolant temperature reduces rapidly to normal range, continue to fly normally and monitor coolant temperature, Cabin Heat.
(4) If coolant temperature does not decrease,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

"WATER LEVEL" LIGHT ILLUMINATES

(1) Increase airspeed and reduce power as quickly as possible
(2) Coolant temperature "CT" check and observe
(3) Oil temperature "OT" check and observe
(4) If coolant temperature and/or oil temperature are rising into amber or red range,
    i) Land as soon as possible
    ii) Be prepared for an emergency landing
    iii) Expect an engine failure

GEARBOX TEMPERATURE TOO HIGH (RED RANGE)**:**

(1) Reduce power to 55% - 75% as quickly as possible
(2) Land as soon as possible

FUEL TEMPERATURE TOO HIGH:

(1) Switch to fuel tank with lower fuel temperature, if this contains sufficient fuel
(2) Reduce engine power, if possible
(3) If fuel temperature remains in Red Range, land as soon as possible

FUEL TEMPERATURE TOO LOW (AMBER RANGE for Diesel Operation, RED RANGE for Kerosine Operation):

(1) Switch to fuel tank with higher fuel temperature, if this contains sufficient fuel
(2) Change to altitude with higher outside air temperature
(3) If use of the non-active tank is intended, switch fuel selector to BOTH

◆ Note: Low fuel temperature may be caused when flying in cold weather with fuel cooler in operation (baffle removed).

PROPELLER RPM TOO HIGH:

With propeller RPM between 2,400 and 2,500 for more than 10 seconds or over 2,500:

(1) Reduce power
(2) Reduce airspeed below 100 KIAS or as appropriate to prevent propeller overspeed
(3) Set power as required to maintain altitude and land as soon as possible.

◆ Note: If the propeller speed control fails, climbs be performed at 65 KIAS and a power setting of 100%.
In case of overspeed the FADEC will reduce the engine power at higher airspeeds to avoid propeller speeds above 2500 rpm.

FLUCTUATIONS IN PROPELLER RPM:

If the propeller RPM fluctuates by more than + / - 100 RPM with a constant Thrust Lever position:

(1) Change the power setting and attempt to find a power setting where the propeller RPM no longer fluctuates.
(2) If this does not work, set the maximum power at an airspeed < 100 KIAS until the propeller speed stabilizes.
(3) If the problem is resolved, continue the flight
(4) If the problem continues, select a power setting where the propeller RPM fluctuations are minimum. Fly at an airspeed below 100 KIAS and land as soon as possible.

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# SECTION 4
# NORMAL PROCEDURES

## PREFLIGHT INSPECTION

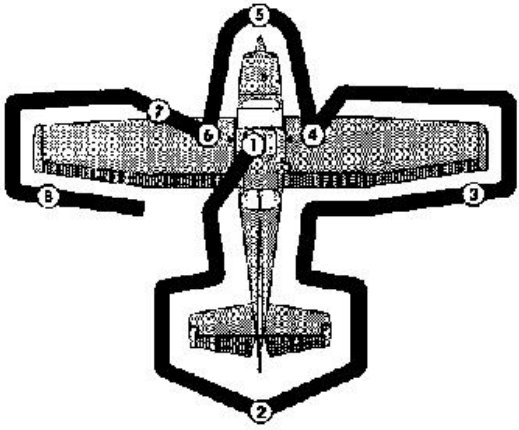



Figure 4-1 Preflight Inspection

◆ Note: Visually check airplane for general condition during walk around inspection. In cold weather, remove even small accumulations of frost, ice or snow from wing, tail and control surfaces. Also, make sure that control surfaces contain no internal accumulations of ice or debris. Prior to flight, check that pitot heater (if installed) is warm to touch within 30 seconds with battery and pitot heat switches on. If a night flight is planned, check operation of all lights, and make sure a flashlight is available.

**(1) CABIN**

(1) Pilot´s Operating Handbook - AVAILABLE IN THE AIRPLANE
(2) Airplane Weight and Balance - CHECKED
(3) Parking Brake - SET
(4) Control Wheel Lock - REMOVE
(5) "Engine Master" - OFF
(6) Avionics Power Switch - OFF
(7) "Shut-off Cabin Heat" - OFF (Push Full Forward

▲ **WARNING:** When turning on the Battery switch, using an external power source, or pulling the propeller through by hand, treat the propeller as if the Engine Master was on.

(8) Battery and Main Bus switches - ON
(9) Fuel Quantity Indicators and Fuel Temperature CHECK
(10) Light "Water Level" - CHECK OFF
(11) Battery and Main Bus switches - OFF
(12) Entry in log-book concerning type of fuel filled - CHECK
(13) Static Pressure Alternate Source Valve - CHECK
(14) Fuel Selector Valve - BOTH
(15) Fuel Shut-off Valve - ON (Push Full In)
(16) Baggage Door - CHECK, lock with key if the child's seat is supposed to be occupied

**(2) EMPENNAGE**

(1) Rudder Gust Lock (if attached) - REMOVE
(2) Tail Tie-Down - DISCONNECT
(3) Control Surfaces - CHECK freedom of movement and security

**(3) RIGHT WING Trailing Edge**

(1) Aileron - CHECK freedom of movement and security

(2) Flap - CHECK for security and condition

**(4) RIGHT WING**

(1) Wing Tie-Down - DISCONNECT

(2) Main Wheel Tire - CHECK for proper inflation and general condition (weather checks, tread depth and wear, etc.).

▲ **WARNING** If, after repeated sampling, evidence of contamination still exists, the airplane should not be flown. Tanks should be drained and system purged by qualified maintenance personnel. All evidence of contamination must be removed before further flight.

(3) Fuel Tank Sump Quick Drain Valves - DRAIN at least a cupful of fuel (using sampler cup) from each sump location to check for water, sediment and the right type of fuel (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling points. Take repeated samples from all fuel drain points until all contamination has been removed. If contaminants are still present, refer to above WARNING and do not fly airplane.

(4) Fuel Quantity - CHECK VISUALLY for desired level not above marking in fuel filler

(5) Fuel Filler Cap - SECURE

**(5) NOSE**

(1) Reservoir Tank Quick Drain Valve - DRAIN at least a cupful of fuel (using sampler cup) from valve to check for water, sediment and proper fuel grade (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling point. Take repeated samples until all contamination has been removed.

◆ Note: The reservoir tank drain is located in the fuselage on the co-pilot side of the aircraft.

(2) Before first flight of the day and after each refueling - DRAIN the Fuel Strainer Quick Drain Valve with the sampler cup to remove water and sediment from the screen. Ensure that the screen drain is properly closed again. If water is discovered, there might be even more water in the fuel system. Therefore, take further samples from fuel strainer and the tank sumps.

◆ Note: The fuel strainer drain is located on the left-hand side of the firewall (flight direction).

(3) Engine Oil Dipstick/Filler Cap:
   a) Oil level - CHECK
   b) Dipstick/filler cap - SECURE

   Do not operate below the minimum dipstick indication.

(4) Engine Air and Cooling Inlets - CLEAR of obstructions

(5) Landing Light - CHECK for condition and cleanliness

(6) Propeller and Spinner - CHECK for nicks and security

(7) Gearbox Oil Level - CHECK the oil has to cover at least half of the inspection glass

(8) Nose Wheel Strut and Tire- CHECK for proper inflation of strut and general condition (weather checks, tread depth and wear, etc.) of tire

(9) Left Static Source Opening - CHECK for blockage

(10) Fuel cooler baffle - CHECK

- REMOVE, if OAT on ground is higher than 20°C (68°F)
- INSTALL, if OAT on ground is lower than 20°C (68°F)

**(6) LEFT WING**

(1) Fuel Quantity - CHECK VISUALLY for desired level not above marking in fuel filler

(2) Fuel Filler Cap - SECURE

(3) Fuel Tank Sump Quick Drain Valves - DRAIN at least a cupful of fuel (using sampler cup) from each sump location to check for water, sediment and the right type of fuel (Diesel or JET-A1) before each flight and after each refueling. If water is observed, take further samples until clear and then gently rock wings and lower tail to the ground to move any additional contaminants to the sampling points. Take repeated samples from all fuel drain points until all contamination has been removed. If contaminants are still present, refer to previous WARNING (see right wing) and do not fly airplane.

(4) Main Wheel Tire - CHECK for proper inflation and general condition (weather checks, tread depth and wear, etc.)

**(7) LEFT WING Leading Edge**

(1) Pitot Tube Cover (if mounted) - REMOVE and CHECK for pitot blockage

(2) Fuel Tank Vent Opening - CHECK for blockage

(3) Stall Warning Opening - CHECK for blockage
To check the system, place a clean handkerchief over the vent opening and apply suction; a sound from the warning horn will confirm system operation.

(4) Wing Tie-Down - DISCONNECT

**(8) LEFT WING Trailing Edge**

(1) Aileron - CHECK freedom of movement and security

(2) Flap - Check for security and conditions

**BEFORE STARTING ENGINE**

(1) Preflight Inspection - COMPLETE

(2) Seats and Seat Belts - ADJUST and LOCK

(3) Brakes - TEST and SET

(4) Avionics Power Switch, Autopilot (if installed) and Electrical Equipment - OFF

■ CAUTION: The Avionics Power Switch must be off during engine start to prevent possible damage to avionics.

(5) Circuit Breakers - CHECK IN

(6) Alternator Switch - CHECK ON

(7) Battery and Main Bus Switches - ON

■ CAUTION: The electronic engine control needs an electrical power source for its operation. For normal operation Battery, Alternator and Main Bus have to be switched on. Separate switching is only allowed for tests and in the event of emergencies.

(8) Fuel Quantity and Temperature - CHECK

(9) Fuel Selector Valve - SET to BOTH position.The fuel temperature limitations must be observed.

(10) Fuel Shut-off Valve - OPEN (Push Full In)

(11) Alternate Air Door - CLOSED

(12) Thrust Lever - CHECK for freedom of movement

(13) Load Display - CHECK 0% at Propeller RPM 0

## PROCEDURES UP TO 5500ft AIRFIELD ELEVATION

STARTING ENGINE

▲ **WARNING:** Do not use ground power unit for engine starts. It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

(1) Electric Fuel Pump - ON
(2) Navigation Lights and Flashing Beacon - ON (as required).
(3) Thrust Lever - IDLE
(4) Area Aircraft / Propeller - CLEAR
(5) "Engine Master" - ON , wait until the Glow Control light extinguishes
(6) Starter - ON, keep starter engaged until min. 500rpm
Release when engine starts, leave Thrust Lever in idle

■ CAUTION: Do not overheat the starter motor. Do not operate the starter motor for more than 10 seconds. After operating the starter motor, let it cool off for 20 seconds. After 6 attempts to start the engine, let the starter cool off for half an hour.

(7) Oil Pressure - CHECK

■ CAUTION: If after 3 seconds the minimum oil pressure of 1 bar is not indicated:
shut down the engine immediately!

(8) CED 125 Test Knob - PRESS (to delete Caution light)
(9) Ammeter - CHECK for positive charging current
(10) Voltmeter - CHECK for green range
(11) FADEC Backup Battery test
  a) Alternator - OFF, engine must operate normally
  b) Battery - OFF, for min. 10 seconds;
  engine must operate normally, the red FADEC lamps must not be illuminated
  c) Battery - ON
  d) Alternator - ON

▲ **WARNING:** It must be ensured that both battery and alternator are ON!
If the guarded alternator switch is installed, the switch guard muts be closed.

(12) Avionic-Power Switch - ON
(13) Radios - ON
(14) Ammeter - Check positive charge, alternator warning light must be OFF
(15) Voltmeter - Check in green range
(16) Electric Fuel Pump - OFF
(17) Flaps - RETRACT

**WARM UP**

(1) Let the engine warm up about 2 minutes at 890 RPM.
(2) Increase RPM to 1400 until oil temperature 50°C (122°F), coolant temperature 60°C (140°F).

**BEFORE TAKE-OFF**

(1) Parking Brake - SET
(2) Cabin Doors and Windows - CLOSED and LOCKED
(3) Flight Controls - FREE and CORRECT
(4) Flight Instruments - CHECK and SET
(5) Fuel quantity - CHECK
(6) Fuel Selector Valve - SET to BOTH
(7) Elevator Trim and Rudder Trim (if installed) - SET for take-off
(8) FADEC and propeller adjustment function check:
    a) Thrust Lever - IDLE (both FADEC lights should be OFF)
    b) FADEC Test Button - PRESS and HOLD button for entire test
    c) Both FADEC lights - ON, RPM increases.

▲ **WARNING:** If the FADEC lights do not come on at this point, it means that the test procedure has failed and take off should not be attempted.

    d) The FADEC automatically switches to B-component (only FADEC B light is ON)
    e) The propeller control is excited, RPM decreases
    f) The FADEC automatically switches to channel A (only FADEC A light is ON), RPM increases
    g) The propeller control is excited, RPM decreases
    h) FADEC A light goes OFF, idle RPM is reached, the test is completed.
    i) FADEC Test Button - RELEASE

(9) Force B Switch - switch to FADEC B

(10) Engine - check running without a change

(11) Force B Switch - switch back to Automatic

| | |
|---|---|
| ▲ **WARNING:** | If there are prolonged engine misfires or the engine shuts down during the test, take off may not be attempted. |
| ▲ **WARNING:** | The whole test procedure has to be performed without any failure. In case the engine shuts down or the FADEC lights are flashing, take-off is prohibited. This applies even if the engine seems to run without failure after the test. |
| ◆ Note: | If the test button is released before the self test is over, the FADEC immediately switches over to normal operation. |
| ◆ Note: | While switching from one FADEC to another, it is normal to hear and feel a momentary surge in the engine. |

(12) Thrust Lever - FULL FORWARD, load display min. 94%, RPM 2240 - 2300

(13) Thrust Lever - IDLE

(14) Engine Instruments and Ammeter - CHECK

(15) Suction gage - CHECK

(16) Wing Flaps - SET for Take-off (0° or 10°).

(17) Electric Fuel Pump - ON

(18) Strobe Lights - AS DESIRED

(19) Radios and Avionics - ON and SET

(20) Autopilot (if installed) - OFF

(21) Air Conditioning (if installed) - OFF

(22) Thrust Lever Friction Control - ADJUST

(23) Brakes - RELEASE

**PROCEDURES OVER 5500ft AIRFIELD ELEVATION**

◆ Note: Due to the increase of the idle speed with increasing pressure altitudes, the FADEC test is only possible to a limited extent from an airfield elevation of approximately 5500ft.
Over 5500ft, the FADEC test is only possible if the load selector lever remains in the idle position after engine start until the FADEC test is starting.
If the load selector lever is moved from the idle position, a FADEC test is no longer possible at pressure altitudes above 5500ft. For this purpose, the engine has to be stopped and re-started to perform the FADEC test.

STARTING ENGINE

▲ **WARNING:** Do not use ground power unit for engine starts. It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

(1) Electric Fuel Pump – ON
(2) Navigation Lights and Flashing Beacon - ON (as required)
(3) Thrust Lever – IDLE
(4) Area Aircraft / Propeller – CLEAR
(5) "Engine Master" – ON, wait until the Glow Control light extinguishes
(6) Starter – ON, keep starter engaged until min. 500rpm
Release when engine starts, leave Thrust Lever in idle

| | |
|---|---|
| ■ CAUTION: | Do not overheat the starter motor. Do not operate the starter motor for more than 10 seconds. After operating the starter motor, let it cool off for 20 seconds. After 6 attempts to start the engine, let the starter cool off for half an hour. |

(7) Oil Pressure – CHECK

| | |
|---|---|
| ■ CAUTION: | If after 3 seconds the minimum oil pressure of 1 bar is not indicated:<br>shut down the engine immediately! |

(8) CED-Test Knob – PRESS (to delete Caution light)

(9) Ammeter – CHECK for positive charging current

(10) Voltmeter – CHECK for green range

(11) FADEC Backup Battery test

a) Alternator – OFF, engine must operate normally

b) Battery – OFF, for min. 10 seconds;
engine must operate normally, the red FADEC lamps must not be illuminated

c) Battery – ON

d) Alternator – ON

| | |
|---|---|
| ▲ **WARNING:** | It must be ensured that both battery and alternator are ON!<br>If the guarded alternator switch is installed, the switch guard must be closed. |

(12) Ammeter – Check positive charge, alternator warning light must be OFF

(13) Voltmeter – Check in green range

(14) Flaps – RETRACT

WARM UP AND FADEC-TEST

(1) Let the engine warm up about 2 minutes at 890 RPM.
(2) Increase RPM to 1,400 until Oil Temperature 50°C, Coolant Temperature 60°C.
(3) Thrust Lever - IDLE
(4) "Engine Master" - OFF
(5) Area Aircraft / Propeller - CLEAR
(6) "Engine Master" - ON, wait until the Glow Control light extinguishes
(7) Starter - ON, keep starter engaged until min. 500rpm Release when engine starts, leave Thrust Lever in idle
(8) Ammeter - CHECK for positive charging current
(9) Voltmeter - CHECK for green range
(10) FADEC and propeller adjustment function check:
    a) Thrust Lever - IDLE (both FADEC lights should be OFF).
    b) FADEC Test Button - PRESS and HOLD button for entire test.
    c) Both FADEC lights - ON, RPM increases

▲ **WARNING:** If the FADEC lights do not come on at this point, it means that the test procedure has failed and take off should not be attempted.

    d) The FADEC automatically switches to B-component (only FADEC B light is ON)
    e) The propeller control is excited, RPM decreases
    f) The FADEC automatically switches to channel A (only FADEC A light is ON), RPM increases
    g) The propeller control is excited, RPM decreases
    h) FADEC A light goes OFF, idle RPM is reached, the test is completed.
    i) FADEC Test Button - RELEASE
(11) Force B Switch - switch to FADEC B
(12) Engine - check running without a change
(13) Force B Switch - switch back to Automatic

| | |
|---|---|
| ▲ **WARNING:** | If there are prolonged engine misfires or the engine shuts down during the test, take off may not be attempted. |

| | |
|---|---|
| ▲ **WARNING:** | The whole test procedure has to be performed without any failure. In case the engine shuts down or the FADEC lights are flashing, take off is prohibited. This applies even if the engine seems to run without failure after the test. |

| | |
|---|---|
| ◆ Note: | If the test button is released before the self test is over, the FADEC immediately switches over to normal operation. |

| | |
|---|---|
| ◆ Note: | While switching from one FADEC to another, it is normal to hear and feel a momentary surge in the engine. |

(14) Avionic Switch – ON
(15) Radios – ON
(16) Electric Fuel Pump – OFF

BEFORE TAKE-OFF

(1) Parking Brake – SET
(2) Cabin Doors and Windows – CLOSED and LOCKED
(3) Flight Controls – FREE and CORRECT
(4) Flight Instruments – CHECK and SET
(5) Fuel quantity – CHECK
(6) Fuel Selector Valve – SET to BOTH position
(7) Elevator Trim and Rudder Trim (if installed) – SET for Takeoff
(8) Thrust Lever – FULL FORWARD, load display min. 94%, RPM 2240 - 2300
(9) Thrust Lever – IDLE
(10) Engine Instruments and Ammeter – CHECK
(11) Suction gage – CHECK
(12) Wing Flaps – SET for Take-off (0° or 10°).
(13) Electric Fuel Pump – ON
(14) Strobe Lights – AS DESIRED
(15) Radios and Avionics – ON and SET
(16) Autopilot (if installed) – OFF
(17) Air Conditioning (if installed) – OFF
(18) Thrust Lever Friction Control – ADJUST
(19) Brakes – RELEASE

**TAKE-OFF**

NORMAL TAKE-OFF

(1) Wing Flaps - 0° or 10°
(2) Thrust Lever - FULL FORWARD
(3) Elevator Control - LIFT NOSE WHEEL at 55 KIAS/63 mph.
(4) Climb Speed - 65 to 80 KIAS/75 to 92 mph

SHORT FIELD TAKE-OFF

(1) Wing Flaps - 10°
(2) Brakes - APPLY
(3) Thrust Lever - FULL FORWARD
(4) Brakes - RELEASE
(5) Airplane Attitude - SLIGHTLY TAIL LOW
(6) Elevator Control - LIFT NOSE WHEEL at 44 KIAS
(7) Climb Speed - 58 KIAS67mph(until all obstacles are cleared)

**AFTER TAKE-OFF**

(1) Altitude about 300 ft, Airspeed more than 65 KIAS/75 mph - Wing Flaps - RETRACT
(2) Electric Fuel Pump - OFF

**CLIMB**

(1) Airspeed - 70 to 85 KIAS/80 to 98 mph.

◆ Note: If a maximum performance climb is necessary, use speeds shown in the "Maximum Rate Of Climb" chart in Section 5. In case that oil temperature and/or coolant temperature are approaching the upper limit, continue at a lower climb angle for better cooling if possible.

◆ Note: The fuel temperatures have to be monitored.

(2) Thrust Lever - FULL FORWARD

**CRUISE**

(1) Power - maximum load 100% (maximum continuous power), 75% or less is recommended.
For economic cruise set load 70% or less.

(2) Elevator trim and Rudder trim (if installed) - ADJUST

(3) Compliance with Limits for oil pressure, oil temperature, coolant temperature and gearbox temperature (CED 125 and Caution light) - MONITOR constantly

(4) Fuel Quantity and Temperature (Display and LOW LEVEL caution lights) - MONITOR.

Whenever possible, the airplane should be flown with the fuel selector in the BOTH position to empty and heat both fuel tanks evenly. However, operation in the LEFT or RIGHT position may be desirable to correct a fuel quantity imbalance or during periods of intentional uncoordinated flight maneuvers. During prolonged operation with the fuel selector in either the LEFT or RIGHT position the fuel balance and temperatures should be closely monitored.

■ CAUTION: Do not use any fuel tank below the minimum permissible fuel temperature!

■ CAUTION: In turbulent air it is strongly recommended to use the BOTH position.

■ CAUTION: With ¼ tank or less prolonged or uncoordinated flight is prohibited when operating on either the left or right tank.

(5) FADEC and Alternator Warning lights - MONITOR

**DESCENT**

(1) Fuel Selector Valve - SELECT BOTH position
(2) Power - AS DESIRED

**BEFORE LANDING**

(1) Pilot and Passenger Seat Backs - MOST UPRIGHT POSITION
(2) Seats and Seat Belts - SECURED and LOCKED
(3) Fuel Selector Valve - SELECT BOTH position
(4) Electric Fuel Pump - ON
(5) Landing / Taxi Lights - ON
(6) Autopilot (if installed) - OFF
(7) Air Conditioning (if installed) - OFF

**LANDING**

NORMAL LANDING

(1) Airspeed - 69 to 80 KIAS/80 to 92 mph (wing flaps UP)
(2) Wing Flaps - AS DESIRED (0°-10° below 110 KIAS/126 mph; 10°-below 85 KIAS/98 mph)
(3) Airspeed in Final Approach:
    - wing flaps 20°: 63 KIAS/72 mph
    - wing flaps 30°: 60 KIAS/69 mph
(4) Touchdown - MAIN WHEELS FIRST
(5) Landing Roll - LOWER NOSE WHEEL GENTLY
(6) Brakes - MINIMUM REQUIRED

SHORT FIELD LANDING

(1) Airspeed - 69 to 80 KIAS/80 to 92 mph (Flaps UP)
(2) Wing Flaps - 30°
(3) Airspeed in Final Approach - 60 KIAS/69 mph (until flare)
(4) Power - REDUCE to idle after clearing obstacles.
(5) Touchdown - MAIN WHEELS FIRST
(6) Brakes - APPLY HEAVILY
(7) Wing Flaps - RETRACT

BALKED LANDING

(1) Thrust Lever - FULL FORWARD
(2) Wing Flaps - RETRACT TO 20° (immediately after Thrust Lever FULL FORWARD)
(3) Climb Speed - 58 KIAS/67 mph
(4) Wing Flaps - 10° (until all obstacles are cleared)
(5) Wing Flaps - RETRACT after reaching a safe altitude and 65 KIAS/75 mph

**AFTER LANDING**

(1) Wing Flaps - RETRACT
(2) Electric Fuel Pump - OFF

**SECURING AIRPLANE**

(1) Parking Brake - SET
(2) Thrust Lever - IDLE
(3) Avionics Power Switch, Electrical Equipment, Autopilot (if installed) - OFF
(4) Main Bus switch - OFF
(5) "Engine Master" - OFF
(6) Switch Battery - OFF
(7) Control Lock - INSTALL
(8) Fuel Selector Valve - LEFT or RIGHT (to prevent crossfeeding between tanks)

## AMPLIFIED PROCEDURES

STARTING ENGINE

The TAE 125-02-114is a direct Diesel injection engine with common-rail technology and a turbocharger. It is controlled automatically by the FADEC, which makes a proper performance of the FADEC test important for safe flight operation. All information relating to the engine are compiled in the CED 125 multifunction instrument.
Potentiometers within the thrust lever transmit the load value selected by the pilot to the FADEC.
If the engine master is switched ON, the preheating relay is actuated by the FADEC and the glow plugs are supplied with power. The glow duration depends on the engine temperature.
If the engine master is switched OFF, the injection valves are not supplied with power and remain closed. The switch/push button "Starter" controls the Starter.

EXTERNAL POWER

External power may be used to charge the battery or for maintenance purposes. To charge the battery with external power the battery switch must be ON.

When using an External Power Source, the Battery Switch must be in the OFF position before connecting the External Power Source to the airplane receptacle.

It is not allowed to start up the engine using external power. If starting the engine is not possible using battery power, the condition of the battery must be verified before flight.

TAXIING

When taxiing, it is important that speed and use of brakes be held to a minimum and that all controls be utilized to maintain directional control and balance.
The alternate air door should always be pushed for ground operation to ensure that no unfiltered air is sucked in.
Taxiing over loose gravel or cinders should be done at low engine speed to avoid abrasion and stone damage to the propeller tips.

BEFORE TAKE-OFF

WARM UP

To warm up the engine, operate the engine for about 2 minutes at 890 RPM.
Let the engine run at propeller RPM of 1400 until it reaches an engine oil temperature of 50°C (122°F) (green range) and a coolant temperature of 60°C (140°F) (green range to ensure normal operation).

MAGNETO CHECK

N/A since this is a Diesel engine.

ALTERNATOR CHECK

Prior to flights where verification of proper alternator and alternator control unit operation is essential (such as night and instrument flights), a positive verification can be made by loading the electrical system momentarily (3 to 5 seconds) with the landing light or by operating the wing flaps during the engine run-up (20% load). The ammeter will remain within a needle width of zero if the alternator and alternator control unit are operating properly.

BATTERY CHECK

If there is doubt regarding the battery conditions or functionality the battery has to be checked after warm-up as follows:

Switch off the alternator while the engine is running (battery remains "ON")"

Perform a 10 sec. engine run. The voltmeter must remain in the green range. If not, the battery has to be charged or, if necessary, exchanged.

After this test the alternator has to be switched on again.

**TAKE-OFF**

POWER CHECK

It is important to check full load engine operation early in the take-off roll. Any signs of rough engine operation or sluggish engine acceleration is good cause for discontinuing the take-off. If this occurs, you are justified in making a thorough full load static run-up before another take-off is attempted.
After full load is applied, adjust the thrust lever friction control to prevent the thrust lever from creeping back from a maximum power position. Similar friction lock adjustments should be made as required in other flight conditions to maintain a fixed thrust lever setting.

WING FLAP SETTINGS

Flap deflections greater than 10° are not approved for normal and short field take-offs. Using 10° wing flaps reduces the ground roll and total distance over a 15 m/50 ft obstacle by approximately 10%.

**CLIMB**

Normal climbs are performed with flaps up and full load and at speeds 5 to 10 knots/7 to 12 mph higher than best rate-of-climb speeds for the best combination of engine cooling, climb speed and visibility. The speed for best climb is about 69 KIAS. If an obstruction dictates the use of a steep climb angle, climb at 62 KIAS/71 mph and flaps up.

◆ Note: Climbs at low speeds should be of short duration to improve engine cooling.

**CRUISE**

As guidance for calculation of the optimum altitude and power setting for a given flight use the tables in Section 5.

**LANDING**

NORMAL LANDING

Remarks in Pilot´s Operating Handbook concerning carburetor pre-heating are **N/A**

BALKED LANDING

In a balked landing (go around) climb, reduce the flap setting to 20° immediately after full power is applied. If obstacles must be cleared during the go-around climb, reduce wing flap setting to 10° and maintain a safe airspeed until the obstacles are cleared. After clearing any obstacles, the flaps may be retracted as the airplane accelerates to the normal flaps up climb speed.

**CARBURETOR ICING**

N/A since this is a Diesel engine.

**FLIGHT IN HEAVY RAIN**

N/A since no special procedures are necessary for heavy rain.

## COLD WEATHER OPERATION

Special attention should be paid to operation of the aircraft and the fuel system in winter or before any flight at low temperatures. Correct preflight draining of the fuel system is particularly important and will prevent the accumulation of water.
The following limitations for cold weather operation are established due to temperature. “Operating limits“ (refer also to Section 2 "Limitations").

| Fuel | Minimum permissible fuel temperature in the fuel tank before Take-off | Minimum permissible fuel temperature in the fuel tank during the flight |
|---|---|---|
| JET A-1, JET-A, Fuel No.3 JP-8 JP8+100 TS-1 | -30°C (-22°F) | -35°C (-31°F) |

Figure 4-1 Minimum fuel temperature limits in the fuel tank

▲ **WARNING:** The fuel temperature of the fuel tank not in use should be observed if it is intended for later use.

▲ **WARNING:** The following applies to Diesel and JET fuel mixtures in the tank:
As soon as the proportion of Diesel in the tank is more than 10% Diesel, the fuel temperature limits for Diesel operation must be monitored. If there is uncertainty about which fuel is in the tank, the assumption should be made that it is Diesel.

◆ Note: It is advisable to refuel before each flight and to enter the type of fuel filled and the additives used in the log-book of the airplane.

If snow or slush covers the take-off surface, allowance must be made for take-off distances which will be increasingly extended as snow or slush depth increases. The depth and consistency of this cover can, in fact, prevent take-off in many instances.

Cold weather starting procedures are the same as the normal starting procedures. Use caution to prevent inadvertent forward movement of the airplane during starting when parked on snow or ice.

**HOT WEATHER OPERATION**

Engine temperatures may rise into the amber range and activate the "Caution" light when operating in hot weather or longer climbs at low speed. This warning gives the pilot the opportunity to keep the engine from possibly overheating by doing the following:

i) decrease rate of climb
ii) increase airspeed
iii) reduce power, if the engine temperatures approach the red range

Should the seldom case occur that the fuel temperature is rising into the amber or red range, switch to the other tank or to the BOTH position

# SECTION 5
# PERFORMANCE

## SAMPLE PROBLEM

The following sample flight problem utilizes information from the various tables and diagrams of this section to determine the predicted performance data for a typical flight.
Assume the following information has already been determined:

AIRPLANE CONFIGURATION

Takeoff Weight ..... 1043 kg
Usable Fuel ..... 127.4 l (33.6 US gal)

TAKEOFF CONDITIONS

Field Pressure Altitude ..... 1000 ft
Temperature ..... 28°C ( ISA +15°C)
Wind Component along Runway ... 12 Knot Headwind
Field Length ..... 1067 m (3500 ft)

CRUISE CONDITIONS

Total Distance ..... 841 km (400 NM)
Pressure Altitude ..... 6000 ft
Temperature ..... 23°C (ISA + 20°C)
Expected Wind Enroute ..... 10 Knot Headwind

LANDING CONDITIONS

Field Pressure Altitude ..... 2000 ft
Temperature ..... 25°C
Field Length ..... 914 m (3000 ft)

## GROUND ROLL AND TAKE-OFF

The ground roll and take-off distance chart, Figure 5-1a ff (Ground Roll and Take-off Distance), should be consulted, keeping in mind that distances shown are based on the short field technique. Conservative distances can be established by reading the chart at the next higher value of weight, temperature and altitude. For example, in this particular sample problem, the takeoff distance information presented for a weight of 1043 kg, pressure altitude of 1000 ft and a temperature of 30°C should be used and results in the following:

Ground Roll.....263 m (863 ft)

Total Distance to clear a 15 m obstacle.....451 m (1478 ft)

These distances are well within the available takeoff field length. However, a correction for the effect of wind may be made based on Note 2 of the takeoff chart. The correction for a 12 Knot Headwind is:

$$\frac{12\ \text{Kt}}{9\ \text{Kt}} \times 10\ \% = 13\ \%\ \text{(Decrease)}$$

This results in the following distances, corrected for wind:

Ground Roll, zero wind ..... 263 m (863 ft)
Decrease at 12 Knot Headwind (263m x 13%)=- 34 m (112 ft)
Corrected Ground Roll ..... 229 m (751 ft)

Total Distance to clear a 15 m obstacle,
zero wind..... 451 m (1478 ft)
Decrease at 12 Knot Headwind (451 m x 13%)= - 58 m (192 ft)
Corrected Total Distance to clear a ..... 393 m (1286 ft)
15 m obstacle

**CRUISE**

The cruising altitude should be selected based on a consideration of trip length, winds aloft and the airplanes performance. A typical cruising altitude and the expected wind enroute have been given for this sample problem. However, the power setting selection for cruise must be determined based on several considerations. These include the cruise performance characteristics presented in Figures 5-4. Considerable fuel savings and longer range result when lower power settings are used.

Figure 5-4a shows a range of 567 NM at zero wind, a power setting of 70% and altitude of 6000 ft.

With an expected headwind of 10 Knot at 6,000 ft altitude the range has to be corrected as follows:

Range at zero wind (standard tanks) ..... 567 NM

Reduction due to Headwind..... (4.5 h x 10 Knots) = - 45NM

Corrected Range..... 522 NM

This shows that the flight can be performed at a power setting of approximately 70% with full tanks without an intermediate fuel stop.

Figure 5-4a is based on ISA conditions. For a temperature of 20°C above ISA temperature, according to Note 3, true airspeed and maximum range are increased by 2 %.

The following values most nearly correspond to the planned altitude and expected temperature conditions. Engine Power setting chosen is 70%.

The resultants are:

Engine Power:.....70%

True Airspeed: .....120 kt + 2% = 122 kt
Fuel Consumption in cruise: .....22.1 l/h (5.8 US gal/h)

**FUEL REQUIRED**

The total fuel requirement for the flight may be estimated using the performance information in Figures 5-3 and 5-4. For this sample problem, Figure 5-3a shows that a climb from 1000 ft to 6,000 ft requires 3.3 l (0.9 US gal) of fuel. The corresponding distance during the climb is 7.6 NM. These values are for a standard temperature and are sufficiently accurate for most flight planning purposes.
However, a further correction for the effect of temperature may be made as noted in Note 2 of the climb chart in Figure 5-3a. An effect of 10°C above the standard temperature is to increase time and distance by 10%.

In this case, assuming a temperature 20°C above standard, the correction would be:

$$\frac{20\ °C}{10\ °C} \times 10\ \% = \quad 20\ \%\ \text{(Increase)}$$

With this factor included, the fuel estimate would be calculated as follows:

Fuel to climb, standard temperature:
3.3 l (0.9 US gal)

Increase due to non-standard temperature:
3.3 l (0.9 US gal) x 20.0% = 0.7 l (0.2 US gal)

Corrected fuel to climb:
4.0 l (1.1 US gal)

Using a similar procedure for the distance to climb results in 9.1 NM.

The resultant cruise distance is:

| | |
|---|---|
| Total Distance | 400.0 NM |
| Climbout Distance | - 9.1 NM |
| Cruise Distance | 390.9 NM |

With an expected 10 Knot headwind, the ground speed for cruise is predicted to be:

| |
|---|
| 122 Knot |
| - 10 Knot |
| 112 Knot |

Therefore, the time required for the cruise portion of the trip is:

$$\frac{390.9 \text{ NM}}{112 \text{ Kt}} = 3.5 \text{ hrs}$$

The fuel required for cruise is:

3.5 h x 22.1 l/h = 77.4 l (20.5 US gal)

The total estimated fuel required is as follows:

| | |
|---|---|
| Engine Start, Taxi and Takeoff | 4.0 l (1.1 US gal) |
| Climb | + 4.0 l (1.1 US gal) |
| Cruise | + 77.4 l (20.5 US gal) |
| Total fuel required | 85.4 l (22.7 US gal) |

This gives with full tanks a reserve of:

| |
|---|
| 127.4 l (33.6 US gal) |
| - 85.4 l (22.7 US gal) |
| 42.0 l (10.9 US gal) |

Once the flight is underway, ground speed checks will provide a more accurate basis for estimating the time enroute and the corresponding fuel required.

**LANDING DISTANCE**

Refer to Pilot´s Operating Handbook

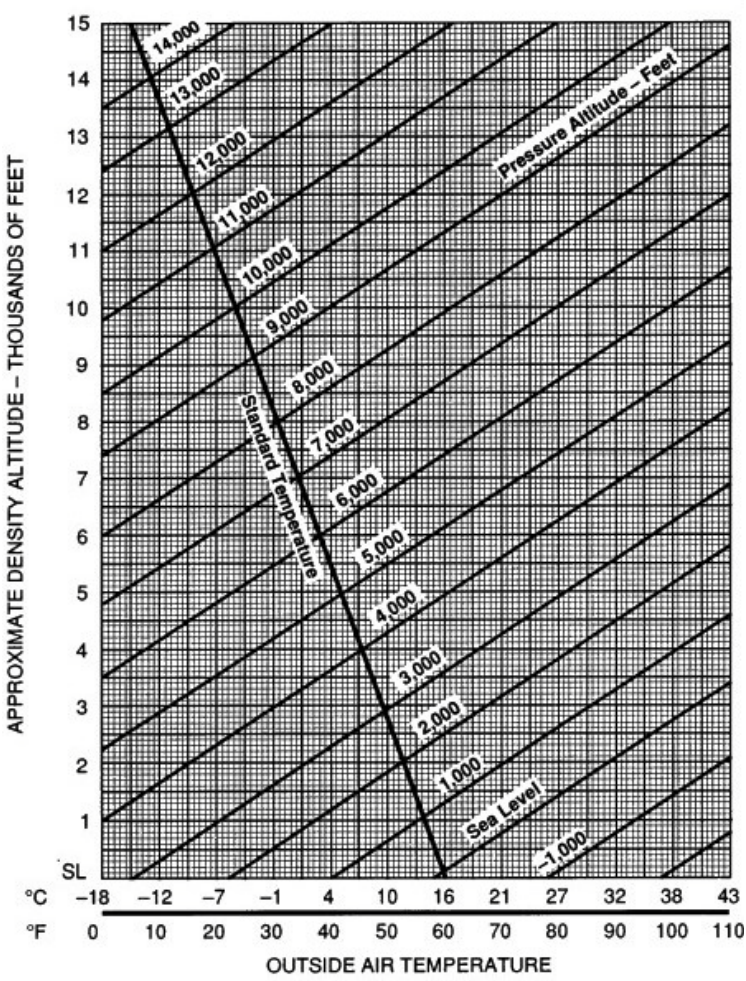



Figure 5-1 Density Altitude Chart

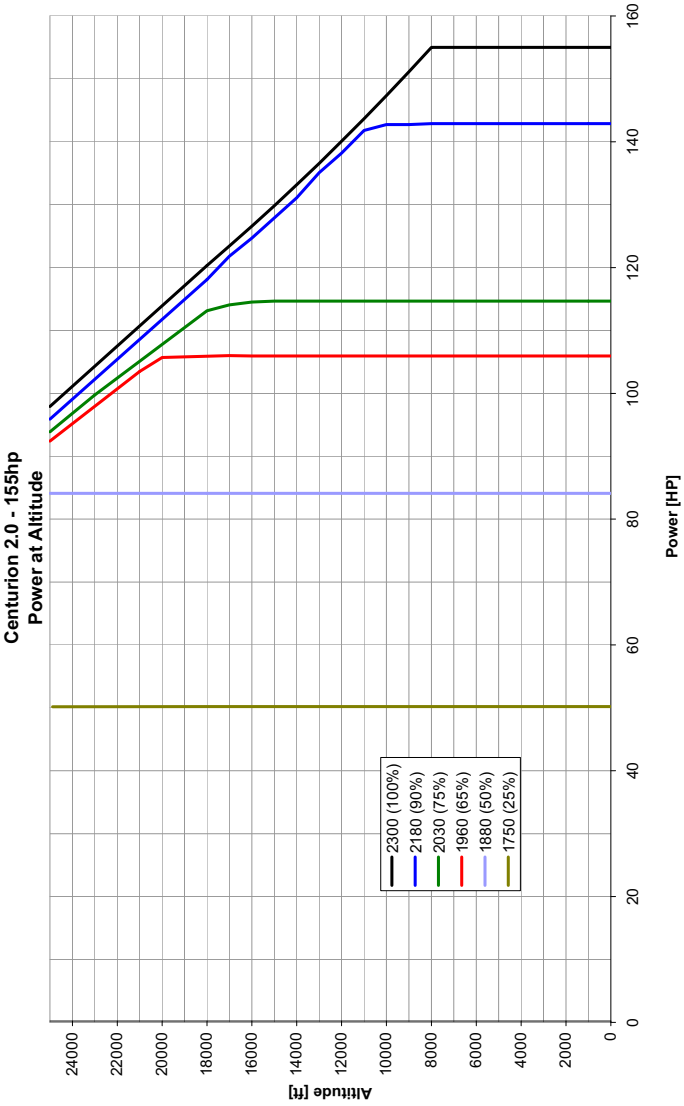



Figure 5-2 Engine Power Over Altitude

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# SECTION 5a
# PERFORMANCE

| | |
|---|---|
| ◆ Note: | This chapter applies to aircraft with propellers **MTV-6-A/187-129**.<br>The correct propeller designation can be found on the blades. |

| | |
|---|---|
| ◆ Note: | The chapter not relevant to the respective propeller can be omitted. |

**GROUND ROLL AND TAKE-OFF DISTANCE**
**at 1043 kg (2300 lbs)**

SHORT FIELD TAKE-OFFS

**Conditions:**
Take-off weight 1043 kg (2300 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: .....48 KIAS/ 55 mph

Speed at 15 m / 50 ft: .....54 KIAS/ 62 mph

**Notes:**

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 172 | 199 | 213 | 227 | 246 | 273 | 307 |
| | 50 ft (15 m) obstacle | 294 | 340 | 364 | 388 | 421 | 469 | 530 |
| 1000 | Gnd Roll | 184 | 213 | 228 | 243 | 263 | 292 | 329 |
| | 50 ft (15 m) obstacle | 315 | 364 | 390 | 416 | 451 | 502 | 568 |
| 2000 | Gnd Roll | 198 | 228 | 244 | 260 | 282 | 313 | 352 |
| | 50 ft (15 m) obstacle | 338 | 391 | 418 | 446 | 483 | 538 | 609 |
| 3000 | Gnd Roll | 212 | 245 | 262 | 279 | 302 | 336 | 378 |
| | 50 ft (15 m) obstacle | 363 | 419 | 448 | 478 | 518 | 577 | 653 |
| 4000 | Gnd Roll | 227 | 263 | 281 | 300 | 324 | 360 | 406 |
| | 50 ft (15 m) obstacle | 389 | 449 | 481 | 513 | 556 | 619 | 701 |
| 5000 | Gnd Roll | 244 | 282 | 302 | 322 | 348 | 387 | 435 |
| | 50 ft (15 m) obstacle | 418 | 482 | 516 | 550 | 597 | 665 | 752 |
| 6000 | Gnd Roll | 262 | 303 | 324 | 345 | 374 | 415 | 468 |
| | 50 ft (15 m) obstacle | 448 | 518 | 554 | 591 | 641 | 714 | 808 |
| 7000 | Gnd Roll | 287 | 332 | 355 | 379 | 410 | 455 | 513 |
| | 50 ft (15 m) obstacle | 492 | 568 | 608 | 648 | 703 | 783 | 886 |
| 8000 | Gnd Roll | 315 | 364 | 390 | 416 | 450 | 500 | 563 |
| | 50 ft (15 m) obstacle | 540 | 624 | 667 | 711 | 771 | 859 | 972 |
| 9000 | Gnd Roll | 350 | 404 | 432 | 461 | 499 | 554 | 624 |
| | 50 ft (15 m) obstacle | 599 | 692 | 741 | 790 | 856 | 954 | 1080 |
| 10000 | Gnd Roll | 389 | 449 | 480 | 512 | 555 | 616 | 694 |
| | 50 ft (15 m) obstacle | 666 | 770 | 824 | 878 | 952 | 1061 | 1201 |

Figure 5-1a Take-Off Distance [m] at take-off weight 1043 kg (2300 lbs)

| PRESS ALT | Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 564 | 652 | 697 | 744 | 805 | 894 | 1007 |
| | 50 ft (15 m) obstacle | 966 | 1116 | 1194 | 1273 | 1380 | 1537 | 1740 |
| 1000 | Gnd Roll | 605 | 698 | 747 | 797 | 863 | 958 | 1079 |
| | 50 ft (15 m) obstacle | 1035 | 1195 | 1279 | 1364 | 1478 | 1647 | 1864 |
| 2000 | Gnd Roll | 648 | 749 | 801 | 854 | 925 | 1027 | 1156 |
| | 50 ft (15 m) obstacle | 1109 | 1281 | 1371 | 1462 | 1585 | 1765 | 1998 |
| 3000 | Gnd Roll | 695 | 803 | 859 | 916 | 992 | 1101 | 1240 |
| | 50 ft (15 m) obstacle | 1189 | 1374 | 1470 | 1567 | 1699 | 1893 | 2142 |
| 4000 | Gnd Roll | 745 | 861 | 921 | 983 | 1064 | 1181 | 1330 |
| | 50 ft (15 m) obstacle | 1276 | 1474 | 1577 | 1682 | 1823 | 2031 | 2299 |
| 5000 | Gnd Roll | 800 | 925 | 989 | 1055 | 1142 | 1268 | 1428 |
| | 50 ft (15 m) obstacle | 1370 | 1582 | 1693 | 1805 | 1957 | 2180 | 2467 |
| 6000 | Gnd Roll | 859 | 993 | 1062 | 1133 | 1227 | 1362 | 1533 |
| | 50 ft (15 m) obstacle | 1471 | 1700 | 1818 | 1939 | 2102 | 2341 | 2650 |
| 7000 | Gnd Roll | 942 | 1089 | 1165 | 1242 | 1345 | 1493 | 1681 |
| | 50 ft (15 m) obstacle | 1613 | 1863 | 1993 | 2126 | 2304 | 2567 | 2905 |
| 8000 | Gnd Roll | 1034 | 1195 | 1278 | 1363 | 1476 | 1638 | 1845 |
| | 50 ft (15 m) obstacle | 1770 | 2045 | 2187 | 2333 | 2529 | 2817 | 3189 |
| 9000 | Gnd Roll | 1148 | 1326 | 1418 | 1513 | 1638 | 1818 | 2048 |
| | 50 ft (15 m) obstacle | 1966 | 2271 | 2429 | 2591 | 2809 | 3129 | 3541 |
| 10000 | Gnd Roll | 1275 | 1473 | 1576 | 1681 | 1820 | 2020 | 2275 |
| | 50 ft (15 m) obstacle | 2186 | 2526 | 2701 | 2881 | 3123 | 3480 | 3938 |

Figure 5-1b Take-Off Distance [ft] at take-off weight 1043 kg (2300 lbs)

**GROUND ROLL AND TAKE-OFF DISTANCE at 1089 kg (2400 lbs) (Cessna 172P only)**

SHORT FIELD TAKE-OFFS

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: .....48 KIAS/ 55 mph

Speed at 15 m / 50 ft: .....54 KIAS/ 62 mph

**Notes:**

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 192 | 222 | 238 | 254 | 275 | 305 | 343 |
| | 50 ft (15 m) obstacle | 329 | 381 | 407 | 434 | 471 | 524 | 593 |
| 1000 | Gnd Roll | 206 | 238 | 255 | 272 | 294 | 327 | 368 |
| | 50 ft (15 m) obstacle | 353 | 408 | 436 | 465 | 504 | 562 | 636 |
| 2000 | Gnd Roll | 221 | 255 | 273 | 291 | 315 | 350 | 394 |
| | 50 ft (15 m) obstacle | 378 | 437 | 467 | 499 | 540 | 602 | 681 |
| 3000 | Gnd Roll | 237 | 274 | 293 | 312 | 338 | 375 | 423 |
| | 50 ft (15 m) obstacle | 406 | 469 | 501 | 535 | 580 | 646 | 731 |
| 4000 | Gnd Roll | 254 | 294 | 314 | 335 | 363 | 403 | 454 |
| | 50 ft (15 m) obstacle | 435 | 503 | 538 | 574 | 622 | 693 | 784 |
| 5000 | Gnd Roll | 273 | 315 | 337 | 360 | 390 | 432 | 487 |
| | 50 ft (15 m) obstacle | 467 | 540 | 577 | 616 | 667 | 744 | 842 |
| 6000 | Gnd Roll | 293 | 339 | 362 | 386 | 418 | 464 | 523 |
| | 50 ft (15 m) obstacle | 502 | 580 | 620 | 661 | 717 | 799 | 904 |
| 7000 | Gnd Roll | 321 | 371 | 397 | 424 | 459 | 509 | 573 |
| | 50 ft (15 m) obstacle | 550 | 636 | 680 | 725 | 786 | 876 | 991 |
| 8000 | Gnd Roll | 353 | 408 | 436 | 465 | 503 | 559 | 629 |
| | 50 ft (15 m) obstacle | 604 | 698 | 746 | 796 | 863 | 961 | 1088 |
| 9000 | Gnd Roll | 391 | 452 | 484 | 516 | 559 | 620 | 698 |
| | 50 ft (15 m) obstacle | 670 | 775 | 829 | 884 | 958 | 1067 | 1208 |
| 10000 | Gnd Roll | 435 | 502 | 537 | 573 | 621 | 689 | 776 |
| | 50 ft (15 m) obstacle | 746 | 861 | 921 | 983 | 1065 | 1187 | 1343 |

Figure 5-1c Take-Off Distance [m] at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 631 | 729 | 780 | 832 | 901 | 1000 | 1126 |
| | 50 ft (15 m) obstacle | 1080 | 1248 | 1335 | 1424 | 1544 | 1720 | 1946 |
| 1000 | Gnd Roll | 676 | 781 | 836 | 892 | 965 | 1072 | 1207 |
| | 50 ft (15 m) obstacle | 1158 | 1337 | 1431 | 1526 | 1654 | 1843 | 2085 |
| 2000 | Gnd Roll | 725 | 838 | 896 | 956 | 1035 | 1149 | 1293 |
| | 50 ft (15 m) obstacle | 1241 | 1434 | 1533 | 1635 | 1773 | 1975 | 2235 |
| 3000 | Gnd Roll | 777 | 898 | 961 | 1025 | 1109 | 1232 | 1387 |
| | 50 ft (15 m) obstacle | 1330 | 1537 | 1644 | 1753 | 1901 | 2118 | 2397 |
| 4000 | Gnd Roll | 834 | 964 | 1031 | 1099 | 1190 | 1321 | 1488 |
| | 50 ft (15 m) obstacle | 1427 | 1649 | 1764 | 1881 | 2039 | 2272 | 2572 |
| 5000 | Gnd Roll | 895 | 1034 | 1106 | 1180 | 1278 | 1418 | 1597 |
| | 50 ft (15 m) obstacle | 1532 | 1770 | 1894 | 2019 | 2189 | 2439 | 2760 |
| 6000 | Gnd Roll | 961 | 1111 | 1188 | 1267 | 1372 | 1523 | 1715 |
| | 50 ft (15 m) obstacle | 1646 | 1901 | 2034 | 2169 | 2351 | 2619 | 2965 |
| 7000 | Gnd Roll | 1054 | 1218 | 1303 | 1390 | 1505 | 1670 | 1881 |
| | 50 ft (15 m) obstacle | 1804 | 2085 | 2230 | 2378 | 2578 | 2872 | 3250 |
| 8000 | Gnd Roll | 1157 | 1337 | 1430 | 1525 | 1651 | 1833 | 2064 |
| | 50 ft (15 m) obstacle | 1980 | 2288 | 2447 | 2610 | 2829 | 3152 | 3567 |
| 9000 | Gnd Roll | 1284 | 1483 | 1587 | 1692 | 1832 | 2034 | 2291 |
| | 50 ft (15 m) obstacle | 2199 | 2541 | 2718 | 2898 | 3142 | 3501 | 3962 |
| 10000 | Gnd Roll | 1426 | 1648 | 1763 | 1880 | 2036 | 2260 | 2545 |
| | 50 ft (15 m) obstacle | 2445 | 2826 | 3022 | 3223 | 3494 | 3893 | 4406 |

Figure 5-1d Take-Off Distance [ft] at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

## MAXIMUM RATE-OF-CLIMB at 1043 kg (2300 lbs)

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power

**Notes:**

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

| PRESS ALT [FT] | Climb speed [KIAS] | Rate of Climb [ft/min] Outside Air Temperature [°C] | | | | |
|---|---|---|---|---|---|---|
| | | -20°C | 0°C | +20°C | +40°C | +50°C |
| 0 | 69 | 879 | 865 | 851 | 706 | 576 |
| 1000 | 69 | 872 | 858 | 844 | 698 | 568 |
| 2000 | 69 | 865 | 851 | 837 | 691 | 561 |
| 3000 | 69 | 858 | 844 | 829 | 683 | 553 |
| 4000 | 69 | 851 | 836 | 822 | 675 | 545 |
| 5000 | 69 | 844 | 829 | 814 | 667 | 536 |
| 6000 | 69 | 836 | 821 | 806 | 658 | 528 |
| 7000 | 69 | 829 | 813 | 797 | 650 | 519 |
| 8000 | 69 | 821 | 805 | 789 | 641 | 510 |
| 9000 | 69 | 788 | 771 | 755 | 610 | 481 |
| 10000 | 69 | 755 | 738 | 721 | 578 | 452 |
| 11000 | 69 | 721 | 704 | 687 | 547 | 423 |
| 12000 | 69 | 688 | 670 | 653 | 515 | 393 |
| 13000 | 69 | 654 | 636 | 619 | 482 | 363 |
| 14000 | 69 | 620 | 601 | 584 | 450 | 333 |
| 15000 | 69 | 585 | 567 | 549 | 417 | 302 |
| 16000 | 69 | 551 | 531 | 513 | 384 | 271 |
| 17000 | 69 | 516 | 496 | 477 | 350 | 240 |
| 18000 | 69 | 480 | 461 | 441 | 317 | 209 |

Figure 5-2a Maximum Rate of Climb at take-off weight 1043 kg (2300 lbs)

**MAXIMUM RATE-OF-CLIMB at 1089 kg (2400 lbs) (Cessna 172P only)**

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up, Full Power

**Notes:**

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

| PRESS ALT [FT] | Climb speed [KIAS] | Rate of Climb [ft/min] Outside Air Temperature [°C] | | | | |
|---|---|---|---|---|---|---|
| | | -20°C | 0°C | +20°C | +40°C | +50°C |
| 0 | 69 | 819 | 804 | 790 | 650 | 525 |
| 1000 | 69 | 812 | 797 | 783 | 642 | 518 |
| 2000 | 69 | 805 | 790 | 776 | 635 | 510 |
| 3000 | 69 | 798 | 783 | 768 | 627 | 502 |
| 4000 | 69 | 790 | 775 | 760 | 619 | 493 |
| 5000 | 69 | 783 | 767 | 752 | 610 | 485 |
| 6000 | 69 | 775 | 759 | 744 | 602 | 476 |
| 7000 | 69 | 767 | 751 | 735 | 593 | 467 |
| 8000 | 69 | 759 | 743 | 727 | 584 | 458 |
| 9000 | 69 | 727 | 710 | 694 | 554 | 430 |
| 10000 | 69 | 695 | 678 | 661 | 523 | 401 |
| 11000 | 69 | 662 | 645 | 628 | 492 | 373 |
| 12000 | 69 | 630 | 612 | 594 | 461 | 344 |
| 13000 | 69 | 596 | 578 | 560 | 429 | 314 |
| 14000 | 69 | 563 | 544 | 526 | 397 | 285 |
| 15000 | 69 | 530 | 511 | 492 | 365 | 255 |
| 16000 | 69 | 496 | 476 | 458 | 333 | 224 |
| 17000 | 69 | 462 | 442 | 423 | 300 | 194 |
| 18000 | 69 | 427 | 407 | 387 | 267 | 163 |

Figure 5-2b Maximum Rate of Climb at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

**TIME, FUEL AND DISTANCE TO CLIMB at 1043 KG (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

**Notes :**

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at $v_y$ = 69 KIAS.

| Press. Alt. | OAT | Vy | ROC | Time | Distance | Fuel used | |
|---|---|---|---|---|---|---|---|
| [ft] | [°C] | [KIAS] | [FPM] | [MIN] | [NM] | [l] | [US Gal] |
| 0 | 15 | 69 | 855 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1000 | 13 | 69 | 849 | 1.2 | 1.4 | 0.7 | 0.2 |
| 2000 | 11 | 69 | 843 | 2.4 | 2.8 | 1.3 | 0.3 |
| 3000 | 9 | 69 | 837 | 3.5 | 4.2 | 2.0 | 0.5 |
| 4000 | 7 | 69 | 831 | 4.7 | 5.8 | 2.6 | 0.7 |
| 5000 | 5 | 69 | 825 | 6.0 | 7.3 | 3.3 | 0.9 |
| 6000 | 3 | 69 | 818 | 7.2 | 9.0 | 4.0 | 1.1 |
| 7000 | 1 | 69 | 812 | 8.4 | 10.7 | 4.7 | 1.2 |
| 8000 | -1 | 69 | 805 | 9.6 | 12.4 | 5.4 | 1.4 |
| 9000 | -3 | 69 | 774 | 10.9 | 14.3 | 5.9 | 1.6 |
| 10000 | -5 | 69 | 742 | 12.2 | 16.3 | 6.5 | 1.7 |
| 11000 | -7 | 69 | 710 | 13.6 | 18.4 | 7.0 | 1.9 |
| 12000 | -9 | 69 | 678 | 15.0 | 20.7 | 7.6 | 2.0 |
| 13000 | -11 | 69 | 645 | 16.6 | 23.1 | 8.1 | 2.1 |
| 14000 | -13 | 69 | 613 | 18.1 | 25.8 | 8.7 | 2.3 |
| 15000 | -15 | 69 | 580 | 19.8 | 28.6 | 9.2 | 2.4 |
| 16000 | -17 | 69 | 547 | 21.6 | 31.7 | 9.8 | 2.6 |
| 17000 | -19 | 69 | 514 | 23.5 | 35.1 | 10.3 | 2.7 |
| 18000 | -21 | 69 | 481 | 25.5 | 38.7 | 10.8 | 2.9 |

Figure 5-3a Time, Fuel and Distance to Climb at 1043 kg (2300 lbs)

**TIME, FUEL AND DISTANCE TO CLIMB at 1089 KG (Cessna 172P only)**

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

**Notes :**

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at $v_y$ = 69 KIAS.

| Press. Alt. | OAT | Vy | ROC | Time | Distance | Fuel used | |
|---|---|---|---|---|---|---|---|
| [ft] | [°C] | [KIAS] | [FPM] | [MIN] | [NM] | [l] | [US Gal] |
| 0 | 15 | 69 | 794 | 0.0 | 0.0 | 0.0 | 0.0 |
| 1000 | 13 | 69 | 788 | 1.3 | 1.5 | 0.7 | 0.2 |
| 2000 | 11 | 69 | 782 | 2.5 | 3.0 | 1.4 | 0.4 |
| 3000 | 9 | 69 | 776 | 3.8 | 4.6 | 2.1 | 0.6 |
| 4000 | 7 | 69 | 770 | 5.1 | 6.2 | 2.9 | 0.8 |
| 5000 | 5 | 69 | 763 | 6.4 | 7.9 | 3.6 | 0.9 |
| 6000 | 3 | 69 | 757 | 7.7 | 9.7 | 4.3 | 1.1 |
| 7000 | 1 | 69 | 750 | 9.1 | 11.5 | 5.1 | 1.3 |
| 8000 | -1 | 69 | 743 | 10.4 | 13.4 | 5.8 | 1.5 |
| 9000 | -3 | 69 | 713 | 11.8 | 15.4 | 6.4 | 1.7 |
| 10000 | -5 | 69 | 682 | 13.2 | 17.6 | 7.0 | 1.9 |
| 11000 | -7 | 69 | 651 | 14.7 | 19.9 | 7.6 | 2.0 |
| 12000 | -9 | 69 | 619 | 16.3 | 22.4 | 8.2 | 2.2 |
| 13000 | -11 | 69 | 588 | 17.9 | 25.1 | 8.8 | 2.3 |
| 14000 | -13 | 69 | 556 | 19.7 | 28.0 | 9.4 | 2.5 |
| 15000 | -15 | 69 | 525 | 21.5 | 31.1 | 10.0 | 2.6 |
| 16000 | -17 | 69 | 493 | 23.5 | 34.5 | 10.6 | 2.8 |
| 17000 | -19 | 69 | 460 | 25.6 | 38.3 | 11.2 | 3.0 |
| 18000 | -21 | 69 | 428 | 27.9 | 42.3 | 11.8 | 3.1 |

Figure 5-3b Time, Fuel and Distance to Climb at 1089 kg (2400 lbs) (Cessna 172P only)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with standard tanks at 1043 kg (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

**Notes:**

1. Endurance information is based on standard tanks with 127.4 l (33.6 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 131 | 151 | 33.6 | 8.9 | 383 | 2.9 |
| SL | 90 | 126 | 145 | 29.6 | 7.8 | 431 | 3.4 |
| SL | 80 | 120 | 138 | 25.8 | 6.8 | 484 | 4.0 |
| SL | 70 | 114 | 131 | 22.1 | 5.8 | 551 | 4.8 |
| SL | 60 | 106 | 122 | 18.6 | 4.9 | 624 | 5.9 |
| SL | 50 | 97 | 112 | 15.3 | 4.0 | 710 | 7.3 |
| | | | | | | | |
| 2000 | 100 | 133 | 154 | 33.6 | 8.9 | 386 | 2.8 |
| 2000 | 90 | 128 | 147 | 29.6 | 7.8 | 435 | 3.3 |
| 2000 | 80 | 122 | 141 | 25.8 | 6.8 | 489 | 3.9 |
| 2000 | 70 | 116 | 133 | 22.1 | 5.8 | 557 | 4.7 |
| 2000 | 60 | 108 | 124 | 18.6 | 4.9 | 631 | 5.8 |
| 2000 | 50 | 99 | 114 | 15.3 | 4.0 | 718 | 7.2 |
| | | | | | | | |
| 4000 | 100 | 136 | 156 | 33.6 | 8.9 | 392 | 2.8 |
| 4000 | 90 | 131 | 150 | 29.6 | 7.8 | 442 | 3.3 |
| 4000 | 80 | 124 | 143 | 25.8 | 6.8 | 493 | 3.9 |
| 4000 | 70 | 118 | 135 | 22.1 | 5.8 | 562 | 4.6 |
| 4000 | 60 | 110 | 126 | 18.6 | 4.9 | 637 | 5.7 |
| 4000 | 50 | 100 | 115 | 15.3 | 4.0 | 720 | 7.1 |
| | | | | | | | |
| 6000 | 100 | 139 | 159 | 33.6 | 8.9 | 398 | 2.7 |
| 6000 | 90 | 133 | 153 | 29.6 | 7.8 | 445 | 3.2 |
| 6000 | 80 | 127 | 146 | 25.8 | 6.8 | 501 | 3.8 |
| 6000 | 70 | 120 | 138 | 22.1 | 5.8 | 567 | 4.5 |
| 6000 | 60 | 111 | 128 | 18.6 | 4.9 | 638 | 5.5 |
| 6000 | 50 | 102 | 117 | 15.3 | 4.0 | 728 | 6.9 |
| | | | | | | | |
| 8000 | 100 | 141 | 162 | 33.6 | 8.9 | 401 | 2.6 |
| 8000 | 90 | 135 | 156 | 29.6 | 7.8 | 449 | 3.1 |
| 8000 | 80 | 129 | 148 | 25.8 | 6.8 | 505 | 3.7 |
| 8000 | 70 | 122 | 140 | 22.1 | 5.8 | 572 | 4.4 |
| 8000 | 60 | 113 | 130 | 18.6 | 4.9 | 644 | 5.4 |
| 8000 | 50 | 103 | 119 | 15.3 | 4.0 | 729 | 6.8 |
| | | | | | | | |
| 10000 | 90 | 138 | 159 | 29.6 | 7.8 | 455 | 3.0 |
| 10000 | 80 | 131 | 151 | 25.8 | 6.8 | 509 | 3.6 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 124 | 142 | 22.1 | 5.8 | 576 | 4.3 |
| 10000 | 60 | 115 | 132 | 18.6 | 4.9 | 650 | 5.3 |
| 10000 | 50 | 105 | 120 | 15.3 | 4.0 | 737 | 6.7 |
| | | | | | | | |
| 12000 | 90 | 140 | 162 | 29.6 | 7.8 | 458 | 2.9 |
| 12000 | 80 | 134 | 154 | 25.8 | 6.8 | 516 | 3.5 |
| 12000 | 70 | 126 | 145 | 22.1 | 5.8 | 581 | 4.2 |
| 12000 | 60 | 117 | 134 | 18.6 | 4.9 | 655 | 5.2 |
| 12000 | 50 | 106 | 122 | 15.3 | 4.0 | 737 | 6.5 |
| | | | | | | | |
| 14000 | 90 | 143 | 165 | 29.6 | 7.8 | 465 | 2.8 |
| 14000 | 80 | 136 | 157 | 25.8 | 6.8 | 520 | 3.4 |
| 14000 | 70 | 128 | 147 | 22.1 | 5.8 | 585 | 4.1 |
| 14000 | 60 | 119 | 137 | 18.6 | 4.9 | 661 | 5.1 |
| 14000 | 50 | 108 | 124 | 15.3 | 4.0 | 744 | 6.4 |
| | | | | | | | |
| 16000 | 80 | 138 | 159 | 25.8 | 6.8 | 524 | 3.2 |
| 16000 | 70 | 130 | 150 | 22.1 | 5.8 | 590 | 4.0 |
| 16000 | 60 | 121 | 139 | 18.6 | 4.9 | 666 | 4.9 |
| 16000 | 50 | 109 | 126 | 15.3 | 4.0 | 745 | 6.2 |
| | | | | | | | |
| 18000 | 80 | 141 | 162 | 25.8 | 6.8 | 531 | 3.1 |
| 18000 | 70 | 132 | 152 | 22.1 | 5.8 | 594 | 3.8 |
| 18000 | 60 | 123 | 141 | 18.6 | 4.9 | 671 | 4.8 |
| 18000 | 50 | 111 | 127 | 15.3 | 4.0 | 751 | 6.1 |
| | | | | | | | |

Figure 5-4a Cruise Performance, Range and Endurance with standard tanks, at 1043 kg (2300 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with long-range tanks (Cessna 172N) at 1043 kg (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

**Notes:**

1. Endurance information is based on long-range tanks with 158.6 l (41.9 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 131 | 151 | 33.6 | 8.9 | 505 | 3.9 |
| SL | 90 | 126 | 145 | 29.6 | 7.8 | 564 | 4.5 |
| SL | 80 | 120 | 138 | 25.8 | 6.8 | 629 | 5.2 |
| SL | 70 | 114 | 131 | 22.1 | 5.8 | 712 | 6.2 |
| SL | 60 | 106 | 122 | 18.6 | 4.9 | 802 | 7.6 |
| SL | 50 | 97 | 112 | 15.3 | 4.0 | 907 | 9.4 |
| | | | | | | | |
| 2000 | 100 | 133 | 154 | 33.6 | 8.9 | 510 | 3.8 |
| 2000 | 90 | 128 | 147 | 29.6 | 7.8 | 570 | 4.4 |
| 2000 | 80 | 122 | 141 | 25.8 | 6.8 | 636 | 5.2 |
| 2000 | 70 | 116 | 133 | 22.1 | 5.8 | 720 | 6.1 |
| 2000 | 60 | 108 | 124 | 18.6 | 4.9 | 812 | 7.5 |
| 2000 | 50 | 99 | 114 | 15.3 | 4.0 | 920 | 9.2 |
| | | | | | | | |
| 4000 | 100 | 136 | 156 | 33.6 | 8.9 | 519 | 3.7 |
| 4000 | 90 | 131 | 150 | 29.6 | 7.8 | 580 | 4.3 |
| 4000 | 80 | 124 | 143 | 25.8 | 6.8 | 643 | 5.1 |
| 4000 | 70 | 118 | 135 | 22.1 | 5.8 | 728 | 6.0 |
| 4000 | 60 | 110 | 126 | 18.6 | 4.9 | 822 | 7.3 |
| 4000 | 50 | 100 | 115 | 15.3 | 4.0 | 924 | 9.1 |
| | | | | | | | |
| 6000 | 100 | 139 | 159 | 33.6 | 8.9 | 527 | 3.6 |
| 6000 | 90 | 133 | 153 | 29.6 | 7.8 | 586 | 4.2 |
| 6000 | 80 | 127 | 146 | 25.8 | 6.8 | 655 | 5.0 |
| 6000 | 70 | 120 | 138 | 22.1 | 5.8 | 736 | 5.9 |
| 6000 | 60 | 111 | 128 | 18.6 | 4.9 | 824 | 7.2 |
| 6000 | 50 | 102 | 117 | 15.3 | 4.0 | 936 | 9.0 |
| | | | | | | | |
| 8000 | 100 | 141 | 162 | 33.6 | 8.9 | 532 | 3.5 |
| 8000 | 90 | 135 | 156 | 29.6 | 7.8 | 591 | 4.1 |
| 8000 | 80 | 129 | 148 | 25.8 | 6.8 | 661 | 4.9 |
| 8000 | 70 | 122 | 140 | 22.1 | 5.8 | 744 | 5.8 |
| 8000 | 60 | 113 | 130 | 18.6 | 4.9 | 834 | 7.1 |
| 8000 | 50 | 103 | 119 | 15.3 | 4.0 | 939 | 8.8 |
| | | | | | | | |
| 10000 | 90 | 138 | 159 | 29.6 | 7.8 | 601 | 4.0 |
| 10000 | 80 | 131 | 151 | 25.8 | 6.8 | 667 | 4.8 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 124 | 142 | 22.1 | 5.8 | 751 | 5.7 |
| 10000 | 60 | 115 | 132 | 18.6 | 4.9 | 843 | 7.0 |
| 10000 | 50 | 105 | 120 | 15.3 | 4.0 | 951 | 8.7 |
| | | | | | | | |
| 12000 | 90 | 140 | 162 | 29.6 | 7.8 | 606 | 3.9 |
| 12000 | 80 | 134 | 154 | 25.8 | 6.8 | 678 | 4.7 |
| 12000 | 70 | 126 | 145 | 22.1 | 5.8 | 759 | 5.6 |
| 12000 | 60 | 117 | 134 | 18.6 | 4.9 | 852 | 6.9 |
| 12000 | 50 | 106 | 122 | 15.3 | 4.0 | 953 | 8.6 |
| | | | | | | | |
| 14000 | 90 | 143 | 165 | 29.6 | 7.8 | 616 | 3.8 |
| 14000 | 80 | 136 | 157 | 25.8 | 6.8 | 685 | 4.6 |
| 14000 | 70 | 128 | 147 | 22.1 | 5.8 | 766 | 5.5 |
| 14000 | 60 | 119 | 137 | 18.6 | 4.9 | 860 | 6.7 |
| 14000 | 50 | 108 | 124 | 15.3 | 4.0 | 964 | 8.4 |
| | | | | | | | |
| 16000 | 80 | 138 | 159 | 25.8 | 6.8 | 691 | 4.4 |
| 16000 | 70 | 130 | 150 | 22.1 | 5.8 | 773 | 5.4 |
| 16000 | 60 | 121 | 139 | 18.6 | 4.9 | 869 | 6.6 |
| 16000 | 50 | 109 | 126 | 15.3 | 4.0 | 967 | 8.3 |
| | | | | | | | |
| 18000 | 80 | 141 | 162 | 25.8 | 6.8 | 701 | 4.3 |
| 18000 | 70 | 132 | 152 | 22.1 | 5.8 | 780 | 5.2 |
| 18000 | 60 | 123 | 141 | 18.6 | 4.9 | 878 | 6.4 |
| 18000 | 50 | 111 | 127 | 15.3 | 4.0 | 977 | 8.1 |
| | | | | | | | |

Figure 5-4b Cruise Performance, Range and Endurance with long-range tanks, at 1043 kg (2300 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with standard tanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on standard tanks with 127.4 l (33.6 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 128 | 147 | 33.6 | 8.9 | 374 | 2.9 |
| SL | 90 | 123 | 141 | 29.6 | 7.8 | 421 | 3.4 |
| SL | 80 | 117 | 135 | 25.8 | 6.8 | 472 | 4.0 |
| SL | 70 | 111 | 127 | 22.1 | 5.8 | 537 | 4.8 |
| SL | 60 | 103 | 119 | 18.6 | 4.9 | 606 | 5.9 |
| SL | 50 | 94 | 108 | 15.3 | 4.0 | 688 | 7.3 |
| | | | | | | | |
| 2000 | 100 | 130 | 150 | 33.6 | 8.9 | 377 | 2.8 |
| 2000 | 90 | 125 | 144 | 29.6 | 7.8 | 424 | 3.3 |
| 2000 | 80 | 119 | 137 | 25.8 | 6.8 | 476 | 3.9 |
| 2000 | 70 | 112 | 129 | 22.1 | 5.8 | 537 | 4.7 |
| 2000 | 60 | 105 | 120 | 18.6 | 4.9 | 613 | 5.8 |
| 2000 | 50 | 96 | 110 | 15.3 | 4.0 | 696 | 7.2 |
| | | | | | | | |
| 4000 | 100 | 133 | 153 | 33.6 | 8.9 | 383 | 2.8 |
| 4000 | 90 | 127 | 146 | 29.6 | 7.8 | 428 | 3.2 |
| 4000 | 80 | 121 | 139 | 25.8 | 6.8 | 481 | 3.8 |
| 4000 | 70 | 114 | 131 | 22.1 | 5.8 | 542 | 4.6 |
| 4000 | 60 | 106 | 122 | 18.6 | 4.9 | 613 | 5.6 |
| 4000 | 50 | 97 | 111 | 15.3 | 4.0 | 698 | 7.0 |
| | | | | | | | |
| 6000 | 100 | 135 | 155 | 33.6 | 8.9 | 386 | 2.7 |
| 6000 | 90 | 129 | 149 | 29.6 | 7.8 | 431 | 3.1 |
| 6000 | 80 | 123 | 142 | 25.8 | 6.8 | 485 | 3.7 |
| 6000 | 70 | 116 | 134 | 22.1 | 5.8 | 547 | 4.5 |
| 6000 | 60 | 108 | 124 | 18.6 | 4.9 | 620 | 5.5 |
| 6000 | 50 | 98 | 113 | 15.3 | 4.0 | 698 | 6.9 |
| | | | | | | | |
| 8000 | 100 | 138 | 158 | 33.6 | 8.9 | 392 | 2.6 |
| 8000 | 90 | 132 | 152 | 29.6 | 7.8 | 438 | 3.0 |
| 8000 | 80 | 125 | 144 | 25.8 | 6.8 | 489 | 3.6 |
| 8000 | 70 | 118 | 136 | 22.1 | 5.8 | 552 | 4.4 |
| 8000 | 60 | 110 | 126 | 18.6 | 4.9 | 626 | 5.4 |
| 8000 | 50 | 100 | 115 | 15.3 | 4.0 | 706 | 6.8 |
| | | | | | | | |
| 10000 | 90 | 134 | 154 | 29.6 | 7.8 | 441 | 2.9 |
| 10000 | 80 | 128 | 147 | 25.8 | 6.8 | 496 | 3.5 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22.1 | 5.8 | 556 | 4.3 |
| 10000 | 60 | 111 | 128 | 18.6 | 4.9 | 626 | 5.3 |
| 10000 | 50 | 101 | 116 | 15.3 | 4.0 | 707 | 6.6 |
| | | | | | | | |
| 12000 | 90 | 137 | 157 | 29.6 | 7.8 | 447 | 2.8 |
| 12000 | 80 | 130 | 149 | 25.8 | 6.8 | 500 | 3.4 |
| 12000 | 70 | 122 | 140 | 22.1 | 5.8 | 561 | 4.2 |
| 12000 | 60 | 113 | 130 | 18.6 | 4.9 | 631 | 5.1 |
| 12000 | 50 | 102 | 118 | 15.3 | 4.0 | 707 | 6.5 |
| | | | | | | | |
| 14000 | 90 | 139 | 160 | 29.6 | 7.8 | 451 | 2.7 |
| 14000 | 80 | 132 | 152 | 25.8 | 6.8 | 503 | 3.3 |
| 14000 | 70 | 124 | 143 | 22.1 | 5.8 | 565 | 4.0 |
| 14000 | 60 | 115 | 132 | 18.6 | 4.9 | 636 | 5.0 |
| 14000 | 50 | 104 | 119 | 15.3 | 4.0 | 714 | 6.3 |
| | | | | | | | |
| 16000 | 80 | 134 | 155 | 25.8 | 6.8 | 507 | 3.2 |
| 16000 | 70 | 126 | 145 | 22.1 | 5.8 | 569 | 3.9 |
| 16000 | 60 | 117 | 134 | 18.6 | 4.9 | 641 | 4.8 |
| 16000 | 50 | 105 | 121 | 15.3 | 4.0 | 714 | 6.1 |
| | | | | | | | |
| 18000 | 80 | 137 | 157 | 25.8 | 6.8 | 514 | 3.0 |
| 18000 | 70 | 128 | 148 | 22.1 | 5.8 | 573 | 3.7 |
| 18000 | 60 | 118 | 136 | 18.6 | 4.9 | 641 | 4.7 |
| 18000 | 50 | 107 | 123 | 15.3 | 4.0 | 720 | 5.9 |
| | | | | | | | |

Figure 5-4c Cruise Performance, Range and Endurance
with standard tanks, Cessna 172P at 1089 kg (2400 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with long-range tanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on long-range tanks with 158.6 l (41.9 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 128 | 147 | 33.6 | 8.9 | 493 | 3.9 |
| SL | 90 | 123 | 141 | 29.6 | 7.8 | 550 | 4.5 |
| SL | 80 | 117 | 135 | 25.8 | 6.8 | 613 | 5.2 |
| SL | 70 | 111 | 127 | 22.1 | 5.8 | 693 | 6.2 |
| SL | 60 | 103 | 119 | 18.6 | 4.9 | 779 | 7.6 |
| SL | 50 | 94 | 108 | 15.3 | 4.0 | 879 | 9.4 |
| | | | | | | | |
| 2000 | 100 | 130 | 150 | 33.6 | 8.9 | 498 | 3.8 |
| 2000 | 90 | 125 | 144 | 29.6 | 7.8 | 556 | 4.4 |
| 2000 | 80 | 119 | 137 | 25.8 | 6.8 | 620 | 5.1 |
| 2000 | 70 | 112 | 129 | 22.1 | 5.8 | 695 | 6.1 |
| 2000 | 60 | 105 | 120 | 18.6 | 4.9 | 789 | 7.4 |
| 2000 | 50 | 96 | 110 | 15.3 | 4.0 | 892 | 9.2 |
| | | | | | | | |
| 4000 | 100 | 133 | 153 | 33.6 | 8.9 | 507 | 3.7 |
| 4000 | 90 | 127 | 146 | 29.6 | 7.8 | 562 | 4.3 |
| 4000 | 80 | 121 | 139 | 25.8 | 6.8 | 627 | 5.0 |
| 4000 | 70 | 114 | 131 | 22.1 | 5.8 | 703 | 6.0 |
| 4000 | 60 | 106 | 122 | 18.6 | 4.9 | 791 | 7.3 |
| 4000 | 50 | 97 | 111 | 15.3 | 4.0 | 895 | 9.1 |
| | | | | | | | |
| 6000 | 100 | 135 | 155 | 33.6 | 8.9 | 512 | 3.6 |
| 6000 | 90 | 129 | 149 | 29.6 | 7.8 | 567 | 4.2 |
| 6000 | 80 | 123 | 142 | 25.8 | 6.8 | 633 | 4.9 |
| 6000 | 70 | 116 | 134 | 22.1 | 5.8 | 711 | 5.9 |
| 6000 | 60 | 108 | 124 | 18.6 | 4.9 | 801 | 7.2 |
| 6000 | 50 | 98 | 113 | 15.3 | 4.0 | 898 | 8.9 |
| | | | | | | | |
| 8000 | 100 | 138 | 158 | 33.6 | 8.9 | 520 | 3.5 |
| 8000 | 90 | 132 | 152 | 29.6 | 7.8 | 577 | 4.1 |
| 8000 | 80 | 125 | 144 | 25.8 | 6.8 | 640 | 4.8 |
| 8000 | 70 | 118 | 136 | 22.1 | 5.8 | 719 | 5.8 |
| 8000 | 60 | 110 | 126 | 18.6 | 4.9 | 810 | 7.1 |
| 8000 | 50 | 100 | 115 | 15.3 | 4.0 | 910 | 8.8 |
| | | | | | | | |
| 10000 | 90 | 134 | 154 | 29.6 | 7.8 | 582 | 4.0 |
| 10000 | 80 | 128 | 147 | 25.8 | 6.8 | 651 | 4.7 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22.1 | 5.8 | 726 | 5.7 |
| 10000 | 60 | 111 | 128 | 18.6 | 4.9 | 812 | 6.9 |
| 10000 | 50 | 101 | 116 | 15.3 | 4.0 | 913 | 8.7 |
| | | | | | | | |
| 12000 | 90 | 137 | 157 | 29.6 | 7.8 | 592 | 3.9 |
| 12000 | 80 | 130 | 149 | 25.8 | 6.8 | 657 | 4.6 |
| 12000 | 70 | 122 | 140 | 22.1 | 5.8 | 733 | 5.6 |
| 12000 | 60 | 113 | 130 | 18.6 | 4.9 | 821 | 6.8 |
| 12000 | 50 | 102 | 118 | 15.3 | 4.0 | 915 | 8.5 |
| | | | | | | | |
| 14000 | 90 | 139 | 160 | 29.6 | 7.8 | 597 | 3.8 |
| 14000 | 80 | 132 | 152 | 25.8 | 6.8 | 663 | 4.5 |
| 14000 | 70 | 124 | 143 | 22.1 | 5.8 | 740 | 5.4 |
| 14000 | 60 | 115 | 132 | 18.6 | 4.9 | 829 | 6.7 |
| 14000 | 50 | 104 | 119 | 15.3 | 4.0 | 926 | 8.3 |
| | | | | | | | |
| 16000 | 80 | 134 | 155 | 25.8 | 6.8 | 669 | 4.4 |
| 16000 | 70 | 126 | 145 | 22.1 | 5.8 | 747 | 5.3 |
| 16000 | 60 | 117 | 134 | 18.6 | 4.9 | 838 | 6.5 |
| 16000 | 50 | 105 | 121 | 15.3 | 4.0 | 928 | 8.2 |
| | | | | | | | |
| 18000 | 80 | 137 | 157 | 25.8 | 6.8 | 679 | 4.2 |
| 18000 | 70 | 128 | 148 | 22.1 | 5.8 | 754 | 5.2 |
| 18000 | 60 | 118 | 136 | 18.6 | 4.9 | 839 | 6.4 |
| 18000 | 50 | 107 | 123 | 15.3 | 4.0 | 938 | 8.0 |
| | | | | | | | |

Figure 5-4d Cruise Performance, Range and Endurance with long-range tanks, Cessna 172P at 1089 kg (2400 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with Integraltanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on integral tanks with 196.8 l (52 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 128 | 147 | 33.6 | 8.9 | 638 | 5.0 |
| SL | 90 | 123 | 141 | 29.6 | 7.8 | 709 | 5.8 |
| SL | 80 | 117 | 135 | 25.8 | 6.8 | 787 | 6.7 |
| SL | 70 | 111 | 127 | 22.1 | 5.8 | 885 | 8.0 |
| SL | 60 | 103 | 119 | 18.6 | 4.9 | 990 | 9.6 |
| SL | 50 | 94 | 108 | 15.3 | 4.0 | 1114 | 11.9 |
| | | | | | | | |
| 2000 | 100 | 130 | 150 | 33.6 | 8.9 | 646 | 4.9 |
| 2000 | 90 | 125 | 144 | 29.6 | 7.8 | 717 | 5.7 |
| 2000 | 80 | 119 | 137 | 25.8 | 6.8 | 796 | 6.6 |
| 2000 | 70 | 112 | 129 | 22.1 | 5.8 | 889 | 7.9 |
| 2000 | 60 | 105 | 120 | 18.6 | 4.9 | 1005 | 9.5 |
| 2000 | 50 | 96 | 110 | 15.3 | 4.0 | 1132 | 11.7 |
| | | | | | | | |
| 4000 | 100 | 133 | 153 | 33.6 | 8.9 | 658 | 4.8 |
| 4000 | 90 | 127 | 146 | 29.6 | 7.8 | 726 | 5.6 |
| 4000 | 80 | 121 | 139 | 25.8 | 6.8 | 806 | 6.5 |
| 4000 | 70 | 114 | 131 | 22.1 | 5.8 | 900 | 7.8 |
| 4000 | 60 | 106 | 122 | 18.6 | 4.9 | 1009 | 9.4 |
| 4000 | 50 | 97 | 111 | 15.3 | 4.0 | 1137 | 11.6 |
| | | | | | | | |
| 6000 | 100 | 135 | 155 | 33.6 | 8.9 | 665 | 4.7 |
| 6000 | 90 | 129 | 149 | 29.6 | 7.8 | 734 | 5.5 |
| 6000 | 80 | 123 | 142 | 25.8 | 6.8 | 816 | 6.4 |
| 6000 | 70 | 116 | 134 | 22.1 | 5.8 | 912 | 7.6 |
| 6000 | 60 | 108 | 124 | 18.6 | 4.9 | 1023 | 9.3 |
| 6000 | 50 | 98 | 113 | 15.3 | 4.0 | 1143 | 11.4 |
| | | | | | | | |
| 8000 | 100 | 138 | 158 | 33.6 | 8.9 | 677 | 4.6 |
| 8000 | 90 | 132 | 152 | 29.6 | 7.8 | 748 | 5.4 |
| 8000 | 80 | 125 | 144 | 25.8 | 6.8 | 825 | 6.3 |
| 8000 | 70 | 118 | 136 | 22.1 | 5.8 | 923 | 7.5 |
| 8000 | 60 | 110 | 126 | 18.6 | 4.9 | 1036 | 9.1 |
| 8000 | 50 | 100 | 115 | 15.3 | 4.0 | 1160 | 11.3 |
| | | | | | | | |
| 10000 | 90 | 134 | 154 | 29.6 | 7.8 | 755 | 5.3 |
| 10000 | 80 | 128 | 147 | 25.8 | 6.8 | 840 | 6.2 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22.1 | 5.8 | 933 | 7.4 |
| 10000 | 60 | 111 | 128 | 18.6 | 4.9 | 1040 | 9.0 |
| 10000 | 50 | 101 | 116 | 15.3 | 4.0 | 1165 | 11.1 |
| | | | | | | | |
| 12000 | 90 | 137 | 157 | 29.6 | 7.8 | 769 | 5.2 |
| 12000 | 80 | 130 | 149 | 25.8 | 6.8 | 849 | 6.1 |
| 12000 | 70 | 122 | 140 | 22.1 | 5.8 | 944 | 7.3 |
| 12000 | 60 | 113 | 130 | 18.6 | 4.9 | 1053 | 8.9 |
| 12000 | 50 | 102 | 118 | 15.3 | 4.0 | 1170 | 11.0 |
| | | | | | | | |
| 14000 | 90 | 139 | 160 | 29.6 | 7.8 | 776 | 5.1 |
| 14000 | 80 | 132 | 152 | 25.8 | 6.8 | 858 | 6.0 |
| 14000 | 70 | 124 | 143 | 22.1 | 5.8 | 954 | 7.2 |
| 14000 | 60 | 115 | 132 | 18.6 | 4.9 | 1065 | 8.7 |
| 14000 | 50 | 104 | 119 | 15.3 | 4.0 | 1186 | 10.8 |
| | | | | | | | |
| 16000 | 80 | 134 | 155 | 25.8 | 6.8 | 867 | 5.9 |
| 16000 | 70 | 126 | 145 | 22.1 | 5.8 | 965 | 7.0 |
| 16000 | 60 | 117 | 134 | 18.6 | 4.9 | 1078 | 8.6 |
| 16000 | 50 | 105 | 121 | 15.3 | 4.0 | 1190 | 10.7 |
| | | | | | | | |
| 18000 | 80 | 137 | 157 | 25.8 | 6.8 | 882 | 5.7 |
| 18000 | 70 | 128 | 148 | 22.1 | 5.8 | 975 | 6.9 |
| 18000 | 60 | 118 | 136 | 18.6 | 4.9 | 1081 | 8.4 |
| 18000 | 50 | 107 | 123 | 15.3 | 4.0 | 1205 | 10.5 |
| | | | | | | | |

Figure 5-4e Cruise Performance, Range and Endurance with Integraltanks, Cessna 172P at 1089 kg (2400 lbs)

# SECTION 5b
# PERFORMANCE

| | |
|---|---|
| ◆ Note: | This chapter applies to aircraft with propellers **MTV-6-A/190-69**.<br>The correct propeller designation can be found on the blades. |

| | |
|---|---|
| ◆ Note: | The chapter not relevant to the respective propeller can be omitted. |

**GROUND ROLL AND TAKE-OFF DISTANCE**
**at 1043 kg (2300 lbs)**

SHORT FIELD TAKE-OFFS

**Conditions:**
Take-off weight 1043 kg (2300 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: .....48 KIAS/ 55 mph

Speed at 15 m / 50 ft: .....54 KIAS/ 62 mph

**Notes:**

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 153 | 177 | 190 | 204 | 217 | 239 | 269 |
| | 50 ft (15 m) obstacle | 240 | 277 | 297 | 319 | 340 | 375 | 424 |
| 1000 | Gnd Roll | 164 | 190 | 203 | 218 | 233 | 256 | 289 |
| | 50 ft (15 m) obstacle | 257 | 297 | 318 | 342 | 365 | 402 | 455 |
| 2000 | Gnd Roll | 176 | 204 | 218 | 234 | 250 | 275 | 309 |
| | 50 ft (15 m) obstacle | 276 | 318 | 341 | 366 | 391 | 431 | 487 |
| 3000 | Gnd Roll | 189 | 218 | 234 | 251 | 268 | 295 | 332 |
| | 50 ft (15 m) obstacle | 295 | 341 | 366 | 393 | 419 | 462 | 522 |
| 4000 | Gnd Roll | 203 | 234 | 251 | 269 | 287 | 316 | 356 |
| | 50 ft (15 m) obstacle | 317 | 366 | 392 | 421 | 450 | 496 | 561 |
| 5000 | Gnd Roll | 218 | 251 | 269 | 289 | 308 | 339 | 382 |
| | 50 ft (15 m) obstacle | 340 | 393 | 421 | 452 | 483 | 532 | 602 |
| 6000 | Gnd Roll | 234 | 270 | 289 | 310 | 331 | 365 | 410 |
| | 50 ft (15 m) obstacle | 365 | 422 | 452 | 486 | 518 | 572 | 646 |
| 7000 | Gnd Roll | 256 | 296 | 317 | 340 | 363 | 400 | 450 |
| | 50 ft (15 m) obstacle | 401 | 463 | 495 | 532 | 568 | 627 | 708 |
| 8000 | Gnd Roll | 281 | 324 | 347 | 373 | 398 | 438 | 493 |
| | 50 ft (15 m) obstacle | 439 | 507 | 543 | 583 | 623 | 687 | 776 |
| 9000 | Gnd Roll | 311 | 359 | 385 | 413 | 441 | 485 | 546 |
| | 50 ft (15 m) obstacle | 487 | 562 | 602 | 646 | 690 | 761 | 860 |
| 10000 | Gnd Roll | 344 | 398 | 426 | 457 | 488 | 537 | 605 |
| | 50 ft (15 m) obstacle | 539 | 623 | 667 | 717 | 765 | 844 | 954 |

Figure 5-1a Take-Off Distance [m] at take-off weight 1043 kg (2300 lbs)

| PRESS ALT | Ground Roll and Take-Off Distance [ft] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 503 | 581 | 622 | 668 | 713 | 785 | 883 |
| | 50 ft (15 m) obstacle | 787 | 909 | 974 | 1046 | 1116 | 1232 | 1392 |
| 1000 | Gnd Roll | 539 | 623 | 667 | 716 | 764 | 841 | 946 |
| | 50 ft (15 m) obstacle | 843 | 974 | 1043 | 1120 | 1196 | 1319 | 1491 |
| 2000 | Gnd Roll | 578 | 668 | 715 | 767 | 819 | 901 | 1014 |
| | 50 ft (15 m) obstacle | 904 | 1044 | 1118 | 1201 | 1282 | 1414 | 1598 |
| 3000 | Gnd Roll | 620 | 716 | 766 | 823 | 878 | 967 | 1088 |
| | 50 ft (15 m) obstacle | 969 | 1120 | 1199 | 1288 | 1374 | 1517 | 1714 |
| 4000 | Gnd Roll | 665 | 768 | 822 | 883 | 942 | 1037 | 1167 |
| | 50 ft (15 m) obstacle | 1040 | 1201 | 1286 | 1381 | 1475 | 1627 | 1839 |
| 5000 | Gnd Roll | 714 | 824 | 883 | 948 | 1011 | 1113 | 1253 |
| | 50 ft (15 m) obstacle | 1116 | 1290 | 1381 | 1483 | 1583 | 1747 | 1973 |
| 6000 | Gnd Roll | 766 | 885 | 948 | 1018 | 1086 | 1196 | 1345 |
| | 50 ft (15 m) obstacle | 1199 | 1385 | 1483 | 1592 | 1700 | 1876 | 2119 |
| 7000 | Gnd Roll | 840 | 970 | 1039 | 1115 | 1191 | 1310 | 1474 |
| | 50 ft (15 m) obstacle | 1314 | 1518 | 1625 | 1745 | 1863 | 2056 | 2323 |
| 8000 | Gnd Roll | 921 | 1064 | 1139 | 1223 | 1305 | 1436 | 1616 |
| | 50 ft (15 m) obstacle | 1440 | 1664 | 1782 | 1913 | 2042 | 2253 | 2546 |
| 9000 | Gnd Roll | 1020 | 1178 | 1261 | 1354 | 1445 | 1591 | 1790 |
| | 50 ft (15 m) obstacle | 1596 | 1844 | 1974 | 2120 | 2263 | 2497 | 2822 |
| 10000 | Gnd Roll | 1130 | 1305 | 1397 | 1500 | 1601 | 1762 | 1983 |
| | 50 ft (15 m) obstacle | 1769 | 2044 | 2189 | 2351 | 2510 | 2769 | 3129 |

Figure 5-1b Take-Off Distance [ft] at take-off weight 1043 kg (2300 lbs)

**GROUND ROLL AND TAKE-OFF DISTANCE at 1089 kg (2400 lbs) (Cessna 172P only)**

SHORT FIELD TAKE-OFFS

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Flaps 10°
Full Power Prior to Brake Release
Paved, level, dry runway
Zero Wind

Lift Off: .....48 KIAS/ 55 mph

Speed at 15 m / 50 ft: .....54 KIAS/ 62 mph

**Notes:**

1. Short field technique
2. Decrease distances 10% for each 9 Knot headwind. For operation with tailwinds up to 10 Knot increase distances by 10% for each 2 Knot.
3. For operation on dry, grass runway, increase distances by 15% of the "ground roll" figure.
4. Consider additional distances (min. 20%) for wet grass runway, softened ground or snow.

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 171 | 198 | 212 | 228 | 243 | 267 | 301 |
| | 50 ft (15 m) obstacle | 268 | 310 | 332 | 356 | 380 | 419 | 474 |
| 1000 | Gnd Roll | 184 | 212 | 227 | 244 | 260 | 286 | 322 |
| | 50 ft (15 m) obstacle | 287 | 332 | 355 | 381 | 407 | 449 | 508 |
| 2000 | Gnd Roll | 197 | 227 | 243 | 261 | 279 | 307 | 345 |
| | 50 ft (15 m) obstacle | 308 | 356 | 381 | 409 | 436 | 482 | 544 |
| 3000 | Gnd Roll | 211 | 244 | 261 | 280 | 299 | 329 | 370 |
| | 50 ft (15 m) obstacle | 330 | 381 | 408 | 438 | 468 | 516 | 583 |
| 4000 | Gnd Roll | 226 | 262 | 280 | 301 | 321 | 353 | 397 |
| | 50 ft (15 m) obstacle | 354 | 409 | 438 | 470 | 502 | 554 | 626 |
| 5000 | Gnd Roll | 243 | 281 | 301 | 323 | 344 | 379 | 427 |
| | 50 ft (15 m) obstacle | 380 | 439 | 470 | 505 | 539 | 595 | 672 |
| 6000 | Gnd Roll | 261 | 301 | 323 | 347 | 370 | 407 | 458 |
| | 50 ft (15 m) obstacle | 408 | 472 | 505 | 542 | 579 | 639 | 722 |
| 7000 | Gnd Roll | 286 | 330 | 354 | 380 | 405 | 446 | 502 |
| | 50 ft (15 m) obstacle | 447 | 517 | 553 | 594 | 634 | 700 | 791 |
| 8000 | Gnd Roll | 313 | 362 | 388 | 416 | 444 | 489 | 550 |
| | 50 ft (15 m) obstacle | 490 | 567 | 607 | 651 | 695 | 767 | 867 |
| 9000 | Gnd Roll | 347 | 401 | 429 | 461 | 492 | 542 | 609 |
| | 50 ft (15 m) obstacle | 543 | 628 | 672 | 722 | 771 | 850 | 961 |
| 10000 | Gnd Roll | 385 | 444 | 476 | 511 | 545 | 600 | 675 |
| | 50 ft (15 m) obstacle | 603 | 696 | 745 | 800 | 855 | 943 | 1065 |

Figure 5-1c Take-Off Distance [m] at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

| PRESS ALT | Ground Roll and Take-Off Distance [m] Outside Air Temperature [°C] | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| [ft] | --- | -20°C | 0°C | 10°C | 20°C | 30°C | 40°C | 50°C |
| 0 | Gnd Roll | 562 | 649 | 695 | 746 | 797 | 877 | 986 |
| | 50 ft (15 m) obstacle | 879 | 1016 | 1087 | 1168 | 1247 | 1375 | 1554 |
| 1000 | Gnd Roll | 602 | 696 | 745 | 800 | 853 | 939 | 1057 |
| | 50 ft (15 m) obstacle | 942 | 1088 | 1165 | 1251 | 1336 | 1474 | 1665 |
| 2000 | Gnd Roll | 645 | 746 | 798 | 857 | 915 | 1007 | 1133 |
| | 50 ft (15 m) obstacle | 1009 | 1166 | 1249 | 1341 | 1431 | 1579 | 1785 |
| 3000 | Gnd Roll | 692 | 799 | 856 | 919 | 981 | 1080 | 1215 |
| | 50 ft (15 m) obstacle | 1082 | 1251 | 1339 | 1438 | 1535 | 1694 | 1914 |
| 4000 | Gnd Roll | 742 | 858 | 918 | 986 | 1052 | 1158 | 1303 |
| | 50 ft (15 m) obstacle | 1161 | 1342 | 1437 | 1543 | 1647 | 1817 | 2053 |
| 5000 | Gnd Roll | 797 | 921 | 986 | 1058 | 1130 | 1243 | 1399 |
| | 50 ft (15 m) obstacle | 1246 | 1440 | 1542 | 1656 | 1768 | 1951 | 2204 |
| 6000 | Gnd Roll | 856 | 989 | 1059 | 1137 | 1213 | 1335 | 1502 |
| | 50 ft (15 m) obstacle | 1339 | 1547 | 1656 | 1779 | 1899 | 2095 | 2367 |
| 7000 | Gnd Roll | 938 | 1084 | 1160 | 1246 | 1330 | 1464 | 1647 |
| | 50 ft (15 m) obstacle | 1467 | 1695 | 1815 | 1949 | 2081 | 2296 | 2594 |
| 8000 | Gnd Roll | 1028 | 1188 | 1272 | 1366 | 1458 | 1604 | 1805 |
| | 50 ft (15 m) obstacle | 1608 | 1858 | 1990 | 2137 | 2281 | 2517 | 2844 |
| 9000 | Gnd Roll | 1139 | 1316 | 1409 | 1512 | 1614 | 1777 | 1999 |
| | 50 ft (15 m) obstacle | 1782 | 2059 | 2205 | 2368 | 2528 | 2789 | 3152 |
| 10000 | Gnd Roll | 1262 | 1458 | 1561 | 1675 | 1788 | 1968 | 2215 |
| | 50 ft (15 m) obstacle | 1976 | 2283 | 2445 | 2626 | 2803 | 3093 | 3495 |

Figure 5-1d Take-Off Distance [ft] at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

## MAXIMUM RATE-OF-CLIMB at 1043 kg (2300 lbs)

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power

**Notes:**

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

| PRESS ALT [FT] | Climb speed [KIAS] | Rate of Climb [ft/min] Outside Air Temperature [°C] | | | | |
|---|---|---|---|---|---|---|
| | | -20°C | 0°C | +20°C | +40°C | +50°C |
| 0 | 69 | 938 | 924 | 911 | 796 | 659 |
| 1000 | 69 | 932 | 917 | 904 | 789 | 651 |
| 2000 | 69 | 925 | 910 | 897 | 782 | 644 |
| 3000 | 69 | 918 | 903 | 889 | 774 | 636 |
| 4000 | 69 | 911 | 896 | 881 | 766 | 628 |
| 5000 | 69 | 904 | 888 | 874 | 758 | 620 |
| 6000 | 69 | 896 | 881 | 866 | 750 | 611 |
| 7000 | 69 | 888 | 873 | 857 | 741 | 603 |
| 8000 | 69 | 881 | 864 | 849 | 732 | 594 |
| 9000 | 69 | 847 | 830 | 814 | 700 | 563 |
| 10000 | 69 | 812 | 796 | 779 | 666 | 533 |
| 11000 | 69 | 778 | 761 | 744 | 633 | 502 |
| 12000 | 69 | 743 | 726 | 709 | 599 | 471 |
| 13000 | 69 | 708 | 691 | 673 | 566 | 439 |
| 14000 | 69 | 673 | 655 | 638 | 531 | 408 |
| 15000 | 69 | 638 | 619 | 601 | 497 | 376 |
| 16000 | 69 | 602 | 583 | 565 | 462 | 343 |
| 17000 | 69 | 566 | 547 | 528 | 427 | 311 |
| 18000 | 69 | 530 | 510 | 491 | 392 | 278 |

Figure 5-2a Maximum Rate of Climb at take-off weight 1043 kg (2300 lbs)

**MAXIMUM RATE-OF-CLIMB at 1089 kg (2400 lbs) (Cessna 172P only)**

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up, Full Power

**Notes:**

1. For operation in air colder than this table provides, use coldest data shown.
2. For operation in air warmer than this table provides, use extreme caution.

| PRESS ALT [FT] | Climb speed [KIAS] | Rate of Climb [ft/min] Outside Air Temperature [°C] | | | | |
|---|---|---|---|---|---|---|
| | | -20°C | 0°C | +20°C | +40°C | +50°C |
| 0 | 69 | 876 | 862 | 849 | 738 | 606 |
| 1000 | 69 | 870 | 855 | 841 | 731 | 598 |
| 2000 | 69 | 863 | 848 | 834 | 723 | 590 |
| 3000 | 69 | 856 | 841 | 826 | 715 | 582 |
| 4000 | 69 | 849 | 833 | 819 | 707 | 574 |
| 5000 | 69 | 841 | 826 | 811 | 699 | 566 |
| 6000 | 69 | 834 | 818 | 802 | 690 | 557 |
| 7000 | 69 | 826 | 810 | 794 | 682 | 548 |
| 8000 | 69 | 818 | 801 | 785 | 673 | 539 |
| 9000 | 69 | 785 | 768 | 752 | 641 | 510 |
| 10000 | 69 | 751 | 734 | 718 | 608 | 480 |
| 11000 | 69 | 718 | 700 | 683 | 576 | 450 |
| 12000 | 69 | 684 | 666 | 649 | 543 | 419 |
| 13000 | 69 | 650 | 632 | 614 | 510 | 388 |
| 14000 | 69 | 616 | 597 | 579 | 476 | 357 |
| 15000 | 69 | 581 | 562 | 544 | 443 | 326 |
| 16000 | 69 | 546 | 527 | 508 | 409 | 294 |
| 17000 | 69 | 511 | 491 | 472 | 374 | 262 |
| 18000 | 69 | 476 | 456 | 436 | 340 | 230 |

Figure 5-2b Maximum Rate of Climb at take-off weight 1089 kg (2400 lbs) (Cessna 172P only)

**TIME, FUEL AND DISTANCE TO CLIMB at 1043 KG (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

**Notes:**

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at $v_y$ = 69 KIAS.

| Press. Alt. | OAT | Vy | ROC | Time | Distance | Fuel used | |
|---|---|---|---|---|---|---|---|
| [ft] | [°C] | [KIAS] | [FPM] | [MIN] | [NM] | [l] | [US Gal] |
| 0 | 15 | 69 | 914 | 0,0 | 0,0 | 0,0 | 0,0 |
| 1000 | 13 | 69 | 908 | 1,1 | 1,3 | 0,6 | 0,2 |
| 2000 | 11 | 69 | 903 | 2,2 | 2,6 | 1,2 | 0,3 |
| 3000 | 9 | 69 | 897 | 3,3 | 4,0 | 1,9 | 0,5 |
| 4000 | 7 | 69 | 891 | 4,4 | 5,4 | 2,5 | 0,7 |
| 5000 | 5 | 69 | 885 | 5,6 | 6,9 | 3,1 | 0,8 |
| 6000 | 3 | 69 | 878 | 6,7 | 8,5 | 3,7 | 1,0 |
| 7000 | 1 | 69 | 872 | 7,8 | 10,1 | 4,4 | 1,2 |
| 8000 | -1 | 69 | 865 | 9,0 | 11,8 | 5,0 | 1,3 |
| 9000 | -3 | 69 | 832 | 10,2 | 13,5 | 5,6 | 1,5 |
| 10000 | -5 | 69 | 800 | 11,4 | 15,4 | 6,1 | 1,6 |
| 11000 | -7 | 69 | 767 | 12,7 | 17,4 | 6,6 | 1,7 |
| 12000 | -9 | 69 | 733 | 14,0 | 19,5 | 7,1 | 1,9 |
| 13000 | -11 | 69 | 700 | 15,4 | 21,8 | 7,6 | 2,0 |
| 14000 | -13 | 69 | 667 | 16,9 | 24,3 | 8,1 | 2,1 |
| 15000 | -15 | 69 | 633 | 18,4 | 27,0 | 8,6 | 2,3 |
| 16000 | -17 | 69 | 599 | 20,0 | 29,8 | 9,1 | 2,4 |
| 17000 | -19 | 69 | 565 | 21,7 | 32,9 | 9,6 | 2,5 |
| 18000 | -21 | 69 | 530 | 23,6 | 36,3 | 10,1 | 2,7 |

Figure 5-3a Time, Fuel and Distance to Climb at 1043 kg (2300 lbs)

**TIME, FUEL AND DISTANCE TO CLIMB at 1089 KG (Cessna 172P only)**

**Conditions:**

Take-off weight 1089 kg (2400 lbs)
Climb speed $v_y$ = 69 KIAS/ 79 mph
Flaps Up
Full Power
Standard Temperature (ISA)

**Notes :**

1. Add 4 l (1.1 US gal) of fuel for engine start, taxi and takeoff allowance.
2. Increase time and distance by 10% for 10°C above standard temperature.
3. Distances shown are based on zero wind.
4. Time, distance and fuel required are only valid from the point where the airplane climbs at $v_y$ = 69 KIAS.

| Press. Alt. | OAT | Vy | ROC | Time | Distance | Fuel used | |
|---|---|---|---|---|---|---|---|
| [ft] | [°C] | [KIAS] | [FPM] | [MIN] | [NM] | [l] | [US Gal] |
| 0 | 15 | 69 | 852 | 0,0 | 0,0 | 0,0 | 0,0 |
| 1000 | 13 | 69 | 846 | 1,2 | 1,4 | 0,7 | 0,2 |
| 2000 | 11 | 69 | 840 | 2,4 | 2,8 | 1,3 | 0,3 |
| 3000 | 9 | 69 | 834 | 3,6 | 4,3 | 2,0 | 0,5 |
| 4000 | 7 | 69 | 828 | 4,8 | 5,9 | 2,7 | 0,7 |
| 5000 | 5 | 69 | 822 | 6,0 | 7,5 | 3,3 | 0,9 |
| 6000 | 3 | 69 | 815 | 7,2 | 9,1 | 4,0 | 1,1 |
| 7000 | 1 | 69 | 809 | 8,4 | 10,9 | 4,7 | 1,2 |
| 8000 | -1 | 69 | 802 | 9,7 | 12,7 | 5,4 | 1,4 |
| 9000 | -3 | 69 | 770 | 10,9 | 14,5 | 6,0 | 1,6 |
| 10000 | -5 | 69 | 738 | 12,3 | 16,6 | 6,5 | 1,7 |
| 11000 | -7 | 69 | 706 | 13,7 | 18,7 | 7,1 | 1,9 |
| 12000 | -9 | 69 | 674 | 15,1 | 21,1 | 7,7 | 2,0 |
| 13000 | -11 | 69 | 641 | 16,6 | 23,6 | 8,2 | 2,2 |
| 14000 | -13 | 69 | 609 | 18,2 | 26,3 | 8,7 | 2,3 |
| 15000 | -15 | 69 | 576 | 19,9 | 29,2 | 9,3 | 2,5 |
| 16000 | -17 | 69 | 543 | 21,7 | 32,3 | 9,8 | 2,6 |
| 17000 | -19 | 69 | 510 | 23,6 | 35,8 | 10,4 | 2,7 |
| 18000 | -21 | 69 | 476 | 25,6 | 39,5 | 10,9 | 2,9 |

Figure 5-3b Time, Fuel and Distance to Climb at 1089 kg (2400 lbs) (Cessna 172P only)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with standard tanks at 1043 kg (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

**Notes:**

1. Endurance information is based on standard tanks with 127.4 l (33.6 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 125 | 144 | 33,6 | 8,9 | 365 | 2,9 |
| SL | 90 | 120 | 139 | 29,6 | 7,8 | 412 | 3,4 |
| SL | 80 | 115 | 133 | 25,8 | 6,8 | 465 | 4,0 |
| SL | 70 | 110 | 126 | 22,1 | 5,8 | 530 | 4,8 |
| SL | 60 | 103 | 119 | 18,6 | 4,9 | 608 | 5,9 |
| SL | 50 | 95 | 110 | 15,3 | 4,0 | 698 | 7,3 |
| | | | | | | | |
| 2000 | 100 | 127 | 147 | 33,6 | 8,9 | 370 | 2,8 |
| 2000 | 90 | 123 | 141 | 29,6 | 7,8 | 416 | 3,3 |
| 2000 | 80 | 118 | 135 | 25,8 | 6,8 | 470 | 3,9 |
| 2000 | 70 | 112 | 129 | 22,1 | 5,8 | 536 | 4,7 |
| 2000 | 60 | 105 | 121 | 18,6 | 4,9 | 613 | 5,8 |
| 2000 | 50 | 97 | 112 | 15,3 | 4,0 | 702 | 7,2 |
| | | | | | | | |
| 4000 | 100 | 130 | 149 | 33,6 | 8,9 | 374 | 2,7 |
| 4000 | 90 | 125 | 144 | 29,6 | 7,8 | 421 | 3,2 |
| 4000 | 80 | 120 | 138 | 25,8 | 6,8 | 475 | 3,8 |
| 4000 | 70 | 114 | 131 | 22,1 | 5,8 | 541 | 4,6 |
| 4000 | 60 | 107 | 123 | 18,6 | 4,9 | 618 | 5,6 |
| 4000 | 50 | 98 | 113 | 15,3 | 4,0 | 707 | 7,0 |
| | | | | | | | |
| 6000 | 100 | 132 | 152 | 33,6 | 8,9 | 379 | 2,6 |
| 6000 | 90 | 127 | 147 | 29,6 | 7,8 | 426 | 3,1 |
| 6000 | 80 | 122 | 140 | 25,8 | 6,8 | 480 | 3,7 |
| 6000 | 70 | 116 | 133 | 22,1 | 5,8 | 546 | 4,5 |
| 6000 | 60 | 109 | 125 | 18,6 | 4,9 | 623 | 5,5 |
| 6000 | 50 | 100 | 115 | 15,3 | 4,0 | 711 | 6,9 |
| | | | | | | | |
| 8000 | 100 | 135 | 155 | 33,6 | 8,9 | 383 | 2,5 |
| 8000 | 90 | 130 | 150 | 29,6 | 7,8 | 431 | 3,0 |
| 8000 | 80 | 124 | 143 | 25,8 | 6,8 | 485 | 3,6 |
| 8000 | 70 | 118 | 136 | 22,1 | 5,8 | 551 | 4,4 |
| 8000 | 60 | 111 | 127 | 18,6 | 4,9 | 628 | 5,4 |
| 8000 | 50 | 101 | 117 | 15,3 | 4,0 | 714 | 6,7 |
| | | | | | | | |
| 10000 | 90 | 133 | 153 | 29,6 | 7,8 | 435 | 2,9 |
| 10000 | 80 | 127 | 146 | 25,8 | 6,8 | 490 | 3,5 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22,1 | 5,8 | 556 | 4,2 |
| 10000 | 60 | 113 | 130 | 18,6 | 4,9 | 632 | 5,2 |
| 10000 | 50 | 103 | 118 | 15,3 | 4,0 | 716 | 6,6 |
| | | | | | | | |
| 12000 | 90 | 135 | 156 | 29,6 | 7,8 | 440 | 2,8 |
| 12000 | 80 | 129 | 149 | 25,8 | 6,8 | 495 | 3,4 |
| 12000 | 70 | 123 | 141 | 22,1 | 5,8 | 560 | 4,1 |
| 12000 | 60 | 115 | 132 | 18,6 | 4,9 | 636 | 5,1 |
| 12000 | 50 | 104 | 120 | 15,3 | 4,0 | 718 | 6,4 |
| | | | | | | | |
| 14000 | 90 | 138 | 159 | 29,6 | 7,8 | 445 | 2,7 |
| 14000 | 80 | 132 | 152 | 25,8 | 6,8 | 500 | 3,2 |
| 14000 | 70 | 125 | 144 | 22,1 | 5,8 | 565 | 3,9 |
| 14000 | 60 | 117 | 134 | 18,6 | 4,9 | 640 | 4,9 |
| 14000 | 50 | 105 | 121 | 15,3 | 4,0 | 718 | 6,2 |
| | | | | | | | |
| 16000 | 80 | 134 | 155 | 25,8 | 6,8 | 505 | 3,1 |
| 16000 | 70 | 127 | 146 | 22,1 | 5,8 | 570 | 3,8 |
| 16000 | 60 | 118 | 136 | 18,6 | 4,9 | 643 | 4,7 |
| 16000 | 50 | 107 | 123 | 15,3 | 4,0 | 717 | 6,0 |
| | | | | | | | |
| 18000 | 80 | 137 | 158 | 25,8 | 6,8 | 512 | 2,9 |
| 18000 | 70 | 130 | 149 | 22,1 | 5,8 | 577 | 3,6 |
| 18000 | 60 | 120 | 139 | 18,6 | 4,9 | 649 | 4,5 |
| 18000 | 50 | 107 | 124 | 15,3 | 4,0 | 718 | 5,8 |
| | | | | | | | |

Figure 5-4a Cruise Performance, Range and Endurance with standard tanks, at 1043 kg (2300 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with long-range tanks (Cessna 172N) at 1043 kg (2300 lbs)**

**Conditions:**

Take-off weight 1043 kg (2300 lbs)
Flaps Up
Zero wind

**Notes:**

1. Endurance information is based on long-range tanks with 158.6 l (41.9 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 125 | 144 | 33,6 | 8,9 | 482 | 3,9 |
| SL | 90 | 120 | 139 | 29,6 | 7,8 | 539 | 4,5 |
| SL | 80 | 115 | 133 | 25,8 | 6,8 | 605 | 5,2 |
| SL | 70 | 110 | 126 | 22,1 | 5,8 | 685 | 6,2 |
| SL | 60 | 103 | 119 | 18,6 | 4,9 | 781 | 7,6 |
| SL | 50 | 95 | 110 | 15,3 | 4,0 | 893 | 9,4 |
| | | | | | | | |
| 2000 | 100 | 127 | 147 | 33,6 | 8,9 | 488 | 3,8 |
| 2000 | 90 | 123 | 141 | 29,6 | 7,8 | 546 | 4,4 |
| 2000 | 80 | 118 | 135 | 25,8 | 6,8 | 612 | 5,1 |
| 2000 | 70 | 112 | 129 | 22,1 | 5,8 | 693 | 6,1 |
| 2000 | 60 | 105 | 121 | 18,6 | 4,9 | 789 | 7,4 |
| 2000 | 50 | 97 | 112 | 15,3 | 4,0 | 900 | 9,2 |
| | | | | | | | |
| 4000 | 100 | 130 | 149 | 33,6 | 8,9 | 495 | 3,7 |
| 4000 | 90 | 125 | 144 | 29,6 | 7,8 | 553 | 4,3 |
| 4000 | 80 | 120 | 138 | 25,8 | 6,8 | 620 | 5,0 |
| 4000 | 70 | 114 | 131 | 22,1 | 5,8 | 701 | 6,0 |
| 4000 | 60 | 107 | 123 | 18,6 | 4,9 | 797 | 7,3 |
| 4000 | 50 | 98 | 113 | 15,3 | 4,0 | 907 | 9,1 |
| | | | | | | | |
| 6000 | 100 | 132 | 152 | 33,6 | 8,9 | 502 | 3,6 |
| 6000 | 90 | 127 | 147 | 29,6 | 7,8 | 560 | 4,2 |
| 6000 | 80 | 122 | 140 | 25,8 | 6,8 | 628 | 4,9 |
| 6000 | 70 | 116 | 133 | 22,1 | 5,8 | 710 | 5,9 |
| 6000 | 60 | 109 | 125 | 18,6 | 4,9 | 806 | 7,2 |
| 6000 | 50 | 100 | 115 | 15,3 | 4,0 | 914 | 8,9 |
| | | | | | | | |
| 8000 | 100 | 135 | 155 | 33,6 | 8,9 | 509 | 3,5 |
| 8000 | 90 | 130 | 150 | 29,6 | 7,8 | 568 | 4,1 |
| 8000 | 80 | 124 | 143 | 25,8 | 6,8 | 635 | 4,8 |
| 8000 | 70 | 118 | 136 | 22,1 | 5,8 | 718 | 5,8 |
| 8000 | 60 | 111 | 127 | 18,6 | 4,9 | 813 | 7,0 |
| 8000 | 50 | 101 | 117 | 15,3 | 4,0 | 920 | 8,8 |
| | | | | | | | |
| 10000 | 90 | 133 | 153 | 29,6 | 7,8 | 575 | 4,0 |
| 10000 | 80 | 127 | 146 | 25,8 | 6,8 | 643 | 4,7 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22,1 | 5,8 | 726 | 5,6 |
| 10000 | 60 | 113 | 130 | 18,6 | 4,9 | 821 | 6,9 |
| 10000 | 50 | 103 | 118 | 15,3 | 4,0 | 926 | 8,6 |
| | | | | | | | |
| 12000 | 90 | 135 | 156 | 29,6 | 7,8 | 583 | 3,8 |
| 12000 | 80 | 129 | 149 | 25,8 | 6,8 | 651 | 4,6 |
| 12000 | 70 | 123 | 141 | 22,1 | 5,8 | 733 | 5,5 |
| 12000 | 60 | 115 | 132 | 18,6 | 4,9 | 828 | 6,7 |
| 12000 | 50 | 104 | 120 | 15,3 | 4,0 | 930 | 8,4 |
| | | | | | | | |
| 14000 | 90 | 138 | 159 | 29,6 | 7,8 | 590 | 3,7 |
| 14000 | 80 | 132 | 152 | 25,8 | 6,8 | 659 | 4,4 |
| 14000 | 70 | 125 | 144 | 22,1 | 5,8 | 741 | 5,4 |
| 14000 | 60 | 117 | 134 | 18,6 | 4,9 | 835 | 6,6 |
| 14000 | 50 | 105 | 121 | 15,3 | 4,0 | 933 | 8,2 |
| | | | | | | | |
| 16000 | 80 | 134 | 155 | 25,8 | 6,8 | 667 | 4,3 |
| 16000 | 70 | 127 | 146 | 22,1 | 5,8 | 749 | 5,2 |
| 16000 | 60 | 118 | 136 | 18,6 | 4,9 | 842 | 6,4 |
| 16000 | 50 | 107 | 123 | 15,3 | 4,0 | 935 | 8,0 |
| | | | | | | | |
| 18000 | 80 | 137 | 158 | 25,8 | 6,8 | 678 | 4,1 |
| 18000 | 70 | 130 | 149 | 22,1 | 5,8 | 760 | 5,0 |
| 18000 | 60 | 120 | 139 | 18,6 | 4,9 | 851 | 6,2 |
| 18000 | 50 | 107 | 124 | 15,3 | 4,0 | 937 | 7,8 |
| | | | | | | | |

Figure 5-4b Cruise Performance, Range and Endurance with long-range tanks, at 1043 kg (2300 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with standard tanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on standard tanks with 127.4 l (33.6 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 125 | 144 | 33,6 | 8,9 | 365 | 2,9 |
| SL | 90 | 120 | 138 | 29,6 | 7,8 | 411 | 3,4 |
| SL | 80 | 115 | 132 | 25,8 | 6,8 | 464 | 4,0 |
| SL | 70 | 109 | 126 | 22,1 | 5,8 | 529 | 4,8 |
| SL | 60 | 103 | 118 | 18,6 | 4,9 | 604 | 5,9 |
| SL | 50 | 95 | 109 | 15,3 | 4,0 | 691 | 7,3 |
| | | | | | | | |
| 2000 | 100 | 127 | 146 | 33,6 | 8,9 | 369 | 2,8 |
| 2000 | 90 | 122 | 141 | 29,6 | 7,8 | 415 | 3,3 |
| 2000 | 80 | 117 | 135 | 25,8 | 6,8 | 469 | 3,9 |
| 2000 | 70 | 111 | 128 | 22,1 | 5,8 | 533 | 4,7 |
| 2000 | 60 | 104 | 120 | 18,6 | 4,9 | 609 | 5,7 |
| 2000 | 50 | 96 | 110 | 15,3 | 4,0 | 695 | 7,2 |
| | | | | | | | |
| 4000 | 100 | 130 | 149 | 33,6 | 8,9 | 373 | 2,7 |
| 4000 | 90 | 125 | 144 | 29,6 | 7,8 | 420 | 3,2 |
| 4000 | 80 | 119 | 137 | 25,8 | 6,8 | 473 | 3,8 |
| 4000 | 70 | 113 | 130 | 22,1 | 5,8 | 538 | 4,6 |
| 4000 | 60 | 106 | 122 | 18,6 | 4,9 | 613 | 5,6 |
| 4000 | 50 | 97 | 112 | 15,3 | 4,0 | 698 | 7,0 |
| | | | | | | | |
| 6000 | 100 | 132 | 152 | 33,6 | 8,9 | 378 | 2,6 |
| 6000 | 90 | 127 | 146 | 29,6 | 7,8 | 424 | 3,1 |
| 6000 | 80 | 122 | 140 | 25,8 | 6,8 | 478 | 3,7 |
| 6000 | 70 | 115 | 133 | 22,1 | 5,8 | 542 | 4,5 |
| 6000 | 60 | 108 | 124 | 18,6 | 4,9 | 617 | 5,5 |
| 6000 | 50 | 99 | 113 | 15,3 | 4,0 | 700 | 6,8 |
| | | | | | | | |
| 8000 | 100 | 135 | 155 | 33,6 | 8,9 | 382 | 2,5 |
| 8000 | 90 | 130 | 149 | 29,6 | 7,8 | 428 | 3,0 |
| 8000 | 80 | 124 | 143 | 25,8 | 6,8 | 482 | 3,6 |
| 8000 | 70 | 118 | 135 | 22,1 | 5,8 | 547 | 4,3 |
| 8000 | 60 | 110 | 126 | 18,6 | 4,9 | 621 | 5,3 |
| 8000 | 50 | 100 | 115 | 15,3 | 4,0 | 701 | 6,7 |
| | | | | | | | |
| 10000 | 90 | 132 | 152 | 29,6 | 7,8 | 433 | 2,9 |
| 10000 | 80 | 126 | 145 | 25,8 | 6,8 | 486 | 3,4 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22,1 | 5,8 | 551 | 4,2 |
| 10000 | 60 | 112 | 128 | 18,6 | 4,9 | 624 | 5,2 |
| 10000 | 50 | 101 | 116 | 15,3 | 4,0 | 701 | 6,5 |
| | | | | | | | |
| 12000 | 90 | 135 | 155 | 29,6 | 7,8 | 437 | 2,7 |
| 12000 | 80 | 129 | 148 | 25,8 | 6,8 | 491 | 3,3 |
| 12000 | 70 | 122 | 140 | 22,1 | 5,8 | 555 | 4,0 |
| 12000 | 60 | 113 | 131 | 18,6 | 4,9 | 627 | 5,0 |
| 12000 | 50 | 102 | 117 | 15,3 | 4,0 | 700 | 6,3 |
| | | | | | | | |
| 14000 | 90 | 138 | 158 | 29,6 | 7,8 | 441 | 2,6 |
| 14000 | 80 | 131 | 151 | 25,8 | 6,8 | 495 | 3,1 |
| 14000 | 70 | 124 | 143 | 22,1 | 5,8 | 558 | 3,9 |
| 14000 | 60 | 115 | 133 | 18,6 | 4,9 | 629 | 4,8 |
| 14000 | 50 | 103 | 118 | 15,3 | 4,0 | 697 | 6,1 |
| | | | | | | | |
| 16000 | 80 | 134 | 154 | 25,8 | 6,8 | 499 | 3,0 |
| 16000 | 70 | 126 | 145 | 22,1 | 5,8 | 562 | 3,7 |
| 16000 | 60 | 117 | 135 | 18,6 | 4,9 | 631 | 4,6 |
| 16000 | 50 | 103 | 119 | 15,3 | 4,0 | 692 | 5,9 |
| | | | | | | | |
| 18000 | 80 | 136 | 157 | 25,8 | 6,8 | 505 | 2,8 |
| 18000 | 70 | 129 | 148 | 22,1 | 5,8 | 568 | 3,5 |
| 18000 | 60 | 119 | 137 | 18,6 | 4,9 | 635 | 4,4 |
| 18000 | 50 | 103 | 119 | 15,3 | 4,0 | 686 | 5,7 |
| | | | | | | | |

Figure 5-4c Cruise Performance, Range and Endurance with standard tanks, Cessna 172P at 1089 kg (2400 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with long-range tanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on long-range tanks with 158.6 l (41.9 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 125 | 144 | 33,6 | 8,9 | 481 | 3,9 |
| SL | 90 | 120 | 138 | 29,6 | 7,8 | 538 | 4,5 |
| SL | 80 | 115 | 132 | 25,8 | 6,8 | 603 | 5,2 |
| SL | 70 | 109 | 126 | 22,1 | 5,8 | 683 | 6,2 |
| SL | 60 | 103 | 118 | 18,6 | 4,9 | 777 | 7,6 |
| SL | 50 | 95 | 109 | 15,3 | 4,0 | 884 | 9,4 |
| | | | | | | | |
| 2000 | 100 | 127 | 146 | 33,6 | 8,9 | 487 | 3,8 |
| 2000 | 90 | 122 | 141 | 29,6 | 7,8 | 544 | 4,4 |
| 2000 | 80 | 117 | 135 | 25,8 | 6,8 | 610 | 5,1 |
| 2000 | 70 | 111 | 128 | 22,1 | 5,8 | 690 | 6,1 |
| 2000 | 60 | 104 | 120 | 18,6 | 4,9 | 784 | 7,4 |
| 2000 | 50 | 96 | 110 | 15,3 | 4,0 | 890 | 9,2 |
| | | | | | | | |
| 4000 | 100 | 130 | 149 | 33,6 | 8,9 | 494 | 3,7 |
| 4000 | 90 | 125 | 144 | 29,6 | 7,8 | 551 | 4,3 |
| 4000 | 80 | 119 | 137 | 25,8 | 6,8 | 618 | 5,0 |
| 4000 | 70 | 113 | 130 | 22,1 | 5,8 | 698 | 6,0 |
| 4000 | 60 | 106 | 122 | 18,6 | 4,9 | 791 | 7,3 |
| 4000 | 50 | 97 | 112 | 15,3 | 4,0 | 896 | 9,0 |
| | | | | | | | |
| 6000 | 100 | 132 | 152 | 33,6 | 8,9 | 500 | 3,6 |
| 6000 | 90 | 127 | 146 | 29,6 | 7,8 | 558 | 4,2 |
| 6000 | 80 | 122 | 140 | 25,8 | 6,8 | 625 | 4,9 |
| 6000 | 70 | 115 | 133 | 22,1 | 5,8 | 705 | 5,9 |
| 6000 | 60 | 108 | 124 | 18,6 | 4,9 | 799 | 7,1 |
| 6000 | 50 | 99 | 113 | 15,3 | 4,0 | 901 | 8,9 |
| | | | | | | | |
| 8000 | 100 | 135 | 155 | 33,6 | 8,9 | 507 | 3,4 |
| 8000 | 90 | 130 | 149 | 29,6 | 7,8 | 565 | 4,0 |
| 8000 | 80 | 124 | 143 | 25,8 | 6,8 | 632 | 4,8 |
| 8000 | 70 | 118 | 135 | 22,1 | 5,8 | 713 | 5,7 |
| 8000 | 60 | 110 | 126 | 18,6 | 4,9 | 805 | 7,0 |
| 8000 | 50 | 100 | 115 | 15,3 | 4,0 | 904 | 8,7 |
| | | | | | | | |
| 10000 | 90 | 132 | 152 | 29,6 | 7,8 | 572 | 3,9 |
| 10000 | 80 | 126 | 145 | 25,8 | 6,8 | 639 | 4,6 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22,1 | 5,8 | 720 | 5,6 |
| 10000 | 60 | 112 | 128 | 18,6 | 4,9 | 812 | 6,8 |
| 10000 | 50 | 101 | 116 | 15,3 | 4,0 | 907 | 8,5 |
| | | | | | | | |
| 12000 | 90 | 135 | 155 | 29,6 | 7,8 | 579 | 3,8 |
| 12000 | 80 | 129 | 148 | 25,8 | 6,8 | 646 | 4,5 |
| 12000 | 70 | 122 | 140 | 22,1 | 5,8 | 727 | 5,4 |
| 12000 | 60 | 113 | 131 | 18,6 | 4,9 | 817 | 6,7 |
| 12000 | 50 | 102 | 117 | 15,3 | 4,0 | 908 | 8,3 |
| | | | | | | | |
| 14000 | 90 | 138 | 158 | 29,6 | 7,8 | 586 | 3,6 |
| 14000 | 80 | 131 | 151 | 25,8 | 6,8 | 654 | 4,4 |
| 14000 | 70 | 124 | 143 | 22,1 | 5,8 | 733 | 5,3 |
| 14000 | 60 | 115 | 133 | 18,6 | 4,9 | 823 | 6,5 |
| 14000 | 50 | 103 | 118 | 15,3 | 4,0 | 907 | 8,1 |
| | | | | | | | |
| 16000 | 80 | 134 | 154 | 25,8 | 6,8 | 661 | 4,2 |
| 16000 | 70 | 126 | 145 | 22,1 | 5,8 | 740 | 5,1 |
| 16000 | 60 | 117 | 135 | 18,6 | 4,9 | 828 | 6,3 |
| 16000 | 50 | 103 | 119 | 15,3 | 4,0 | 903 | 7,9 |
| | | | | | | | |
| 18000 | 80 | 136 | 157 | 25,8 | 6,8 | 670 | 4,0 |
| 18000 | 70 | 129 | 148 | 22,1 | 5,8 | 749 | 4,9 |
| 18000 | 60 | 119 | 137 | 18,6 | 4,9 | 834 | 6,1 |
| 18000 | 50 | 103 | 119 | 15,3 | 4,0 | 897 | 7,7 |
| | | | | | | | |

Figure 5-4d Cruise Performance, Range and Endurance with long-range tanks, Cessna 172P at 1089 kg (2400 lbs)

**CRUISE PERFORMANCE, RANGE AND ENDURANCE with Integraltanks at 1089 kg (2400 lbs) (Cessna 172P)**

**Conditions:**
Take-off weight 1089 kg (2400 lbs)
Flaps Up
Zero wind

**Notes:**
1. Endurance information is based on integral tanks with 196.8 l (52 US gal) usable fuel.
2. The table assumes 4 l (1.1 US gal) for startup and taxi; time, fuel and distance to climb and 45 min. reserve.
3. Increase true airspeed (KTAS) and maximum range (NM) by 1% per 10°C above ISA temperature.
4. Cruise Power above 75% not recommended. For economic cruise set load 70% or less.

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| SL | 100 | 125 | 144 | 33,6 | 8,9 | 623 | 5,0 |
| SL | 90 | 120 | 138 | 29,6 | 7,8 | 693 | 5,8 |
| SL | 80 | 115 | 132 | 25,8 | 6,8 | 774 | 6,7 |
| SL | 70 | 109 | 126 | 22,1 | 5,8 | 872 | 8,0 |
| SL | 60 | 103 | 118 | 18,6 | 4,9 | 988 | 9,6 |
| SL | 50 | 95 | 109 | 15,3 | 4,0 | 1120 | 11,9 |
| | | | | | | | |
| 2000 | 100 | 127 | 146 | 33,6 | 8,9 | 632 | 4,9 |
| 2000 | 90 | 122 | 141 | 29,6 | 7,8 | 702 | 5,7 |
| 2000 | 80 | 117 | 135 | 25,8 | 6,8 | 784 | 6,6 |
| 2000 | 70 | 111 | 128 | 22,1 | 5,8 | 883 | 7,9 |
| 2000 | 60 | 104 | 120 | 18,6 | 4,9 | 999 | 9,5 |
| 2000 | 50 | 96 | 110 | 15,3 | 4,0 | 1130 | 11,7 |
| | | | | | | | |
| 4000 | 100 | 130 | 149 | 33,6 | 8,9 | 641 | 4,8 |
| 4000 | 90 | 125 | 144 | 29,6 | 7,8 | 712 | 5,6 |
| 4000 | 80 | 119 | 137 | 25,8 | 6,8 | 794 | 6,5 |
| 4000 | 70 | 113 | 130 | 22,1 | 5,8 | 894 | 7,7 |
| 4000 | 60 | 106 | 122 | 18,6 | 4,9 | 1010 | 9,3 |
| 4000 | 50 | 97 | 112 | 15,3 | 4,0 | 1139 | 11,5 |
| | | | | | | | |
| 6000 | 100 | 132 | 152 | 33,6 | 8,9 | 651 | 4,7 |
| 6000 | 90 | 127 | 146 | 29,6 | 7,8 | 722 | 5,4 |
| 6000 | 80 | 122 | 140 | 25,8 | 6,8 | 805 | 6,4 |
| 6000 | 70 | 115 | 133 | 22,1 | 5,8 | 905 | 7,6 |
| 6000 | 60 | 108 | 124 | 18,6 | 4,9 | 1020 | 9,2 |
| 6000 | 50 | 99 | 113 | 15,3 | 4,0 | 1147 | 11,4 |
| | | | | | | | |
| 8000 | 100 | 135 | 155 | 33,6 | 8,9 | 660 | 4,6 |
| 8000 | 90 | 130 | 149 | 29,6 | 7,8 | 732 | 5,3 |
| 8000 | 80 | 124 | 143 | 25,8 | 6,8 | 816 | 6,3 |
| 8000 | 70 | 118 | 135 | 22,1 | 5,8 | 916 | 7,5 |
| 8000 | 60 | 110 | 126 | 18,6 | 4,9 | 1031 | 9,0 |
| 8000 | 50 | 100 | 115 | 15,3 | 4,0 | 1154 | 11,2 |
| | | | | | | | |
| 10000 | 90 | 132 | 152 | 29,6 | 7,8 | 743 | 5,2 |
| 10000 | 80 | 126 | 145 | 25,8 | 6,8 | 826 | 6,1 |

| Press. Alt. | Load | Speed | | Fuel Flow | | Distance | Endu-rance Time |
|---|---|---|---|---|---|---|---|
| [ft] | [%] | [KTAS] | [mph] | [l/h] | [US Gal/h] | [NM] | [Hrs] |
| 10000 | 70 | 120 | 138 | 22,1 | 5,8 | 926 | 7,3 |
| 10000 | 60 | 112 | 128 | 18,6 | 4,9 | 1041 | 8,9 |
| 10000 | 50 | 101 | 116 | 15,3 | 4,0 | 1159 | 11,0 |
| | | | | | | | |
| 12000 | 90 | 135 | 155 | 29,6 | 7,8 | 753 | 5,1 |
| 12000 | 80 | 129 | 148 | 25,8 | 6,8 | 837 | 6,0 |
| 12000 | 70 | 122 | 140 | 22,1 | 5,8 | 937 | 7,2 |
| 12000 | 60 | 113 | 131 | 18,6 | 4,9 | 1051 | 8,7 |
| 12000 | 50 | 102 | 117 | 15,3 | 4,0 | 1163 | 10,8 |
| | | | | | | | |
| 14000 | 90 | 138 | 158 | 29,6 | 7,8 | 764 | 4,9 |
| 14000 | 80 | 131 | 151 | 25,8 | 6,8 | 848 | 5,8 |
| 14000 | 70 | 124 | 143 | 22,1 | 5,8 | 948 | 7,0 |
| 14000 | 60 | 115 | 133 | 18,6 | 4,9 | 1060 | 8,5 |
| 14000 | 50 | 103 | 118 | 15,3 | 4,0 | 1164 | 10,6 |
| | | | | | | | |
| 16000 | 80 | 134 | 154 | 25,8 | 6,8 | 859 | 5,7 |
| 16000 | 70 | 126 | 145 | 22,1 | 5,8 | 958 | 6,8 |
| 16000 | 60 | 117 | 135 | 18,6 | 4,9 | 1068 | 8,3 |
| 16000 | 50 | 103 | 119 | 15,3 | 4,0 | 1161 | 10,4 |
| | | | | | | | |
| 18000 | 80 | 136 | 157 | 25,8 | 6,8 | 872 | 5,5 |
| 18000 | 70 | 129 | 148 | 22,1 | 5,8 | 971 | 6,7 |
| 18000 | 60 | 119 | 137 | 18,6 | 4,9 | 1078 | 8,1 |
| 18000 | 50 | 103 | 119 | 15.3 | 4.0 | 1154 | 10.2 |
| | | | | | | | |

Figure 5-4e Cruise Performance, Range and Endurance
with Integraltanks, Cessna 172P at 1089 kg (2400 lbs)

# SECTION 6
# HANDLING ON GROUND & MAINTENANCE

| | | |
|---|---|---|
| ▲ | **WARNING:** | Do not start the engine in any case when filling levels are below the corresponding minimum marking. |

| | | |
|---|---|---|
| ■ | CAUTION: | Normally, a refill of coolant or gearbox oil between service intervals is not necessary. In case of low coolant or gearbox oil levels, inform the maintenance company immediately. |

**ENGINE OIL**

The TAE 125-02-114 engine variants are filled with 4.5 - 6 l engine oil (refer to section 1 of this supplement for specification).
A dip stick is used to check the oil level. It is accessible by a flap on the upper right-hand side of the engine cowling.
Notice that on warm engines 5 minutes after engine shut-off there are 80% of the entire engine oil in the oil pan and therefore visible o the oil dip stick. On warm engines oil should be added if the oil dip stick shows oil levels below 50%. After 30 minutes the real oil level is visible on the dip stick.
The drain screw is located on the lower left-hand outside of the oil pan, the oil filter is on the upper left-hand side of the housing.
The oil system has to be checked for sealing after the first 5 operating hours (visual inspection).
Checks and changes of oil and oil filter have to be performed regularly according to the engine Operation and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

## GEARBOX OIL

To ensure the necessary propeller speed, the engine is equipped with a reduction gearbox filled with gearbox oil. (refer to section 1 of this supplement for specification).
The level can be checked through a viewing glass on the lower leading edge of the gearbox. To do so open the flap on the left front side of the engine cowling.
The drain screw is located at the lowest point of the gearbox. A filter is installed upstream of the pump, as well as microfilter in the Constant Speed Unit. Check the gearbox for sealing after the first 5 hours of operation (visual inspection). Regular checks as well as oil and filter changes have to be performed in according with the engine Operation and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

| | | |
|---|---|---|
| ▲ | **WARNING:** | It is not allowed to start the engine with low gearbox oil level. |

| | | |
|---|---|---|
| ■ | CAUTION: | Between scheduled maintenance topping up gearbox oil should not be necessary. If low gearbox oil level is detected, inform your service centre immediately. |

**FUEL**

The TAE 125-02-114 engine can be operated with kerosene(JET A-1, Jet A, Fuel No.3, JP-8, TS-1) or Diesel fuel. Due to the higher specific density of JET A-1 or Diesel in comparison to aviation gasoline (AVGAS) the permissible capacity for standard tanks was reduced as mentioned in Section 1.
Appropriate placards are attached near the fuel filler connections.
For temperature limitations refer to Section 2 "Limitations" and Section 4 "Normal Operation".

It is recommended to refuel before each flight and to enter the type of fuel into the log-book.

**COOLANT**

To cool the engine a liquid cooling system was installed with a water/approved radiator protection mixture at a ratio of 1:1.
A heat exchanger for cabin heating is part of the cooling system.
Check the cooling system for sealing after the first 5 hours of operation (visual inspection).
The coolant has to be changed in accordance with the engine Operations and Maintenance Manual, see OM-02-02. The Supplement of the Aircraft Maintenance Manual has to be considered as well, see AMM-20-02.

| | | |
|---|---|---|
| ▲ | **WARNING:** | It is not allowed to start the engine with low coolant level. |

| | | |
|---|---|---|
| ■ | CAUTION: | Between scheduled maintenance topping-up coolant should not be necessary. If low coolant level is detected, inform your service centre immediately. |

| | |
|---|---|
| ■ CAUTION: | The water has to satisfy the following requirements:<br>1. visual appearance: colorless, clear and no deposits allowed<br>2. pH-value: 6.5 to 8.5<br>3. maximum water hardness: 2.7 mmol/l<br>4. maximum hydrogen carbonate concentration: 100 mg/l<br>5. maximum chloride concentration: 100 mg/l<br>6. maximum sulfate concentration: 100 mg/l |
| ◆ Note: | The freezing point of the coolant is -36°C. |
| ◆ Note: | The waterworks also provide information.<br>In general, tap water may be diluted with distilled water. Pure distilled water may not be used to mix with approved radiator protection. |

# SECTION 7
# WEIGHT & BALANCE

| Item | Weight x Arm = Moment (kg) | (m) | (mkp) |
|---|---|---|---|
| Empty Weight | | | |
| plus Engine Oil (6 l to 0.9 kg/l) | | -0.31 | |
| plus Gearbox Oil (1 l to 0.9 kg/l) | | -0.69 | |
| plus unusable fuel standard tanks (11.4 l to 0.84 kg/l) | | 1.17 | |
| long-range tanks (15.0 l to 0.84 kg/l) | | 1.17 | |
| integraltanks (22.8 l to 0.84 kg/l) | | 1.17 | |
| plus Coolant (4 l to 1.0 kg/l) | | -0.26 | |
| Changes in Equipment | | | |
| Basic Empty Weight | | | |

Figure 7-1 Calculating the Basic Empty Weight

| | | Your aircraft | |
|---|---|---|---|
| | | Mass kg | Moment mkp |
| Calculation of the loaded condition | **1. Basic Empty Weight:** Use the values for your airplane with the present equipment. Unusable fuel, engine oil, gearbox oil and coolant are included. | | |
| | **2. Usable Fuel (at 0.84 kg/l),** Standard tanks (127.4 l max.) Long-range tanks (158.6 l max.) Integral tanks (196.8 l max.) | | |
| | **3. Pilot and Front Passenger** (Station 0.86 to 1.17 m) | | |
| | **4. Rear Passenger** | | |
| | **5. *Baggage Area 1 or Passenger on the children's seat** (Station 2.08 to 2.74; max.54kg) | | |
| | **6. *Baggage Area 2** (Station 2.74 to 3.61; max. 23kg) | | |
| | **7. Ramp Weight and Moment** | | |
| | **8. Fuel allowance for engine start, taxi and runup** | | |
| | **9. Take-off Weight and Moment** (Subtract Step 8 from Step 7) | | |
| | **10.Locate this point in the weight and balance envelope in the original POH.** Check if its within the envelope. *Maximum allowable combinded weight capacity for Baggage Areas 1 and 2 is 54 kg. | | |

Figure 7-2 Calculating Weight and Moment

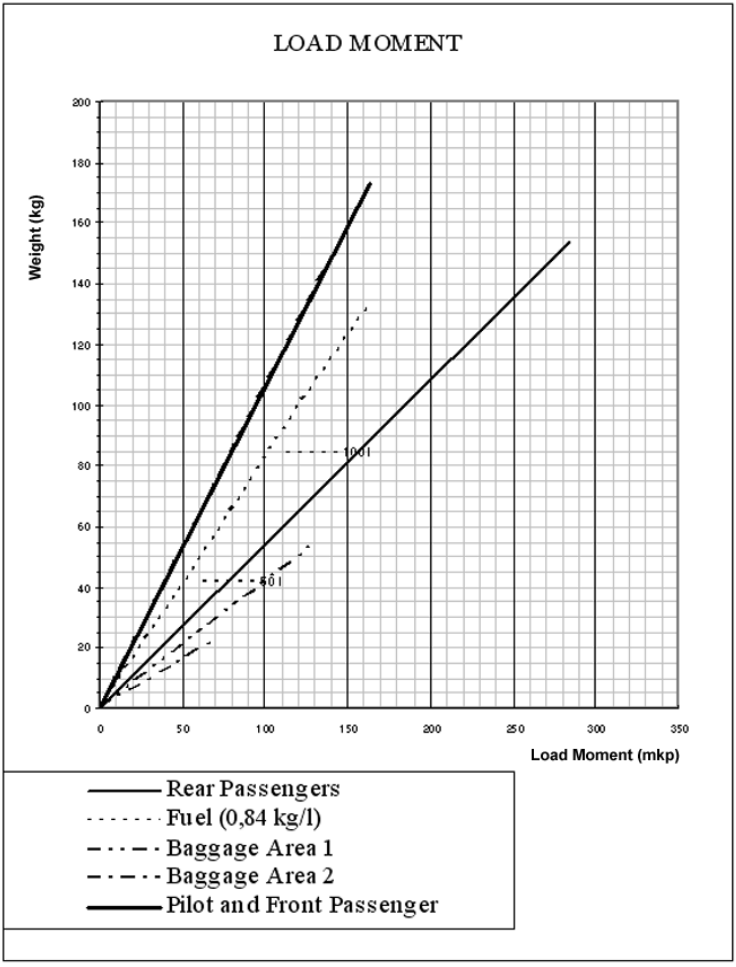



Figure 7-3 Load Moment

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# SECTION 8
# SPECIAL EQUIPMENT
# EQUIPMENT LIST

**CARBURETOR AIR TEMPERATURE GAGE**

N/A

**QUICK OIL DRAIN VALVE**

N/A

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# SECTION 9
# SUPPLEMENTS

**TABLE OF CONTENTS**

No supplements

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